

A MERGED RESEARCH REPORT OF THE SURREAL PROJECT

Surcomplex Dynamics

Linearization Thresholds, Periods, and Normal Forms

21 September 2026; extended 22 September 2026

Abstract

Ten independently written manuscripts developed local and global dynamics over complex Hahn fields $K_\Gamma = \mathbb{C}((t^\Gamma))$ realized inside $\mathbf{No}[i]$, with Γ a set-sized ordered abelian group of arbitrary rank. This report is their *union*. Every result any one of them reached survives; results all of them reached are printed once; results proved by genuinely different routes keep each route, marked; and every honest limitation is preserved.

Nine of the ten are germ and fixed-point dynamics: multipliers, return iterates, periods, Hamiltonians at the origin of a phase halo. Source 07 is the *quasi-periodic* half of the same subject — a cohomological equation on a real torus whose frequency vector has surreal components, with divisors indexed by an integer lattice — and it is not a duplicate of any of the others: none of them contains a resonance lattice, a cohomological equation, a Fourier divisor or a conjugacy to a constant vector field.

Five inequivalent conditions travelled under the single word *resonance* across the sources, and six inequivalent conditions travelled under the words *small divisor*; worse, two families of sources used the word *radius* with opposite monotonicity. [Section 2](#) therefore opens the report with a notion table that gives each variant its own name and symbol, and every threshold statement below names which variant its hypothesis uses and in which coefficient category it holds. A threshold proved under one notion is never quoted under another.

The mathematical content: a sharp five-row coefficient-category classification of universal linearization for an exact nonresonant diagonal unitary multiplier, keyed to one exponential divisor rate, with necessity witnessed at a single infinitesimal weight; for an exact *scalar* multiplier λI_d , in every dimension, one common polydisk of the sharp radius $Re^{-\tau}$ for all coefficients of the normalized conjugacy and its inverse, which answers the common-domain germ question positively there; for an exact *nonscalar* diagonal unitary multiplier, one common polydisk of radius $Re^{-r_{\text{ang}}\tau}$, where r_{ang} is the rational rank of the multiplier's angles, sharp when $r_{\text{ang}} = 1$, so that the common-domain question is answered positively for every such multiplier and only the optimal radius for $r_{\text{ang}} \geq 2$ remains open; an exact radius-collapse law $\text{rad}(h_{k,1}) = Re^{-(k+1)\sigma}$ in the *drifted*-multiplier category, which answers the same question negatively there, already over the cyclic value group $\mathbb{Z}$ with a single Hahn monomial; an exact maximal centered valuation ball read off a *finite return iterate*, with its shell cycles, its cancellation phase diagram and its common-domain lifting; an algebraicity dichotomy and an exact coefficientwise monodromy exponent for parabolic Euler maps, whose infinitesimal part no finite branched cover removes; a complete period classification, explicit moduli and beyond-all-orders invisible moduli for global near-identity maps; a gauge-normalized symplectic Hamiltonian normal form, its full Poisson centralizer, and support-certified domains reaching infinite phase magnitudes; and, on the quasi-periodic side, the theorem that a frequency vector with d surreal components has **at most** d arithmetically visible divisor valuations, identified by a canonical descending flag of integer resonance lattices, together with the exact arithmetic criterion for universal solvability of the cohomological equation on one fixed analytic strip, an arbitrary-order analytic lifting obstruction, a unique infinitesimal conjugacy to a constant vector field proved by marked support words rather than by sequential valuation convergence, and its canonical invariant density.

Status. This is a merged report assembled from ten AI-assisted, unrefereed research drafts. The full report is not proof-assistant verified. The repository's Lean development checks the finite-word support lemma and the input-relative ancestry bounds; its coverage ledger is `docs/FORMALIZATION.md`. Priority is not certified, and no named published conjecture is claimed solved. The ten sources' own verification programs supply finite exact symbolic checks only; [section 24](#) records exactly what each one does and does not establish, and [section 26](#) carries every limitation stated by every source.

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1 What this report is, and how to read it

1.1 Ten manuscripts, one body of machinery

The first seven source manuscripts were all written independently on 21 September 2026 against a pinned repository snapshot: sources 01–06 against commit `39f2be6667ade51bca2b45daa47e289d69c09764` and source 07 against commit `aa846271b4dcae2c055b216126a87210292ec19b`. Sources 08 and 09 were written independently on 22 September 2026 against commit `a3124af79f66b8b9c196d76b4cbc5ac3938907c4`. Source 10 was written independently on 22 September 2026 against commit `048b72cf7cbfc8ab246e4f73788c10460cb3f6e0`, which already contained sources 08 and 09 as merged here. All ten share a single body of machinery — the set-sized Hahn workspace inside $\mathbf{No}[i]$, strong summability by well-ordered support and local finiteness, the positive-support lemma of Neumann in finite-word form, halo evaluation by coefficientwise double Taylor sums, Taylor substitution and its near-identity inverse — and each of the ten states that machinery from scratch. The positive-support lemma is stated in all ten. Between 45 and 55 per cent of the combined text of sources 01–06 is the same theory, and sources 07–10 restate the same foundation again.

Sources 01–06 are germ and fixed-point dynamics and were merged first; source 07 is the quasi-periodic half of the subject and was integrated afterwards, as [sections 17](#) to [19](#). It is genuinely additive rather than a seventh treatment of the same material: grepping sources 01–06 for a resonance lattice, a cohomological equation, a Fourier divisor or a mean-zero conjugacy to a constant vector field returns nothing, and conversely source 07 contains no multiplier, no return iterate and no period. What the two halves do share is this machinery, which is therefore stated once for both, and the coefficient-category architecture of [remark 2.2](#), which is stated once and instantiated twice ([remark 18.2](#)).

Sources 08 and 09 belong to the germ half and were integrated last, as [section 7](#) and [section 8.6](#). Both answer [question 8.20](#), which sources 01–07 had left open for an exact multiplier. Source 08 answers it for a scalar multiplier λI_d in every dimension, and source 09 in one variable; both obtain the same sharp radius $Re^{-\tau}$. Source 08 also answers [question 25.2](#), negatively, in the one-variable drifted setting in which it was posed. The two manuscripts prove the same theorem by different routes, and they contradict neither each other nor any earlier source. The positive answer lies in the exact category $\mathbf{SD}_{\text{ex}}$ and the negative one in the drifted category $\mathbf{SD}_{\text{dr}}$, which [section 2](#) keeps apart. Source 08 is the base of their merge, because it has the weaker hypotheses (every d) and also carries the drifted theorem. Source 09 contributes the sharper tree bound [theorem 7.5](#), the forward plane-tree route of [section 7.4](#), the parameter and higher-rank remarks, and its hypothesis audit. Where their statements coincide, the result is printed once and both sources are credited.

Source 10 also belongs to the germ half and was integrated after sources 08 and 09, as [section 7.7](#) and the five subsections after it, at the end of [section 7](#). It answers the case of [question 8.20](#) that sources 08 and 09 had left open, a *nonscalar* exact diagonal unitary multiplier. For every such multiplier with $\tau < \infty$ it gives one common polydisk of radius $Re^{-r_{\text{ang}}\tau}$, where r_{ang} is the rational rank of the angles ([theorem 7.23](#)); this is sharp when $r_{\text{ang}} = 1$, which includes every scalar multiplier, and for $r_{\text{ang}} \geq 2$ the optimal radius is left open ([theorem 7.34](#)). Source 10 read this report after sources 08 and 09 had been merged, cites their scalar theorem rather than claiming it, and contradicts no earlier source. Its proof is a genuinely different route; in the scalar case it is kept as a third route to the rate of [corollary 7.6\(ii\)](#) ([remark 7.29](#)).

They also disagree, in the specific and dangerous way that independent runs at one *foundation* disagree: not by contradicting each other's theorems, but by attaching the same words to different hypotheses. That is what [section 2](#) exists to prevent.

Convention 1.1 (The merge rule). A merge is a union, not a selection.

- (i) A statement proved by every source is printed once.
- (ii) A statement proved by genuinely different routes keeps every route, explicitly marked, because the routes carry different resources and different by-products.
- (iii) Anything one source reached that the others did not survives in full.
- (iv) Every honest limitation survives. [Section 26](#) is the consolidated record; dropping one of its items would silently upgrade the report, and that is the worst failure available in an assembly of this kind.
- (v) Nothing is strengthened beyond what a source proves. Where a source records a hypothesis as load-bearing, the hypothesis travels with the statement everywhere it is quoted.

1.2 Discipline inherited from companion reports

Three constructions here are inherited rather than restated, and are named by the title and repository path of the companion report that owns them. [Section 23](#) gives the full inheritance ledger. In brief:

- the no-topological-convergence discipline — every set of surcomplex numbers is closed and discrete in the fine order topology, so $\sum_n t^n = (1 - t)^{-1}$ is a Hahn identity about a summable family and never a limit of partial sums — is the foundation fixed by *Surcomplex analysis: holomorphic functions on $K = \mathbf{No}[i]$* ([docs/surcomplex/analysis/](#)). All ten sources restate it; we cite it and state it once, in [definition 3.1](#).
- the coefficient-ring taxonomy is the four-ring table of *Infinitesimal Analytic Geometry over the Surcomplex Numbers* ([docs/surcomplex/analytic-geometry/](#)), whose names we adopt verbatim. We add exactly two rows to it, and they are new *rows*, not new *rings*: see [section 3.2](#).
- the warning that a Hahn constant for a coordinate derivative need not be a constant for the scalar Berarducci–Mantova derivation is the cross-category warning of *Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers* ([docs/surcomplex/differential-equations/](#)). All ten sources derive it independently — source 07 in the form that its torus derivatives are external derivations annihilating K_Γ — and we adopt that report’s wording once, in [warning 3.2](#).

1.3 What is offered as new, and what is not

Not new, and credited as classical throughout: the positive-support lemma (Neumann, via Higman’s word well-quasi-order) [\[2, 3\]](#); algebraic closure of a maximally complete Hahn field [\[4\]](#); formal logarithms of tangent-to-identity maps, formal flow embedding and semisimple–unipotent splitting [\[6, 7\]](#); the exponential correspondence for strongly summable derivations of generalized-series algebras [\[8\]](#); the ordinary homological equation and the Brjuno–Yoccoz theory of small divisors [\[11, 13, 14, 12\]](#); tree expansions for linearization and Davie-type small-divisor counting [\[16, 18, 12\]](#); rank-one ultrametric linearization and its disk estimates [\[22, 23\]](#); the classical iterative residue and modified-equation coefficients [\[24, 10\]](#); and classical formal Birkhoff normalization with its convergence–divergence theory [\[25, 26\]](#).

Offered for scrutiny as candidate contributions, in the exact form proved below and with no priority claim: the coefficient-category classification [theorem 6.1](#) with its first-weight necessity mechanism; the exact drifted radius-collapse law [theorem 8.2](#) and the trichotomy [theorem 8.5](#) it yields; from sources 08 and 09, the fixed-complexity tree bound [theorem 7.5](#), the sharp common radius [theorem 7.13](#) for an exact scalar multiplier, and the single-exponent collapse [theorem 8.15](#) with its collision noncancellation [proposition 8.14](#); the exact finite-return threshold [theorem 9.5](#) with

theorems 9.7, 10.1 and 13.1; the algebraicity dichotomy theorem 14.15 with the monodromy formula theorem 14.11 and its torsion obstruction; the period classification theorem 15.13 with theorems 15.16, 15.18 and 15.26; the Hamiltonian package theorems 16.7, 16.12, 16.16 and 16.21; and, from source 07, the finite resonance flag with its single common reciprocal support theorem 17.5, the exact fixed-strip solvability criterion theorem 18.3, the arbitrary-order analytic lifting obstruction theorem 18.7, and the torus normal-form package theorems 19.1, 19.4 and 19.8 and proposition 19.6; and, from source 10, the rate-count lemma lemma 7.24, the rational-rank product bound theorem 7.27, and their consequence, the nonscalar common domain theorem 7.23 with its rank-one sharpness theorem 7.34 and the two-mode example proposition 7.35. Each source’s own bounded novelty assessment is reproduced in section 26. Source 07 adds to the classical list the ordinary Fourier division mechanism with its subexponential criterion, the Liouville construction method, and positive-support Hahn inversion [19, 20, 2, 4], none of which it claims. Sources 08 and 09 credit the Davie estimates discussed by Marmi and the tree expansions of Fauvet, Menous and Sauzin [12, 16, 18]. Their clustering-and-packing estimate is of the same kind as that classical counting. Source 08 inspected [16] at abstract level only, and both sources disclaim priority (remark 7.8; N94, N105). The priority of their contributions is correspondingly uncertain. Source 10 credits the same tree expansions, inspecting the HTML text of their revision of 13 September 2026, including their preservation of admissible cuts [18, Prop. 5.6], and Carletti’s tree formulas for inversion and linearization over non-Archimedean formal-series structures [17], which it saw at abstract level only. It claims only the rank count and its common-domain consequence, and disclaims priority (N111).

2 The notion table: five resonances, six small divisors, two radii

This section is the load-bearing apparatus of the merge. Read it before anything else. Every threshold statement in this report carries tags from the tables below, and section 2.5 lists, theorem by theorem, which tags it actually uses.

2.1 Why unifying the notation without this table would invert results

The sources use the word *resonance* for five inequivalent conditions and the words *small divisor* for six, and two groups of sources use the word *radius* with opposite monotonicity. A merge that harmonized the symbols without separating the notions would produce statements that read correctly and are false: it would apply an exact maximal-ball formula to a multiplier its source explicitly excludes, attach an arithmetic threshold to a theorem proved with no arithmetic hypothesis at all, and reverse the direction of every domain inequality in half the report.

2.2 Four notions of resonance

Tag	Condition	Where it is the hypothesis, and what it is not
RES_q <i>residue-resonant of order q</i>	$\lambda \in K_\Gamma^\times$ with $v(\lambda) = 0$, λ is never a root of unity, and its residue $\alpha = \bar{\lambda} \in \mathbb{C}^\times$ is a root of unity of exact order $q \geq 1$. Then $c = \lambda^q - 1 \neq 0$ and $\delta = v(c) > 0$.	This is the hypothesis of the exact finite-return threshold theorem 9.5 , the shell theorem theorem 9.7 , the phase diagram theorem 13.1 and the common-domain lifting theorem 10.1 . It is not λ itself being a root of unity — that case is excluded — and it is not a divisor size condition of any kind.
MR / NR <i>multiplier-resonant</i>	For $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ with $ \lambda_j = 1$: $\lambda^\alpha = \lambda_j$ for some $ \alpha \geq 2$ and some j . NR is its negation, nonresonance. In one variable NR reads $\lambda^n \neq \lambda$ for $n \geq 2$.	This is the hypothesis of theorems 6.1, 6.3, 7.13, 7.23, 8.5 and 8.15 and of everything built on the homological operator $\mathcal{L}$ of section 5 . The multiplier is <i>exactly</i> Λ in theorems 6.1, 7.13 and 7.23 and <i>drifted</i> in theorems 8.2 and 8.15 ; see warning 2.6 .
FR <i>frequency-resonant</i>	For a frequency vector $\omega \in \mathbb{C}^d$ of ordinary complex constants: $\omega \cdot k = 0$ for some $0 \neq k \in \mathbb{Z}^d$; equivalently $\omega \cdot (\alpha - \beta) = 0$ for a monomial $q^\alpha p^\beta$. Its negation is the hypothesis of the Hamiltonian package.	FR is the exact Hamiltonian analogue of MR, and its negation is what makes the kernel of $L = \{\cdot, H_0\}$ exactly the diagonal monomials $\alpha = \beta$. It is a condition on <i>ordinary complex</i> numbers, never on Hahn or surreal data.
PAR <i>parabolic</i>	The leading multiplier is exactly 1, perturbed infinitesimally: $F = z + h$ with $v(h) > 0$, or $G_\varepsilon = w + \varepsilon V(w)$ with $\varepsilon = t^\delta$, $\delta > 0$.	This is the hypothesis of the whole of sections 14 and 15 . No divisor hypothesis of any kind is imposed anywhere under PAR. Source 04 says plainly that the word in its own title is not a small-divisor condition, and source 05 says it has no resonance and no small-divisor condition at all because its maps are unipotent. A reader who reads PAR as a small-divisor hypothesis will misattribute every theorem in those two sections.

Tag	Condition	Where it is the hypothesis, and what it is not
QR <i>lattice- resonant, for a Hahn frequency</i>	For a frequency vector $\Upsilon \in F_{\Gamma}^d = \mathbb{R}((t^{\Gamma}))^d$ whose components are Hahn (hence surreal) scalars : the exact resonance lattice $N_{\Upsilon} = \{k \in \mathbb{Z}^d : k \cdot \Upsilon = 0\}$ is nonzero. Its negation $\neg$ QR is $N_{\Upsilon} = \{0\}$.	The flag theorem 17.5 and the linear theorem theorem 18.3 <i>permit</i> QR; the torus normal form theorem 19.1 and all of section 19 require $\neg$ QR. This is exact vanishing of a Hahn scalar , never smallness of anything. It is not FR: that one is a condition on ordinary complex $\omega \in \mathbb{C}^d$ with divisors graded by Taylor degree, whereas QR concerns surreal components with divisors $k \cdot \Upsilon$ graded by the Fourier index $ k $ and of <i>nonzero</i> valuation (warning 2.9).

Warning 2.1 (The four germ notions do not intersect in the way the word suggests). RES_q requires λ *not* to be a root of unity while its residue is one; MR is about exact equality of ordinary unit multipliers; FR is a linear condition on an integer vector; PAR is about the multiplier being 1 up to an infinitesimal. In particular:

- the exact maximal-ball formula of [theorem 9.5](#) needs RES_q and must never be applied under PAR or under MR;
- the resonant projection and the retained resonant Hamiltonian algebra of [section 16](#) need FR;
- the parabolic results of [sections 14](#) and [15](#) need none of the three others, and acquire no arithmetic hypothesis by being printed next to them;
- the finite-order splitting of [theorem 20.1](#) needs $\Lambda^q = I$ — the multiplier itself of finite order — which is *neither* RES_q (where λ is not a root of unity) nor mere MR (where resonances exist but need not have finite order).

The fifth notion QR intersects none of these four: it is a condition on an integer lattice attached to a vector of *Hahn* scalars, it has no multiplier and no fixed point, and no theorem stated under RES_q , MR, FR or PAR is quoted under it or vice versa.

2.3 Six notions of small divisor

The first five concern *ordinary complex* divisors. Every nonzero ordinary complex number, however tiny, has Hahn valuation zero ([remark 2.3](#)), so none of the five is visible to the Hahn valuation, and a proof that keeps only the valuation cannot distinguish them. The sixth, SD_{ss} , is different in one respect and identical in another: its divisors $k \cdot \Upsilon$ are Hahn scalars of *nonzero* valuation, but the arithmetic condition it imposes is again a condition on ordinary real numbers only — the leading real forms — so that here too the valuation cannot see the arithmetic ([warning 2.11](#)).

Tag	Condition	Scope
SD ₀ <i>support only</i>	No arithmetic condition whatsoever. Control is by well-ordered positive support and local finiteness alone.	Sources 01, and the polynomial and formal rows of theorem 6.1 , and the polynomial-layer case (a) of theorem 16.7 . Source 01 insists on the point: the Schröder denominators $\nu - \nu^n$ have nonzero ordinary constant terms $\alpha - \alpha^n$, which may be arbitrarily small in the ordinary modulus and still all have Hahn valuation zero, so no Diophantine, Brjuno or Siegel condition on surreal data appears in section 9 at all.
SD _{dr} <i>drifted exponential rate</i>	$\sigma(\lambda) = \limsup_{n \rightarrow \infty} \frac{\log^+ \lambda^n - \lambda ^{-1}}{n} \in [0, \infty]$, together with multiplier drift $F'(0) = \lambda + u$, $v(u) > 0$.	Sources 02 and 08. The drift is not decoration: it is what creates the higher reciprocal jets $c_{n,k}$, $k \geq 1$, whose poles of order $k + 1$ drive the collapse law. Warning 2.6 records that removing drift removes the mechanism. Source 08's theorem 8.15 needs the drift but not a second independent exponent.
SD _{ex} <i>exact-multiplier exponential rate</i>	$\tau(\Lambda) = \limsup_{m \rightarrow \infty} \frac{1}{m} \log M_m(\Lambda)$, $M_m = \max(1, \max_{ \alpha =m, j} \lambda^\alpha - \lambda_j ^{-1})$, with the multiplier exactly Λ : every f_γ has $f_\gamma(0) = 0$ and $df_\gamma(0) = 0$.	Sources 03, 08, 09 and 10. No infinitesimal linear part is permitted in theorem 6.1 or in section 7 . Source 10 adds a second invariant of the same multiplier, the rational angular rank r_{ang} of (7.15), which bounds how many distinct exponential rates the divisors can show at fixed tree complexity; it is an arithmetic invariant of the angles, not a divisor condition. Infinitesimal linear terms <i>are</i> permitted in the identity-reducing flow theorem theorem 4.3 .

Tag	Condition	Scope
SD _{fr} <i>frequency</i> <i>exponential</i> <i>rate</i>	$\Theta(\omega) = \limsup_{D \rightarrow \infty} M_D(\omega)^{1/D}$, $M_D = \max(1, \max_{ \alpha + \beta =D, \alpha \neq \beta} \omega \cdot (\alpha - \beta) ^{-1})$, so $\Theta \in [1, \infty]$.	Source 06, where this invariant is called $\Lambda(\omega)$; we write $\Theta(\omega)$ to free the symbol Λ for the diagonal multiplier matrix of sources 02 and 03. Note the different normalization : σ and τ are logarithmic rates in $[0, \infty]$ while Θ is their exponential. $\Theta(\omega) < \infty$ is the exact analogue of $\tau < \infty$; $\Theta(\omega) = 1$ would be the analogue of $\tau = 0$ and is nowhere required , because source 06's entire coefficients already have the common domain $\mathbb{C}^{2d}$ and so lose nothing to a finite radius factor.
SD _∅ <i>none</i>	No divisor condition, and no divisor family to impose one on.	Sources 04 and 05. Under PAR the multiplier is $1 + \mu\varepsilon$ or 1, so $\lambda^n - \lambda$ is automatically nonzero for every $n \geq 2$, formal uniqueness is free, and the entire difficulty is well-ordered support.
SD _{ss} <i>stratified</i> <i>subexponential</i>	For each flag stratum Σ_j of theorem 17.5 and every real $\varepsilon > 0$ there is $C_{j,\varepsilon} > 0$ with $ \mathbf{a}_j(k) ^{-1} \leq C_{j,\varepsilon} e^{\varepsilon k }$ for $k \in \Sigma_j$, where $\mathbf{a}_j(k) = k \cdot \varpi_{\gamma_j}$ is the leading <i>real</i> form of the divisor $k \cdot \Upsilon$ on that stratum; equivalently $\limsup_{k \in \Sigma_j} \log^+(1/ \mathbf{a}_j(k))/ k = 0$ for every j .	Source 07, where it is called (SS). It is the vanishing-rate condition of remark 2.2 on a <i>fourth</i> divisor family (remark 18.2), imposed stratum by stratum on the finitely many leading real forms. It therefore corresponds to $\tau = 0$ and $\sigma = 0$, not to $\tau < \infty$; and it says nothing whatever about the surreal size $v(k \cdot \Upsilon) = \gamma_j$ of the divisor itself (warning 2.11).

Remark 2.2 (One invariant, three names, one category conclusion). Fix a family $(d_i)_{i \in I}$ of nonzero ordinary complex divisors graded by a degree $\deg i \in \mathbb{N}$, and set

$$\kappa = \limsup_{m \rightarrow \infty} \frac{1}{m} \log \max \left(1, \max_{\deg i = m} |d_i|^{-1} \right).$$

Then σ , τ and $\log \Theta$ are this one quantity evaluated on three different divisor families: $\lambda^n - \lambda$ graded by n ; $\lambda^\alpha - \lambda_j$ graded by $|\alpha|$; and $\omega \cdot (\alpha - \beta)$ graded by $|\alpha| + |\beta|$. The *category conclusion* is the same in all three instances, and is the single most quotable fact in this report:

Coefficient category	Universal criterion
polynomial coefficients	nonresonance only (SD ₀)
formal coefficients	nonresonance only (SD ₀)
radius-free convergent germs	$\kappa < \infty$
entire coefficients	$\kappa < \infty$
one fixed unchanged polydisk	$\kappa = 0$

The three sources reach this table on three divisor families, in three notations, with three different proofs of necessity. What they do *not* share is the sixth category, common-domain germs. At

the pins of sources 01–07 it was decided only on the drifted family. Sources 08 and 09 have since decided it on the exact family for a *scalar* multiplier, and source 10 for every diagonal unitary exact multiplier; the exact and drifted answers are different:

Sixth category	exact family $\lambda^\alpha - \lambda_j$ (SD_{ex})	drifted family $\lambda^n - \lambda$ (SD_{dr})
common-domain germs $\mathcal{R}_n$	$\kappa < \infty$, with universal radius $Re^{-\kappa}$, for $\Lambda = \lambda I_d$ (theorem 7.13); $\kappa < \infty$ also for nonscalar Λ , with universal radius between $Re^{-r_{\text{ang}}\kappa}$ and $Re^{-\kappa}$, equal to $Re^{-\kappa}$ when $r_{\text{ang}} = 1$ (theorem 7.34); the optimal radius for $r_{\text{ang}} \geq 2$ is open (question 8.20)	$\kappa = 0$ (theorem 8.5), already over any fixed nonzero Γ , including $\mathbb{Z}$ (corollary 8.17)

These two entries belong to different categories and must never be merged into one row ([warning 8.21](#)). No source decides the sixth category on the frequency family.

A *fourth* divisor family, $k \cdot \Upsilon$ for a Hahn frequency vector graded by the Fourier index $|k|$, is added by source 07, and the first row of the table above is instantiated on it as [theorem 18.3](#): “one fixed unchanged polydisk” becomes “one fixed unchanged strip” and “ $\kappa = 0$ ” becomes (SD_{ss}), the same vanishing rate imposed on each of the finitely many strata. The table is stated here once and instantiated exactly twice; the two instantiations, and the rows that are **not** instantiated on the quasi-periodic side, are set out in [remark 18.2](#). Reading an uninstantiated row across would assert a theorem no source proves.

Remark 2.3 (Arithmetic invisible to the Hahn valuation). For every nonzero ordinary divisor, $v(\lambda^\alpha - \lambda_j) = 0$ and $v(\omega \cdot (\alpha - \beta)) = 0$. The distinction among SD_0 , SD_{dr} , SD_{ex} and SD_{fr} therefore comes entirely from the *ordinary complex Taylor growth inside each coefficient function*, not from the valuation. This is exactly why formal and polynomial coefficients always permit normalized linearization under nonresonance while the analytic categories have sharp arithmetic thresholds, and it is why the negative results below never prohibit a conjugacy on an infinitesimal monad: the formal-coefficient conjugacy exists and can be evaluated there ([proposition 6.14](#)).

2.4 Two notions of radius, running in opposite directions

Object	Definition	Monotonicity and sources
valuation ball B_r , shell S_r	$B_r = \{z \in K_\Gamma : v(z) > r\} = t^r \mathfrak{m}$ and $S_r = \{z : v(z) = r\}$, for $r \in \Gamma$.	Increasing r makes the ball smaller. Sources 01 (B_r , $r = \max(\delta - v(b_n))/(n - 1)$) and 04 ($D = \{v(z) > \delta/q\}$, so larger δ/q is a smaller disk). S_r is called a bounding shell and is not the topological boundary (valuation balls are clopen) and not a circle for the ordered-surreal modulus.

Object	Definition	Monotonicity and sources
valuation polydisk D_ρ	$D_\rho = (t^\rho \mathcal{O}_\Gamma)^{2d} = \{z : v(z_j) \geq \rho \ \forall j\}$.	Increasing ρ makes the polydisk smaller ; $\rho < 0$ gives domains containing phase points of <i>infinite</i> magnitude. Source 06. Same direction as B_r .
ordinary disk / polydisk $\mathbb{D}_R$, $\mathbb{D}_R^d$, D	$\mathbb{D}_R^d = \{z \in \mathbb{C}^d : z_j < R\}$; D an ordinary connected open subset of $\mathbb{C}$.	Increasing R makes the domain larger . Sources 02, 03, 05, and the coefficient domains of 04 and 06. This is a domain for the ordinary <i>coefficient functions</i> , never a surcomplex ball.
Taylor radius rad, rad _{iso}	$\text{rad}(g)^{-1} = \limsup_n g_n ^{1/n}$; $\text{rad}_{\text{iso}}(g)^{-1} =$ $\limsup_m (\max_{ \alpha =m, j} g_{j,\alpha})^{1/m}$.	Larger is better . Sources 02 (rad) and 03 (written ρ there; we write rad _{iso} , freeing ρ for source 06's rescaling exponent).

Warning 2.4 (Do not identify a valuation ball with an ordinary disk). B_r and $\mathbb{D}_R$ are objects of different kinds and their inequalities run in opposite directions. Source 01 states the point sharply: all points of the normalized monad have ordinary residue zero, whereas the ordinary disk D of [theorem 10.1](#) contains many nonzero ordinary points; D is a domain of holomorphy for coefficient *functions* and not a claim of a larger centered surcomplex linearization ball. Likewise a Lindahl-type ultrametric disk radius in a fixed rank-one workspace and source 01's maximal centered valuation ball at arbitrary rank are different categories; see [section 23.5](#).

Warning 2.5 (Support orientation). Throughout this report supports are well ordered in the *increasing* order of Γ and $v = \min \text{supp}$, so that $t^\gamma = \omega^{-\gamma}$ converts increasing Hahn exponents to decreasing surreal monomials. A threshold rewritten using the Conway growth exponent $\alpha = -\gamma$ requires negating its exponent and reversing the corresponding exponent inequality. Ordinary coefficient estimates are unchanged. Silently transferring a valuation threshold without this conversion inverts it.

2.5 Tag ledger: which notion each threshold statement uses

Statement	Resonance	Divisor	Coefficient category	Domain notion
theorem 9.5 exact finite-return ball	RES _q	SD ₀	formal, $\mathbb{C}[[X]]((t^\Gamma))$	B_r (valuation)
theorem 9.7 shell cycles	RES _q	SD ₀	ordinary polynomial $P \in \mathbb{C}[X^q]$	S_r (valuation shell)
theorem 13.1 phase diagram	RES _q	SD ₀	formal	B_r
theorem 10.1 common-domain lifting	RES _q	SD ₀	fixed-polydisk $\mathcal{O}(D)((t^\Gamma))$	$\mathbb{D}$ (ordinary)

Statement	Resonance	Divisor	Coefficient category	Domain notion
proposition 11.1 no common domain	NR	SD_0 hypothesis, Brjuno used externally	fails radius-free germ	ordinary
theorem 9.10 return perturbation certificate	RES_q	SD_0	return coefficients only	B_r
theorem 6.1 five-row classification	NR, exact Λ	SD_{ex}	all five rows	$\mathbb{D}_R^d$, germ, entire, poly, formal
lemma 5.1 single-step loss	NR	SD_{ex}	isotropic Taylor radius	rad_{iso}
lemma 5.2 sharpness	NR	SD_{ex}	$\mathcal{O}(\mathbb{D}_R^d)$ and entire	rad_{iso}
theorem 6.9 word-depth radius bound	NR	SD_{ex}	germ / fixed polydisk	rad_{iso} , <i>lower bound only</i> ; superseded for $\Lambda = \lambda I_d$ by theorem 7.13 , and improved to $\min(r_{\text{ang}}, \ell_S(\gamma))\tau$ by corollary 7.32
theorem 7.5 tree product bound	NR, one rotation	arithmetic only; usable under SD_{ex}	none	none
theorem 7.13 sharp common radius	NR, exact λI_d	SD_{ex}	common-domain germs $\mathcal{R}_d$ (sixth category)	$\mathbb{D}_{Re^{-\tau}}^d$, rad_{iso}
proposition 7.19 route D	NR, exact, $d = 1$	SD_{ex}	$\mathcal{R}_1$	rad
lemma 7.24 and corollary 7.25 rate count	none	arithmetic only: angles and integer heights	none	none
theorem 7.27 colored product bound	NR, diagonal unitary	SD_{ex} rate, with r_{ang}	none	none
theorem 7.23 nonscalar common radius	NR, exact diagonal Λ	SD_{ex}	common-domain germs $\mathcal{R}_d$	$\mathbb{D}_{Re^{-r_{\text{ang}}\tau}}^d$, rad_{iso}
theorem 7.34 criterion, rank-one sharpness	NR, exact diagonal Λ	SD_{ex} , iff $\tau < \infty$	$\mathcal{R}_d$	radius in $[Re^{-r_{\text{ang}}\tau}, Re^{-\tau}]$; exact if $r_{\text{ang}} = 1$

Statement	Resonance	Divisor	Coefficient category	Domain notion
corollary 7.32 input-relative radius	NR, exact diagonal Λ	SD_{ex}	germ	rad_{iso} , lower bound
theorem 6.3 support-controlled lifting	NR	SD_{dr} allowed (drift permitted)	four rows	$\mathbb{D}_R$
theorem 8.2 exact radius collapse	NR	SD_{dr} , drift required	germ yes, common domain no	rad
theorem 8.15 single-exponent collapse	NR	SD_{dr} , drift required ; one monomial over $\mathbb{Z}$	germ yes, common domain no	rad
theorem 8.5 trichotomy	NR	SD_{dr}	common domain vs germ; over Γ_* , and over $\mathbb{Z}$ by corollary 8.17	$\mathbb{D}_R$
corollary 8.17 fixed-group criterion	NR	SD_{dr}	common domain iff $\sigma = 0$, any fixed Γ	$\mathbb{D}_R$
theorem 8.7 several-variable collapse	NR	SD_{dr} , mul- tiplicative detuning	polydisk germ; over Γ_* only	rad_{iso}
theorem 14.15 algebraicity dichotomy	PAR	$SD_{\emptyset}$	$\mathbb{C}(w)((t^\Gamma))$ base	ordinary universal cover
theorem 14.11 finite-cover obstruction	PAR	$SD_{\emptyset}$	rational-coefficient	ordinary loops
theorem 14.22 sharp summability disk	PAR	$SD_{\emptyset}$	formal Taylor family	$B_{\delta/q}$ (valuation)
theorem 15.13 period classification	PAR	$SD_{\emptyset}$	$\mathcal{O}(D)((t^\Gamma))$, no shrinking	ordinary D
theorem 15.16 moduli $\mathfrak{m}_K^r$	PAR	$SD_{\emptyset}$	$\mathcal{O}(D_r)((t^\Gamma))$	ordinary punctured plane
theorem 15.18 invisible moduli	PAR	$SD_{\emptyset}$	$\mathcal{O}(D_r)((t^\Gamma))$	ordinary; needs $J_\delta \neq 0$
theorem 16.5 sharp homological criterion	$\neg\text{FR}$	SD_{fr}	entire / germ	$\mathbb{C}^{2d}$ / germ radius
theorem 16.7(a) polynomial layers	$\neg\text{FR}$	SD_0	$\mathcal{P}_{2d}$ polynomial	finite halo $\mathcal{O}_\Gamma^{2d}$
theorem 16.7(b) entire layers	$\neg\text{FR}$	SD_{fr}	$\mathcal{E}_{2d} = \mathcal{O}(\mathbb{C}^{2d})$	finite halo
theorem 16.16 universal criterion	$\neg\text{FR}$	SD_{fr}	entire; germ at first order	finite halo

Statement	Resonance	Divisor	Coefficient category	Domain notion
theorem 16.12 Poisson centralizer	$\neg\text{FR}$	SD_{fr} (through 06(b))	entire-coefficient	finite halo
theorem 16.21 certified domains	$\neg\text{FR}$	SD_0 (polynomial input)	polynomial	D_ρ (valuation)
theorem 20.1 finite-order splitting	$\Lambda^q = I$	SD_0	all five rows	$\mathbb{D}_R^d$
theorem 4.3 exp/log correspondence	PAR	$\text{SD}_\emptyset$	$\mathcal{O}(U)$, germ, entire, poly, formal	arbitrary open U
theorem 4.12 centralizer	PAR	$\text{SD}_\emptyset$	$\mathcal{O}(U)$, $U \subset \mathbb{C}$ connected	ordinary U
theorem 4.10 fixed-point ideals	PAR	$\text{SD}_\emptyset$	any of the five	finite halo
theorem 17.5 finite resonance flag	QR allowed	none used	<i>no coefficient functions:</i> a statement about F_Γ^d and $\mathbb{Z}^d$	none
proposition 17.7 inverse coefficients	QR allowed	none used	scalar, in K_Γ	none
proposition 17.9 arithmetic-free bound	QR allowed	none: no divisor estimate at all	fixed strip $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$	$\mathbb{T}_\rho^d$ (ordinary)
theorem 18.3 fixed-strip solvability	QR allowed	SD_{ss} , iff	fixed strip	$\mathbb{T}_\rho^d$ (ordinary)
corollary 18.4 loss κ_Υ attained	QR allowed	SD_{ss}	fixed strip	$\mathbb{T}_\rho^d$
corollary 18.6 higher tails	$\neg\text{QR}$ for the last clause	SD_{ss} preserved	fixed strip	$\mathbb{T}_\rho^d$
theorem 18.7 arbitrary-order lifting obstruction	$\neg\text{QR}$	SD_{ss} <i>fails</i> ; Liouville	entire input, fixed-strip target	$\mathbb{T}_\rho^2$, all ρ
theorem 19.1 torus normal form	$\neg\text{QR}$	SD_{ss} ; also $v(\mathcal{V}) > \kappa_\Upsilon$	fixed strip	$\mathbb{T}_\rho^d$
lemma 19.3 coordinate inverses	—	none used	fixed strip	$\mathbb{T}_\rho^d$
theorem 19.4 necessity beyond any scale	$\neg\text{QR}$	SD_{ss} <i>fails</i>	fixed strip	$\mathbb{T}_\rho^d$
proposition 19.6 failure at $v(\mathcal{V}) = \kappa_\Upsilon$	$\neg\text{QR}$	SD_{ss} <i>holds</i>	fixed strip	$\mathbb{T}_\rho^d$

Statement	Resonance	Divisor	Coefficient category	Domain notion
theorem 19.8 invariant density	$\neg$ QR	SD_{ss} (through theo-rem 19.1)	fixed strip	$\mathbb{T}_\rho^d$

Three entries of this ledger deserve to be read twice.

Warning 2.6 (The drift hypothesis travels with the radius law). [Theorem 8.2](#) and its consequences [corollary 8.4](#) and [theorems 8.5](#) and [8.7](#) require SD_{dr} : the multiplier drifts, $F'(0) = \lambda + u$ with $v(u) > 0$, and two rationally independent positive exponents are used ($\Gamma_* = \mathbb{Q} + \sqrt{2}\mathbb{Q}$, $u = t$, $v = t^{\sqrt{2}}$). Source 02 records that eliminating the drift removes the direct reciprocal-jet mechanism and that it does not settle the fixed-multiplier problem for $0 < \sigma < \infty$. Source 08 later removed the second exponent but not the drift: [theorem 8.15](#) and [corollaries 8.17](#) and [8.18](#) work over $\mathbb{Z}$ with one Hahn monomial and still **require** drift. The exact laws $\text{rad}(h_{k,1}) = R e^{-(k+1)\sigma}$ and $\text{rad}(h_m^{\text{cyc}}) = R e^{-m\sigma}$ must therefore never be quoted without the drift hypothesis. In particular they must not be quoted as an answer in the exact-multiplier category SD_{ex} of [theorem 6.1](#), where for a scalar multiplier the answer is now known to be the opposite: one common radius $R e^{-\tau}$ ([theorem 7.13](#)). For a nonscalar diagonal unitary multiplier with $\tau < \infty$ it is also the opposite: one common radius of at least $R e^{-\tau_{\text{ang}}\tau}$ ([theorem 7.23](#)). See [section 8.7](#) for the resolution of this apparent tension with [question 8.20](#).

Warning 2.7 (RES_q and equal characteristic zero). The divisor cancellation behind [theorem 9.5](#) leaves a factor $n - 1$ in the denominator ([lemma 9.3](#)). In equal characteristic zero every such nonzero integer is a valuation unit; in mixed characteristic it is not. Source 01 states outright that the exact finite-return formula and the shell-periodicity conclusion must not be transported to $\mathbb{C}_p$, and cites p -adic quadratic-like examples whose linearization bounding sphere carries no periodic point at all [[23](#), Cor. C]. That contrast identifies a hypothesis; it is not a contradiction.

Warning 2.8 (PAR acquires no arithmetic by proximity). [Sections 14](#) and [15](#) sit between two sections full of divisor thresholds. They have none. Source 04's title word "resonance" means only that the leading multiplier is 1 perturbed by an infinitesimal; source 05 has no eigenvalue ratios at all. Their analogue of a linearization domain is a fixed common ordinary domain with no shrinking whatever, and their analogue of a normal form is a rational time form. A merge of either with a framework carrying a resonance or small-divisor hypothesis would identify two different frameworks.

Warning 2.9 (Two frequency vectors, two divisor families, two κ 's). This report now contains two frequency vectors and they are objects of different kinds. Conflating them would attach a Hahn-scale statement to an ordinary-complex hypothesis, or the reverse.

- $\omega \in \mathbb{C}^d$ of [section 16](#) is a vector of **ordinary complex constants**. Its divisors are $\omega \cdot (\alpha - \beta)$ for Taylor multi-indices, graded by the *Taylor degree* $|\alpha| + |\beta|$, and they all have Hahn valuation zero ([remark 2.3](#)). Its resonance notion is FR and its divisor notion SD_{fr} .
- $\Upsilon \in F_\Gamma^d = \mathbb{R}((t^\Gamma))^d$ of [sections 17](#) to [19](#) is a vector of **Hahn (hence surreal) scalars**. Its divisors are $k \cdot \Upsilon$ for $k \in \mathbb{Z}^d$, graded by the *Fourier index* $|k| = |k_1| + \cdots + |k_d|$, and they have *nonzero* Hahn valuations — that is the entire point of [theorem 17.5](#). Its resonance notion is QR and its divisor notion SD_{ss} ([sections 2.2](#) and [2.3](#)).

Further, κ in [remark 2.2](#) is a generic *logarithmic divisor rate* in $[0, \infty]$, whereas κ_Υ in [sections 17](#) to [19](#) is an *element of* Γ : the valuation of the last visible flag level. They are never compared. The symbol Ω continues to denote an infinite element of Γ ([examples 3.7](#) and [13.3](#)) and $\Omega_F = dz/W_F$ a

time form (section 15); the frequency vector is Υ and never Ω . The coordinate change $\Phi_h = \text{id} + h$ of theorem 19.1 is a change of torus coordinates, *not* the flow map Φ_X or Φ_c of theorem 4.3 and corollary 4.8.

Warning 2.10 (The visible levels are not a spectrum). The finitely many visible divisor valuations $\gamma_1 < \dots < \gamma_r$ of theorem 17.5 read like a spectrum of scales and are not one in any sense used in this collection: not an eigenvalue list, not a multiset of purely infinite phases, and not the prime spectrum of a ring. They are values of the valuation v on a family of Hahn scalars. The word *spectrum* is not used for them anywhere below.

Warning 2.11 (Arithmetic size and valuation size are incomparable). In (17.9) the quantity $|\mathfrak{a}_j(k)|$ is an *ordinary real absolute value* while $v(k \cdot \Upsilon) = \gamma_j$ lies in Γ . Neither bounds the other, and no inequality in one kind of size is used as an estimate for the other anywhere below without the explicit leading-coefficient formula (17.9). A divisor can be infinitesimal (large γ_j) with completely harmless arithmetic, and a divisor of scale one ($\gamma_j = 0$) can have disastrous arithmetic. This is the torus form of remark 2.3: the arithmetic lives in the ordinary complex coefficient, not in the valuation — with the one difference that here the *divisors themselves* do have nonzero valuations, whereas the divisors of remark 2.3 all have valuation zero.

3 Workspaces, supports, coefficient categories, and evaluation

All ten sources set up this material independently. It is stated once here.

3.1 The set-sized workspace inside $\mathbf{No}[i]$

Let Γ be a nonzero ordered abelian group which is a *set*; divisibility is assumed only where it is used and is flagged there. A Hahn series over an ordinary commutative $\mathbb{C}$ -algebra B is a formal sum $A = \sum_{\gamma \in S} t^\gamma A_\gamma$ with $A_\gamma \in B$ and $S \subset \Gamma$ well ordered in the increasing order (warning 2.5), with the standard convolution product. For $A \neq 0$ put $v(A) = \min \text{supp } A$ and $v(0) = +\infty$; for a vector or matrix take the minimum over entries. Write

$$\mathcal{H}_\Gamma(B) = B((t^\Gamma)), \quad K_\Gamma = \mathcal{H}_\Gamma(\mathbb{C}) = \mathbb{C}((t^\Gamma)), \quad \mathcal{O}_\Gamma = \{v \geq 0\}, \quad \mathfrak{m}_\Gamma = \{v > 0\},$$

and append a subscript $+$ or > 0 for strictly positive support. Positive support is a condition on *exponents*, never on the signs of the ordinary complex coefficients. The residue field of K_Γ is canonically $\mathbb{C}$, and a bar denotes reduction modulo $\mathfrak{m}_\Gamma$; in particular every nonzero ordinary complex number, including every nonzero integer, has valuation zero. When Γ is divisible, K_Γ is algebraically closed [4, Cor. 4], and algebraic closure is used only in lemma 9.4 and theorem 12.1; the Hamiltonian package of section 16 needs neither closure nor divisibility.

For a set-sized ordered subgroup $\Gamma \subset \mathbf{No}$ the Conway normal form and the substitution

$$t^\gamma \longmapsto \omega^{-\gamma} \tag{3.1}$$

embed K_Γ as a subfield of $\mathbf{SC} = \mathbf{No}[i]$, using Gonshor's normal-form arithmetic [5]; the minus sign is what converts increasing Hahn supports to decreasing surreal monomial supports. Every *set* of surcomplex constants lies in one such workspace: collect the exponents of the real and imaginary Conway normal forms of those constants, change sign, and take the additive (if needed $\mathbb{Q}$ -linear) span. The proper-class field $\mathbf{No}[i]$ is never formed as a single object; every conclusion about it is read pointwise through a workspace enlarged to contain the exponents of the point and of the input,

and all constructions commute with such enlargements (proposition 4.14 and theorem 12.1). No sum in this report is indexed by a proper class.

Definition 3.1 (Strong summability). A *set-indexed* family $(A_i)_{i \in I}$ of Hahn series is *strongly summable* if $\bigcup_i \text{supp } A_i$ is well ordered and each exponent lies in the support of only finitely many members; its sum is then formed coefficientwise by finite sums. A map is *strongly linear* if it is linear and preserves such families and their sums.

There is no limiting operation in definition 3.1. The identity $\sum_{j \geq 0} c^j = (1 - c)^{-1}$ for $c \in \mathfrak{m}_\Gamma$ is a statement about a summable family and multiplication, not about convergence of partial sums in the fine order topology of $\mathbf{No}$; this is the discipline established by the companion report *Surcomplex analysis: holomorphic functions on $K = \mathbf{No}[i]$* (docs/surcomplex/analysis/), which we inherit rather than re-derive. No Banach contraction, Cauchy limit, compactness principle or surcomplex path occurs in any proof below. Ordinary Taylor series of coefficient functions, by contrast, use ordinary complex convergence, and the two summations are never interchanged without a support argument. Cancellation may not rescue an inadmissible family: infinitely many nonzero terms at one exponent already violates definition 3.1 even if some other summation method would assign a value to their complex coefficients.

Warning 3.2 (The derivative in this report; the cross-category pair). The symbols ∂_j , ∂_z , ∂_w differentiate the *ordinary* coordinate and act coefficientwise; every t^γ is a constant for them. They are not a derivation of the surreal coefficient field, and in particular not the Berarducci–Mantova derivation [27]. No compatibility with a global composition operator on $\mathbf{No}$ is asserted. We adopt here, once, the wording of the cross-category warning of the companion report *Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers* (docs/surcomplex/differential-equations/): a Hahn constant for a coordinate derivative need not be a constant for the scalar Berarducci–Mantova derivation, so $\varepsilon = \omega^{-1}$ is permitted as a Hahn coefficient of a coordinate problem and prohibited as an infinitesimal phase residue of the scalar derivation. All ten sources derive this distinction independently; it is not restated again below.

3.2 The coefficient categories, in the collection’s ring vocabulary

The companion report *Infinitesimal Analytic Geometry over the Surcomplex Numbers* (docs/surcomplex/analytic-geometry/) fixes a four-ring table and a ring discipline: every numbered statement names the ring over which it is asserted, and no theorem proved of one ring is transferred to another. We adopt its names:

$$\mathcal{O}(D)((t^\Gamma)) \longrightarrow \mathcal{R}_n \subsetneq \mathcal{A}_n \subsetneq \mathcal{F}_n, \quad (3.2)$$

where

- $\mathcal{O}(D)((t^\Gamma))$ is the *fixed-polydisk* ring: one specified, unchanged ordinary domain for every coefficient;
- $\mathcal{R}_n = \varinjlim_{U \ni 0} \mathcal{H}_\Gamma(\mathcal{O}(U))$ is the *common-domain germ* ring: one common domain, then allowed to shrink;
- $\mathcal{A}_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$ is the *radius-free* ring: individually convergent germs, with no common radius required;
- $\mathcal{F}_n = \mathbb{C}[[z_1, \dots, z_n]]((t^\Gamma))$ is the *formal-coefficient* ring.

We add exactly two rows to that table, and they are new *rows*, not new rings: the *entire-coefficient* row $\mathcal{O}(\mathbb{C}^n)((t^\Gamma))$ (source 06’s $\mathcal{E}_n$, also the third row of theorem 6.1) and the *polynomial-coefficient*

row $\mathbb{C}[z_1, \dots, z_n]((t^\Gamma))$ (source 06's $\mathcal{P}_n$, also the fourth row of [theorem 6.1](#)). Both sit between the fixed-polydisk ring and $\mathcal{F}_n$ in the obvious way, and neither carries a uniform bound: in the entire row *no* uniform norm bound is imposed as γ varies, and in the polynomial row *no* uniform degree bound.

We do not reprint the witness that separates the rings; it is the element $F(z) = \sum_{r \geq 1} t^{r\eta}/(1 - rz)$ of the analytic-geometry report, whose coefficient at $r\eta$ has a pole at $1/r$. Sources 02, 03 and 04 each re-erect that separation from scratch, with $\eta = 1$ in source 02 ($\sum_{m \geq 1} t^m/(1 - mz)$) and $\eta = \delta$ in source 04 ($\sum_{n \geq 1} t^{n\delta}/(1 - nw)$). Those are the same series.

Convention 3.3 (Ring discipline, inherited). Every numbered statement below names its coefficient category in the vocabulary of [\(3.2\)](#) and the two added rows, and the tag ledger [section 2.5](#) records it. No statement proved of one category is transferred to another. In particular the fixed-polydisk row of [theorem 6.1](#) concerns preservation of a *specified unchanged* polydisk, the radius-free row concerns separate germs, and neither row, nor their combination, classifies $\mathcal{R}_n$ in the intermediate regime. For a scalar multiplier λI_d that is done by [theorem 7.13](#); for a nonscalar diagonal one by [theorem 7.34](#), except for the value of the optimal radius when $r_{\text{ang}} \geq 2$, which is [question 8.20](#).

3.3 The support certificate: finite words, not sequential limits

For well ordered $S \subset \Gamma_{>0}$ let $S^* = M(S)$ be its additive monoid, including the empty sum 0, and for $\gamma \in \Gamma$ let

$$\mathcal{W}_S(\gamma) = \{(s_1, \dots, s_k) : k \geq 0, s_i \in S, s_1 + \dots + s_k = \gamma\}$$

be its set of *ordered* words of total weight γ , permutations counted separately.

Lemma 3.4 (Positive-support lemma of Neumann, in finite-word form). *Let $S \subset \Gamma_{>0}$ be well ordered. Then S^* is well ordered, and $\mathcal{W}_S(\gamma)$ is finite for every $\gamma \in \Gamma$; in particular*

$$\ell_S(\gamma) = \max\{k : (s_1, \dots, s_k) \in \mathcal{W}_S(\gamma)\} \quad (3.3)$$

is finite for $\gamma \in S^$, with $\ell_S(0) = 0$. Moreover, if $A \subset \Gamma$ is well ordered then $A + S^*$ is well ordered, and for fixed γ there are only finitely many pairs consisting of $a \in A$ and a word in S with $a + s_1 + \dots + s_k = \gamma$.*

Proof. Order finite words over S by subsequence embedding with coordinatewise domination. Higman's lemma says this is a well-quasi-order: every infinite sequence of words contains indices $i < j$ with the earlier word embedded in the later [\[3\]](#); the special case needed is reproved from scratch in [section A](#). Because all letters are positive, an embedding weakly increases the sum of the letters, and increases it strictly unless the two words are identical: an unused positive letter contributes positively, and a strictly increased matched letter increases the sum. An infinite strictly decreasing sequence in S^* , after choosing one word per element, would contradict the weak increase; infinitely many distinct words of one fixed weight would contradict the strict increase. This proves the first two assertions.

For the last, assume $A \neq \emptyset$, subtract $a_0 = \min A$, and apply the first part to the well-ordered positive set $T = S \cup ((A - a_0) \setminus \{0\})$ at weight $\gamma - a_0$: a pair in the assertion is encoded by a word with one distinguished initial letter from $A - a_0$, or none if $a = a_0$. Well-ordering follows from $A + S^* \subset a_0 + T^*$. $\square$

This is classical Hahn–Neumann mathematics [\[2, 4\]](#) and is not claimed as a contribution. All ten sources state it; sources 02, 05 and 09 each reproduce a minimal-bad-sequence proof of the underlying

word order, which is the same argument as [section A](#) and is therefore not printed again. Source 10 derives it from Higman's lemma, cited as a classical theorem. Source 08 instead reduces the lemma to Poonen's positive-support theorem over a free associative algebra ([remark 7.15](#)). Source 07 uses the *full* finiteness assertion, across all word lengths, together with a *labelled* refinement of it ([section 19.2](#)); both are already contained in [lemma 3.4](#).

For $\gamma \in S^*$ let $D_S(\gamma)$ be the finite set of letters occurring in the words of $\mathcal{W}_S(\gamma)$.

For a well-ordered input support $B \subset \Gamma$, retain the starting exponent as well as the positive word. Define

$$\begin{aligned}\mathcal{W}_{B,S}(\gamma) &= \{(b, w) : b \in B, w \in \mathcal{W}_S(\gamma - b)\}, \\ \ell_{B,S}(\gamma) &= \max(\{0\} \cup \{|w| : (b, w) \in \mathcal{W}_{B,S}(\gamma)\}).\end{aligned}$$

Here $|w|$ is the number of letters of w . Let $E_{B,S}(\gamma)$ be the set of starting exponents b in these pairs, and $D_{B,S}(\gamma)$ the set of positive letters occurring in their words. All three sets are finite by [lemma 3.4](#); the depth is zero if there is no pair. When $B = \{0\}$ and $\gamma \in S^*$, this depth is $\ell_S(\gamma)$.

Corollary 3.5 (Finite ancestry relative to the input support). *Let $S \subset \Gamma_{>0}$ be well ordered, and suppose a $\mathbb{C}$ -linear operator T on Hahn coefficient families is given coefficientwise by*

$$(TA)_\gamma = \sum_{b \in \text{supp } A} \sum_{\substack{w \in \mathcal{W}_S(\gamma - b) \\ |w| \geq 1}} T_{b,w}(A_b).$$

Each $T_{b,w}$ is a fixed linear coefficient operator, expressible by finitely many operations in the chosen coefficient category. For any Hahn input A with $B = \text{supp } A$ and any ordinary scalars c_k , the family $(c_k T^k A)_{k \geq 0}$ is strongly summable and its sum has support in $B + S^$. At an output exponent γ , contributions use at most $\ell_{B,S}(\gamma)$ applications of T , only input coefficients A_b with $b \in E_{B,S}(\gamma)$, and only positive support letters in $D_{B,S}(\gamma)$. Only finitely many coefficient operations contribute.*

These conclusions hold for finite-dimensional vector and matrix coefficient families and for operators with Hahn coefficients whenever the displayed positive-word hypothesis holds. If $(A_i)_{i \in I}$ is strongly summable, the family $(c_k T^k A_i)_{(i,k) \in I \times \mathbb{N}}$ is strongly summable too, using $B = \bigcup_i \text{supp } A_i$. Formal power-series identities in one operator may therefore be evaluated coefficientwise; this applies to the geometric, logarithmic and exponential identities used below.

Proof. The displayed definition of T is meaningful: at a fixed output exponent, [lemma 3.4](#) gives only finitely many input-exponent/word pairs. Its support is contained in $B + (S^* \setminus \{0\})$, a subset of the well-ordered set $B + S^*$, so iteration is legitimate.

Expand a contribution to $(T^k A)_\gamma$ by its successive nonempty words $w_1, \dots, w_k$, starting at $b \in B$. Their concatenation w satisfies $b + \sum w = \gamma$, hence $(b, w) \in \mathcal{W}_{B,S}(\gamma)$. Conversely, once (b, w) and its decomposition into k consecutive nonempty blocks are specified, all intermediate exponents are determined by the partial sums. A word of length $n \geq 1$ has only 2^{n-1} such block decompositions across all possible k , with $k \leq n$; the empty word occurs only for $k = 0$. Since $\mathcal{W}_{B,S}(\gamma)$ is finite and every block uses finitely many coefficient operations, only finitely many expanded contributions occur at γ , across all iterates. This also proves the stated depth and input-data bounds. Their supports all lie in $B + S^*$, proving both requirements of strong summability.

For a strongly summable input family, its union of supports B is well ordered. There are finitely many relevant pairs (b, w) , and for each such b only finitely many i have $(A_i)_b \neq 0$. The same argument thus proves the joint assertion and permits commuting T with the input sum.

Finite-dimensional vector and matrix operations introduce only finitely many component choices and do not change the argument.

Finally, fix an input and an output exponent. All contributing powers of T have bounded degree; in a product or a composition of scalar formal power series, each such degree has only finitely many decompositions. Consequently the formal identity reduces to a finite coefficient identity. Compositions are understood in their usual formal domain (in particular the inner series has zero constant term), as for $\log(1 + T)$ and $\exp(T) - I$. $\square$

Why the input support cannot be omitted. Take $\Gamma = \mathbb{Z}$, $S = \{1\}$ and $T(A) = tA$. For every integer $n \geq 0$,

$$T^n(t^{-n}) = 1, \quad \ell_S(0) = 0, \quad \ell_{\{-n\}, S}(0) = n.$$

Thus no bound depending only on S and the output exponent can control all Hahn inputs, and input exponents need not be letters of S . The earlier version of [corollary 3.5](#) used $\ell_S(\gamma)$ and $D_S(\gamma)$ for arbitrary input; the relative sets above correct that statement. The sharper unshifted bound remains valid in [proposition 6.2](#): its initial input $\mathcal{L}^{-1}f$ is already supported in S , so its starting exponent itself supplies a positive letter. The original source manuscripts remain available in repository history.

Warning 3.6 (Why a valuation-contraction argument would be incomplete). Writing $v(T^k A) \geq v(A) + k \min S$ does *not* imply that these lower bounds become cofinal in Γ : for example $k < \Omega$ for every integer k . The finite-word certificate of [lemma 3.4](#) and [corollary 3.5](#) establishes summability even when that sequence of bounds never passes a specified higher rank, and every infinite operator series in this report rests on the stronger certificate. Likewise “only finitely many terms below each cutoff” is *not* the criterion: in $\Gamma = \mathbb{Z}^2$ lexicographic with $\eta = (0, 1)$ and $\theta = (1, 0)$ one has $n\eta < \theta$ for every integer n , yet $\sum_{n \geq 0} t^{n\eta}$ is a valid Hahn series with $(1 - t^\eta) \sum_{n \geq 0} t^{n\eta} = 1$, while the partial sums do not converge in the valuation topology. Sources 03, 05 and 06 each record this independently.

Example 3.7 (Infinitely many earlier weights, one relevant word). Let $\Gamma = \mathbb{Q} + \mathbb{Q}\Omega$ with Ω greater than every rational, and $S = \{1\} \cup \{2 - 1/n : n \geq 2\} \cup \{\Omega\}$. This S is well ordered and lies in no finitely generated additive grid, and infinitely many of its members are below Ω . Nevertheless $\mathcal{W}_S(\Omega) = \{(\Omega)\}$, $\ell_S(\Omega) = 1$ and $\ell_S(\Omega + 2) = 3$, the words at $\Omega + 2$ being the three permutations of $(\Omega, 1, 1)$: any finite sum of rational letters is rational, and any sum involving two Ω ’s exceeds $\Omega + 2$. An algorithm computing a coefficient at Ω needs the input at Ω , not infinitely many rational predecessors. [Section 22.5](#) computes with this S .

3.4 A uniform-monoid support algebra

The finite-return arguments of [section 9](#) need a slightly stronger object than [corollary 3.5](#): one monoid uniform across all Taylor degrees.

Definition 3.8. For $\mathbf{M} = \langle S \rangle$ with $S \subset \Gamma_{>0}$ well ordered, put

$$\mathcal{B}_{\mathbf{M}} = \left\{ \sum_{n \geq 0} h_n X^n \in K_{\Gamma}[[X]] : \text{supp}(h_n) \subseteq \mathbf{M} \text{ for every } n \right\},$$

a subring of $\mathcal{F}_1 = \mathbb{C}[[X]]((t^{\Gamma}))$ after regrouping as $\sum_{\gamma \in \mathbf{M}} H_{\gamma}(X)t^{\gamma}$ with $H_{\gamma} \in \mathbb{C}[[X]]$. Coefficients need not converge as ordinary complex power series.

The uniformity of the single monoid across all Taylor degrees is load-bearing: an arbitrary member of $\mathcal{O}_{\Gamma}[[X]]$ need not lie in any such algebra.

Lemma 3.9 (Evaluation, inversion, isometry). *Every $H \in \mathcal{B}_M$ is strongly Hahn-evaluable at every $u \in \mathfrak{m}_\Gamma$. If $H = X + O(X^2)$ then its formal inverse also lies in $\mathcal{B}_M$, and the evaluated maps are mutually inverse isometric bijections of $\mathfrak{m}_\Gamma$, isometry meaning $v(H(u) - H(w)) = v(u - w)$. Formal composition identities between zero-fixing series with integral unit linear coefficients and a common positive-support certificate remain valid after evaluation.*

Proof. For $u \neq 0$ with $U = \text{supp } u \subset \Gamma_{>0}$, the supports of $h_n u^n$ lie in $\langle S \cup U \rangle$, and every monomial of u^n uses at least n generators from U , so lemma 3.4 leaves finitely many n at a fixed exponent: both clauses of definition 3.1 hold. The coefficients of the formal inverse of $X + \sum_{n \geq 2} h_n X^n$ are integer polynomials in finitely many h_n , hence supported in M , and the substitutions in the inverse identities are summable by the same count. For $u \neq w$ in $\mathfrak{m}_\Gamma$,

$$\frac{H(u) - H(w)}{u - w} = 1 + \sum_{n \geq 2} h_n \sum_{j=0}^{n-1} u^j w^{n-1-j},$$

whose nonconstant summands all have positive valuation and whose total has residue one and valuation zero; this is the isometry. For compositions, expand both levels into products: at a fixed exponent the number of positive generators is finite, and at a fixed formal degree the number of zero-exponent contributions is finite because the substituted series has zero constant term. A unit linear coefficient and its inverse are adjoined by separating their nonzero complex constant terms and expanding the inverse in its positive part. $\square$

Example 3.10 (Coefficient-valuation bounds are insufficient at rank > 1). Let $\Gamma = \mathbb{Q} + \mathbb{Q}\varepsilon_0$ with $0 < \varepsilon_0 < 1/n$ for every positive integer n . All coefficients of $A(X) = \sum_{n \geq 2} t^{1/n} X^n$ have positive valuation. Yet at $u = t^{\varepsilon_0}$ the proposed Taylor sum has exponents $1/n + n\varepsilon_0$, which *strictly decrease* as n grows: it is not a Hahn series. Hence an assertion that “all normalized coefficients are integral” would leave a real gap in an arbitrary-rank linearization proof. The set $\{1/n : n \geq 2\}$ is not well ordered, so this example does not satisfy the uniform-support hypothesis of lemma 3.9; it is the reason that hypothesis is stated. This is source 01’s observation, and no other source records it.

3.5 Halos, monads, and evaluation as a ring homomorphism

For an ordinary open $U \subset \mathbb{C}^d$ put

$$U_\Gamma^\# = \{a + \xi : a \in U, \xi \in \mathfrak{m}_\Gamma^d\},$$

the *finite halo*; the representation is unique and $a = \text{st}(a + \xi)$ is the standard part. For $A = \sum_\gamma t^\gamma A_\gamma$ with all A_γ holomorphic on U , define

$$A^\#(a + \xi) = \sum_{\gamma, \alpha} t^\gamma \frac{\partial^\alpha A_\gamma(a)}{\alpha!} \xi^\alpha. \quad (3.4)$$

Proposition 3.11 (Faithful analytic realization). *The family in (3.4) is strongly summable, and (3.4) defines a faithful K_Γ -algebra homomorphism from $\mathcal{H}_\Gamma(\mathcal{O}(U))$ to functions on $U_\Gamma^\#$. It respects coordinate derivatives, substitution by positive near-identity Hahn maps, and infinitesimal Taylor expansions, with the expected coefficients $(\partial^\alpha A)^\#(z)/\alpha!$. For germ or formal coefficients use the Taylor coefficients at 0 and restrict the domain to $\mathfrak{m}_\Gamma^d$; polynomial and entire coefficients evaluate on every finite halo, but not automatically at infinite surcomplex points.*

Proof. Let $T = \text{supp } A$ and let S be the union of the positive supports of the components of ξ , a finite union of well-ordered positive sets. Every monomial contribution has exponent in $T + S^*$, and at a fixed exponent [lemma 3.4](#) permits only finitely many choices of γ and of a word from S ; the word length bounds $|\alpha|$ and a finite word has finitely many coordinate assignments. Hence the expanded family is strongly summable. Ordinary Taylor identities for products and derivatives then apply, every regrouping at a Hahn exponent being finite; adding the positive support of a near-identity substitution to S gives the composition assertion from the ordinary Taylor chain rule. For faithfulness, let $\gamma_0 = \min \text{supp } A$ for $A \neq 0$ and pick an ordinary $a \in U$ with $A_{\gamma_0}(a) \neq 0$: the coefficient of t^{γ_0} in $A^\#(a)$ is then nonzero. $\square$

At a fixed ordinary Taylor center the numbers $n!$, or extremely large inverse small divisors, remain *ordinary complex constants* of valuation zero. Strong summability imposes no real growth condition on them. This single fact explains both the strength of the theorems below and the necessity of stating their domains exactly.

Lemma 3.12 (Faithfulness of infinitesimal evaluation). *If $0 \neq A \in \mathcal{F}_1$ and $\Gamma \neq \{0\}$ is divisible, then $A(u) \neq 0$ for some $u \in \mathfrak{m}_\Gamma$; evaluation on the whole monad is injective on the formal-coefficient ring.*

Proof. Let $\gamma_0 = \min \text{supp } A$ and let n be the least degree of the nonzero formal series A_{γ_0} . If another Hahn exponent occurs, let $\Delta > 0$ be the gap from γ_0 to the next one and choose $s > 0$ with $ns < \Delta$; divisibility supplies such an s when $n > 0$, and any positive s works when $n = 0$. If no other exponent occurs, take any $s > 0$. At $u = t^s$ the exponent $\gamma_0 + ns$ appears with nonzero coefficient, while higher degrees at γ_0 and all other exponents contribute only larger exponents. Evaluation is legitimate by [lemma 3.9](#) after factoring out t^{γ_0} . $\square$

[Lemma 3.12](#) blocks the evasion of representing a divergent conjugacy by a *different* common-domain family agreeing on all infinitesimals; it is used in [proposition 11.1](#).

4 Positive substitution, logarithms, flows, and centralizers

4.1 The substitution group

For $U \subset \mathbb{C}^d$ open set

$$\mathcal{G}_+(U) = \{F(z) = z + u(z) : u \in \mathcal{H}_\Gamma(\mathcal{O}(U))_+^d\}.$$

No condition at $z = 0$ is imposed, and 0 need not lie in U . The pullback of $F = \text{id} + u$ on $A \in \mathcal{H}_\Gamma(\mathcal{O}(U))$ is

$$C_F A = A \circ F = \sum_{\alpha \in \mathbb{N}^d} \frac{1}{\alpha!} (\partial^\alpha A) u^\alpha, \quad (4.1)$$

every coefficient of which is a finite sum of products and derivatives of functions holomorphic on U , hence holomorphic on U regardless of its ordinary norm; the support lies in $\text{supp } A + S^*$ for S the union of the supports of the components of u .

Proposition 4.1 (Substitution group). *(4.1) defines an associative group law on $\mathcal{G}_+(U)$. Every inverse again lies in $\mathcal{G}_+(U)$, on the same U , with support in $S^* \setminus \{0\}$. Evaluation gives mutually inverse bijections of $U_\Gamma^\#$ preserving standard parts and commutes with composition. The same*

statements hold in the common-domain germ, radius-free germ, entire, polynomial and formal categories, with the evaluation domains of [section 3.5](#). The order convention is

$$C_F C_G = C_{G \circ F} : \quad (4.2)$$

pullback reverses composition.

Proof. Strong summability is [lemma 3.4](#) and [corollary 3.5](#), applied with the support of each Hahn input retained. At a fixed Hahn exponent only finitely many input coefficients, derivatives and products contribute. Keep those input coefficients separately, introduce a formal parameter for each relevant positive support letter, and discard parameter monomials beyond the corresponding relative word depth. Taylor composition in that finite nilpotent parameter algebra is associative and multiplicative, and each asserted coefficient identity is one of those identities. For the inverse put $\Delta = C_F - I$: every term of Δ adds a nonempty word from S , so $C_F^{-1} = \sum_{k \geq 0} (-\Delta)^k$ is defined coefficientwise and is a two-sided multiplicative inverse. Let $G = C_F^{-1} z$, identity plus positive support with support in S^* . On coordinate polynomials multiplicativity gives $C_F^{-1} = C_G$; at each Hahn weight both operators are finite-order differential operators on an ordinary input, so agreement on polynomial jets extends to all holomorphic inputs, and strong linearity extends it to Hahn inputs. For formal coefficients the finite-jet argument is coefficientwise in z . Substituting $z = a + \xi$ and adding $\text{supp } \xi$ to the control set preserves every identity, and $\text{st } F(a + \xi) = a$ keeps the evaluated map in $U_\Gamma^\#$. Iterates of a single map are unaffected by (4.2), and commuting maps have commuting pullbacks; the convention is retained explicitly to avoid a sign or order ambiguity in logarithms. $\square$

For maps $F = \Lambda z + f$ on $\mathbb{D}_R^d$ with Λ diagonal unitary and f positive, use instead

$$A \circ F = \sum_{\alpha \in \mathbb{N}^d} \frac{1}{\alpha!} (\partial^\alpha A)(\Lambda z) f^\alpha, \quad (4.3)$$

legitimate because $\Lambda \mathbb{D}_R^d = \mathbb{D}_R^d$. For formal coefficients or germs, $F(0) = 0$ is imposed whenever composition with the nonidentity reduction is used. [Lemma 4.2](#) records the near-identity inverse in the form source 02 uses.

Lemma 4.2 (Near-identity inverses). *Let $H = z + U$ be normalized with coefficients in any one of $\mathcal{O}(\mathbb{D}_R)$, $\mathbb{C}\{z\}$, $\mathbb{C}[[z]]$, $\mathcal{O}(\mathbb{C})$, $\mathbb{C}[z]$, and positive Hahn support B . There is a unique normalized inverse $J = z + V$ with coefficient support in $B^* \setminus \{0\}$ and coefficients in the same algebra, with $H \circ J = J \circ H = \text{id}$.*

Proof. In $H(J) = J + U(J)$ the coefficient v_γ occurs with coefficient one, and every other contribution at exponent γ uses only v_β with $\beta < \gamma$; set v_γ to minus that finite sum and recurse on B^* . Derivatives, products and finite sums preserve each algebra and the order- z^2 normalization. The identical triangular argument gives a left inverse, and associativity, which is coefficientwise a finite Taylor identity, identifies the two. $\square$

4.2 Logarithms are actual coefficientwise vector fields

A positive Hahn vector field on U is $X = a_1 \partial_1 + \cdots + a_d \partial_d$ with $a \in \mathcal{H}_\Gamma(\mathcal{O}(U))_+^d$, and

$$\exp X = \sum_{k \geq 0} \frac{X^k}{k!}, \quad \log C_F = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} (C_F - I)^k \quad (4.4)$$

are operators on the Hahn algebra, not numerical logarithms of values of F .

Theorem 4.3 (Same-domain exponential–logarithm correspondence). *For every open $U \subset \mathbb{C}^d$ the assignments $F \mapsto X_F = \log C_F$ and $X \mapsto \Phi_X = (\exp X)z$ are inverse bijections between $\mathcal{G}_+(U)$ and positive Hahn vector fields on U . All coefficient functions stay holomorphic on the original U : no shrinking, and no uniform bound on the coefficient family. If S contains the positive support of the input then the output support lies in $S^* \setminus \{0\}$, and each output coefficient has a finite word certificate. For $F \neq \text{id}$,*

$$v(X_F) = v(F - \text{id}), \quad (4.5)$$

and the leading coefficient of $X_F z$ equals that of $F - \text{id}$. The inverse of F is $(\exp(-X_F))z$. The same assertions hold with common-domain germ, radius-free germ, entire, polynomial or formal coefficients.

Proof. Summability. Put $\Delta = C_F - I$ and $S = \text{supp}(F - \text{id})$. By (4.1), Δ adds nonempty words from S , so both series in (4.4) are strongly summable by corollary 3.5 for each fixed Hahn input A , with depth $\ell_{\text{supp } A, S}(\gamma)$ at output γ . Every output coefficient is a finite differential expression with coefficients holomorphic on U . For the coordinate input z , the input support is $\{0\}$, so the unshifted depth $\ell_S(\gamma)$ applies for $\gamma \in S^*$.

The derivation identity. Introduce an ordinary indeterminate s and put $C_F^s = \sum_{k \geq 0} \binom{s}{k} \Delta^k$. For fixed inputs and a fixed Hahn exponent this is a *polynomial* in s , not a power series. At every nonnegative integer s it is the usual integer power of the multiplicative operator C_F , so the coefficientwise polynomial identity $C_F^s(AB) = C_F^s(A)C_F^s(B)$ holds identically in s . Differentiating at $s = 0$ and using $\frac{d}{ds} \binom{s}{k} \big|_{s=0} = (-1)^{k+1}/k$ for $k \geq 1$ gives $X_F(AB) = X_F(A)B + AX_F(B)$.

Why the derivation is a vector field. At each Hahn weight γ the restriction of X_F to ordinary coefficient functions is a finite-order differential operator D_γ , and the preceding identity makes it a derivation. Put $a_{j,\gamma} = D_\gamma z_j$; by Leibniz, D_γ agrees with $\sum_j a_{j,\gamma} \partial_j$ on every coordinate polynomial, and at an ordinary point a finite-order differential operator is determined by a finite Taylor jet, so the two agree on every holomorphic function. For germs or formal series use finite jets at the origin; for polynomials it is immediate. This finite-order step is essential: no classification of arbitrary abstract algebraic derivations of a function ring is invoked.

Conversely. If $X = a \cdot \partial$ is positive, each application adds a positive letter, so $\exp X$ is strongly summable, and repeated Leibniz $X^k(AB) = \sum_j \binom{k}{j} X^j(A)X^{k-j}(B)$ makes it multiplicative with inverse $\exp(-X)$. Put $\Phi_X = (\exp X)z = \text{id} + \text{positive}$. On coordinate polynomials multiplicativity gives $\exp X = C_{\Phi_X}$, and the finite-jet argument extends the equality to the whole coefficient algebra and then to Hahn inputs.

Inverse identities and valuation. $\exp(\log(I + T)) = I + T$ and $\log(\exp T) = T$ hold in a nilpotent operator algebra, and at a specified Hahn coefficient only finitely many words occur, so they hold coefficientwise; this is the bijection. Finally $X_F z = (F - \text{id})z + \text{terms with at least two support letters}$, whose valuation is at least $2v(F - \text{id})$, which gives (4.5) and the leading-coefficient assertion. Derivatives and finite products preserve every stated category. $\square$

Remark 4.4 (Three sources, one route, three by-products). Sources 03, 04 and 05 each prove theorem 4.3, and all three use the same essential device: the binomial operator family $\sum_k \binom{s}{k} \Delta^k$, polynomiality in s at each fixed Hahn exponent, and a finite-jet identification of the resulting derivation with a coordinate vector field. Being one route, it is printed once, in source 03's several-variable form on an arbitrary open $U \subset \mathbb{C}^d$. The routes differ only in by-products, and each by-product survives separately below: source 05 adds that $v(W_F) = v(h)$ with equal leading coefficient and that inverses are $\exp(-W_F \partial_z)z$ (both included in theorem 4.3); source 04 adds the exact fixed-point identity lemma 4.5, which is what determines the complete monodromy exponent

of [section 14](#); and source 05 adds the Julia equation [corollary 4.6](#) and the intertwining criterion [proposition 4.7](#), which are what make [section 15](#) possible. Source 05 also notes explicitly that the general algebraic mechanism for exponentiating strongly summable derivations of generalized-series algebras is established work [8] and that the existence of a formal logarithm is not claimed new.

Lemma 4.5 (Derivative of the logarithmic generator at an ordinary fixed point). *Let $F = w + \Delta$ with $\Delta \in \mathcal{H}_\Gamma(\mathcal{O}(U))_{>0}$ in one variable, $X_F = A(w)\partial_w$. If an ordinary $a \in U$ satisfies $F(a) = a$, then*

$$A(a) = 0, \quad A'(a) = \text{Log } F'(a), \quad (4.6)$$

where $\text{Log}(1 + u) = \sum_{n \geq 1} (-1)^{n+1} u^n / n$ for $v(u) > 0$.

Proof. Evaluation at a annihilates $N\Phi = (C_F - I)\Phi$, hence annihilates $A = Dw$. With $\lambda_a = F'(a)$, the chain rule for Taylor substitution gives $(N\Phi)'(a) = (\lambda_a - 1)\Phi'(a)$, so $(N^r w)'(a) = (\lambda_a - 1)^r$; apply the logarithm series, which is a strong sum because $v(\lambda_a - 1) > 0$ (the identity is immediate if $\lambda_a = 1$). $\square$

This *exact* identity, and not a truncated modified differential equation, is what fixes the complete monodromy exponent in [section 14](#).

Corollary 4.6 (Julia equation). *For every $F \in \mathcal{G}_+(U)$ in one variable, with $W_F = X_F z$, $W_F(F(z)) = F'(z)W_F(z)$.*

Proof. C_F commutes with its logarithm; apply $C_F(W_F \partial_z) = (W_F \partial_z)C_F$ to z . $\square$

Proposition 4.7 (Conjugacy and vector fields). *For $F_1, F_2, H \in \mathcal{G}_+(U)$ in one variable, $H \circ F_1 = F_2 \circ H$ if and only if $W_{F_2}(H) = H'W_{F_1}$.*

Proof. The conjugacy relation is $C_{F_1}C_H = C_H C_{F_2}$; conjugating the summable logarithm gives $(W_{F_1} \partial_z)C_H = C_H(W_{F_2} \partial_z)$, and evaluation on z gives the displayed identity. Conversely that identity and the chain rule give the operator intertwining on the whole algebra, and exponentiation recovers the conjugacy. $\square$

4.3 Times, flows, and unique roots

Corollary 4.8 (Exact flow law and admissible Hahn times). *Let $X = a \cdot \partial$ be positive. For every ordinary $s \in \mathbb{C}$, $\Phi_s = (\exp(sX))z \in \mathcal{G}_+(U)$ and*

$$\Phi_0 = \text{id}, \quad \Phi_{s+r} = \Phi_s \circ \Phi_r, \quad \frac{d}{ds} \Phi_s = a \circ \Phi_s, \quad (4.7)$$

each Hahn coefficient of Φ_s being a polynomial in s . For $c \in K_\Gamma$ with $c = 0$ or

$$v(c) + v(a) > 0, \quad (4.8)$$

the map $\Phi_c = (\exp(cX))z$ is also defined and positive, and the additive flow law holds for such times.

Proof. All ordinary scalars have valuation zero, each coefficient involves finitely many powers of s , and commuting exponentials of multiples of X satisfy the addition law. Coefficientwise differentiation and commuting X with its exponential give the last identity. For a Hahn time, (4.8) says exactly that cX has positive support, since c is a scalar independent of z ; apply [theorem 4.3](#) to cX , and note that the positive supports of two admissible multiples have well-ordered union. $\square$

Condition (4.8) characterizes when this scalar multiple of a nonzero vector field is positive; it is *not* asserted to be a maximal possible lifetime in a larger class of maps (a constant vector field, for instance, has translations beyond the positive subgroup). Likewise (4.7) is coefficientwise analytic time dependence, not a continuous path in the fine surreal topology.

Corollary 4.9 (Unique fractional iteration; torsion-freeness). *For every $F \in \mathcal{G}_+(U)$ and integer $N \geq 1$ there is a unique $G \in \mathcal{G}_+(U)$ with $G^{\circ N} = F$, namely $G = (\exp(X_F/N))z$. The group $\mathcal{G}_+(U)$ is uniquely divisible and torsion-free.*

Proof. Existence is [corollary 4.8](#). If $G^{\circ N} = F$ then $C_G^N = C_F$, and taking logarithms gives $NX_G = X_F$, whence uniqueness. If $F^{\circ N} = \text{id}$ then $NX_F = 0$, and characteristic zero gives $F = \text{id}$. $\square$

4.4 Fixed-point ideals, and two routes to iteration invariance

For a coefficient map F let $I_{\text{Fix}(F)} = (F_1 - z_1, \dots, F_d - z_d)$ be its fixed-point ideal in the chosen Hahn coefficient algebra. No Noetherianity and no Nullstellensatz is needed to compare such ideals, and none is invoked: ideals are compared through an explicit invertible matrix.

Theorem 4.10 (Fixed-point ideal of a positive flow). *Let $X = a \cdot \partial$ be a positive Hahn vector field on U and let $c \neq 0$ be an admissible Hahn time. Then*

$$I_{\text{Fix}(\Phi_c)} = (a_1, \dots, a_d) \quad \text{in } \mathcal{H}_\Gamma(\mathcal{O}(U)), \quad (4.9)$$

and likewise in the other coefficient categories and after localization. In particular all nonzero integer iterates of $F \in \mathcal{G}_+(U)$ have identical fixed-point ideals; on the evaluated halo there are no nonfixed points of finite period; and whenever a fixed-point quotient or local algebra has finite length, that length is unchanged by nonzero integer iteration.

Proof. Set $E = cX = b \cdot \partial$ with $b = ca$ positive, so that $\Phi_c - z = b + \sum_{k \geq 2} E^k z / k!$. Since $E^k z_j = \sum_\ell b_\ell \partial_\ell (E^{k-1} z_j)$, define

$$B_{j\ell} = \delta_{j\ell} + \sum_{k \geq 2} \frac{\partial_\ell (E^{k-1} z_j)}{k!}, \quad (4.10)$$

strongly summable by [corollary 3.5](#), of the form $I + \text{positive}$, hence invertible by its matrix geometric series, and

$$\Phi_c - z = Bb. \quad (4.11)$$

The two coordinate ideals are therefore equal, and since the scalar c is a unit of K_Γ , the ideal generated by b is that generated by a . On evaluation at any halo point B remains invertible by [proposition 4.1](#), so a point is fixed by Φ_c exactly when all components of a vanish there; apply this with $c = 1$ and any nonzero integer $c = n$. Identical ideals give identical quotient algebras. $\square$

This theorem concerns maps whose ordinary reduction is the identity. It makes no claim that a general surcomplex map lacks periodic points; in particular rotations of finite order lie outside $\mathcal{G}_+(U)$ unless they are the identity, and [theorem 9.7](#) exhibits genuine period- q points for a map that is *not* in $\mathcal{G}_+(U)$.

Remark 4.11 (Two routes). Source 03 proves iteration invariance by the explicit matrix (4.10)–(4.11), obtaining ideal equality — scheme structure and finite length — rather than inferring it from equality of zero sets. Source 05 obtains the same invariance as a corollary of unique n -th roots and the centralizer (corollary 4.9 and theorem 4.12), which gives less (no ideal statement) by a shorter argument but also gives torsion-freeness of the whole group. Both are kept: the ideal statement is not available from the root argument, and the group statement is not available from the matrix.

Theorem 4.12 (The full positive centralizer in one variable). *Let $U \subset \mathbb{C}$ be connected, let $F \in \mathcal{G}_+(U)$ be nonidentity, and write $X_F = a(z)\partial_z$ with $\alpha = v(a) > 0$. Then*

$$\text{Cent}_{\mathcal{G}_+(U)}(F) = \{(\exp(ca\partial_z))z : c \in K_\Gamma, c = 0 \text{ or } v(c) + \alpha > 0\}, \quad (4.12)$$

and the parametrization is an isomorphism from that additive group of Hahn times onto the centralizer. Zeros of a need not be removed from U . No nowhere-zero hypothesis on the leading coefficient of a is required.

Proof. Let G be positive with $X_G = b\partial_z$. Commuting maps have commuting pullbacks, hence commuting logarithms, all double regroupings being valid over the union of the two positive supports; conversely commuting logarithms have commuting exponentials. Thus $FG = GF$ iff $[a\partial_z, b\partial_z] = 0$ iff $ab' - ba' = 0$. Embed the coefficient algebra into $\mathcal{M}(U)((t^\Gamma))$ with $\mathcal{M}(U)$ the field of ordinary meromorphic functions on the connected U ; the nonzero series a is invertible there (factor its leading term and use a positive geometric series), so $(b/a)' = (ab' - ba')/a^2 = 0$. Every Hahn coefficient of b/a is a meromorphic function with zero derivative, hence an ordinary complex constant by connectedness, so $b/a = c \in K_\Gamma$. Conversely $b = ca$ commutes with a for every scalar c , and has positive support exactly when $c = 0$ or $v(c) + \alpha > 0$; theorem 4.3 gives (4.12). Equal maps give $(c_1 - c_2)a = 0$ and $a \neq 0$, so $c_1 = c_2$; corollary 4.8 gives the group assertion. Meromorphic division was used only to identify the ratio: the resulting coefficient ca is holomorphic on the original U , including at the zeros of a . $\square$

Sources 03 and 05 state this theorem with the same hypotheses, the same conclusion — source 05 writes $v(c) + \delta > 0$ for $\delta = v(W_F)$, which is (4.12) verbatim — the same meromorphic-ratio proof and the same remark that zeros of a need not be removed. It is therefore printed once, per convention 1.1(i). Two consequences belong to source 05 and survive separately: if the leading coefficient of W_F is nonzero at z_* , no nonidentity centralizer element fixes z_* , so the basepoint normalization in theorem 15.13 removes exactly the centralizer freedom, and the unnormalized conjugacies between two conjugate maps form a torsor under either centralizer. The admissible times c may have negative valuation provided cW_F stays positive; the positivity threshold is part of the conclusion, not an unrestricted large-time claim.

Remark 4.13 (Limits of the centralizer statement). Connectedness is necessary: otherwise one obtains independent scalar series on different ordinary components. In dimension $d > 1$ the correct general statement is only that commuting positive maps have commuting positive logarithmic vector fields, and those need *not* be scalar multiples of one another; source 03 states that the one-variable meromorphic-ratio argument cannot classify higher-dimensional Lie centralizers and that additional first integrals or commuting geometric structures can occur, and source 05 states that in several variables there is no single reciprocal time one-form encoding a vector field, so the centralizer proof and the finite-period normal form must not be transported. The Hamiltonian centralizer theorem 16.12 is a genuinely different several-variable statement, proved by a different argument in a different algebra.

Proposition 4.14 (Workspace invariance). *Let $\Gamma \hookrightarrow \Gamma'$ be an ordered group inclusion. The substitution inverse, the positive logarithm and exponential, and the normalized nonresonant linearizer of [theorem 6.1](#) are unchanged when their inputs are regarded in the larger workspace; their support certificates remain valid; a normalized positive solution constructed in the larger workspace acquires no exponents outside Γ ; and evaluation identities remain valid after adjoining the supports of new halo points.*

Proof. The support words use only input exponents and finite addition, and the ordinary coefficient operations are independent of Γ , so the explicit formulas compute the same coefficients in Γ and Γ' , with support already inside Γ . Uniqueness of substitution inverses, of the exponential–logarithm correspondence and of normalized linearization identifies these with the corresponding positive solutions in the larger workspace. Evaluation follows by including the new positive point supports in [lemma 3.4](#). $\square$

Warning 4.15 (One recorded exception: the centralizer can grow). [Proposition 4.14](#) does *not* say that the centralizer [\(4.12\)](#) is unchanged by scalar extension: a larger workspace can supply new admissible times c . Source 03 records this exception explicitly. What is invariant is the particular normalized construction from the original data.

5 The homological operator and the radius of one inversion

Fix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ with $|\lambda_j| = 1$ and assume NR: $\lambda^\alpha \neq \lambda_j$ for $|\alpha| \geq 2$ and all j . Let B_2^d denote coefficient vectors whose Taylor series have no terms of total degree 0 or 1, and put

$$\mathcal{L}h = h(\Lambda z) - \Lambda h(z), \quad (\mathcal{L}h)_{j,\alpha} = (\lambda^\alpha - \lambda_j)h_{j,\alpha}, \quad (5.1)$$

so that under NR the inverse on formal coefficients is

$$(\mathcal{L}^{-1}g)_j(z) = \sum_{|\alpha| \geq 2} \frac{g_{j,\alpha}}{\lambda^\alpha - \lambda_j} z^\alpha. \quad (5.2)$$

It preserves total polynomial degree and acts coefficientwise on Hahn families without changing Hahn exponents. Recall $\text{rad}_{\text{iso}}(g)^{-1} = \limsup_m (\max_{|\alpha|=m,j} |g_{j,\alpha}|)^{1/m}$, the radius of absolute convergence on centered equal-radius polydisks (the count $\binom{m+d-1}{d-1}$ of multi-indices of degree m is polynomial in m and does not affect the root test).

Lemma 5.1 (Single-step analytic loss). *If $\tau = \tau(\Lambda) < \infty$ then*

$$\text{rad}_{\text{iso}}(\mathcal{L}^{-1}g) \geq e^{-\tau} \text{rad}_{\text{iso}}(g). \quad (5.3)$$

Hence $\mathcal{L}^{-1}$ preserves $\mathcal{O}(\mathbb{D}_R^d)_2^d$ when $\tau = 0$, preserves radius-free convergent germs and entire coefficients when $\tau < \infty$, and always preserves polynomial and formal coefficients under NR alone.

Proof. By [\(5.2\)](#) the maximal degree- m coefficient of $\mathcal{L}^{-1}g$ is at most M_m times that of g ; take m -th roots and limsups. For $\text{rad}_{\text{iso}}(g) = \infty$ the root growth of g is zero while $\limsup M_m^{1/m} = e^\tau < \infty$, so the product again has root growth zero. Polynomial support in the Taylor degree is not changed at all, and for formal coefficients [\(5.2\)](#) divides by nonzero constants. $\square$

Lemma 5.2 (Sharpness and analytic failure). *If $\tau > 0$ and $0 < R < \infty$, there is $\varphi \in \mathcal{O}(\mathbb{D}_R^d)_2^d$ for which $\mathcal{L}^{-1}\varphi$ is not holomorphic on $\mathbb{D}_R^d$; it can be chosen with Taylor radius exactly $Re^{-\tau}$, read as zero when $\tau = \infty$. If $\tau = \infty$ there is even an entire $\varphi \in \mathcal{O}(\mathbb{C}^d)_2^d$ for which $\mathcal{L}^{-1}\varphi$ has zero radius.*

Proof. Choose increasing degrees m_k with $m_k^{-1} \log M_{m_k} \rightarrow \tau$ and for each k a pair (α_k, j_k) , $|\alpha_k| = m_k$, realizing the maximal inverse divisor; discard finitely many degrees so that $M_{m_k} > 1$. Then

$$\varphi(z) = \sum_{k \geq 1} R^{-m_k} z^{\alpha_k} e_{j_k} \quad (5.4)$$

is holomorphic on $\mathbb{D}_R^d$, and its inverse has degree- m_k coefficient of modulus $R^{-m_k} M_{m_k}$ and no other nonzero coefficient, so its radius is exactly $Re^{-\tau}$. For the stronger assertion choose m_k with $M_{m_k} \geq \exp(k^2 m_k)$ and put

$$\varphi(z) = \sum_{k \geq 1} \exp(-km_k) z^{\alpha_k} e_{j_k}, \quad (5.5)$$

which converges normally on every closed polydisk once $R \exp(-k) \rightarrow 0$ and is therefore entire, while its inverse has a degree- m_k coefficient of modulus at least $\exp((k^2 - k)m_k)$ and Taylor radius zero. $\square$

These witnesses use only ordinary analytic coefficient functions; no infinitesimal support manipulation is hidden in them. Sources 02, 03 and 06 each construct such a witness; in one variable source 02's forcing is the explicit rational function

$$P_R(z) = \frac{z^2}{1 - z/R} = \sum_{n \geq 2} R^{2-n} z^n, \quad (5.6)$$

for which $\text{rad}(\mathcal{L}^{-1}P_R) = Re^{-\sigma}$ exactly — all its Taylor coefficients being R^{2-n} , so no choice of a sparse subsequence is needed — and source 02's entire witness sets $q_{n_j} = |d_{n_j}|^{1/2}$ along a sequence with $|d_{n_j}|^{-1/n_j} \rightarrow \infty$. Source 06's version, [theorem 16.5](#), is on frequency divisors and is stated as a sharp *iff*; source 06's explicit super-Liouville instance is [proposition 16.17](#).

Proposition 5.3 (A compact-disk estimate, offered only as a starting point). *Suppose $d = 1$ and $\sigma = 0$, and let $0 < r < s < R$. Then*

$$\|\mathcal{L}^{-1}q\|_r \leq \|q\|_s \sum_{n \geq 2} \frac{(r/s)^n}{|\lambda^n - \lambda|}, \quad \|q\|_s = \max_{|z| \leq s} |q(z)|, \quad (5.7)$$

and the numerical series converges.

Proof. Cauchy's estimate gives $|q_n| \leq \|q\|_s s^{-n}$; sum the absolutely convergent Taylor majorant on $|z| \leq r$, convergence following from $\limsup |\lambda^n - \lambda|^{-1/n} = e^\sigma = 1$. $\square$

Source 02 offers [\(5.7\)](#) explicitly as a *starting point* for normed questions and not as part of any theorem: it shows the inverse is continuous for the compact-open Fréchet topology, at the cost of a slightly larger compact disk, while the domain of holomorphy remains the full open $\mathbb{D}_R$. None of the theorems below imposes or produces such an estimate.

6 The coefficient-category classification

6.1 The sharp five-row table, for an exact multiplier

Consider

$$F(z) = \Lambda z + f(z), \quad f \in \mathcal{H}_\Gamma(B)_+^d, \quad f_\gamma(0) = 0, \quad df_\gamma(0) = 0, \quad (6.1)$$

the last condition being the exactness of the multiplier: no infinitesimal linear part is permitted at any Hahn exponent (tag SD_{ex} , [section 2.3](#)).

Theorem 6.1 (Universal coefficient-category classification). *For every nonzero Γ and every Λ satisfying NR, the following table gives necessary and sufficient conditions for every map (6.1) in the indicated category to admit a normalized conjugacy*

$$H(z) = z + h(z), \quad h \in \mathcal{H}_\Gamma(B)_+^d, \quad h_\gamma(0) = 0, \quad dh_\gamma(0) = 0, \quad H \circ F = \Lambda H. \quad (6.2)$$

Whenever it exists, that conjugacy is unique among normalized positive-support conjugacies and its inverse lies in the same category.

Coefficient category B	Exact condition
$\mathcal{O}(\mathbb{D}_R^d)$, on the <i>same fixed</i> polydisk (fixed-polydisk row)	$\tau(\Lambda) = 0$
$\mathbb{C}\{z_1, \dots, z_d\}$, no common radius required (radius-free row)	$\tau(\Lambda) < \infty$
$\mathcal{O}(\mathbb{C}^d)$, entire coefficients (new row)	$\tau(\Lambda) < \infty$
$\mathbb{C}[z_1, \dots, z_d]$, polynomial coefficients (new row)	NR only
$\mathbb{C}[[z_1, \dots, z_d]]$, formal coefficients	NR only

If $S = \text{supp } f \subset \Gamma_{>0}$ then $\text{supp } h \subseteq S^ \setminus \{0\}$. Each h_γ depends on only finitely many coefficient functions of f — those at letters in $D_S(\gamma)$ — and on at most $\ell_S(\gamma)$ ordinary homological inversions. The result is compatible with enlargement of the Hahn workspace (proposition 4.14).*

Necessity is already visible in an input with support $\{\eta\}$ for any single $\eta > 0$. Sufficiency is a genuinely nonlinear arbitrary-support statement: no countability, finite rank, or finitely generated grid assumption is made on S .

6.2 Sufficiency: a support-controlled inverse of the nonlinear operator

Write $F = \Lambda z + f$, seek $H = z + h$, and set

$$\mathcal{N}_f h = h(\Lambda z + f(z)) - h(\Lambda z) = \sum_{|\beta| \geq 1} \frac{(\partial^\beta h)(\Lambda z)}{\beta!} f^\beta, \quad (6.3)$$

so that Schröder's equation becomes

$$(\mathcal{L} + \mathcal{N}_f)h = -f. \quad (6.4)$$

The map F is nonlinear, but with F fixed this equation is *linear* in h , and in the Hahn setting that observation is decisive: $\mathcal{L}$ does not alter Hahn weights while $\mathcal{N}_f$ adds positive support words, and both preserve the absence of coordinate terms of degrees zero and one. Whenever $\mathcal{L}^{-1}$ acts on the category, put $T = \mathcal{L}^{-1}\mathcal{N}_f$ and

$$h = - \sum_{k \geq 0} (-T)^k \mathcal{L}^{-1} f. \quad (6.5)$$

Proposition 6.2 (Support-controlled inverse; route A). *Let B_2^d be one of the five categories of theorem 6.1 and suppose $\mathcal{L}^{-1}$ preserves it. Then (6.5) defines the unique normalized positive conjugacy; its support lies in $S^* \setminus \{0\}$; at weight γ only terms with*

$$k + 1 \leq \ell_S(\gamma) \quad (6.6)$$

contribute, and only input functions f_σ with $\sigma \in D_S(\gamma)$ are involved.

Proof. Each term of $\mathcal{N}_f$ contains at least one coefficient of f , hence adds a nonempty word in S , and the initial input $\mathcal{L}^{-1}f$ already contains one letter; a term with k copies of T therefore contains at least $k + 1$ letters. More explicitly, prefix the initial support exponent to the concatenated operator words: the resulting word belongs to $\mathcal{W}_S(\gamma)$ and has length at least $k + 1$. This gives (6.6) and puts every participating coefficient of f at a letter of $D_S(\gamma)$. Corollary 3.5, with input support $\text{supp}(\mathcal{L}^{-1}f) \subseteq S$, gives strong summability and the support inclusion. This use of the unshifted depth relies on the initial positive letter, not on an input-independent operator bound. Every contributing expression is assembled from finitely many derivatives, products, compositions with the fixed diagonal rotation, and applications of $\mathcal{L}^{-1}$, all of which preserve the assumed category, so $h_\gamma \in B_2^d$; the operator geometric identity gives $(I+T)h = -\mathcal{L}^{-1}f$, which is (6.4), hence $H \circ F = \Lambda H$. For uniqueness let $h, \tilde{h}$ be normalized positive solutions, not necessarily with the constructed support, and let γ_0 be the least weight of $q = h - \tilde{h} \neq 0$; since $\mathcal{N}_f$ adds positive weights, the coefficient of $\mathcal{N}_f q$ at γ_0 vanishes, so $\mathcal{L}q_{\gamma_0} = 0$ and NR gives $q_{\gamma_0} = 0$, a contradiction. Finally H has a positive inverse in the same category by proposition 4.1, normalized by differentiating the inverse identity. $\square$

Applying lemma 5.1 gives sufficiency in all five rows. Note that the polynomial row requires no bound on the inverse divisors at all: a polynomial uses only finitely many of them at any one Hahn coefficient, and the infinitely many nonlinear corrections are separated into different positive Hahn weights.

Theorem 6.3 (Support-controlled lifting; route B, drift permitted). *Let λ be an irrational unit multiplier ($d = 1$), let $A \subset \Gamma_{>0}$ be well ordered, and write*

$$F(z) = \lambda z + P(z), \quad P = \sum_{a \in A} t^a f_a, \quad \delta_a = f'_a(0), \quad \delta = \sum_a t^a \delta_a, \quad \Lambda_{\text{dr}} = \lambda + \delta, \quad N = P - \delta z, \quad (6.7)$$

with $f_a \in z\mathbb{C}[[z]]$, so that every coefficient of N vanishes to order at least two and the multiplier is permitted to drift. There is a unique normalized positive-support formal solution $H = z + U$ of $H(F(z)) = \Lambda_{\text{dr}} H(z)$, with

$$\text{supp}(H - z) \subseteq A^* \setminus \{0\}, \quad (6.8)$$

and a normalized compositional inverse with the same bound. Moreover: (i) if $\sigma = 0$ and all f_a are holomorphic on $\mathbb{D}_R$, all coefficients of H and H^{-1} are holomorphic on that same $\mathbb{D}_R$; (ii) if $\sigma < \infty$ and all f_a are convergent germs, the coefficients are convergent germs; (iii) if $\sigma < \infty$ and all f_a are entire, the coefficients are entire; (iv) if all f_a are polynomials, all coefficients are polynomials, with no arithmetic assumption beyond NR. No uniform coefficient bound, no lower Archimedean bound on the positive exponents, and no convergence in a complex parameter is assumed.

Proof. Substituting $H = z + U$ and expanding at λz gives, after cancelling $\Lambda_{\text{dr}} z$,

$$\mathcal{L}U = -N + \delta U - \sum_{\beta > 0} \sum_{k \geq 1} t^\beta \frac{h_\beta^{(k)}(\lambda z)}{k!} P(z)^k, \quad (6.9)$$

whose coefficient at $\gamma \in A^* \setminus \{0\}$ is

$$\mathcal{L}h_\gamma = -N_\gamma + \sum_{\substack{a \in A, \beta \in A_{>0}^* \\ a + \beta = \gamma}} \delta_a h_\beta - \sum_{\substack{k \geq 1, \beta \in A_{>0}^* \\ a_1, \dots, a_k \in A \\ \beta + a_1 + \dots + a_k = \gamma}} \frac{h_\beta^{(k)}(\lambda z)}{k!} f_{a_1}(z) \cdots f_{a_k}(z). \quad (6.10)$$

Every sum is finite even for a non-Archimedean order: represent β by a word in A and append the a_i , then apply [lemma 3.4](#); and every occurring β is strictly less than γ . The right side vanishes to order at least two — N_γ by definition, $\delta_a h_\beta$ by induction, and in the last line $h_\beta^{(k)}$ vanishes to order at least $\max(2 - k, 0)$ while the product of k functions f_{a_i} vanishes to order at least k . Since A^* is well ordered and $\mathcal{L}$ is invertible on $z^2\mathbb{C}[[z]]$, [\(6.10\)](#) defines h_γ by well-founded recursion, with all coefficients outside A^* set to zero. For (i)–(iv), induct on γ : differentiation, rotation, multiplication and finite addition preserve each stated algebra, and [lemma 5.1](#) does the same for the homological inverse — and in case (i) the induction retains *exactly* $\mathbb{D}_R$ at every exponent rather than replacing it by a sequence of smaller disks. For uniqueness compare two normalized positive-support solutions at the least exponent where they differ, as in [proposition 6.2](#). [Lemma 4.2](#) constructs the inverse in the same algebra, and $H \circ F \circ H^{-1}(z) = \Lambda_{\text{dr}} z$ is then a Hahn identity. $\square$

Remark 6.4 (Two genuinely different routes to a support-controlled linearizer). [Proposition 6.2](#) and [theorem 6.3](#) prove overlapping statements by different means, and both are kept because each carries something the other does not. Route A (source 03) inverts the nonlinear Schröder operator as an *operator geometric series* [\(6.5\)](#), which is what yields the finite word-depth bound [\(6.6\)](#) and hence the radius and degree certificates [theorems 6.9](#) and [6.11](#); it requires the multiplier to be exactly Λ and works in several variables. Route B (source 02) is an exponent-by-exponent well-founded recursion [\(6.10\)](#), which is what accommodates the *drift* term $\delta_a h_\beta$; source 02 remarks that omitting that term gives the wrong recurrence and misses the obstruction of [section 8](#) entirely. A third route, source 01’s Taylor-degree recurrence with a pre-cancelled infinitesimal divisor, appears as [lemmas 9.1](#) and [9.3](#) and is what makes the *finite-return* threshold [theorem 9.5](#) exact. A fourth route, source 09’s forward plane-tree expansion (route D, [section 7.4](#)), computes the inverse coordinate H^{-1} directly as a sum over full plane trees. It exposes the branching case of the tree bound [theorem 7.5](#), and it shares that bound’s arithmetic lemmas with route A as route A is used in [section 7](#). Source 10 extends route D to several variables and nonscalar Λ by colored trees ([proposition 7.30](#)); its arithmetic is its own ([remark 7.29](#)). Source 02 also records the route-A statement in category-free form: [proposition 6.5](#).

Proposition 6.5 (Reusable abstract form). *Let B be any ordinary coefficient algebra closed under differentiation, finite products and rotation by λ , and such that the nonresonant homological inverse preserves $z^2 B$. Then every positive Hahn perturbation $F = \lambda z + \sum_a t^a f_a$ with $f_a \in zB$ admits a unique normalized linearizer with coefficients in B and support in the additive monoid generated by the input support, and its inverse has the same properties. No norm on B is required.*

Proof. This is the proof of [theorem 6.3](#) with the coefficient algebra left abstract. $\square$

The common-disk, germ, entire and polynomial rows are then obtained by checking one closure property. [Theorem 8.2](#) shows that such a closure check is sufficient but that passing to an *inductive limit* of domains cannot simply be interchanged with taking a full Hahn family: for $0 < \sigma < \infty$ every coefficient belongs to the germ inductive limit while the entire Hahn family does not belong to the common-domain inductive limit. That noncommutation is the algebraic form of the exact radius-collapse phenomenon. In the exact scalar category the interchange does hold, at the cost of exactly one factor $e^{-\tau}$ of radius ([theorem 7.13](#)), and for every exact diagonal unitary multiplier it holds at the cost of at most $e^{-r_{\text{ang}} \tau}$ ([theorem 7.23](#)); the noncommutation is a phenomenon of drift.

6.3 Necessity: the first-weight obstruction

Proposition 6.6 (The first-weight obstruction). *Fix $\eta > 0$ and an ordinary coefficient vector φ vanishing to order at least two, and let $F = \Lambda z + t^\eta \varphi(z)$. Any normalized positive conjugacy in any of the stated analytic categories has no nonzero coefficients of weight below η , and necessarily*

$$h_\eta = -\mathcal{L}^{-1}\varphi. \quad (6.11)$$

Later Hahn weights, additional scales, and enlargement of Γ cannot repair this.

Proof. If a coefficient below η were nonzero, take the least nonzero weight of h : there the right side of (6.4) vanishes and $\mathcal{N}_f h$ also vanishes because it adds at least η , leaving $\mathcal{L}h_\gamma = 0$, contradicting NR. At weight η the same reasoning eliminates $\mathcal{N}_f h$, whose input would need weight at most zero, and the equation is exactly (6.11). $\square$

Completion of the proof of theorem 6.1. For the fixed-polydisk row with $\tau > 0$, take (5.4) and $f = t^\eta \varphi$: proposition 6.6 forces a coefficient not holomorphic on the specified $\mathbb{D}_R^d$, so universal preservation of that polydisk fails. For the radius-free and entire rows with $\tau = \infty$, take the entire witness (5.5): the forced h_η has zero complex radius, so it is not even a convergent germ and certainly not entire. Higher Hahn coefficients, extra scales and enlargement of Γ cannot change the equation at this first weight. Polynomial and formal sufficiency was proved without arithmetic restrictions beyond NR, and the support and uniqueness claims are those of proposition 6.2; workspace compatibility is proposition 4.14. $\square$

Remark 6.7 (Nonresonance is not cosmetic). If $\lambda^\alpha = \lambda_j$ for some $|\alpha| \geq 2$, take $\varphi = z^\alpha e_j$: (6.11) would require $0 = -1$ in that Taylor coefficient. A universal normalized linearization statement therefore fails under MR even for a polynomial with one infinitesimal weight. What is needed is a different target equation, not division by a zero divisor; theorem 20.1 supplies one for multipliers of finite order, and question 25.3 records that the general infinite-order resonant case is open. The several-variable version of the same remark is source 02's: a first-order forcing $tz^\alpha e_j$ lies outside the range of $\mathcal{L}$ at that monomial and cannot be removed by a normalized near-identity change of variable.

Corollary 6.8 (The entire-coefficient threshold; polynomial universality). *Universal linearization of entire-coefficient perturbations by entire-coefficient normalized coordinate changes holds exactly when the relevant divisor rate is finite. Universal polynomial-coefficient linearization holds for every nonresonant multiplier.*

Proof. Positive assertions: theorem 6.1 and theorem 6.3(iii), (iv). For the negative direction with an infinite rate, take the entire forcing q of lemma 5.2 and $F = \lambda z + tq(z)$: the coefficient of t in the normalized linearizer is $-\mathcal{L}^{-1}q$, which is not even a convergent germ. $\square$

6.4 Quantitative certificates at a specified exponent

Theorem 6.9 (Radius at a specified Hahn exponent). *Suppose $\tau < \infty$ and $f = \sum_{\sigma \in S} t^\sigma f_\sigma$ has convergent-germ coefficients. For $\gamma \in S^* \setminus \{0\}$ choose positive $r_\sigma \leq \text{rad}_{\text{iso}}(f_\sigma)$ for the finite set $\sigma \in D_S(\gamma)$ and put $r_{\text{in}}(\gamma) = \min_{\sigma \in D_S(\gamma)} r_\sigma$. Then*

$$\text{rad}_{\text{iso}}(h_\gamma) \geq r_{\text{in}}(\gamma) \exp(-\ell_S(\gamma) \tau), \quad (6.12)$$

which for inputs on one $\mathbb{D}_R^d$ reads $\text{rad}_{\text{iso}}(h_\gamma) \geq R \exp(-\ell_S(\gamma) \tau)$; entire input gives entire h_γ .

Proof. Consider one term of (6.5) contributing at γ : its input letters lie in $D_S(\gamma)$; derivatives, products and composition with a diagonal unitary map do not reduce the minimum ordinary Taylor radius; each application of $\mathcal{L}^{-1}$ costs at most $e^{-\tau}$ by lemma 5.1; and a term with k copies of T uses $k + 1$ inversions and has at least $k + 1$ letters, hence at most $\ell_S(\gamma)$ inversions. Finitely many terms occur at this weight, so the minimum is preserved under their sum. $\square$

Warning 6.10 ((6.12) is a lower bound only). The single-inversion factor is sharp by lemma 5.2, but it is *not* asserted that the exponent $\ell_S(\gamma)\tau$ is ever attained by a nonlinear example; cancellations can improve the radius substantially. In particular (6.12) alone does **not** decide whether one common positive radius survives when $0 < \tau < \infty$ and word depths are unbounded, and source 03 states explicitly that its coefficientwise lower bound is not a proof that radius collapse occurs. That question is question 8.20. At the pins of sources 01–07 the only source that answered it did so in a different category, the drifted one, and section 8 keeps the two apart. For a scalar multiplier $\Lambda = \lambda I_d$, sources 08 and 09 have since shown that the depth-dependent factor is never needed: every coefficient has isotropic radius at least $Re^{-\tau}$ (theorem 7.13). For a nonscalar Λ the question remained open at the pins of sources 08 and 09, and this paragraph said so. Source 10 has since shown that every coefficient has isotropic radius at least $Re^{-r_{\text{ang}}\tau}$, with r_{ang} the rational angular rank (theorem 7.23), and that (6.12) holds with $\ell_S(\gamma)$ replaced by $\min(r_{\text{ang}}, \ell_S(\gamma))$ (corollary 7.32). What remains open is the optimal radius when $r_{\text{ang}} \geq 2$ (question 8.20).

Theorem 6.11 (Polynomial coefficient degrees). *Suppose every f_σ is polynomial and put $d_\sigma = \max_j \deg(f_{\sigma,j}) \geq 2$. Then*

$$\deg h_\gamma \leq 1 + \max_{(\sigma_1, \dots, \sigma_k) \in \mathcal{W}_S(\gamma)} \sum_{i=1}^k (d_{\sigma_i} - 1), \quad (6.13)$$

and if all $d_\sigma \leq D$ then $\deg h_\gamma \leq 1 + \ell_S(\gamma)(D - 1)$ uniformly. No small-divisor bound is assumed. For polynomial and formal coefficients the theorem and the linearization construction also hold for any diagonal invertible nonresonant Λ , dropping $|\lambda_j| = 1$.

Proof. $\mathcal{L}^{-1}$ preserves total degree; the initial term at one letter σ has degree at most $d_\sigma = 1 + (d_\sigma - 1)$. If a term for the word w has degree at most $1 + \sum_{\sigma \in w} (d_\sigma - 1)$, a term of $\mathcal{N}_f$ differentiates it $|\beta|$ times and multiplies by $|\beta|$ further coefficient polynomials, giving degree at most $1 + \sum_{\sigma \in w} (d_\sigma - 1) + \sum_{i \leq |\beta|} (d_{\sigma_i} - 1)$; a vanishing derivative contributes no term. Induct through (6.5) and take the finite maximum at γ . For polynomial and formal coefficients, composition with any invertible diagonal matrix preserves the category and the same coefficient equations apply. $\square$

Warning 6.12 (What “finite dependence” does and does not mean). Proposition 6.2 and theorems 6.9 and 6.11 bound the number of input *coefficient functions* and analytic operators needed at a specified Hahn weight. This is not an unconditional finite-time algorithm for arbitrary analytic inputs: one application of $\mathcal{L}^{-1}$ to an arbitrary analytic function can involve infinitely many of its ordinary Taylor coefficients. Only with polynomial input and an effective support certificate does each requested Hahn coefficient become a finite exact algebraic calculation; section C states exactly what must then be certified. Source 02 adds the practical corollary that a domain certificate cannot be a single global Boolean flag: for the $u^k v$ coefficient of theorem 8.2 the appropriate certificate is the exact radius $Re^{-(k+1)\sigma}$, and reporting merely “each coefficient converges” would erase the decisive failure.

6.5 Actual conjugacies on halos and on the monad

Proposition 6.13 (Actual conjugacy on a finite halo). *Under [theorem 6.1](#) or [theorem 6.3\(i\)](#), $H^\#$ is a bijection of $(\mathbb{D}_R^d)^\#_\Gamma$ with inverse $(H^{-1})^\#$; both $F^\#$ and multiplication by the multiplier are bijections of that halo; and $H^\#(F^\#(w)) = \Lambda H^\#(w)$ there. With entire or polynomial coefficients the same holds on $\mathbb{C}^\# = \mathbb{C}^d + \mathfrak{m}_\Gamma^d$.*

Proof. Hahn–Taylor evaluation respects composition whenever the reductions map the base domains into one another, by expanding both sides and regrouping at a fixed exponent ([lemma 3.4](#) reduces this to finite Taylor identities). The reductions of H, H^{-1} are the identity and preserve $\mathbb{D}_R^d$; the reduction of F and of the linear normal form is $z \mapsto \Lambda z$, which preserves $\mathbb{D}_R^d$ because $|\lambda_j| = 1$. Positive perturbations of the invertible Λ have a Hahn inverse, so the conjugate maps are bijections of the halo. $\square$

Proposition 6.14 (The monad needs no arithmetic). *For every irrational unit multiplier, the formal-coefficient solution of [theorem 6.1](#)/[theorem 6.3](#) evaluates on $\mathfrak{m}_\Gamma^d$ and yields a bijective conjugacy there, with **no** arithmetic hypothesis, even when some ordinary coefficient series have radius zero.*

Proof. At $\xi \in \mathfrak{m}_\Gamma^d$ use the formal Taylor coefficients at 0 in [\(3.4\)](#): a coefficient may be an arbitrary element of $\mathbb{C}[[z]]$, since its scalar coefficients all have valuation zero while powers of ξ carry positive-support factors, and [lemma 3.4](#) makes the family summable. The condition $F(0) = 0$ keeps $F(\xi) \in \mathfrak{m}_\Gamma^d$, and the normalized inverse and conjugacy identities survive evaluation. $\square$

[Proposition 6.14](#) is the exact scope limit of every negative result in this report: they concern analyticity of the ordinary coefficient *functions on neighborhoods*, not the absence of every form of surcomplex linearization at infinitesimal arguments. There is no assertion that every algebraic conjugacy of evaluated functions must arise from a coefficient family, and failure in one analytic category prohibits no set-theoretic conjugacy.

6.6 Several variables

For $d \geq 1$, $L = \text{diag}(\lambda_1, \dots, \lambda_d)$ with $|\lambda_j| = 1$ and NR, put

$$\Delta_n = \min_{|\alpha|=n, 1 \leq j \leq d} |\lambda^\alpha - \lambda_j| > 0, \quad \sigma_d(L) = \limsup_{n \rightarrow \infty} \frac{\log^+ \Delta_n^{-1}}{n}. \quad (6.14)$$

All several-variable assertions below are about the specified *equal-radius* polydisk geometry $\mathbb{D}_R^d$; no claim about an optimal general Reinhardt domain is made or needed.

Lemma 6.15 (Multivariable homological inverse). *$\mathcal{L}_L h = h(Lz) - Lh(z)$ is invertible on vector formal series of order at least two, with $(\mathcal{L}_L^{-1} q)_{\alpha,j} = q_{\alpha,j}/(\lambda^\alpha - \lambda_j)$. If $\sigma_d = 0$ it preserves holomorphy on every $\mathbb{D}_R^d$; if $\sigma_d < \infty$ it preserves radius-free germs and entire functions, a finite nonzero isotropic radius losing at most the factor $e^{-\sigma_d}$; and it preserves polynomial vectors under NR alone.*

Proof. $z^\alpha e_j$ is an eigenvector with eigenvalue $\lambda^\alpha - \lambda_j$. Taking maxima at total degree n gives $\max_{|\alpha|=n,j} |(\mathcal{L}_L^{-1} q)_{\alpha,j}| \leq \Delta_n^{-1} \max_{|\alpha|=n,j} |q_{\alpha,j}|$; apply the root test, the count of degree- n monomials having only polynomial growth. $\square$

Theorem 6.16 (Multivariable support-controlled lifting). *Let $F(z) = Lz + \sum_{a \in A} t^a f_a(z)$ with $f_a(0) = 0$, $A \subset \Gamma_{>0}$ well ordered, and $M = DF(0) = L + \sum_a t^a Df_a(0)$. Under NR there is a unique normalized formal coordinate change $H = z + U$ with $DU(0) = 0$ coefficientwise and*

$$H(F(z)) = MH(z), \quad \text{supp}(H - z) \subseteq A^* \setminus \{0\}. \quad (6.15)$$

If $\sigma_d = 0$ and all f_a are holomorphic on $\mathbb{D}_R^d$, then H and its inverse have coefficients holomorphic on that same polydisk; for finite σ_d , radius-free germs and entire coefficients are preserved; polynomial coefficients are preserved for every nonresonant L . The infinitesimal matrix $M - L$ need not be diagonal.

Proof. With $P = F - Lz$, $E = M - L$, $N = P - Ez$, the vector form of (6.9) is

$$\mathcal{L}_L U = -N + EU - \sum_{\beta > 0} \sum_{k \geq 1} \frac{t^\beta}{k!} D^k h_\beta(Lz)[P(z), \dots, P(z)], \quad (6.16)$$

with $D^k h$ the symmetric k -linear derivative. At a fixed Hahn exponent only finitely many terms occur by lemma 3.4, every h_β occurring has strictly smaller exponent, and the right side has total Taylor order at least two by the order count of theorem 6.3. Apply lemma 6.15 by well-founded recursion; the operations preserve the common polydisk because L does. The least-exponent argument proves uniqueness, and the inverse-coordinate recursion is vectorial with coefficient one on the new unknown, so lemma 4.2 applies with multilinear derivatives. $\square$

As in proposition 6.13, common-polydisk solutions evaluate to actual conjugacies on $(\mathbb{D}_R^d)_\Gamma^\#$.

Theorem 6.17 (A common coordinate for the centralizer). *Suppose $d = 1$, $\sigma(\lambda) = 0$, and F satisfies theorem 6.3(i) on $\mathbb{D}_R$ with normalized linearizer H . Let $G(z) = \mu z + \sum_{b \in B} t^b g_b(z)$ with $|\mu| = 1$, $g_b \in z\mathcal{O}(\mathbb{D}_R)$, $B \subset \Gamma_{>0}$ well ordered, and suppose $G \circ F = F \circ G$. Then, with $M = G'(0)$,*

$$H \circ G \circ H^{-1}(z) = Mz, \quad (6.17)$$

*with **no** nonresonance or small-divisor assumption on μ . Consequently every such G commuting with F is linearized by the same H , any two such maps commute with each other, and $G'(0)$ determines G uniquely — in particular $G'(0) = 1$ forces $G = \text{id}$.*

Proof. All compositions are legitimate on $\mathbb{D}_R$ by lemma 3.4 and $|\lambda| = |\mu| = 1$. Put $K = H \circ G \circ H^{-1}$: it commutes with $z \mapsto \Lambda_{\text{dr}} z$, fixes zero and has derivative M there. Expanding $K(z) = \sum_{n \geq 1} k_n z^n$ and comparing coefficients in $K(\Lambda_{\text{dr}} z) = \Lambda_{\text{dr}} K(z)$ gives $(\Lambda_{\text{dr}}^n - \Lambda_{\text{dr}})k_n = 0$ for $n \geq 2$; the factor is a nonzero Hahn-field element because its exponent-zero coefficient is $\lambda^n - \lambda \neq 0$. Hence $k_n = 0$ for $n \geq 2$ and $k_1 = M$; at each Hahn exponent the coefficient function has the Taylor series of a linear function, so the identity theorem on $\mathbb{D}_R$ yields $K(z) = Mz$ as a common-domain section. Conjugating back and using that two scalar multiplications commute gives the remaining assertions. $\square$

The same proof works in the radius-free germ category when $\sigma < \infty$, and in the polynomial setting whenever the constructed linearizers have polynomial coefficients; in those statements equality is interpreted in the indicated category rather than on a disk that has not been shown to exist. A set-indexed commuting family is treated one member at a time with the same H ; no unsupported sum over the family is taken.

7 The exact scalar multiplier: one radius loss

This section is contributed by sources 08 and 09, written on 22 September 2026 against repository commit `a3124af79f66b8b9c196d76b4cbc5ac3938907c4`, after sources 01–07 had been merged. It answers [question 8.20](#) for a *scalar* exact multiplier $\Lambda = \lambda I_d$: source 08 in every finite dimension d , source 09 in one variable. The answer is positive and comes with a sharp constant. The common-domain germ category, which [theorem 6.1](#) left undecided for $0 < \tau < \infty$, costs exactly one factor $e^{-\tau}$ of radius, however deep the Hahn words are. Sources 08 and 09 do *not* settle the nonscalar diagonal case of [question 8.20](#) ([section 7.6](#)). Source 10, merged later, settles it up to the optimal radius, in [section 7.7](#) and the five subsections after it, which close this section.

Tags. Resonance NR, with the multiplier **exactly** λI_d ; divisor notion SD_{ex} ; category the common-domain germ ring $\mathcal{R}_d$ of [\(3.2\)](#), with a specified radius; domain notion $\mathbb{D}_R^d$ and rad_{iso} . No drift is permitted anywhere in this section. The drifted counterpart, which has the opposite answer, is [section 8](#), and in particular [section 8.6](#).

Convention 7.1 (Setting, rate, and the two coordinate conventions). (i) Let $\lambda = e^{2\pi i\theta}$ with θ irrational. For the scalar matrix $\Lambda = \lambda I_d$ this is exactly NR, because $\lambda^\alpha - \lambda = \lambda(|\lambda|^\alpha - 1)$. Write

$$\mathfrak{d}_n = |\lambda^n - 1| \quad (n \geq 1), \quad \tau = \tau(\lambda I_d) = \limsup_{n \rightarrow \infty} \frac{\log^+ \mathfrak{d}_n^{-1}}{n} \in [0, \infty]. \quad (7.1)$$

Since $|\lambda^\alpha - \lambda_j| = \mathfrak{d}_{|\alpha|-1}$ when $\Lambda = \lambda I_d$, this is the rate $\tau(\Lambda)$ of [section 2.3](#); the index shift does not change the limsup. In one variable [proposition 8.8](#) makes it numerically equal to $\sigma(\lambda)$, but the hypotheses that accompany the two rates remain different ([warning 2.6](#)). The divisor $d_n = \lambda^n - \lambda$ of [section 8](#) satisfies $|d_{n+1}| = \mathfrak{d}_n$.

- (ii) The value group Γ is any nonzero set-sized ordered abelian group. No divisibility, rank, countability or algebraic-closure hypothesis is made: source 08 notes that algebraic closedness of the workspace is used nowhere, and source 09 excludes Archimedean, divisibility, countability and finite-rank hypotheses. With $S \subset \Gamma_{>0}$ well ordered, the input is

$$F(z) = \lambda z + f(z), \quad f = \sum_{s \in S} t^s f_s, \quad f_s \in \mathcal{O}(\mathbb{D}_R^d)^d, \quad f_s(0) = 0, \quad Df_s(0) = 0, \quad (7.2)$$

the last two conditions holding coefficientwise. This is [\(6.1\)](#) with $\Lambda = \lambda I_d$ and $B = \mathcal{O}(\mathbb{D}_R^d)$. No common bound on the sizes of the f_s is assumed.

- (iii) *Conjugacy convention.* As everywhere in this report, $H = z + h$ and $H \circ F = \lambda H$ ([\(6.2\)](#)); source 08 uses this convention. Source 09 works in the *forward* convention $F \circ \check{H} = \check{H}(\lambda \cdot)$. Its forward coordinate is our $\check{H} = H^{-1} = z + \check{h}$, and its inverse coordinate is our H . Every formula taken from source 09 below has been translated into this notation. In particular its first sharp coefficient is our $\check{h}_\eta = -h_\eta$, and what it calls the linearizer is our H^{-1} .
- (iv) *Local notation, chosen to avoid collisions.* $\|x\|_{\mathbb{Z}}$ is the distance from x to the nearest integer. q_* and p_*/q_* are the cluster denominator and fraction of [lemma 7.2](#); they are neither the order q of RES_q nor a convergent denominator q_j . A forest has m vertices, positive integer weights w_v , subtree sums s_v and total weight n , so that $n+1$ is a Taylor degree; n is *not* the nonlinear part N of [theorem 6.3](#). The numbers $a > \tau$ and C_a are those of [\(7.5\)](#), and $r < R$ is an auxiliary ordinary radius, not the finite-return threshold r of [section 9](#). The radius $Re^{-\tau}$ is always written out; ρ keeps its meaning from [section 2.4](#).

7.1 Exceptionally small divisors share one rational direction

As in the proof of [proposition 8.8](#),

$$4\|n\theta\|_{\mathbb{Z}} \leq \mathfrak{d}_n \leq 2\pi\|n\theta\|_{\mathbb{Z}}. \quad (7.3)$$

Lemma 7.2 (One rational cluster; sources 08 and 09). *Let $n \geq 1$, and call $s \in \{1, \dots, n\}$ marked if $\|s\theta\|_{\mathbb{Z}} < 1/(4n)$. If some s is marked, there is a reduced fraction p_*/q_* with $1 \leq q_* \leq n$ such that every marked s is a multiple $s = kq_*$ whose nearest integer to $s\theta$ is kp_* , and*

$$\|q_*\theta\|_{\mathbb{Z}} < \frac{1}{4n}, \quad \|s\theta\|_{\mathbb{Z}} = k\|q_*\theta\|_{\mathbb{Z}}, \quad \mathfrak{d}_s \geq \mathfrak{d}_{q_*}.$$

Proof. Let p_s be the nearest integer to $s\theta$. For marked s, s' the integer $p_s s' - p_{s'} s$ has absolute value at most $s'|p_s - s\theta| + s|p_{s'} - s'\theta| < (s + s')/(4n) \leq \frac{1}{2}$, so it vanishes. All the fractions p_s/s therefore equal one reduced fraction p_*/q_* , and $s = kq_*$, $p_s = kp_*$. Then $|q_*\theta - p_*| = |s\theta - p_s|/k < 1/(4n)$, so p_* is the nearest integer to $q_*\theta$ and $\|s\theta\|_{\mathbb{Z}} = k\|q_*\theta\|_{\mathbb{Z}} \geq \|q_*\theta\|_{\mathbb{Z}}$. Both distances are below $\frac{1}{4}$, where $x \mapsto |e^{2\pi i x} - 1| = 2\sin(\pi x)$ is increasing, so $\mathfrak{d}_s \geq \mathfrak{d}_{q_*}$. $\square$

Source 09 proves the lemma in this form. Source 08 uses the cutoff $\mathfrak{d}_s < n^{-3}$ with the same determinant argument and obtains $\mathfrak{d}_{kq_*} \geq (2/\pi)k\mathfrak{d}_{q_*}$. By its own account that cutoff is convenient rather than optimal, and it costs a factor n^3 for each unmarked divisor where the cutoff above costs n . It is one argument, printed once.

Lemma 7.3 (Divisible-subtree packing; sources 08 and 09). *Let a nonempty finite rooted forest carry positive integer vertex weights w_v , with subtree sums s_v (the subtree of v contains v) and total weight n . For every integer $q \geq 1$ and every set A of vertices with $q \mid s_v$ for all $v \in A$,*

$$q|A| \leq n. \quad (7.4)$$

Proof. For $v \in A$ let $C(v)$ be the set of closest strict descendants of v that lie in A , and let B_v be the vertex set of the subtree of v with the subtrees of the members of $C(v)$ removed. B_v contains v , so its weight $b_v = s_v - \sum_{u \in C(v)} s_u$ is positive; it is divisible by q , hence $b_v \geq q$. A vertex lies in B_v only for its closest ancestor-or-self in A , so the sets B_v are pairwise disjoint and $\sum_{v \in A} b_v \leq n$. $\square$

Both sources give this proof; source 08 phrases it through the forest on the marked vertices. It is printed once.

Warning 7.4 (Strict positivity is exactly what drift destroys). With a zero weight [lemma 7.3](#) fails. A two-vertex chain with weights 0 and q has both subtree sums equal to q and total weight q (source 09). A chain of m vertices whose only positive weight n sits at its leaf has every subtree sum equal to n , so its divisor product is $\mathfrak{d}_n^{-m}$, of exponential rate $m\tau$ (source 08). Both sources identify this as the combinatorial model of a *linear* multiplier perturbation, since a vertex of arity one consumes no Taylor degree. It is the mechanism behind [theorem 8.2](#) and [lemma 8.11](#).

For $\tau < \infty$ and $a > \tau$ put

$$C_a = \max\left(1, \sup_{s \geq 1} e^{-as} \mathfrak{d}_s^{-1}\right) < \infty, \quad \text{so that} \quad \mathfrak{d}_s^{-1} \leq C_a e^{as}. \quad (7.5)$$

Theorem 7.5 (Tree product bound; source 09). *Let $\tau < \infty$ and $a > \tau$. For every nonempty finite rooted forest with m vertices, positive integer weights and total weight n ,*

$$\prod_v \mathfrak{d}_{s_v}^{-1} \leq (C_a n)^m e^{an}. \quad (7.6)$$

Proof. Mark v when $\|s_v\theta\|_{\mathbb{Z}} < 1/(4n)$; since $1 \leq s_v \leq n$ this is the marking of [lemma 7.2](#). An unmarked factor is at most n by [\(7.3\)](#). If $M \geq 1$ vertices are marked, [lemma 7.2](#) supplies q_* dividing all their s_v and bounds each marked factor by $\mathfrak{d}_{q_*}^{-1} \leq C_a e^{aq_*}$, and [lemma 7.3](#) gives $Mq_* \leq n$. Hence the product is at most $n^{m-M} C_a^M e^{aMq_*} \leq (C_a n)^m e^{an}$. $\square$

The small divisors are paid for once, by disjoint blocks of Taylor weight, and not once per nested subtree along a path. That is the gain over applying [lemma 5.1](#) at every inversion. The inequality holds for every m , but $(C_a n)^m$ is polynomial in n only for fixed m . Source 09 makes no assertion that it is harmless uniformly in m .

Corollary 7.6 (Strict chains and the fixed-complexity rate; sources 08 and 09). *(i) If $\tau < \infty$ and $a > \tau$, every strictly increasing chain $1 \leq k_1 < \dots < k_r \leq n$ satisfies*

$$\prod_{j=1}^r \max(1, \mathfrak{d}_{k_j}^{-1}) \leq (C_a n)^r e^{an}. \quad (7.7)$$

(ii) Let $\tau \in [0, \infty]$ and fix $m \geq 1$. Let $A_m(n)$ be the maximum of $\prod_v \mathfrak{d}_{s_v}^{-1}$ over rooted trees of total weight n with at most m vertices. Then, in the extended sense,

$$\limsup_{n \rightarrow \infty} \frac{\log^+ A_m(n)}{n} = \tau. \quad (7.8)$$

The same holds for trees with exactly m vertices and $n \geq m$, and for the maximum over chains of length at most m in [\(7.7\)](#). The exponential rate at fixed complexity is τ , not $m\tau$.

Proof. (i) The factors exceeding one form a subchain $k_{j_1} < \dots < k_{j_{r'}}$. It is the list of subtree sums of the path whose leaf has weight k_{j_1} and whose successive vertices towards the root have weights $k_{j_{i+1}} - k_{j_i} > 0$. Its total weight is at most n , so [theorem 7.5](#) bounds the product by $(C_a n)^{r'} e^{an} \leq (C_a n)^r e^{an}$. (ii) If $\tau < \infty$, [theorem 7.5](#) gives $n^{-1} \log A_m(n) \leq a + m \log(C_a n)/n$; let $n \rightarrow \infty$ and then $a \downarrow \tau$. The one-vertex tree of weight n has product $\mathfrak{d}_n^{-1}$, whose exponential rate is τ . The star with $m-1$ leaves of weight one and root weight $n-m+1$ has product $\mathfrak{d}_n^{-1} \mathfrak{d}_1^{-(m-1)}$, with the same rate. These lower bounds also cover $\tau = \infty$. The one-element chain $k_1 = n$ does the same for chains. $\square$

Source 08 proves (i) with a different constant and the polynomial factor n^{3r} , which comes from its cutoff, and proves a tree version with repeated subtree sums allowed. The forest form and the sharper polynomial factor are source 09's.

Remark 7.7 (Which uniformity is obtained, and which is not). The exponential constant a does not depend on the number of vertices; the polynomial factor does. That suits Hahn coefficients, where the number of vertices is finite at each exponent but unbounded over the support. It gives no estimate for summing all depths in an ordinary complex parameter; both sources say so. Source 09 makes the point concrete. For $\lambda z + t^\eta z^2$ every internal vertex of the tree expansion of [proposition 7.17](#) has weight one, so $m = n$ and $(C_a n)^m = (C_a n)^n$. The estimate then gives no ordinary analytic convergence, and no hidden uniform estimate lets the two quantifier orders be exchanged. This agrees with [proposition 21.1](#).

Remark 7.8 (Antecedents of the arithmetic). Grouping the exceptionally small divisors under one rational approximant, and paying for repeated small divisors with disjoint pieces of degree, is the same kind of argument as classical small-divisor counting. Examples are the Davie estimates

discussed by Marmi [12] and the tree expansions of Fauvet, Menous and Sauzin [16, 18] with their Bruno estimates. Both sources credit these. Neither claims the framework or the tree method, and neither certifies priority. Source 08 inspected [16] at abstract and metadata level only, and both describe their literature comparison as targeted (N94, N105). What they offer is the fixed-complexity form above. Polynomial losses are allowed there because Hahn support keeps the number of vertices finite before any ordinary Taylor convergence is considered.

7.2 Why the divisors are distinct: strict degree chains

Use route A, [proposition 6.2](#), with $\Lambda = \lambda I_d$. On a coordinate monomial the homological operator acts by $\mathcal{L}(z^\beta e_j) = (\lambda^{|\beta|} - \lambda)z^\beta e_j$. So an inversion at total degree $n + 1$ divides by a number of modulus $\mathfrak{d}_n$, and this number depends only on n . An *expanded term* of (6.5) consists of one choice of each of the following: a support word; a grouping of it into the Taylor blocks of $\mathcal{N}_f$ ((6.3)); the coordinate component of every perturbation factor; and one ordinary Taylor monomial of each such factor.

Lemma 7.9 (Strict degree chains; source 08). *Consider an expanded term with m perturbation factors that contributes to a coordinate monomial of total degree $n + 1$.*

- (i) *Its input Taylor degrees $n_1, \dots, n_m$ are at least two and satisfy $\sum_j n_j = n + m$.*
- (ii) *Its total order of differentiation is $m - 1$.*
- (iii) *Its homological divisors have moduli $\mathfrak{d}_{k_1}, \dots, \mathfrak{d}_{k_r}$ along a strictly increasing chain $1 \leq k_1 < \dots < k_r = n$ with $r \leq m$.*

Proof. Initially the degree is n_1 and one divisor is used. A Taylor block of order $k = |\kappa|$ differentiates the current monomial k times and multiplies it by k new perturbation monomials of degrees $n_{i+1}, \dots, n_{i+k}$. The net change of degree is $\sum_{l=1}^k (n_{i+l} - 1) \geq k \geq 1$, so the next inversion takes place at a strictly larger degree. At the end $n + 1 = n_1 + \sum_{j \geq 2} (n_j - 1) = 1 + \sum_{j=1}^m (n_j - 1)$, which is (i). The orders of all blocks add up to $m - 1$, which is (ii). Each new inversion comes with at least one new letter, which is (iii). $\square$

The hypothesis $Df_s(0) = 0$ is used exactly in $n_j - 1 \geq 1$ (source 08; source 09's audit, [table 5](#)). With $f_s(0) = 0$ alone, factors of degree one give zero increments and the chain is no longer strict. That is [warning 7.4](#) again.

Example 7.10 (Two divisors at the second order; sources 08 and 09). Let $d = 1$, $F = \lambda z + t^n g$ with $g = \sum_{k \geq 2} g_k z^k$, and $H = z + \sum_{j \geq 1} t^j u_j$. Then $\mathcal{L}u_1 = -g$, $\mathcal{L}u_2 = -g u'_1(\lambda z)$, and

$$u_2(z) = \sum_{i, k \geq 2} \frac{k \lambda^{k-1} g_i g_k}{(\lambda^k - \lambda)(\lambda^{i+k-1} - \lambda)} z^{i+k-1}. \quad (7.9)$$

The two divisors have net indices $k - 1$ and $n = i + k - 2$, which are distinct because $i \geq 2$. If both are marked at total weight n , [lemma 7.2](#) gives a q_* dividing both, and hence dividing $i - 1 \geq 1$ as well; so $2q_* \leq (k - 1) + (i - 1) = n$. The dangerous pair is paid for by the two positive pieces $k - 1$ and $i - 1$, not twice by n . Iterating one-step estimates would lose $e^{-\tau}$ twice; the chain estimate loses it once. Source 09 makes the analogous computation for the forward coordinate $\tilde{H}$, and [section 22.1](#) records both sources' quadratic coefficients.

7.3 One common radius at every Hahn depth

Proposition 7.11 (A depth-uniform exponential rate; source 08). *Assume (7.2) and $\tau < \infty$. For every $\gamma \in S^* \setminus \{0\}$, every $0 < r < R$ and every $a > \tau$ there is a finite $A_{\gamma,r,a}$ with*

$$\max_{\substack{|\beta|=n+1 \\ 1 \leq j \leq d}} |[z^\beta]h_{\gamma,j}| \leq A_{\gamma,r,a} (n+1)^{(d+4)\ell_S(\gamma)} r^{-n} e^{an} \quad (n \geq 1). \quad (7.10)$$

The exponent a does not depend on γ or on its word depth.

Proof. Only the finitely many words of $\mathcal{W}_S(\gamma)$ contribute (proposition 6.2). For each letter s occurring in them put $M_s(r) = \max_j \sup_{\mathbb{D}_r^d} |f_{s,j}| < \infty$, so that $|[z^\nu]f_{s,j}| \leq M_s(r)r^{-|\nu|}$ by Cauchy's estimate. Fix a word of length m and a grouping into Taylor blocks. By lemma 7.9(i), the input coefficients of a contribution of final degree $n+1$ have product at most $r^{-(n+m)} \prod_s M_s(r)$, the product running over the letters of the word with multiplicity. Every intermediate degree is at most $n+1$, so a derivative of order k together with its factorial denominator contributes at most $(n+1)^k$; by (ii) these factors multiply to at most $(n+1)^{m-1}$. Composition with λz contributes factors of modulus one. The m input multi-indices have dm entries summing to $n+m$, so there are at most $\binom{n+m+dm-1}{dm-1} \leq C_{d,m}(n+1)^{dm-1}$ of them. The coordinate choices and the groupings are finite in number, independently of n . By (iii) and corollary 7.6(i), the inverse divisors multiply to at most $(C_a n)^m e^{an} \leq C_a^m n^{3m} e^{an}$; the right-hand form is the one source 08 uses. The total polynomial exponent is therefore at most $(m-1) + (dm-1) + 3m = (d+4)m - 2$. Summing over the finitely many words and groupings gives (7.10). The bound applies to the sum of absolute values, so it covers cancellations too. $\square$

Corollary 7.12 (One open polydisk for every coefficient; source 08). *Every h_γ lies in $\mathcal{O}(\mathbb{D}_{Re^{-\tau}}^d)$.*

Proof. Fix $r < R$, $a > \tau$ and $0 < b < re^{-a}$. There are $\binom{n+d}{d-1}$ monomials of total degree $n+1$, polynomially many, so (7.10) gives a normally convergent Taylor series on the closed polydisk of radius b . Letting $r \uparrow R$ and $a \downarrow \tau$ exhausts $\mathbb{D}_{Re^{-\tau}}^d$. The Taylor coefficients do not change, so they define one holomorphic function on the union, and the union does not depend on γ . $\square$

Only the *open* polydisk is claimed. There is no boundedness and no continuation across its boundary, and the constants and polynomial exponents in (7.10) may grow without bound as γ varies (source 08). A common domain of holomorphy is not a common norm bound (source 09; compare warning 10.3).

Theorem 7.13 (Sharp common domain for an exact scalar multiplier; sources 08 and 09). *Let Γ be any nonzero set-sized ordered abelian group, $d \geq 1$, $\lambda = e^{2\pi i \theta}$ with θ irrational, and $\tau = \tau(\lambda I_d) < \infty$. For every input (7.2), the unique normalized positive conjugacy $H = z + h$ of theorem 6.1, $H \circ F = \lambda H$, and its inverse $H^{-1} = z + \check{h}$ satisfy*

$$h_\gamma, \check{h}_\gamma \in \mathcal{O}(\mathbb{D}_{Re^{-\tau}}^d)^d, \quad \text{ord}_0 h_\gamma, \text{ord}_0 \check{h}_\gamma \geq 2, \quad \text{supp } h, \text{supp } \check{h} \subseteq S^* \setminus \{0\}. \quad (7.11)$$

The radius $Re^{-\tau}$ is the optimal universal common radius. For every $\eta \in \Gamma_{>0}$ the input $\lambda z + t^\eta P_R(z_1)e_1$, with P_R as in (5.6), has coefficient h_η of isotropic Taylor radius exactly $Re^{-\tau}$. If $\tau = \infty$ the same input has h_η of radius zero, so it has no normalized conjugacy with even separately convergent coefficient germs. When $\tau < \infty$, H and H^{-1} lie in the common-domain germ ring $\mathcal{R}_d$ and evaluate to mutually inverse bijective conjugacies of the finite halo $(\mathbb{D}_{Re^{-\tau}}^d)_\Gamma^\#$.

Proof. [Proposition 6.2](#), applied in the formal row, gives the unique normalized positive conjugacy with support in $S^* \setminus \{0\}$. [Corollary 7.12](#) shows that each of its coefficients is holomorphic on $\mathbb{D}_{Re^{-\tau}}^d$. At a fixed Hahn exponent, $H \circ F$ is computed by (4.3) as a finite expression ([lemma 3.4](#)) in functions holomorphic on that polydisk, which λ preserves. Its Taylor series agrees with the formal one, so the conjugacy identity holds there by the identity theorem. [Proposition 4.1](#), with $U = \mathbb{D}_{Re^{-\tau}}^d$, gives the inverse on the same connected domain, with no second loss of radius and with support in $S^* \setminus \{0\}$; it is normalized as in [proposition 6.2](#). Source 08's inverse lemma and source 09's same-domain inversion are this statement and are not reprinted. Uniqueness holds in the formal category, and therefore a fortiori in the analytic one.

For optimality, [proposition 6.6](#) gives

$$h_\eta = -\mathcal{L}^{-1}(P_R(z_1)e_1) = -\sum_{n \geq 2} \frac{R^{2-n}}{\lambda^n - \lambda} z_1^n e_1,$$

whose degree- n coefficient has modulus $R^{2-n}\mathfrak{d}_{n-1}^{-1}$. Because $\mathfrak{d} \leq 2$, the negative parts of $\log \mathfrak{d}^{-1}$ are bounded, and the root test gives radius $Re^{-\tau}$, including the value 0 when $\tau = \infty$. A coefficient holomorphic on a larger common polydisk would restrict on the z_1 -axis to an extension of this series with a larger Taylor radius, which is impossible. The halo statement follows from the proof of [proposition 6.13](#) with $\mathbb{D}_{Re^{-\tau}}^d$ in place of $\mathbb{D}_R^d$: the reductions of H and H^{-1} are the identity, and that of F is $z \mapsto \lambda z$, which preserves the polydisk. $\square$

“Unique” means unique among tangent-to-identity normalized formal coordinates and their coherent coefficientwise analytic realizations. It does not refer to arbitrary set-theoretic maps (source 09). In source 09's normalization $g_R = P_R/R = z^2/(R - z)$, the one-variable witness has forward coefficient

$$\check{h}_\eta = \lambda^{-1} \sum_{n \geq 1} \frac{R^{-n} z^{n+1}}{\lambda^n - 1} = -h_\eta,$$

with the same radius $Re^{-\tau}$. The minus sign is the convention flip of [convention 7.1\(iii\)](#). For unequal input radii $R_1, \dots, R_d$, a diagonal change of ordinary coordinates commutes with λI_d , and the same proof gives the common polydisk with radii $R_j e^{-\tau}$ (source 08). This is a rescaling corollary. It says nothing about optimal Reinhardt domains.

Corollary 7.14 (The sixth category for a scalar exact multiplier; sources 08 and 09). *Fix a nonzero Γ , $\lambda = e^{2\pi i \theta}$ with θ irrational, $d \geq 1$ and $0 < R < \infty$. Consider inputs (6.1) with $\Lambda = \lambda I_d$ whose coefficients lie in the indicated category; for the second row, holomorphic on one common polydisk $\mathbb{D}_R^d$ as in (7.2). The following conditions are necessary and sufficient for the normalized conjugacy and its inverse of every such input to lie in that category.*

Coefficient category	Exact condition
the same fixed $\mathcal{O}(\mathbb{D}_R^d)$ (row one of theorem 6.1)	$\tau = 0$
some common polydisk, i.e. $\mathcal{R}_d$ (the sixth category)	$\tau < \infty$; largest universal radius $Re^{-\tau}$
separate convergent germs, $\mathcal{A}_d$ (row two)	$\tau < \infty$
entire coefficients (row three)	$\tau < \infty$
polynomial and formal coefficients (rows four and five)	NR only

Proof. The second row is [theorem 7.13](#): sufficiency and the radius come from the theorem, and necessity from its $\tau = \infty$ witness, whose first coefficient is not even a germ. The other rows are [theorem 6.1](#) for $\Lambda = \lambda I_d$. $\square$

Only the second row is new, and both sources say so. Source 08 prints the first three rows in dimension d . Source 09 prints a six-row table in one variable that adds the entire, polynomial and formal rows. Its necessity witness for the entire row is the one in [lemma 5.2](#), and its polynomial-row argument is that of [theorem 6.1](#). Neither is reprinted. Source 08's corollary on entire and polynomial layers restates rows three and four. The table shows that, for a scalar exact multiplier, requiring one common germ imposes *no stronger* arithmetic condition than requiring separate germs (source 08). Under drift the answer is the opposite ([corollary 8.17](#)), and the two answers are kept in separate rows ([remark 2.2](#) and [table 7](#)).

Remark 7.15 (What is cited rather than reprinted). Both sources restate the machinery of [sections 3](#) and [6](#), and it is cited here. That covers their strong-summability definitions and coefficient-ring tables ([definition 3.1](#), [section 3.2](#)); the positive-support lemma ([lemma 3.4](#)); the formal existence and uniqueness of the normalized conjugacy ([proposition 6.2](#)); same-domain inversion ([proposition 4.1](#) and [lemma 4.2](#)); halo evaluation and the bijective halo conjugacy ([propositions 3.11](#) and [6.13](#)); workspace invariance ([proposition 4.14](#)); and evaluation on the monad ([proposition 6.14](#)). Source 09 proves the positive-support lemma by the minimal-bad-sequence argument of [section A](#). Source 08 takes a second route: it reduces the lemma to Poonen's strong geometric-series theorem [[4](#), Thm. 4.1] over the free associative algebra $\mathbb{Z}\langle X_s : s \in S \rangle$. There every ordered word of weight γ occurs in $\sum_m a^m$, $a = \sum_s X_s t^s$, with a positive integer coefficient, so no cancellation can hide infinitely many words. The continued-fraction material of both sources is [proposition 8.8](#) and [theorem 8.9\(b\)–\(d\)](#). Source 09's rate-zero non-Bruno rotation, $a_{j+1} = \lceil e^{q_j/(j+1)} \rceil$, is [theorem 8.9\(b\)](#) exactly.

Remark 7.16 (No conflict with Brjuno–Yoccoz or with [proposition 11.1](#)). The Brjuno condition implies $\tau = 0$, whereas [theorem 7.13](#) allows every finite τ . The two statements concern different kinds of convergence: Brjuno–Yoccoz concerns an ordinary germ whose whole nonlinear Taylor series is present at once. Here each Hahn coefficient is constructed before any attempt to sum over perturbation orders as ordinary numbers, and its norm may grow arbitrarily fast with the exponent (sources 08 and 09; [remark 8.10](#)). [Proposition 11.1](#) is not a counterexample either. Its nonlinear part X^2 sits at Hahn exponent zero, whereas [\(7.2\)](#) requires strictly positive support. Neither source reproves or alters Brjuno–Yoccoz theory, and both cite it as external.

7.4 Route D: plane trees in the forward coordinate

Source 09's main proof is a *fourth* route to a support-controlled linearizer, kept and marked under [convention 1.1\(ii\)](#) because it is genuinely different from routes A–C ([remark 6.4](#)). It works in one variable. It computes the forward coordinate $\check{H} = H^{-1}$ directly, organizes it by full plane trees with slot multiplicities, and uses the branching (forest) case of [theorem 7.5](#), whereas route A uses only the chain case. Source 09 also gives an operator-chain proof in the inverse convention (its Appendix B). That proof is route A combined with the chain case of [theorem 7.5](#): for $d = 1$ it gives the bound of [proposition 7.11](#) with polynomial exponent $3m - 2$ and combinatorial constant 2^{m-1} . It is not printed again. The two routes **share** the arithmetic of [lemmas 7.2](#) and [7.3](#), so they are not two independent proofs of it ([N107](#)). Their value is that they expose different possible errors: slot multiplicities in the forward tree formula, and derivative factors and signs in the inverse operator expansion. Source 09's program compares both ([section 24](#)).

In one variable, $F \circ \check{H} = \check{H}(\lambda z)$ with $\check{H} = z + \check{h}$ is equivalent to

$$\mathcal{L}\check{h} = f(z + \check{h}). \quad (7.12)$$

At Taylor degree n the right side involves only Taylor coefficients of $\check{h}$ of degree below n . Under NR, $\check{h}$ therefore exists and is unique, with support in $S^* \setminus \{0\}$ (source 09's formal proposition, whose

Hahn-support part is [lemma 3.4](#)). Expand $f_s(z) = \sum_{k \geq 2} a_{s,k} z^k$. A *full plane tree* is a finite rooted tree with ordered children in which every internal vertex has at least two children; its leaves stand for z . An internal vertex v with k_v children carries a label $\sigma_v \in S$. Put $w_v = k_v - 1 \geq 1$, let s_v be the sum of w_u over the internal vertices u of the subtree of v , and set $n = \sum_v w_v$ and $\gamma(T) = \sum_v \sigma_v$. The subtree at v has $s_v + 1$ leaves.

Proposition 7.17 (Plane-tree coefficient formula; source 09).

$$[t^\gamma z^{n+1}] \check{h} = \sum_{\substack{T: \gamma(T)=\gamma \\ \sum_v w_v=n}} \prod_{v \text{ internal}} \frac{a_{\sigma_v, k_v}}{\lambda^{s_v+1} - \lambda}. \quad (7.13)$$

For fixed γ and n the sum is finite, over trees with at most $\ell_S(\gamma)$ internal vertices.

Proof. A term $a_{\sigma,k} t^\sigma (z + \check{h})^k$ on the right of (7.12) is a root with k ordered children, each of which is either the leaf z or a previously constructed contribution of $\check{h}$. The inversion at the resulting degree $s + 1$ contributes $1/(\lambda^{s+1} - \lambda)$. Recursion on the nontrivial children gives the product. A plane tree has a unique root decomposition, so each tree is counted exactly once; the children are ordered, so there are no symmetry factors. Reading the internal labels in preorder gives a word in $\mathcal{W}_S(\gamma)$. By [lemma 3.4](#) there are finitely many such words, of length at most $\ell_S(\gamma)$, and there are finitely many shapes with $n + 1$ leaves. $\square$

This is a direct form of the classical tree formulas [16]. Source 09 claims only their regrouping by Hahn exponent and the estimate below, not the tree representation.

Lemma 7.18 (Polynomial tree count; source 09). *For $1 \leq m \leq n$, the number of unlabeled full plane trees of weight n with m internal vertices is at most $4^{m-1} n^{m-1} (n+1)^{m-1}$. With labels summing to γ it is at most that number times $W_{\gamma,m} = \#\{w \in \mathcal{W}_S(\gamma) : |w| = m\}$.*

Proof. Deleting the leaves leaves a plane skeleton on the m internal vertices. There are at most 4^{m-1} such skeletons: encode the depth-first traversal of their $m - 1$ edges by $2(m - 1)$ up and down symbols. There are $\binom{n-1}{m-1} \leq n^{m-1}$ positive weight assignments summing to n . Given the weights, $k_v = w_v + 1$, and the d_v internal children of v occupy one of at most $\binom{w_v+1}{d_v} \leq (n+1)^{d_v}$ sets of slots, with $\sum_v d_v = m - 1$. A label word in preorder assigns the labels in one way. $\square$

Only polynomiality at fixed m matters. An overcount exponential in n would lose the sharp radius (source 09).

Proposition 7.19 (Finite-depth estimate for the forward coordinate; source 09). *Let $d = 1$, $\tau < \infty$, $\gamma \in S^* \setminus \{0\}$, $0 < r < R$ and $a > \tau$, and put $B_\gamma(r) = \max_{\sigma \in D_S(\gamma)} \max_{|z| \leq r} |f_\sigma(z)|/r$. For $n \geq 1$,*

$$|[z^{n+1}] \check{h}_\gamma| \leq \left(\frac{e^a}{r}\right)^n \sum_{1 \leq m \leq \min(\ell_S(\gamma), n)} W_{\gamma,m} 4^{m-1} (C_a B_\gamma(r))^m (n+1)^{3m-2} \leq K_{\gamma,r,a} (n+1)^{3\ell_S(\gamma)} \left(\frac{e^a}{r}\right)^n, \quad (7.14)$$

where $K_{\gamma,r,a} = \sum_{m=1}^{\ell_S(\gamma)} W_{\gamma,m} 4^{m-1} (C_a B_\gamma(r))^m$. Consequently $\text{rad}(\check{h}_\gamma) \geq Re^{-\tau}$ for every γ , and $H = \check{H}^{-1}$ has its coefficients on the same disk ([lemma 4.2](#)). This is a second proof of [theorem 7.13](#) for $d = 1$.

Proof. By Cauchy's estimate, $\prod_v |a_{\sigma_v, k_v}| \leq \prod_v r^{-k_v} \max_{|z| \leq r} |f_{\sigma_v}| \leq r^{-n} B_\gamma(r)^m$, because $\sum_v k_v = n + m$. Also $|\lambda^{s_v+1} - \lambda| = \mathfrak{d}_{s_v}$, so [theorem 7.5](#) bounds the divisor product by $(C_a n)^m e^{an}$. Multiply by the count of [lemma 7.18](#) and use $n^{2m-1}(n+1)^{m-1} \leq (n+1)^{3m-2}$. The root test at fixed γ , followed by $r \uparrow R$ and $a \downarrow \tau$, gives the radius. $\square$

The same r and a serve at every exponent, while the constants and the polynomial degrees depend on γ . That is exactly the difference between a common domain of holomorphy and a common norm bound. The root test is taken *after* a Hahn exponent and its finite word certificate have been fixed. No bound uniform in all words is summed as an ordinary series (source 09).

7.5 Formal parameters, higher rank, and the hypothesis audit

Corollary 7.20 (Several formal parameters, one dynamical variable; source 09). *Let $x = (x_1, \dots, x_e)$ be finitely many formal parameters and $F(z, x) = \lambda z + \sum_{\nu \in \mathbb{N}^e \setminus \{0\}} x^\nu f_\nu(z)$ with $f_\nu \in z^2 \mathcal{O}(\mathbb{D}_R)$. If $\tau < \infty$, all the normalized parameter coefficients of the linearizing coordinate and of its inverse are holomorphic on $\mathbb{D}_{Re^{-\tau}}$, whatever $|\nu|$ is.*

Proof. Embed the monoid $\mathbb{N}^e$ into the lexicographically ordered group $\mathbb{Z}^e$. Its nonzero elements are positive, and it is well ordered, so [theorem 7.13](#) applies to $\Gamma = \mathbb{Z}^e$. Source 09 also gives a direct tree argument: at a fixed ν only finitely many nonzero multi-indices, at most $|\nu|$ of them, can label vertices. $\square$

“Several” here refers to parameters, not to dynamical variables: there is still one variable z (source 09). Source 08's theorem applies verbatim to the same embedded input in d variables with multiplier λI_d . No ordinary convergence of the parameter series is asserted.

Example 7.21 (A higher-rank support; source 09). Take $\Gamma = \mathbb{Z}^2$ in lexicographic order, $u = t^{(0,1)}$, $v = t^{(1,0)}$, and $F(z) = \lambda z + ug(z) + vk(z)$ with $g, k \in z^2 \mathcal{O}(\mathbb{D}_R)$. Every power u^n has smaller valuation than v , so an iteration that waits for powers of u to pass the scale v is inappropriate ([warning 3.6](#)). Yet the coefficient at $u^i v^j$ uses exactly i labels of the first kind and j of the second, in finitely many orders, and for $\tau < \infty$ all these coefficients share the radius $Re^{-\tau}$. The theorem is therefore not an argument confined to rank one or to a cofinal sequence of powers.

Table 5: Where each hypothesis of [theorem 7.13](#) is used (source 09's audit, translated; the first row also records source 08's several-variable scope).

Hypothesis	Precise role
One dynamical variable (source 09), or a scalar multiplier λI_d (source 08) $ \lambda = 1$	All divisors are indexed by one integer $ \beta - 1$ for one irrational rotation, and the determinant argument of lemma 7.2 produces one common reduced rational approximation. Identifies divisor moduli with $\mathfrak{d}_n$, removes rotation growth from the operator route, and preserves centered polydisks.
λ not a root of unity (NR)	Makes every homological divisor nonzero and the normalized formal coordinates unique.
No coefficientwise constant term	Fixes the ordinary origin and allows tangent-to-identity formal composition at zero.

Hypothesis	Precise role
No coefficientwise linear term (exactness, SD_{ex})	Makes every tree weight $k_v - 1$ and every operator degree increment (lemma 7.9) strictly positive. Dropping it is the drift of section 8.
Well-ordered positive Hahn support	Makes the output monoid well ordered and bounds all input word lengths at each fixed exponent (lemma 3.4).
A common input polydisk	Supplies one ordinary $r < R$ for all relevant input coefficients, whatever the output exponent.
$\tau < \infty$	Supplies $C_a < \infty$ for every $a > \tau$.
No uniform coefficient-norm requirement	Lets the finite constants and the polynomial powers depend on the exponent. It does not weaken the common-radius conclusion.
Set-sized workspace	Makes all support families and recursions sets. Embeddings (3.1) connect the result to surcomplex values.

The optimality witness satisfies every structural hypothesis and needs no delicate cancellation. Source 09 lists three substantive limitations, which are not hidden exceptions to the statement: the absence of ordinary parameter convergence, the remaining open problem (in this report the nonscalar case, since answered by source 10 except for the optimal radius when $r_{\text{ang}} \geq 2$), and the absence of machine-checked proofs. The first row of table 5 is exactly the hypothesis that source 10 removes, at the price of the factor r_{ang} in the exponent (theorem 7.23).

7.6 What this section does not settle

The nonscalar boundary. For $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ the homological divisor on $z^\beta e_j$ is $\lambda^\beta - \lambda_j$. It is no longer controlled by a single integer $|\beta| - 1$ and a single rotation number, so the determinant argument of lemma 7.2 does not apply as written. Very small values need not reduce to one rational number p_*/q_* . Positivity still constrains the degree bookkeeping, but that alone does not prove the same nonaccumulation estimate for these families (source 08). A replacement would have to respect vector resonance relations and the geometry of decorated trees. The one-dimensional constant $e^{-\tau}$ may not be imported unchanged, nor may one simply substitute the minimum divisor Δ_n of (6.14) into the scalar proof (sources 08 and 09). Source 09 names the most direct target: a vector analogue of theorem 7.5 whose exponential loss stays independent of fixed tree complexity. Establishing it, or constructing a counterexample, would address the remaining case of question 8.20. Neither source asserts success or failure there.

Status after source 10. The analysis above is that of sources 08 and 09 and is kept as they state it. Source 10 supplies the vector analogue that source 09 named, in a weaker form than a single $e^{-\tau}$: at fixed tree complexity m the exponential loss is at most $\min(r_{\text{ang}}, m)\tau$, which never exceeds $r_{\text{ang}}\tau$ however large m is (theorem 7.27). It does not import the one-dimensional constant and does not substitute Δ_n into the scalar proof; it bounds the number of distinct exponential rates by r_{ang} (corollary 7.25) and pays for each rate once with contracted blocks that stay inside the nonlinear cone (lemma 7.26). The constant $e^{-\tau}$ is recovered, and is sharp, when $r_{\text{ang}} = 1$ (theorem 7.34); whether it holds when $r_{\text{ang}} \geq 2$ is open.

Individual inputs. $Re^{-\tau}$ is a universal guarantee, not a formula for each input; polynomial inputs already do much better. Criteria for input-specific improvement, from vanishing coefficients or algebraic structure, are a different question, and any such criterion must survive the nonlinear

degree interactions of the conjugacy equation (source 08; [question 25.7](#)).

Depth-uniform and ordinary estimates. The estimates allow the constants to depend arbitrarily on a fixed support word. So they do not give ordinary specialization of a Hahn parameter, Gevrey bounds across depth, or analytic convergence jointly in a complex parameter and the coordinate (sources 08 and 09; [question 25.8](#)). Even in the quadratic example, where every coefficient is a polynomial, no estimate on the growth of its scalar coefficient with the depth is asserted ([section 22.1](#)). For a non-Bruno λ , specializing t^η to a nonzero complex number in $\lambda z + t^\eta z^2$ gives a germ linearly conjugate to $\lambda w + w^2$, whose linearizer diverges. The Hahn family does not justify the specialization ([proposition 21.1](#)).

The drifted category. [Theorem 7.13](#) is not a statement that all small-divisor radius collapse is illusory. It identifies the role of exactness in preventing collapse (source 09). Admitting one infinitesimal linear term already brings collapse back over the same cyclic workspace ([theorem 8.15](#) and [example 8.19](#)).

7.7 The nonscalar diagonal multiplier: rational angular rank

This subsection and the five after it are contributed by source 10, written on 22 September 2026 against repository commit 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0. That commit already contained sources 08 and 09 as merged above, and at that commit this report's source file was byte-identical to its state immediately before source 10 was added. Source 10 answers the case of [question 8.20](#) that [section 7.6](#) leaves open: a *nonscalar* exact diagonal unitary multiplier. The answer is again positive. Every Hahn coefficient of the normalized conjugacy and of its inverse is holomorphic on one polydisk, whose radius loses at most the factor $e^{-r_{\text{ang}}\tau}$, where r_{ang} is the rational rank of the multiplier's angles. When $r_{\text{ang}} = 1$ the loss is exactly $e^{-\tau}$ and is sharp, in particular for every scalar λI_d ([theorem 7.13](#)) and for many nonscalar multipliers. For $r_{\text{ang}} \geq 2$ and $0 < \tau < \infty$ the optimal radius is not determined ([section 7.13](#)). Source 10 does not claim the scalar theorem or the scalar fixed-complexity rate as new; it cites both.

Tags. Resonance NR, with the multiplier **exactly** $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$, $|\lambda_j| = 1$; divisor notion SD_{ex} ; category the common-domain germ ring $\mathcal{R}_d$ of [\(3.2\)](#), with a specified radius; domain notion $\mathbb{D}_R^d$ and rad_{iso} . No drift is permitted anywhere in these subsections.

Convention 7.22 (Setting and local notation of source 10). (i) Write $\lambda_j = e^{2\pi i \theta_j}$ with $\theta_j \in \mathbb{R}$ and $\theta = (\theta_1, \dots, \theta_d)$, and assume NR. Repeated eigenvalues are allowed: NR is not a simple-spectrum hypothesis. The rate is $\tau = \tau(\Lambda)$ of SD_{ex} ([section 2.3](#)); it equals $\sigma_d(\Lambda)$ of [\(6.14\)](#), whose minimal divisor $\Delta_n = \min_{|\alpha|=n, j} |\lambda^\alpha - \lambda_j|$ is used below.

(ii) The *rational angular rank* of Λ is

$$r_{\text{ang}} = r_{\text{ang}}(\Lambda) = \dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}}(1, \theta_1, \dots, \theta_d) - 1. \quad (7.15)$$

It does not change when an angle changes by an integer, so it is an invariant of the eigenvalue list of Λ . Under NR, $1 \leq r_{\text{ang}} \leq d$: if $r_{\text{ang}} = 0$ every λ_j would be a root of unity, and $\lambda_j^{q+1} = \lambda_j$ for an order q of λ_j would violate NR at $\alpha = (q+1)e_j$. For $\Lambda = \lambda I_d$ with θ irrational, $r_{\text{ang}} = 1$. Source 10 writes r ; this report writes r_{ang} , because r is already the finite-return threshold of [section 9](#), the auxiliary radius of [convention 7.1\(iv\)](#), the chain length of [corollary 7.6\(i\)](#) and the flag length of [theorem 17.5](#).

(iii) r_{ang} is **not** the valuation rank of Γ and has nothing to do with the Hahn support: it is a property of d ordinary real angles. Nor is it the flag length of [theorem 17.5](#). Both are bounded by d through a rank count of integer vectors, but the flag concerns valuations of the Hahn divisors

- $k \cdot \Upsilon$ and r_{ang} concerns exponential rates of ordinary divisors; no statement is transferred between them.
- (iv) *Conjugacy convention.* As everywhere in this report, $H = z + h$ and $H \circ F = \Lambda H$. Source 10 computes the forward coordinate first. It writes $G = z + g$ with $F \circ G = G \circ \Lambda$; this is our $\check{H} = H^{-1} = z + \check{h}$ of [convention 7.1\(iii\)](#). Its H is ours. Every formula below has been translated.
- (v) *Other local symbols.* rad_{iso} is source 10's rad_{iso} . An auxiliary ordinary radius is written $r < R$, as in [convention 7.1\(iv\)](#); source 10 writes ρ , which this report reserves ([section 2.4](#)). The divisor at a tree vertex is $\mathfrak{d}_v$ (source 10: δ_v); for $\Lambda = \lambda I_d$ it is $\mathfrak{d}_{s_v}$ of [\(7.1\)](#). For $a > \tau$, $C_a(\Lambda)$ is the least constant ≥ 1 with

$$\max(1, |\lambda^\alpha - \lambda_j|^{-1}) \leq C_a(\Lambda) e^{a|\alpha|} \quad (|\alpha| \geq 2, 1 \leq j \leq d), \quad (7.16)$$

the vector form of [\(7.5\)](#). Source 10 writes $A_m(n)$ and $P(T)$ for the quantities called $A_m^\Lambda(n)$ and $P(T)$ below, to keep them apart from $A_m(n)$ of [corollary 7.6\(ii\)](#) and the shell polynomial P .

Theorem 7.23 (Common-domain linearization for a nonscalar exact multiplier; source 10). *Let Γ be any nonzero set-sized ordered abelian group, $d \geq 1$, Λ diagonal unitary with NR, and $\tau = \tau(\Lambda) < \infty$. Let*

$$F(z) = \Lambda z + \sum_{s \in S} t^s f_s(z), \quad f_s \in \mathcal{O}(\mathbb{D}_R^d)^d, \quad f_s(0) = 0, \quad Df_s(0) = 0, \quad (7.17)$$

with $S \subset \Gamma_{>0}$ well ordered and no common bound on the sizes of the f_s . Then the unique normalized positive conjugacy $H = z + h$ of [theorem 6.1](#), $H \circ F = \Lambda H$, and its inverse $\check{H} = H^{-1} = z + \check{h}$ satisfy

$$h_\gamma, \check{h}_\gamma \in \mathcal{O}(\mathbb{D}_{Re^{-r_{\text{ang}}\tau}}^d)^d, \quad \text{ord}_0 h_\gamma, \text{ord}_0 \check{h}_\gamma \geq 2, \quad \text{supp } h, \text{supp } \check{h} \subseteq S^* \setminus \{0\} \quad (7.18)$$

for every Hahn exponent γ . The domain is independent of the exponent, of the word depth $\ell_S(\gamma)$, of the rank of Γ and of the sizes of the input coefficients. H and H^{-1} lie in $\mathcal{R}_d$ and evaluate to mutually inverse bijective conjugacies of the finite halo $(\mathbb{D}_{Re^{-r_{\text{ang}}\tau}}^d)_\Gamma^\#$, on which $F^\#$ is also a bijection. The construction is unchanged under ordered enlargement of Γ .

The polydisk is *open*; nothing is asserted about the boundary. Uniqueness is uniqueness in the normalized formal positive-support class ([proposition 6.2](#)), so the analytic coordinate is not chosen afresh at each smaller radius. When $r_{\text{ang}} = 1$ the radius is $Re^{-\tau}$, which [theorem 7.34](#) shows is optimal. For a scalar multiplier this is [theorem 7.13](#) again. The proof below is then a third route to it ([remark 7.29](#)), and it gives nothing new in that case.

7.8 Exponential approximation rates are bounded by angular rank

The arithmetic input is elementary. A fixed finite family of integer vectors whose heights grow only linearly cannot exhibit more distinct exponentially small values of one real linear form than the rank allows. The count is of distinct *rates*, not of vectors or of their multiplicities.

Lemma 7.24 (Exponential rates and integer height; source 10). *Fix $q \geq 1$ and $\psi = (\psi_1, \dots, \psi_q) \in \mathbb{R}^q$. Let $N_k \rightarrow \infty$ be integers, and for v in a fixed finite set V let $a_v(k) = (u_v(k), p_v(k)) \in \mathbb{Z}^{q+1}$ satisfy $\|a_v(k)\|_\infty \leq CN_k$. Suppose that $e_v(k) = \psi \cdot u_v(k) - p_v(k) \neq 0$ and that*

$$b_v = \lim_{k \rightarrow \infty} -\frac{1}{N_k} \log |e_v(k)| \in (0, \infty) \quad (v \in V).$$

Then $\{b_v : v \in V\}$ has at most q elements.

Proof. The vectors eventually span at most q rational dimensions. Let $b_* = \min_v b_v > 0$; eventually $|e_v(k)| \leq e^{-b_* N_k/2}$ for every v . Take any $q + 1$ of the vectors as the rows of a square integer matrix and subtract $\psi_1, \dots, \psi_q$ times its first q columns from its last column. The determinant does not change, and the last column becomes $(-e_v(k))_v$. Expanding along it bounds the determinant by $C_1 N_k^q e^{-b_* N_k/2}$, with C_1 independent of k . An integer of modulus less than one is zero, so for large k every such determinant vanishes. There are finitely many choices of rows, so the whole family has rational rank at most q for large k .

Each strictly smaller positive rate adds a dimension. Pass to a subsequence along which the rank of every subset of the family is constant; there are finitely many subsets and ranks. Suppose a vector $a_v(k)$ with rate b lay, along this subsequence, in the rational span of vectors whose rates all exceed b . Choose a basis among those faster vectors, passing to a further subsequence to fix its indices and a nonsingular coordinate minor. Cramer's rule writes $a_v(k)$ in that basis with coefficients of modulus at most $C_2 N_k^q$: the denominator is a nonzero integer, of modulus at least one, and the numerators are polynomially bounded in N_k . Applying the linear form $(u, p) \mapsto \psi \cdot u - p$ gives $|e_v(k)| \leq C_3 N_k^q \max_w |e_w(k)|$, the maximum over the faster basis. Polynomial factors do not change a positive exponential rate, so the rate of e_v would exceed b , a contradiction. Order the distinct rates from the largest down. The fastest nonzero vector spans a nonzero space, and a vector at each later rate lies outside the span of all faster ones, so each rate raises the rank by at least one. The first step bounds that rank by q . $\square$

No coordinate of ψ need be algebraic or Diophantine; the argument is an integer-determinant argument, not a subspace theorem. Source 10 records two limits of the lemma. The finiteness of the family is essential: nothing is asserted for a number of vector sequences growing with N_k . And the height bound must be linear, or at least polynomial; exponentially large heights would destroy the polynomial Cramer bound.

Corollary 7.25 (Angular-rank form; source 10). *Let $k_v(N) \in \mathbb{Z}^d$, for finitely many indices v , satisfy $\|k_v(N)\|_\infty = O(N)$, and suppose the nonzero circle divisors have limits*

$$b_v = \lim_{N \rightarrow \infty} \frac{1}{N} \log |e^{2\pi i \theta \cdot k_v(N)} - 1|^{-1} \in (0, \infty).$$

Then there are at most r_{ang} distinct b_v . The same holds along any subsequence $N_k \rightarrow \infty$.

Proof. Choose $\psi_1, \dots, \psi_{r_{\text{ang}}}$ so that $1, \psi_1, \dots, \psi_{r_{\text{ang}}}$ is a rational basis of $\text{span}_{\mathbb{Q}}(1, \theta_1, \dots, \theta_d)$, write $\theta_j = a_{j0} + \sum_\ell a_{j\ell} \psi_\ell$ with rational $a_{j\ell}$, and let $Q \geq 1$ clear all denominators. With $p_v(N)$ a nearest integer to $\theta \cdot k_v(N)$, the integers $u_{v,\ell} = \sum_j Q a_{j\ell} k_{v,j}$ and $p'_v = Q p_v - \sum_j Q a_{j0} k_{v,j}$ have height $O(N)$ and satisfy $\psi \cdot u_v - p'_v = Q(\theta \cdot k_v - p_v)$. By (7.3), $4\|x\|_{\mathbb{Z}} \leq |e^{2\pi i x} - 1| \leq 2\pi\|x\|_{\mathbb{Z}}$, so these real errors have the same exponential rates as the circle divisors. Apply lemma 7.24 with $q = r_{\text{ang}}$. $\square$

7.9 Colored trees, block contractions, and the product bound

An *admissible colored tree* is a finite rooted substitution tree with internal vertices and external leaves. Each internal vertex v has an output color $j_v \in \{1, \dots, d\}$ and $a_v \geq 2$ input slots, each with a coordinate color, and each slot holds either an external leaf of that color or a child internal vertex of that output color. The internal skeleton is the tree left after forgetting the leaves. For an internal vertex v let T_v be the set of internal vertices of its subtree and $\alpha_v \in \mathbb{N}^d$ the vector counting the

leaves of each color below v , and put

$$\begin{aligned} w_v = a_v - 1 \geq 1, \quad s_v = \sum_{u \in T_v} w_u = |\alpha_v| - 1, \quad k_v = \alpha_v - e_{j_v}, \\ \mathfrak{d}_v = |\lambda^{\alpha_v} - \lambda_{j_v}| = |e^{2\pi i \theta \cdot k_v} - 1| > 0. \end{aligned} \quad (7.19)$$

A tree with m internal vertices and total weight $n = \sum_v w_v$ has $n + 1$ leaves and $\sum_v a_v = n + m$, by counting its $m - 1$ internal edges. In several variables a difference of two homological exponent vectors need not be a homological exponent vector again. The contraction below produces exactly the differences that are, so the proof never strengthens NR to a condition on all of $\mathbb{Z}^d$. Preservation of admissible cuts in operator trees has a precedent in [18, Prop. 5.6]; source 10 claims the rank estimate, not the use of tree cuts.

Lemma 7.26 (Marked-block contraction; source 10). *Mark any set A of internal vertices. For $v \in A$ let $C_A(v)$ be its nearest strict marked descendants and $B_v = T_v \setminus \bigcup_{u \in C_A(v)} T_u$. The B_v are pairwise disjoint and $v \in B_v$. Contracting each removed subtree to one leaf of its root's output color leaves a tree at v with leaf vector*

$$\beta_v = \alpha_v - \sum_{u \in C_A(v)} \alpha_u + \sum_{u \in C_A(v)} e_{j_u} \in \mathbb{N}^d, \quad (7.20)$$

and

$$|\beta_v| = 1 + \sum_{u \in B_v} w_u \geq 2, \quad \beta_v - e_{j_v} = k_v - \sum_{u \in C_A(v)} k_u, \quad \sum_{v \in A} |\beta_v| \leq n + |A|. \quad (7.21)$$

In particular $|\lambda^{\beta_v} - \lambda_{j_v}|$ is a nonzero divisor of degree $|\beta_v| \geq 2$, covered by NR and by (7.16).

Proof. The removed subtrees are disjoint and their leaves form submultisets of the leaves below v ; replacing each by one leaf of the stated color gives (7.20) and nonnegativity. A vertex lies in B_v exactly when v is its closest marked ancestor-or-self, so the blocks are disjoint, and $v \in B_v$ because only strict descendants were removed. The remaining internal vertices keep their arities, the edge to a removed root having become a leaf, and counting leaves of the contracted tree gives the degree identity. Subtracting e_{j_v} from (7.20) gives the second identity. The positive weights in disjoint blocks sum to at most n , and adding one per block gives the third. $\square$

In two coordinates, let the root have output color 1 and input monomial x^2y , and fill its y slot with a child of output color 2 and monomial xy^2 . Then $\alpha_{\text{root}} = (3, 2)$, $\alpha_{\text{child}} = (1, 2)$ and both weights are 2. With both marked, $\beta_{\text{root}} = (3, 2) - (1, 2) + (0, 1) = (2, 1)$, of degree $3 = 1 + w_{\text{root}}$, and $(2, 2) - (1, 1) = (1, 1) = \beta_{\text{root}} - e_1$; the contracted degrees sum to $6 = n + 2$. The subtraction has produced a nonlinear monomial, not an arbitrary signed lattice vector (source 10's example).

The weight $w_v \geq 1$ is where exactness enters. An arity-one vertex, which is what an infinitesimal linear term creates, would give a block of weight zero and contracted degree one. Its divisor could be an exact linear resonance, and small divisors could then be incurred repeatedly without spending nonlinear degree. This is [warning 7.4](#) in colored form. The drifted collapse of [section 8](#) lies exactly outside this hypothesis and is not a counterexample to [theorem 7.23](#) (source 10).

For an admissible colored tree T put $P(T) = \prod_v \max(1, \mathfrak{d}_v^{-1})$, and for integers $m, n \geq 1$ let $A_m^\Lambda(n)$ be the maximum of $P(T)$ over trees with at most m internal vertices and total weight n . Such trees exist (a single vertex of arity $n + 1$), and after a canonical slot convention there are finitely many of them, so the maximum is attained. The factor $\max(1, \cdot)$ only enlarges the divisor product of an actual tree expansion.

Theorem 7.27 (Rational-rank divisor-product bound; source 10). *Assume NR and $\tau < \infty$. For every fixed $m \geq 1$,*

$$\limsup_{n \rightarrow \infty} \frac{\log A_m^\Lambda(n)}{n} \leq \min(r_{\text{ang}}, m) \tau. \quad (7.22)$$

Hence for every $\varepsilon > 0$ there is a finite $C_{m,\varepsilon,\Lambda}$ with

$$P(T) \leq C_{m,\varepsilon,\Lambda} \exp((\min(r_{\text{ang}}, m)\tau + \varepsilon)n) \quad (7.23)$$

for all admissible colored trees with at most m internal vertices and total weight n . No bound uniform in m is asserted for the prefactor.

Proof. Fix $a > \tau$, so that (7.16) holds. Choose maximizing trees T_k of weights $n_k \rightarrow \infty$ along which the limsup in (7.22) is attained, and pass to a subsequence with a fixed number $m' \leq m$ of internal vertices, a fixed rooted internal skeleton and fixed output colors; there are finitely many choices. Input arities and leaf counts still vary with k . Pass to a further subsequence along which

$$b_v = \lim_{k \rightarrow \infty} \frac{1}{n_k} \log^+(\mathfrak{d}_v(k)^{-1})$$

exists for every internal vertex; the limits are finite and nonnegative by (7.16) and $|\alpha_v| \leq n_k + 1$. The rate of $P(T_k)$ along the subsequence is $\sum_v b_v$.

At most r_{ang} positive rates. For the vertices with $b_v > 0$ take $k_v(k) = \alpha_v(k) - e_{j_v}$, of height $O(n_k)$. Their circle divisors have positive rates b_v , so by corollary 7.25 there are at most r_{ang} distinct positive values among the b_v . There are also at most m' .

Each rate value is charged at most τ in total. Fix a positive value b and mark $A_b = \{v : b_v \geq b\}$. For $0 < \zeta < b$ and all large k , with $p_v(k)$ a nearest integer to $\theta \cdot k_v(k)$, $|\theta \cdot k_v(k) - p_v(k)| \leq e^{-(b-\zeta)n_k}$ for $v \in A_b$, after fixed constants are absorbed by starting later. Apply lemma 7.26 to A_b . By the second identity of (7.21), the real number $\theta \cdot (\beta_v - e_{j_v})$ lies within $|A_b|e^{-(b-\zeta)n_k}$ of an integer, so the contracted divisor is at most $2\pi|A_b|e^{-(b-\zeta)n_k}$. Cancellation can only help this upper bound, and the divisor is nonzero because $|\beta_v| \geq 2$. Applying (7.16) to it gives $(b - \zeta)n_k \leq a|\beta_v| + O(1)$, with the constant independent of v and k . Summing over A_b and using the third identity of (7.21), $|A_b|(b - \zeta)n_k \leq a(n_k + |A_b|) + O(1)$. Divide by n_k and let $k \rightarrow \infty$, $\zeta \downarrow 0$ and $a \downarrow \tau$:

$$|A_b|b \leq \tau. \quad (7.24)$$

Exact cancellation to zero would contradict NR; no lower estimate for a signed lattice divisor was used.

Summing the rate values. Let $b^{(1)}, \dots, b^{(q)}$ be the distinct positive rates, $q \leq \min(r_{\text{ang}}, m')$, and let V_i be the set of vertices with rate exactly $b^{(i)}$. Then $|V_i| \leq |A_{b^{(i)}}|$, and (7.24) gives $\sum_v b_v = \sum_i |V_i|b^{(i)} \leq \sum_i |A_{b^{(i)}}|b^{(i)} \leq q\tau \leq \min(r_{\text{ang}}, m)\tau$. This is (7.22). All sufficiently large n then satisfy (7.23) with prefactor one, and the finitely many remaining maxima are absorbed into $C_{m,\varepsilon,\Lambda}$. $\square$

Corollary 7.28 (The fixed-complexity rate in angular rank one; source 10). *If $r_{\text{ang}} = 1$, then for every $m \geq 1$*

$$\limsup_{n \rightarrow \infty} \frac{\log A_m^\Lambda(n)}{n} = \tau.$$

Proof. The upper bound is theorem 7.27. For each n choose a monomial and output color realizing Δ_{n+1} ; its one-vertex tree has weight n and is admissible for $A_m^\Lambda(n)$. The resulting lower bound has limsup τ , since replacing $n + 1$ by n does not change the rate. $\square$

The argument extracts subsequences, so the prefactor $C_{m,\varepsilon,\Lambda}$ is finite but not given by a formula. That suffices for the holomorphy of a fixed Hahn coefficient. A bound uniform as $m \rightarrow \infty$ would be a stronger theorem and does not follow; in particular, replacing m by n in (7.23) and discarding the prefactor is invalid (source 10). Source 10's appendix restates [theorem 7.27](#) in a form that mentions no Hahn field; it is the same statement and is not reprinted.

Remark 7.29 (The scalar case, and how the routes differ). For $\Lambda = \lambda I_d$ every colored tree has $\mathfrak{d}_v = \mathfrak{d}_{s_v}$, so its internal skeleton is a positively weighted rooted tree with the same subtree sums, and $\max(1, \mathfrak{d}^{-1}) \leq 2\mathfrak{d}^{-1}$ because $\mathfrak{d} \leq 2$. Hence $A_m^{\lambda I_d}(n) \leq 2^m A_m(n)$, and [corollary 7.28](#) for a scalar multiplier already follows from the rate statement of [corollary 7.6\(ii\)](#). The two proofs are different. Sources 08 and 09 bound each tree explicitly, by $(C_a n)^m e^{an}$ ([theorem 7.5](#)), using one rational cluster ([lemma 7.2](#)) and divisible-subtree packing ([lemma 7.3](#)). Source 10 bounds only the limsup along subsequences, by the rank count [corollary 7.25](#) and the block budget (7.24), and its constant is not explicit. The blocks B_v of [lemma 7.26](#) are those of [lemma 7.3](#); what is new is that the contracted leaf vector β_v keeps each block's divisor inside the nonlinear cone in several colors. Both routes are kept under [convention 1.1\(ii\)](#). For a scalar multiplier the explicit bound is strictly more information; for a nonscalar one only source 10's route is available.

7.10 Route D in several variables, and the proof of the common domain

Source 10 works with the forward coordinate, as source 09 does in one variable ([section 7.4](#)). With $\check{H} = z + \check{h}$ and $F = \Lambda z + f$, the equation $F \circ \check{H} = \check{H} \circ \Lambda$ reads

$$\mathcal{L}\check{h} = f(z + \check{h}), \quad \mathcal{L}\check{h} = \check{h}(\Lambda z) - \Lambda\check{h}(z), \quad (7.25)$$

the several-variable form of (7.12), with $\mathcal{L}$ of (5.1).

Proposition 7.30 (Colored tree expansion of the forward coordinate; source 10). *For formal input coefficients of order at least two and well-ordered $S \subset \Gamma_{>0}$, (7.25) has a unique normalized positive-support formal solution, supported in $S^* \setminus \{0\}$. At a fixed exponent γ its expansion uses trees with at most $\ell_S(\gamma)$ internal vertices, labelled only by letters in $D_S(\gamma)$, and at a fixed exponent and spatial multi-index it is a finite sum. Each vertex v contributes one Taylor coefficient f_{s_v, j_v, μ_v} of an input, with $|\mu_v| = a_v$, and the divisor $\lambda^{\alpha_v} - \lambda_{j_v}$ of modulus $\mathfrak{d}_v$.*

Proof. Existence and uniqueness are [proposition 6.2](#) for H , with $\check{H} = H^{-1}$ from [proposition 4.1](#); source 10 also proves them directly, by the least-difference argument for (7.25), which does not assume that a second solution is supported in S^* . For the expansion, iterate $\check{h} = \mathcal{L}^{-1}f(z + \check{h})$ formally. A coefficient $f_{s,j,\mu} z^\mu e_j$ supplies an internal vertex of output color j , arity $|\mu| \geq 2$ and Hahn letter s . Expanding a slot by a further copy of $\check{h}$ supplies a child, and an unexpanded slot becomes an external leaf of its coordinate color. The subtree of v carries the monomial $z^{\alpha_v} e_{j_v}$, so by (5.1) the divisor at v is $\lambda^{\alpha_v} - \lambda_{j_v}$. The Hahn exponent is the sum of the letters, which form a word in $\mathcal{W}_S(\gamma)$, so there are at most $\ell_S(\gamma)$ vertices ([lemma 3.4](#)). At a fixed spatial degree every arity is bounded and there are finitely many trees, monomials and slot choices. $\square$

This is the several-variable, colored form of source 09's route D ([proposition 7.17](#)); each slot-labelled term occurs once, so there are no symmetry factors.

Lemma 7.31 (Polynomial diagram count; source 10). *Fix $m, d \geq 1$ and a finite set of input labels. There is a constant C such that the number of slot-labelled expansion terms of total spatial degree $n + 1$ with at most m internal vertices is at most $C(n + m + 1)^{dm+m}$. A term with m' vertices*

has total input monomial degree $n + m'$, and the count includes the multinomial multiplicities of substitution.

Proof. There are finitely many rooted plane skeletons with $m' \leq m$ vertices, output colorings and label assignments. Each vertex's input multi-index has coordinates at most $n + m$, giving at most $(n + m + 1)^{dm'}$ choices. After that, each of the $m' - 1$ children is assigned to a compatible slot of its parent, in at most $(n + m + 1)^{m'-1}$ ways. The remaining slots are leaves whose colors are already fixed by the input monomial, so there is *no* independent factor d^{n+1} for them. Ordering children and slots only overcounts and absorbs the binomial coefficients of the repeated factors $(z_i + \check{h}_i)^{\mu_i}$. The degree assertion is $\sum_v a_v = n + m'$. $\square$

Only polynomiality in n at fixed m matters, as in [lemma 7.18](#). Treating leaf colorings as independent would cost a spurious factor exponential in n and a smaller radius (source 10).

Proof of theorem 7.23. Construct $\check{H}$ by [proposition 7.30](#). Fix γ , put $m = \ell_S(\gamma)$, and fix $0 < r < R$. By Cauchy's estimate there is a finite $M_\gamma(r) \geq 1$ with $|f_{s,j,\mu}| \leq M_\gamma(r)r^{-|\mu|}$ for the finitely many $s \in D_S(\gamma)$ and all j, μ . A term of total spatial degree $n + 1$ with $m' \leq m$ vertices has input coefficients of product at most $M_\gamma(r)^{m'}r^{-(n+m')} \leq C_{\gamma,r}r^{-n}$, and by [theorem 7.27](#) its divisor product is at most $C_{m,\varepsilon,\Lambda}e^{(\min(r_{\text{ang}},m)\tau+\varepsilon)n}$. With [lemma 7.31](#), and the finitely many letter assignments summing to γ absorbed into the constant,

$$\max_{j, |\mu|=n+1} |[z^\mu]\check{h}_{\gamma,j}| \leq C'(n + m + 1)^{dm+m} r^{-n} e^{(\min(r_{\text{ang}},m)\tau+\varepsilon)n}. \quad (7.26)$$

The root test gives $\text{rad}_{\text{iso}}(\check{h}_\gamma) \geq re^{-\min(r_{\text{ang}},m)\tau-\varepsilon}$; letting $r \uparrow R$ and $\varepsilon \downarrow 0$,

$$\text{rad}_{\text{iso}}(\check{h}_\gamma) \geq Re^{-\min(r_{\text{ang}},\ell_S(\gamma))\tau} \geq Re^{-r_{\text{ang}}\tau}. \quad (7.27)$$

The last bound does not depend on γ , so every $\check{h}_\gamma$ is holomorphic on $U = \mathbb{D}_{Re^{-r_{\text{ang}}\tau}}^d$. [Proposition 4.1](#), with this U , gives the inverse H on the same domain, with no second loss of radius and support in $S^* \setminus \{0\}$; source 10's own positive inverse lemma is this statement and is not reprinted. The formal identities transport (7.25) to $H \circ F = \Lambda H$, and uniqueness is formal. The halo statement is the proof of [proposition 6.13](#) with U in place of $\mathbb{D}_R^d$: the reductions of H, H^{-1} are the identity and that of F is $z \mapsto \Lambda z$, which preserves U because $|\lambda_j| = 1$. Workspace invariance is [proposition 4.14](#): every formula uses only the input exponents and finite sums of them. $\square$

Corollary 7.32 (Input-relative radius at a specified exponent; source 10). *Assume NR, $\tau < \infty$ and exactness, with input coefficients convergent germs. For $\gamma \in S^* \setminus \{0\}$ let $R_\gamma > 0$ be the minimum of $\text{rad}_{\text{iso}}(f_s)$ over the finite set $s \in D_S(\gamma)$. Then*

$$\text{rad}_{\text{iso}}(h_\gamma), \text{rad}_{\text{iso}}(\check{h}_\gamma) \geq R_\gamma e^{-\min(r_{\text{ang}},\ell_S(\gamma))\tau}. \quad (7.28)$$

Proof. For $\check{h}_\gamma$ use the Cauchy bound with any $r < R_\gamma$ in the proof above. For h_γ expand the inverse recursion. Every $\check{h}_\beta$ that occurs has an input word that extends to one at γ , so its letters lie in $D_S(\gamma)$ and $\ell_S(\beta) \leq \ell_S(\gamma)$ (concatenate a longest word at β with a word for the remaining weight). Each such $\check{h}_\beta$ is holomorphic on the polydisk of (7.28), and finite products and derivatives keep it there. $\square$

Since $\min(r_{\text{ang}}, \ell) \leq \ell$, (7.28) improves the word-depth bound (6.12) of [theorem 6.9](#) whenever $\ell_S(\gamma) > r_{\text{ang}}$, and reproduces it otherwise. Arbitrary radius-free input germs thus acquire a finite input-relative certificate even when there is no common input radius.

Corollary 7.33 (Rectangular polydisks and ordinary parameters; source 10). *If the input coefficients are holomorphic on $\prod_j \mathbb{D}_{R_j}$, all coefficients of H and H^{-1} are holomorphic on $\prod_j \mathbb{D}_{R_j e^{-r_{\text{ang}} \tau}}$. If the f_s also depend holomorphically on ordinary parameters w in a complex domain V , with Λ independent of w , the coefficients are jointly holomorphic on that polydisk times V .*

Proof. The rescaling $z_j = R_j x_j$ commutes with the diagonal Λ and does not change its divisors; apply [theorem 7.23](#) with radius one and undo it. For parameters, fix a relatively compact subset of V . The finitely many inputs used at one exponent have Cauchy bounds uniform on it, so [\(7.26\)](#) gives normal convergence on compact subsets of the polydisk, uniformly in the parameter. The Taylor coefficients are holomorphic in w , hence so is the sum; exhaust V . $\square$

The rescaling part is the nonscalar form of source 08’s unequal-radius statement after [theorem 7.13](#); like that one, it says nothing about optimal Reinhardt domains. The joint holomorphy in ordinary parameters is new to this report. The constants in [\(7.26\)](#) may grow arbitrarily fast from one exponent to another; only their finiteness at each exponent is used. An ordinary coefficient of size $10^{10^{10}}$ still has valuation zero ([remark 2.3](#)), and a common domain of holomorphy is not a common norm bound ([warning 10.3](#)).

Source 10 lists the places where a proof of [theorem 7.23](#) could fail and how its proof avoids each.

Potential failure	How it is handled
Exponent differences leave the nonlinear cone	Only contracted tree characters are used; lemma 7.26 gives a nonnegative leaf vector of degree at least two.
A rational relation with huge coefficients	Integer minors have modulus at least one, and Cramer numerators grow polynomially with the height.
Exponential cancellation changes a rate	The block argument needs only an upper bound on the angular error. Cancellation helps, and exact resonance is excluded by NR.
Exponentially many leaf colorings	Leaf colors are fixed by the input multi-indices; only the $m - 1$ internal edges choose slots (lemma 7.31).
A different radius at each Hahn exponent	The prefactor varies with the exponent, but the exponential rate is bounded by the same $r_{\text{ang}} \tau$ everywhere.
Inverting the conjugacy loses another factor	Positive Hahn inversion uses only differentiation and multiplication, no homological inversion (proposition 4.1).

7.11 The exact criterion, and sharpness in angular rank one

Theorem 7.34 (Exact criterion and rank-one sharpness; source 10). *Fix a diagonal unitary Λ with NR, a nonzero Γ and $0 < R < \infty$. The following are equivalent:*

- (i) $\tau(\Lambda) < \infty$;
- (ii) every input [\(7.17\)](#) with coefficients on $\mathbb{D}_R^d$ has a normalized conjugacy and inverse whose coefficients are all holomorphic on one common polydisk of positive radius;
- (iii) every input with one positive exponent and entire coefficient functions has a normalized conjugacy whose coefficients are convergent germs.

For $\tau < \infty$ let R_{univ} be the supremum of the $q > 0$ such that every input on $\mathbb{D}_R^d$ has all coefficients of its normalized conjugacy and inverse holomorphic on $\mathbb{D}_q^d$. Then

$$Re^{-r_{\text{ang}} \tau} \leq R_{\text{univ}} \leq Re^{-\tau}, \quad (7.29)$$

the lower value being attained as a guaranteed open domain. If $r_{\text{ang}} = 1$, then $R_{\text{univ}} = Re^{-\tau}$ exactly.

Proof. (i) $\Rightarrow$ (ii) and the lower bound are [theorem 7.23](#). (ii) implies (iii) by restricting an entire one-exponent input to $\mathbb{D}_R^d$. (iii) $\Rightarrow$ (i) is the necessity half of rows two and three of [theorem 6.1](#): if $\tau = \infty$, the entire forcing (5.5) of [lemma 5.2](#) placed at one weight η has, by [proposition 6.6](#), a forced coefficient $h_\eta = -\mathcal{L}^{-1}\varphi$ of radius zero. For the upper bound, take the witness (5.4) at one weight η ; its h_η has isotropic radius exactly $Re^{-\tau}$, so no larger q is guaranteed. Source 10 notes that this witness needs no case distinction at $\tau = 0$: choose degrees n_k with $n_k^{-1} \log \Delta_{n_k}^{-1} \rightarrow \tau$, which exist also when $\tau = 0$ because $\Delta_n \leq 2$ bounds the negative parts, and use the coefficients R^{-n_k} ; the root test gives the radius $Re^{-\tau}$ directly. When $r_{\text{ang}} = 1$ the two bounds of (7.29) coincide. $\square$

Source 10 includes the necessity and upper-bound halves only to make the criterion complete; they are the first-weight obstructions of [lemma 5.2](#) and [proposition 6.6](#), printed once. For $\tau = 0$, (7.29) gives $R_{\text{univ}} = R$, which is row one of [theorem 6.1](#). For $r_{\text{ang}} \geq 2$ and $0 < \tau < \infty$ the theorem decides the *existence* of a common positive radius, not its optimal value.

“Sharp” refers to the largest *universally guaranteed centered isotropic polydisk* for the ordinary coefficient functions. A particular input may have much larger domains, or be polynomial at every exponent, and a coefficient may continue analytically beyond its centered Taylor polydisk; none of this contradicts sharpness (source 10). The proof also gives no classification of nonnormalized, non-Hahn-coherent or purely set-theoretic conjugacies.

7.12 Nonscalar examples

Proposition 7.35 (Two modes in one irrational direction; source 10). *Let $\lambda = e^{2\pi i\theta}$ with θ irrational and $\Lambda = \text{diag}(\lambda^2, \lambda^3)$. Then Λ is nonscalar, satisfies NR, has $r_{\text{ang}}(\Lambda) = 1$, and*

$$\tau(\Lambda) = 3\tau(\lambda), \quad \tau(\lambda) = \limsup_{k \rightarrow \infty} \frac{1}{k} \log^+ \mathfrak{d}_k^{-1}, \quad (7.30)$$

in the extended sense when $\tau(\lambda) = \infty$; here $\tau(\lambda)$ is (7.1) and (8.15). Consequently, when $\tau(\lambda) < \infty$, every exact input (7.17) for this Λ , with arbitrary mixed monomials and arbitrary well-ordered positive support, has all coefficients on the polydisk of radius $Re^{-3\tau(\lambda)}$, and this radius is the optimal universal one.

Proof. If $a + b \geq 2$ then $2a + 3b \geq 4$, so $2a + 3b \notin \{2, 3\}$ and irrationality excludes $\lambda^{2a+3b} = \lambda^c$ for $c \in \{2, 3\}$; this is NR. The angles $2\theta, 3\theta$ span the same rational space with 1 as θ does, so $r_{\text{ang}} = 1$. A divisor of degree $n = a + b \geq 2$ is $|\lambda^k - 1|$ with $k = 2a + 3b - c$, $c \in \{2, 3\}$, and $1 \leq k \leq 3n - 2$. If $\tau(\lambda) < \infty$ and $u > \tau(\lambda)$, then $|\lambda^k - 1|^{-1} \leq C_u e^{uk} \leq C_u e^{3un}$, so $\tau(\Lambda) \leq 3u$. Conversely, for large k choose $a \in \{0, 1, 2\}$ with $2a \equiv k \pmod{3}$ and $b = (k + 3 - 2a)/3$: then $2a + 3b - 3 = k$, so $|\lambda^k - 1|$ occurs in component 2 at degree $n = a + b = k/3 + O(1)$. Along a sequence realizing $\tau(\lambda)$ this gives $\tau(\Lambda) \geq 3\tau(\lambda)$, and $\tau(\Lambda) = \infty$ when $\tau(\lambda) = \infty$. The last assertion is [theorems 7.23](#) and [7.34](#) with $r_{\text{ang}} = 1$. $\square$

The hypothesis is not vacuous at positive rate. [Theorem 8.9\(c\)](#) gives, for each $s > 0$, the explicit continued fraction $a_{j+1} = \lceil e^{sq_j} \rceil$ with $\tau(\lambda) = s$ ([proposition 8.8](#)). Source 10 constructs the same family, starting from $a_1 = 1$, and proves its rate by the same continued-fraction estimates; it is printed once, and source 10 does not claim it. For [proposition 7.35](#) it gives a nonscalar spectrum with $\tau(\Lambda) = 3s$ and sharp common radius Re^{-3s} , so for any $0 < \kappa < 1$ the choice $s = -\frac{1}{3} \log \kappa$ gives universal radius exactly κR . The theorem therefore covers genuinely exponential small divisors, not

merely rate zero, and the input may couple the two coordinates arbitrarily: the example is not two uncoupled scalar maps.

Example 7.36 (Angular rank two with zero loss; source 10). Let $\theta = (\sqrt{2}, \sqrt{3})$. Since $1, \sqrt{2}, \sqrt{3}$ are rationally independent, Λ satisfies NR and $r_{\text{ang}} = 2$. For integers u, v, p with $|u| + |v| = O(n)$ and $(u, v) \neq 0$, the algebraic integer $u\sqrt{2} + v\sqrt{3} - p$ is nonzero, its norm from $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is a nonzero integer, and its three other conjugates are $O(n+1)$ when p is a nearest integer to $u\sqrt{2} + v\sqrt{3}$. Hence $|u\sqrt{2} + v\sqrt{3} - p| \geq C(n+1)^{-3}$. With $(u, v) = \alpha - e_j$ this bounds every homological divisor polynomially from below, so $\tau = 0$ and the original polydisk is kept. This is row one of [theorem 6.1](#); source 10 gives it only to show that $r_{\text{ang}} \geq 2$ is compatible with zero loss, and it is a familiar algebraic-number estimate, not a new Diophantine statement.

For an arbitrary well-ordered support, source 10 uses exactly the set S and group $\Gamma = \mathbb{Q} + \mathbb{Q}\Omega$ of [example 3.7](#), with any of the spectra above and independently chosen order-two coefficients f_s on $\mathbb{D}_R^d$, with no bound uniform in s . [Theorem 7.23](#) gives one common domain for all coefficients; $\check{h}_\Omega = \mathcal{L}^{-1}f_\Omega$ and $h_\Omega = -\check{h}_\Omega$, since the only word at Ω is (Ω) , and at $\Omega+2$ only the three permutations of $(\Omega, 1, 1)$ occur. The infinitely many input exponents below Ω do not enter the coefficient at Ω . The construction is local in the support, not a sequential limit through all smaller scales ([warning 3.6](#)). Source 10's lexicographic $\mathbb{Z}^2$ caution is [example 7.21](#), and its surcomplex realization through (3.1), with the sign $t^\gamma = \omega^{-\gamma}$ and set-sized workspaces, is [section 3.5](#) and [proposition 4.14](#); neither is reprinted. All derivatives are coordinate derivatives of coefficient functions, not the Berarducci–Mantova derivation ([warning 3.2](#)), and strong summation is the support operation of [definition 3.1](#), not a limit in the fine topology. The failure at $\tau = \infty$ concerns ordinary coefficient holomorphy only; the formal conjugacy still evaluates on the monad ([proposition 6.14](#)).

7.13 What source 10 does not settle

The optimal radius in angular rank at least two. For $r_{\text{ang}} \geq 2$ and $0 < \tau < \infty$ the universal radius lies between the two bounds of (7.29) and is not determined. The proof charges up to τ to each distinct exponential rate. It does not show that several rate values can saturate their charges simultaneously, nor that saturation would survive the coefficient cancellations of an actual conjugacy. A sharper count might reduce the factor r_{ang} ; a matching lower example would need more than large individual tree factors (source 10). This is what remains of [question 8.20](#).

Other multipliers. No theorem is asserted for infinitesimally drifting multipliers in several variables, resonant spectra, nontrivial Jordan blocks, nonunitary spectra in analytic categories, or infinite-dimensional coordinate spaces. The support method is compatible with many of these settings, but the admissible-divisor and polydisk-geometry arguments above do not transfer automatically.

Ordinary specialization and depth uniformity. Nothing is asserted after substituting a nonzero complex number for t . A bound on each coefficient's spatial radius gives no control of its size as the Hahn or parameter order grows, and the prefactor of [theorem 7.27](#) is not uniform in m . So the theorem neither replaces a Bruno-type theorem for ordinary germs nor contradicts the known failures of classical analytic linearization ([remark 7.16](#) and [question 25.8](#)).

8 The drifted category: exact radius collapse and a trichotomy

This section decides the common-domain germ category in the drifted category SD_{dr} , negatively. At the pins of sources 01–07 it was the only place in the report where that category was decided at

all.

applies to everything in this section, including the single-exponent theorem of [section 8.6](#).

8.1 Two independent monomials in one Hahn field

Work in the divisible ordered group

$$\Gamma_* = \mathbb{Q} + \sqrt{2}\mathbb{Q} \subset \mathbb{R}, \quad u = t, \quad v = t^{\sqrt{2}}.$$

The map $(k, \ell) \mapsto k + \ell\sqrt{2}$ is injective on $\mathbb{N}^2$ and only finitely many such pairs lie below any fixed real bound, so the image of $\mathbb{N}^2$ is a well-ordered additive monoid and a double formal series $\sum_{k, \ell \geq 0} b_{k, \ell} u^k v^\ell$ is an unambiguous Hahn series with that support. This is a use of two independent *exponents*, not of two ordinary complex parameter values. For a fixed finite $R > 0$ put

$$F_{u,v}(z) = (\lambda + u)z + vP_R(z), \quad P_R \text{ as in (5.6)}, \quad (8.1)$$

so that both perturbation coefficients are holomorphic on $\mathbb{D}_R$ and the multiplier is $\Lambda_{\text{dr}} = \lambda + u$. [Theorem 6.3](#) gives a unique formal linearizer with support in $\mathbb{N} + \sqrt{2}\mathbb{N}$. At $v = 0$ the map is already linear and its normalized linearizer is the identity — either by setting $v = 0$ in [\(6.10\)](#) or by ordinary formal uniqueness for the multiplier $\lambda + u$, every $(\lambda + u)^n - (\lambda + u)$ having nonzero constant term in u . Hence its first nontrivial v -block is

$$H(z) = z + vQ(u, z) + O(v^2), \quad Q(u, z) = \sum_{k \geq 0} u^k h_{k,1}(z), \quad (8.2)$$

where $O(v^2)$ means only the blocks of v -degree at least two and is *not* an asymptotic assertion in an ordinary parameter. Extracting the coefficient of v in $H(F) = \Lambda_{\text{dr}}H$ gives $Q(u, (\lambda + u)z) - (\lambda + u)Q(u, z) = -P_R(z)$ and hence

$$h_{k,1}(z) = - \sum_{n \geq 2} R^{2-n} c_{n,k} z^n, \quad c_{n,k} = [u^k] \frac{1}{(\lambda + u)^n - (\lambda + u)}. \quad (8.3)$$

The radius problem is thereby reduced to an exact estimate of the *reciprocal jets* $c_{n,k}$ for fixed k and increasing n .

8.2 Reciprocal jets and dominance of the highest pole

With $a_{n,1} = n\lambda^{n-1} - 1$ and $a_{n,j} = \binom{n}{j}\lambda^{n-j}$ for $j \geq 2$ (and $\binom{n}{j} = 0$ for $j > n$), one has $(\lambda + u)^n - (\lambda + u) = d_n + \sum_{j \geq 1} a_{n,j} u^j$ with $d_n = \lambda^n - \lambda$, so that for $k \geq 1$

$$c_{n,k} = \sum_{r=1}^k (-1)^r d_n^{-r-1} \sum_{\substack{j_1 + \dots + j_r = k \\ j_i \geq 1}} a_{n,j_1} \dots a_{n,j_r}, \quad c_{n,0} = d_n^{-1}, \quad (8.4)$$

and the term with $r = k$ is unique and equals

$$(-1)^k a_{n,1}^k d_n^{-k-1}. \quad (8.5)$$

It is this *highest* pole in the small divisor, and not merely its first power, that drives the successive radius losses. These are exactly the objects source 02's program checks.

Lemma 8.1 (Uniform reciprocal-jet estimates). *For each fixed $k \geq 1$ there is $C_k > 0$ independent of n with*

$$|c_{n,k}| \leq C_k n^k \max\{1, |d_n|^{-k-1}\}, \quad (8.6)$$

$$|c_{n,k} - (-1)^k a_{n,1}^k d_n^{-k-1}| \leq C_k n^k |d_n|^{-k} \quad \text{when } |d_n| \leq 1. \quad (8.7)$$

Also $|a_{n,1}| \geq n - 1$. Consequently, along any sequence with $n \rightarrow \infty$ and $d_n \rightarrow 0$,

$$c_{n,k} = (-1)^k a_{n,1}^k d_n^{-k-1} (1 + o(1)), \quad (8.8)$$

which in particular excludes cancellation of the highest-pole contribution along the subsequences that control exponential growth.

Proof. The reverse triangle inequality gives $|n\lambda^{n-1} - 1| \geq n - 1$ and $|a_{n,1}| \leq n + 1$; for $j \geq 2$, $|a_{n,j}| \leq n^j/j!$. Any product indexed by a composition $j_1 + \dots + j_r = k$ is therefore bounded by a constant depending only on k times n^k , and there are finitely many compositions of k ; summing (8.4) gives (8.6), since $|d_n| \leq 2$ and every pole order is at most $k + 1$. Removing the $r = k$ term leaves pole orders at most k , giving (8.7) for $|d_n| \leq 1$. The removed term has modulus at least $(n - 1)^k |d_n|^{-k-1}$, and the ratio of the remainder bound to that lower bound is at most $C_k(n/(n - 1))^k |d_n| \rightarrow 0$ when $d_n \rightarrow 0$. $\square$

8.3 The exact collapse law and the trichotomy

Theorem 8.2 (Exact coefficient-radius collapse; requires drift). *For the family (8.1) with $\sigma = \sigma(\lambda) > 0$ ($\sigma = \infty$ allowed), for every $k \geq 0$,*

$$\limsup_{n \rightarrow \infty} |c_{n,k}|^{1/n} = e^{(k+1)\sigma}, \quad (8.9)$$

and the coefficient of $u^k v$ in the normalized formal linearizer has exact radius

$$\boxed{\text{rad}(h_{k,1}) = R e^{-(k+1)\sigma}}. \quad (8.10)$$

If $0 < \sigma < \infty$ the full normalized linearizer lies in the radius-free ring $\mathcal{A}_1$ over Γ_* but **not** in the common-domain germ ring $\mathcal{R}_1$; if $\sigma = \infty$ it does not even lie in $\mathcal{A}_1$.

Proof. For $k = 0$, (8.9) is $\limsup |d_n|^{-1/n} = e^\sigma$, which follows from the definition of σ because $|d_n| \leq 2$, so replacing $\log |d_n|^{-1}$ by its positive part does not change the limsup after division by n . For fixed $k \geq 1$ and finite positive σ , (8.6) gives the upper bound after n -th roots since $(C_k n^k)^{1/n} \rightarrow 1$. Choose a subsequence n_j with $n_j^{-1} \log |d_{n_j}|^{-1} \rightarrow \sigma$; since $\sigma > 0$, $d_{n_j} \rightarrow 0$, so (8.8) applies and

$$|c_{n_j,k}|^{1/n_j} = |a_{n_j,1}|^{k/n_j} |d_{n_j}|^{-(k+1)/n_j} |1 + o(1)|^{1/n_j} \rightarrow e^{(k+1)\sigma},$$

using $n_j - 1 \leq |a_{n_j,1}| \leq n_j + 1$. If $\sigma = \infty$, choose n_j with $|d_{n_j}|^{-1/n_j} \rightarrow \infty$: again $d_{n_j} \rightarrow 0$ and the same asymptotic gives $|c_{n_j,k}|^{1/n_j} \rightarrow \infty$. The Cauchy–Hadamard formula in (8.3) now gives (8.10). For $0 < \sigma < \infty$, theorem 6.3(ii) says every coefficient of the full nonlinear linearizer is a convergent germ, while the radii (8.10) have infimum zero, so its coefficients cannot all be holomorphic on any $\mathbb{D}_r$ with $r > 0$. For $\sigma = \infty$ already $h_{0,1}$ has radius zero. Formal uniqueness in theorem 6.3 shows that no alternative normalized solution, even with additional positive Hahn exponents, can avoid these coefficients. $\square$

Remark 8.3 (Why two exponents, and why the first one does not suffice). The first coefficient $h_{0,1}$ alone loses the radius only by $e^{-\sigma}$, a loss compatible with the existence of *some* smaller common disk; it is the higher multiplier jets $h_{k,1}$ that force losses by every power $e^{-(k+1)\sigma}$ and thereby exclude such a disk. The argument is therefore strictly stronger than a first-order homological counterexample. The rational independence of 1 and $\sqrt{2}$ prevents different pairs (k, ℓ) from merging into the same Hahn coefficient and lets the v -linear block be isolated without contamination by higher v -powers. Source 02 records explicitly that replacing v by an integer power of u would merge blocks and could introduce cancellation, so that the proof would no longer establish the claimed radius for a particular Hahn coefficient, and that it does not silently make that replacement. It also records that the same necessity statement is *not* asserted for a fixed cyclic value group such as $\mathbb{Z}$, where different parameter blocks can collide at one Hahn exponent; resolving the effect of those collisions is a separate problem. Source 08 later resolved it in one variable. Over $\mathbb{Z}$ the colliding blocks cannot cancel the highest pole, and the coefficient radii are $Re^{-m\sigma}$ ([theorem 8.15](#)).

Corollary 8.4 (A finite-support instability). *For every irrational multiplier with $0 < \sigma < \infty$, two nonzero positive Hahn coefficients already suffice for a holomorphic perturbation of a rotation to have a radius-free germ linearizer and no common-domain linearizer; the input coefficients can be taken to be rational functions of z , namely z at exponent 1 and $P_R(z)$ at exponent $\sqrt{2}$.*

This is source 02's statement. Source 08 sharpens it to *one* nonzero positive Hahn coefficient over $\mathbb{Z}$ ([corollary 8.18](#)).

Theorem 8.5 (The common-domain / radius-free trichotomy). *Let $\text{CD}(\lambda)$ assert that for every set-sized ordered abelian Γ , every $R > 0$ and every normalized input (6.7) with coefficients on $\mathbb{D}_R$ there is a normalized linearizer in $\mathcal{R}_1$; let $\text{CG}(\lambda)$ assert that every such input with radius-free germ coefficients has a normalized linearizer in $\mathcal{A}_1$. Then for an irrational unit multiplier*

$$\text{CD}(\lambda) \iff \sigma(\lambda) = 0, \quad \text{CG}(\lambda) \iff \sigma(\lambda) < \infty, \quad (8.11)$$

and both equivalences survive replacing the universal quantifier over Γ by the single group Γ_ . In the positive common-domain case the linearizer and its inverse preserve the original disk, not merely an unspecified smaller one.*

Proof. Positive implications: [theorem 6.3](#)(i), (ii). For $0 < \sigma < \infty$, [theorem 8.2](#) disproves CD within Γ_* while the radius-free implication still holds; for $\sigma = \infty$ the same example has a nonconvergent coefficient and disproves both. $\square$

Remark 8.6 (One cyclic scale suffices; source 08). Both equivalences of [theorem 8.5](#) also survive replacing the universal quantifier over Γ by *any* single nonzero group, in particular by $\mathbb{Z}$. See [corollary 8.17](#) and the paragraph after it. The negative assertions then use one Hahn monomial instead of the two exponents of Γ_* . The drift hypothesis is still required.

The distinction between the middle two columns of [table 6](#) is not a contradiction: for entire input coefficients a finite exponential loss of Taylor coefficient growth leaves every coefficient entire, whereas for a finite input radius repeated losses can exhaust every positive common radius. The fixed coefficient algebra, and not the mere existence of Hahn sums, determines the answer.

Arithmetic (SD_{dr})	Common ordinary disk ($\mathcal{R}_1$)	Radius-free germ / entire	Polynomial coefficients
$\sigma = 0$	Always, with the <i>same</i> $\mathbb{D}_R$	Always / always	Always
$0 < \sigma < \infty$	Not universally; explicit radii collapse to zero	Always / always	Always
$\sigma = \infty$	Not universally	Neither universally	Always

Table 6: Universal normalized linearization in the drifted category. Negative common-domain assertions use $\Gamma_* = \mathbb{Q} + \sqrt{2}\mathbb{Q}$ here; by source 08 they already hold over $\mathbb{Z}$ with a single Hahn monomial (theorem 8.15). Positive assertions hold for every set-sized ordered abelian group. The polynomial column means that each Hahn coefficient is a polynomial, not that the whole coordinate change is a finite polynomial over the Hahn field.

8.4 The several-variable collapse

Theorem 8.7 (Exact multivariable radius collapse). *Put $P_R^{(d)}(z) = \prod_{j=1}^d (1 - z_j/R)^{-1} - 1 - \sum_j z_j/R = \sum_{|\alpha| \geq 2} R^{-|\alpha|} z^\alpha$, let $\mathbf{1} = (1, \dots, 1)^T$, and in the same two-monomial workspace consider*

$$F_{u,v}(z) = (1+u)Lz + vP_R^{(d)}(z)\mathbf{1}, \quad (8.12)$$

the multiplicative detuning $(1+u)L$ being what makes the highest reciprocal pole uniform over every component and every multi-index. Assume NR and $\sigma_d(L) > 0$. Then for every $k \geq 0$

$$\text{rad}_{\text{iso}}(h_{k,1}) = R e^{-(k+1)\sigma_d(L)}. \quad (8.13)$$

Consequently the universal common-polydisk-germ property holds exactly when $\sigma_d = 0$, and the universal radius-free germ property exactly when $\sigma_d < \infty$, when the workspaces include Γ_ .*

Proof. At $v = 0$ the map is linear with identity normalized coordinate change, so the coefficient of v satisfies the homological equation for the diagonal multiplier $(1+u)L$: in component j at multi-index α with $n = |\alpha|$ it is $-R^{-n}/((1+u)^n \lambda^\alpha - (1+u)\lambda_j)$. Write $D_{\alpha,j} = \lambda^\alpha - \lambda_j$, $A_{\alpha,j,1} = n\lambda^\alpha - \lambda_j$ and $A_{\alpha,j,r} = \binom{n}{r} \lambda^\alpha$ for $r \geq 2$; (8.4) remains valid with these replacements, and since all phases have modulus one, $|A_{\alpha,j,1}| \geq n-1$ and $|A_{\alpha,j,r}| \leq n^r/r!$, so the estimates of lemma 8.1 hold with constants independent not only of n but also of α and j . The upper estimate gives $\max_{|\alpha|=n,j} |C_{\alpha,j,k}| \leq C_k n^k \max\{1, \Delta_n^{-k-1}\}$; for the lower bound choose n along a subsequence realizing the limsup in (6.14) and at each such degree a pair (α, j) attaining the finite minimum Δ_n , positivity of σ_d ensuring $\Delta_n \rightarrow 0$, so that the highest-pole term dominates its remainder uniformly in the chosen multi-indices. Multiplication by R^{-n} and the definition of rad_{iso} give (8.13). For finite positive σ_d , theorem 6.16 makes every coefficient a convergent germ while the displayed isotropic radii have infimum zero, and every neighborhood of 0 in $\mathbb{C}^d$ contains an equal-radius polydisk; for infinite σ_d the first v coefficient already fails to be a germ. $\square$

8.5 Continued fractions: every regime occurs, and none is Brjuno

Let θ be irrational with continued-fraction convergents p_j/q_j , and $\|x\|_{\mathbb{Z}}$ the distance to the nearest integer. The classical estimates

$$\frac{1}{q_j + q_{j+1}} < \|q_j \theta\|_{\mathbb{Z}} < \frac{1}{q_{j+1}}, \quad \|m\theta\|_{\mathbb{Z}} \geq \|q_j \theta\|_{\mathbb{Z}} \quad (q_j \leq m < q_{j+1}) \quad (8.14)$$

are standard [12, 11].

Proposition 8.8 (Continued-fraction formula). *For $\lambda = e^{2\pi i\theta}$,*

$$\sigma(\lambda) = \tau(\lambda) = \limsup_{j \rightarrow \infty} \frac{\log q_{j+1}}{q_j}. \quad (8.15)$$

Proof. For real x , $4\|x\|_{\mathbb{Z}} \leq |1 - e^{2\pi i x}| \leq 2\pi\|x\|_{\mathbb{Z}}$ — the upper bound from $|\sin y| \leq |y|$, the lower from concavity of $\sin$ on $[0, \pi/2]$ — and constants do not affect a logarithmic limsup divided by m . Hence both invariants equal $\limsup_m m^{-1} \log \|m\theta\|_{\mathbb{Z}}^{-1}$, with $m = n - 1$ and $m/(m + 1) \rightarrow 1$. At $m = q_j$, (8.14) gives the lower bound $\log q_{j+1}/q_j$; for $q_j \leq m < q_{j+1}$ the other half gives $m^{-1} \log \|m\theta\|_{\mathbb{Z}}^{-1} \leq \log(q_j + q_{j+1})/q_j \leq \log q_{j+1}/q_j + (\log 2)/q_j$, and $(\log 2)/q_j \rightarrow 0$. $\square$

Equation (8.15) is the point at which *one* continued-fraction quantity governs both of the one-variable divisor rates SD_{dr} and SD_{ex} ; sources 02 and 03 prove it independently with the same estimates, so it is printed once. It does *not* identify the two notions: the hypotheses that accompany them, drift versus exactness, remain different (warning 2.6).

The classical Brjuno condition is

$$B(\theta) = \sum_{j \geq 0} \frac{\log q_{j+1}}{q_j} < \infty, \quad (8.16)$$

strictly stronger than the mere convergence of those summands to zero, which is exactly the vanishing of the rate.

Theorem 8.9 (Explicit arithmetic separations). *The following continued fractions are explicitly specified, not numerically approximated.*

- (a) Rate zero, not Brjuno (source 03's form). With $a_1 = 16$ and $a_{k+1} = \lceil \exp(q_k/k) \rceil$, $q_{k+1} = a_{k+1}q_k + q_{k-1}$, $q_0 = 1$, $q_1 = 16$, one has rate 0 and $B(\theta) = \infty$.
- (b) Rate zero, not Brjuno (source 02's form). With $a_1 = 2$ and $a_{j+1} = \lceil \exp(q_j/(j+1)) \rceil$, the same conclusion holds.
- (c) Any prescribed finite rate $s > 0$. $a_{j+1} = \lceil e^{sq_j} \rceil$ gives rate exactly s .
- (d) Infinite rate. $a_{j+1} = \lceil e^{q_j^2} \rceil$, or $a_{k+1} = \lceil \exp(kq_k) \rceil$, gives rate ∞ .
- (e) Source 01's non-Brjuno witness. With $q_{-1} = 0$, $q_0 = 1$ and $a_{n+1} = 2^{q_n^2}$, one has $(\log q_{n+1})/q_n \geq q_n \log 2$, so $B(\theta) = \infty$ (this one is used in proposition 11.1, where only divergence of the Brjuno sum is needed).

Consequently every row of theorem 6.1 and every regime of theorem 8.5 is a nonempty arithmetic possibility, and the rate-zero rows apply to rotations that are not Brjuno.

Proof. In (a), every partial quotient is a finite positive integer, so θ is a well-specified irrational and $q_k \rightarrow \infty$; from the recursion $q_k \exp(q_k/k) \leq q_{k+1} \leq q_k(\exp(q_k/k) + 2)$, whence

$$\frac{1}{k} + \frac{\log q_k}{q_k} \leq \frac{\log q_{k+1}}{q_k} \leq \frac{1}{k} + \frac{\log q_k}{q_k} + \frac{\log(1 + 2e^{-q_k/k})}{q_k}.$$

Both outer expressions tend to zero, so the rate is zero by proposition 8.8, while the middle expression is at least $1/k$, so (8.16) diverges. Case (b) is identical with $j + 1$ in place of k and $\log 3$ in place of $\log(1 + 2e^{-q_k/k})$. In (c) and (d) the same estimates give the quotient tending to s , respectively to $q_j + o(1)$ or $k + o(1)$. In (e) the recursion gives $q_{n+1} \geq a_{n+1}q_n = 2^{q_n^2}q_n$, so $\log q_{n+1} \geq q_n^2 \log 2$. $\square$

Remark 8.10 (No contradiction with Brjuno–Yoccoz). By the Brjuno–Yoccoz theorem the ordinary quadratic $Q_\lambda(w) = \lambda w + w^2$ is locally analytically linearizable exactly for Brjuno rotations [13, 14]. For the multipliers of [theorem 8.9\(a\)](#), (b) it is therefore *not* analytically linearizable, while every positive Hahn perturbation with coefficients on a fixed disk *is* linearizable on that same disk. This weakens nothing: the ordinary nonlinear part in Q_λ is of size one, whereas the nonlinear part in [theorem 6.1](#) has strictly positive Hahn support and zero ordinary reduction, and the resulting conjugacy is a Hahn family, not a convergent ordinary family under a numerical specialization of t . Sources 01, 02 and 03 all invoke Brjuno–Yoccoz for exactly this separation, as an external classical theorem which none of them reproves.

8.6 A single cyclic exponent: collisions cannot cancel the highest pole

This subsection is source 08's, and **everything in it requires drift** (SD_{dr} , [warning 2.6](#)). It answers [question 25.2](#) in the one-variable setting in which that question is posed. The negative direction of [theorem 8.2](#), proved above with two rationally independent exponents, already holds over the cyclic value group $\mathbb{Z}$ with a single Hahn monomial. Rates are written $\sigma = \sigma(\lambda)$, never τ , because the multiplier drifts. The numerical equality $\sigma = \tau$ of [proposition 8.8](#) does not move a statement from SD_{dr} to SD_{ex} .

Work over $K_{\mathbb{Z}} = \mathbb{C}((t))$, $v(t) = 1$, and for a fixed finite $R > 0$ put

$$F_t(z) = (\lambda + t)z + tP_R(z), \quad (8.17)$$

with P_R as in (5.6). One Hahn monomial, t , carries both the drift z and the nonlinearity $P_R(z)$. [Theorem 6.3](#), with $A = \{1\}$ and $f_1 = z + P_R$, gives a unique normalized positive-support formal linearizer

$$H_t(z) = z + \sum_{m \geq 1} t^m h_m^{\text{cyc}}(z), \quad H_t \circ F_t = (\lambda + t)H_t. \quad (8.18)$$

Bookkeeping by two formal parameters. Let u, v be independent formal parameters and let $F_{u,v}$ be as in (8.1), now over $\mathbb{C}[[u, v]]$ rather than over Γ_* . Write $H_{u,v} = z + W$ with $W = \sum_{k \geq 0, \ell \geq 1} u^k v^\ell h_{k,\ell}(z)$. The conjugacy equation $H_{u,v} \circ F_{u,v} = (\lambda + u)H_{u,v}$ becomes

$$\mathcal{L}W = -vP_R + uW - \mathcal{N}_{u,v}W, \quad \mathcal{N}_{u,v}W = W(\lambda z + uz + vP_R(z)) - W(\lambda z). \quad (8.19)$$

The operator $\mathcal{L}^{-1}(u \text{ id} - \mathcal{N}_{u,v})$ raises the total parameter degree by at least one. So (8.19) has a unique normalized solution: the geometric series in that operator applied to $-\mathcal{L}^{-1}(vP_R)$. Each parameter degree is a finite sum of operator patterns, and every term is divisible by v . The map $(k, \ell) \mapsto k + \ell\sqrt{2}$ is injective on $\mathbb{N}^2$, and the coefficient recursion has a unique solution; so the $h_{k,\ell}$ are exactly the coefficients at $u^k v^\ell$ of the linearizer of (8.1) over Γ_* , and in particular $h_{k,1}$ is the function of (8.3). The parameters u, v are bookkeeping only. They do not assume that the final value group contains two independent positive elements.

The fold. Only finitely many monomials $u^k v^\ell$ have a given total degree. So $u = v = t$ defines a homomorphism from formal series in (u, v) to formal series in t that respects the Taylor substitutions in (8.19). It carries $F_{u,v}$ to F_t and the conjugacy equation to that of (8.18), so by uniqueness

$$h_m^{\text{cyc}} = \sum_{\ell=1}^m h_{m-\ell,\ell}. \quad (8.20)$$

The whole sum (8.20) must be analyzed, not one of its summands. Over $\mathbb{Z}$ the blocks that [remark 8.3](#) keeps apart collide at a single Hahn exponent. The summand with $\ell = 1$ is $h_{m-1,1}$, whose radius is

$Re^{-m\sigma}$ by (8.10); its highest pole is (8.5) with $k = m - 1$. The question is whether the summands with $\ell \geq 2$ can cancel that pole.

Lemma 8.11 (Pole multiplicity after collisions; source 08). *At total parameter degree m , an expanded term of the geometric-series solution of (8.19) that contributes to z^n contains at most m homological inversions. Their Taylor degrees form a nondecreasing chain in $\{2, \dots, n\}$ whose last entry is n . If the term contains at least two factors P_R , at least one inversion has Taylor degree strictly below n . In particular the power of d_n^{-1} in such a term is at most $m - 1$.*

Proof. The initial forcing uses one parameter v , one P_R and one inversion. The term uW adds one parameter and one inversion without changing the coordinate degree. A Taylor block of order k in $\mathcal{N}_{u,v}$ differentiates k times and multiplies by k factors, each either uz or vP_R . A factor uz has degree one, so its net increment is zero. A monomial of P_R has degree at least two, so its net increment is positive. Degrees therefore never decrease, and any block that contains a further P_R strictly increases them. Each new inversion costs at least one more parameter degree, so at most m inversions occur. With two or more factors P_R , a strict increase happens after the initial inversion, so some inversion sits below the final degree n . $\square$

The argument applies term by term, before any cancellation. It is therefore unaffected by the many parameter blocks that merge when $u = v = t$.

Lemma 8.12 (Polynomial numerators at fixed parameter degree; source 08). *Fix $m \geq 1$ and $R > 0$. In degree z^n , the expanded terms of total parameter degree m are bounded in number by a polynomial in n , and so are the moduli of their derivative and combinatorial factors, with constants depending only on m . If a term contains ℓ factors P_R , the product of their Taylor coefficients is exactly $R^{\ell+1-n}$.*

Proof. There are finitely many Taylor-block patterns of total parameter degree m , independently of n . Coordinate degrees in a nonzero pattern are at most n , so its derivative factors grow polynomially. If the ℓ nonlinear factors have degrees $n_1, \dots, n_\ell$, the linear factors contribute nothing to the degree, so $\sum_{j=1}^{\ell} (n_j - 1) = n - 1$. There are at most $n^{\ell-1}$ such degree lists, and the coefficients R^{2-n_j} of P_R multiply to $R^{2\ell-(n-1+\ell)} = R^{\ell+1-n}$. The operations uW have coefficient one, and $1 \leq \ell \leq m$, so the powers R^ℓ are absorbed into constants depending on m and R . $\square$

Proposition 8.13 (The upper bound under drift; source 08). *If $\sigma < \infty$, then for every $a > \sigma$ and every fixed m there are a constant $A_{m,a,R}$ and an integer B_m with*

$$|[z^n]h_m^{\text{cyc}}| \leq A_{m,a,R} n^{B_m} R^{-n} e^{man}. \quad (8.21)$$

Consequently $\text{rad}(h_m^{\text{cyc}}) \geq Re^{-m\sigma}$.

Proof. By lemma 8.11 each of the at most m divisors is some $d_{n'}$ with $n' \leq n$. Each reciprocal is at most $A_a e^{an}$, for a constant A_a supplied by the definition of σ . So their product is at most $A_a^m e^{man}$. Combine this with lemma 8.12, sum over the finitely many patterns, apply the root test, and let $a \downarrow \sigma$. $\square$

Repeated indices prevent replacing ma by a here, unlike in the exact case (proposition 7.11). The next proposition shows that this worst loss really occurs, after all the collisions.

At a large convergent denominator q_j of θ , every earlier divisor is only polynomially small. The best-approximation property $\|s\theta\|_{\mathbb{Z}} \geq \|q_{j-1}\theta\|_{\mathbb{Z}}$ for $0 < s < q_j$, combined with (8.14) at $j-1$ and (7.3), gives [12, 15]

$$\mathfrak{d}_s \geq 4\|q_{j-1}\theta\|_{\mathbb{Z}} > \frac{4}{q_{j-1} + q_j} > \frac{2}{q_j} \quad (1 \leq s < q_j), \quad (8.22)$$

for all sufficiently large j . This is source 08's lemma on convergents. Its other assertions are (8.14) and proposition 8.8, and its corollary on prescribed rates is theorem 8.9(c),(d); they are not reprinted.

Proposition 8.14 (Collision noncancellation at convergent degrees; source 08). *Fix $m = k + 1 \geq 1$. Let $n - 1 = q_j$ be a sufficiently large convergent denominator with $|d_n| < 1$. Then*

$$[z^n]h_{k+1}^{\text{cyc}} = -R^{2-n}(-1)^k \frac{a_{n,1}^k}{d_n^{k+1}} + E_{k,n}, \quad |E_{k,n}| \leq C_{k,R} n^{B_k} R^{-n} |d_n|^{-k}, \quad (8.23)$$

with $a_{n,1} = n\lambda^{n-1} - 1$ as in lemma 8.1, and with finite constants $C_{k,R}$ and integers B_k . For $k = 0$ the error is zero.

Proof. Separate the summand $h_{k,1}$ in (8.20). By (8.3), $[z^n]h_{k,1} = -R^{2-n}c_{n,k}$. By (8.7), $c_{n,k}$ is the highest-pole term (8.5) plus an error of modulus at most $C_k n^k |d_n|^{-k}$; this is inside the stated bound. Every other summand has at least two factors P_R . By lemma 8.11, its final divisor d_n occurs to power at most k , and all its other divisors are $d_{n'}$ with $2 \leq n' < n$. By (8.22) each of those has $|d_{n'}| = \mathfrak{d}_{n'-1} > 2/q_j$, so its reciprocal is below $n/2$. There are at most m of them, so they cost only a polynomial factor, not an exponential one. Since $|d_n| < 1$, a power $|d_n|^{-i}$ with $i \leq k$ is at most $|d_n|^{-k}$. Lemma 8.12 then gives the error bound after summing all colliding contributions. The argument bounds their absolute values and assumes nothing about their signs or phases. $\square$

Theorem 8.15 (Sharp collapse with one cyclic exponent; **requires drift**; source 08). *Let $\lambda = e^{2\pi i\theta}$ with θ irrational, $R > 0$, and $0 < \sigma = \sigma(\lambda) < \infty$. The coefficients of the normalized formal linearizer (8.18) of (8.17) satisfy*

$$\text{rad}(h_m^{\text{cyc}}) = R e^{-m\sigma} \quad (m \geq 1). \quad (8.24)$$

So every h_m^{cyc} is a convergent germ, and H_t lies in the radius-free ring $\mathcal{A}_1$ over $\mathbb{Z}$. But no positive disk carries all of them, so H_t is not in the common-domain germ ring $\mathcal{R}_1$. If $\sigma = \infty$, every h_m^{cyc} has radius zero. The value group is $\mathbb{Z}$ itself, and the input uses one Hahn monomial.

Proof. First let $0 < \sigma < \infty$. By (8.14) and (7.3), $2/q_{j+1} \leq 4/(q_j + q_{j+1}) < \mathfrak{d}_{q_j} < 2\pi/q_{j+1}$, so $\log \mathfrak{d}_{q_j}^{-1}$ and $\log q_{j+1}$ differ by less than $\log(2\pi)$. Proposition 8.8 therefore gives convergent denominators q_j along which $q_j^{-1} \log \mathfrak{d}_{q_j}^{-1} \rightarrow \sigma$. Put $n = q_j + 1$, so that $|d_n| = \mathfrak{d}_{q_j}$; then $|d_n| \leq e^{-cn}$ eventually on this subsequence for some $c > 0$. By lemma 8.1, $|a_{n,1}| \geq n - 1$. Dividing the error bound of (8.23) by the modulus of the leading term gives at most $C'_{k,R} n^{B'_k} |d_n| \rightarrow 0$. Along the subsequence the full coefficient is therefore the leading term times $1 + o(1)$, and

$$\frac{\log|[z^n]h_m^{\text{cyc}}|}{n} = -\log R + m \frac{\log |d_n|^{-1}}{n} + o(1) \rightarrow -\log R + m\sigma.$$

With proposition 8.13 this gives (8.24). The radii tend to zero, so no positive disk works for every coefficient, even though each coefficient converges separately. When $\sigma = \infty$, choose convergent denominators with $q_j^{-1} \log \mathfrak{d}_{q_j}^{-1} \rightarrow \infty$. The same ratio tends to zero, and the displayed growth tends to infinity. $\square$

Remark 8.16 (What the lower-bound proof actually excludes; source 08). The error may contain many terms, and some of them may be large. No termwise positivity is claimed. Cancellation is excluded because the leading contribution carries a full extra power of an exponentially small divisor, while every earlier divisor at a convergent degree is at worst polynomially small. A statement about independent parameter blocks alone would not establish this. At $m = 2$ the mechanism is visible directly. The drift block contributes $R^{2-n}a_{n,1}/d_n^2$ to $[z^n]h_2^{\text{cyc}}$; this is $[z^n]h_{1,1}$. The block with two factors P_R has two distinct divisor indices, and at a convergent degree one of them is not exceptional. Higher m need the full bookkeeping above.

Corollary 8.17 (No independent scales are needed; source 08). *Work in one variable. Fix any nonzero set-sized ordered abelian group Γ , in particular $\mathbb{Z}$, and an irrational unit multiplier λ . Allow positive Hahn perturbations $F = \lambda z + P$ whose coefficients are holomorphic on a common $\mathbb{D}_R$ and vanish at 0, with infinitesimal linear terms permitted, and normalize the target multiplier to be $F'(0)$. Then every such input has a normalized linearizer with a common positive coefficient disk if and only if $\sigma(\lambda) = 0$. At $\sigma = 0$ the original disk $\mathbb{D}_R$ is preserved.*

Proof. Necessity for $\sigma > 0$ is [theorem 8.15](#), already over $\mathbb{Z}$. For another nonzero Γ , choose $\eta > 0$ and substitute $t \mapsto t^\eta$. Since $n \mapsto n\eta$ is a strictly increasing additive embedding $\mathbb{Z} \rightarrow \Gamma$, the linearizer of $(\lambda + t^\eta)z + t^\eta P_R$ is $z + \sum_m t^{m\eta} h_m^{\text{cyc}}$, with the same coefficient functions. No competing normalized solution with extra exponents exists. At the least exponent of the difference of two solutions, all perturbative terms have larger weight, since the difference of the target multipliers has positive weight too. So $\mathcal{L}$ annihilates the leading coefficient, and NR makes it zero. This is source 08's workspace statement for the drifting equation, the argument of [proposition 6.2](#) and [theorem 8.2](#). Sufficiency at $\sigma = 0$ is [theorem 6.3\(i\)](#), source 02's route B. Source 08 records that this mechanism was already in the repository and that its new content is the single-scale witness. $\square$

Combined with [theorem 6.3\(ii\)](#) and the $\sigma = \infty$ clause of [theorem 8.15](#), the corollary shows that both equivalences of [theorem 8.5](#) survive replacing the universal quantifier over Γ by any single nonzero group, in particular by $\mathbb{Z}$. Extending the workspace cannot turn the formal linearizer of (8.17) into one with a common positive coefficient domain. And collapse is not the absence of every surcomplex linearization: the formal conjugacy still evaluates bijectively on the monad ([proposition 6.14](#), which source 08 restates for the drifting setting). Source 08 constructs no globally Hahn-entire nonpolynomial map on $\mathbf{No}[i]$.

Corollary 8.18 (One positive Hahn coefficient suffices; source 08). *For every irrational multiplier with $0 < \sigma < \infty$, one nonzero positive Hahn coefficient already suffices for a holomorphic perturbation of the rotation to have a radius-free germ linearizer and no common-domain linearizer. The perturbation can be taken to be $t(z + P_R(z))$ over $\mathbb{Z}$. This sharpens [corollary 8.4](#), which uses two coefficients over Γ_* .*

Source 08 draws the main qualitative conclusion itself. Exactness of the multiplier changes the universal arithmetic threshold for a common coefficient germ, even when the Hahn value group is held fixed.

Example 8.19 (The same workspace, two answers; source 08). Take $R = 1$ and the multiplier of [theorem 8.9\(c\)](#) with $s = \log 2$, so that $\tau = \sigma = \log 2$. Over the same $\mathbb{C}((t))$ compare

$$F_{\text{ex}}(z) = \lambda z + t \frac{z^2}{1-z}, \quad F_{\text{dr}}(z) = (\lambda + t)z + t \frac{z^2}{1-z}.$$

One-variable category	Divisor tag	Universal common-germ condition	Sharp behavior for a rate in $(0, \infty)$
Exact multiplier λ	SD_{ex}	$\tau < \infty$	one common radius $Re^{-\tau}$ (theorem 7.13)
Drifting multiplier $\lambda + u$	SD_{dr} , drift required	$\sigma = 0$	one cyclic example already has radii $Re^{-m\sigma}$ (theorem 8.15)

Table 7: The sixth category in one variable, by source 08. The two rows are different categories and must never be merged into one “intermediate case” ([warning 8.21](#)). Positive assertions hold over every nonzero Γ , and negative ones already over $\mathbb{Z}$. The nonscalar exact case in several variables, open when source 08 wrote, is settled by source 10 except for the optimal radius when $r_{\text{ang}} \geq 2$ ([theorem 7.34](#) and [question 8.20](#)).

For F_{ex} every coefficient of the linearizer and of its inverse is holomorphic on $\mathbb{D}_{1/2}$, and the first one has radius exactly $\frac{1}{2}$ ([theorem 7.13](#)). For F_{dr} the coefficient of t^m has radius exactly 2^{-m} ([theorem 8.15](#)). The only change in the input is the infinitesimal linear term tz . The difference is a common-domain obstruction. It is not a change in the availability of a formal conjugacy, or of its evaluation on the monad. The infinite continued fraction specifies λ ; no floating-point approximation to it is involved.

What this subsection does not reach. Source 08 does not address the several-variable multiplicative-detuning collapse [theorem 8.7](#) over a fixed cyclic group. That statement is still asserted only over Γ_* ([question 25.2](#)). Neither [theorem 8.15](#) nor [theorem 7.13](#) gives ordinary analytic convergence after t is replaced by a nonzero complex number ([N83](#)).

8.7 Resolution: which category answers which question

Question 8.20 (Common-domain germs for a finite positive rate; *settled* for every exact diagonal unitary multiplier, the optimal radius *open* in angular rank at least two). Assume $0 < \tau(\Lambda) < \infty$ and an *exact* multiplier (([6.1](#)), tag SD_{ex}). Characterize the positive perturbations whose coefficients are holomorphic on one common ordinary polydisk and whose unique radius-free linearizer has coefficients holomorphic on *some* common, possibly smaller, polydisk.

Status. For a scalar multiplier $\Lambda = \lambda I_d$ the answer is: all of them, with the sharp universal radius $Re^{-\tau}$ ([theorem 7.13](#); source 08 for every d , source 09 for $d = 1$). For a nonscalar diagonal unitary Λ the answer is again: all of them ([theorem 7.23](#); source 10). Every coefficient of the normalized conjugacy and its inverse is holomorphic on the polydisk of radius $Re^{-r_{\text{ang}}\tau}$, where r_{ang} is the rational angular rank ([7.15](#)), and the largest universal radius R_{univ} satisfies $Re^{-r_{\text{ang}}\tau} \leq R_{\text{univ}} \leq Re^{-\tau}$, with equality $R_{\text{univ}} = Re^{-\tau}$ when $r_{\text{ang}} = 1$ ([theorem 7.34](#)). Still **open**: the value of R_{univ} when $r_{\text{ang}} \geq 2$ and $0 < \tau < \infty$ ([section 7.13](#)). At the pins of sources 01–07 the question was open throughout SD_{ex} ; at the pins of sources 08 and 09 it was open for every nonscalar Λ .

Warning 8.21 (The sources answer different questions; do not average them). Source 03 poses [question 8.20](#) as open and proves only the word-depth lower bound ([6.12](#)), stating explicitly that this does not prove radius collapse ([warning 6.10](#)). Source 02 proves outright *failure*: $\text{CD}(\lambda)$ holds iff $\sigma(\lambda) = 0$, with the exact law ([8.10](#)), so for $0 < \sigma < \infty$ the linearizer is a radius-free coefficient germ with no common domain whatsoever.

This is not a contradiction, and each source says so. Source 02’s negative direction **requires** multiplier drift $F'(0) = \lambda + u$ and two rationally independent positive exponents, both of which

source 03 excludes: source 03's multiplier is *exactly* Λ , with no infinitesimal linear part at any Hahn exponent. The resolution adopted here, and the only one consistent with both proofs:

- Theorems 8.2 and 8.5 are the answer in the drifted category SD_{dr} . There question 8.20 is settled, negatively, and the law (8.10) is exact. Source 08's theorem 8.15 shows the same failure over $\mathbb{Z}$ with one Hahn monomial.
- Sources 01–07 left question 8.20 open in the fixed-multiplier category SD_{ex} . Source 02 itself records that eliminating drift removes the direct reciprocal-jet mechanism and that a different counterexample, or a positive cancellation theorem, would be needed. Sources 08 and 09 supply a positive theorem for a scalar multiplier $\Lambda = \lambda I_d$: there every input has a common-domain linearizer, of the sharp radius $Re^{-\tau}$ (theorem 7.13). The theorem needs no cancellation. Positive nonlinear degree makes the divisor indices strictly increase (lemma 7.9), and a single rational cluster pays for all the exceptional divisors at once (theorem 7.5). Source 10 supplies the positive theorem for every nonscalar diagonal unitary Λ , with radius at least $Re^{-r_{\text{ang}}\tau}$ (theorem 7.23); again no cancellation is needed, and the number of distinct exponential rates, at most r_{ang} , is what is paid for (theorem 7.27). Only the optimal radius for $r_{\text{ang}} \geq 2$ remains open.
- The two categories must not be averaged into a single “intermediate case” claim (table 7). Neither (8.10) nor (8.24) may ever be quoted without its drift hypothesis.

For the weaker requirement of retaining the *original* disk, one positive parameter is already enough: lemma 5.2 and the first-order equation give failure whenever the rate is positive. The issue that sources 01–07 left unresolved concerns the existence of *some* smaller common neighborhood for every input. It is now resolved in the drifted category over any fixed value group (no such neighborhood exists in general), for a scalar exact multiplier (it always exists, with radius $Re^{-\tau}$), and for a nonscalar exact diagonal unitary multiplier (it always exists, with radius at least $Re^{-r_{\text{ang}}\tau}$, and exactly $Re^{-\tau}$ when $r_{\text{ang}} = 1$). The optimal radius when $r_{\text{ang}} \geq 2$ is still open.

9 The exact finite-return threshold and its shell cycles

Everything in this section is under RES_q and SD_0 : λ is *never* a root of unity, its residue $\alpha = \bar{\lambda}$ has exact finite order q , and **no** arithmetic condition of any kind is imposed. Warnings 2.1, 2.4 and 2.7 apply throughout. The ambient Γ is divisible here, so that K_Γ is algebraically closed and rational division of exponents is available.

9.1 Two support-controlled linearization lemmas; route C

For $T(X) = \nu X + \sum_{j=2}^D d_j X^j$ and $H(X) = X + \sum_{n \geq 2} h_n X^n$ with $\nu \neq 0$ not a root of unity, comparison of degree n in $H(T(X)) = \nu H(X)$ gives the triangular recurrence

$$(\nu - \nu^n)h_n = \sum_{j=1}^{n-1} h_j [X^n]T(X)^j, \quad h_1 = 1, \quad (9.1)$$

whose denominators are nonzero, so (9.1) defines a unique normalized formal conjugacy over any field of characteristic zero. What remains is to control all of its coefficients *simultaneously*. This is route C of remark 6.4.

Lemma 9.1 (Nonresonant residue). *Suppose $T \in \mathcal{O}_\Gamma[X]$, $v(\nu) = 0$, and $\alpha = \bar{\nu}$ is not a root of unity in $\mathbb{C}$. Then the normalized formal conjugacy and its inverse belong to $\mathcal{B}_\mathbf{M}$ for a positive-support monoid $\mathbf{M}$ depending only on the finitely many coefficients of T .*

Proof. Write $\nu = \alpha + e$ with $e \in \mathfrak{m}_\Gamma$ and let S be the finite union of the positive supports of e and of the nonlinear coefficients d_j . For $n \geq 2$ the element $\nu - \nu^n$ has nonzero constant term $\alpha - \alpha^n$, so its inverse is a formal power series in e with ordinary complex coefficients and support in $\langle \text{supp } e \rangle$. Then (9.1) proves inductively that every h_n is integral with support in $\mathbf{M} = \langle S \rangle$, and the formal inverse has the same property by lemma 3.9. $\square$

Remark 9.2 (No Diophantine bound, and why that is a real statement). There is *no* lower bound on the ordinary complex numbers $\alpha - \alpha^n$ anywhere in lemma 9.1. Those constants may be arbitrarily small in the ordinary modulus and still all have Hahn valuation zero (remark 2.3), which is exactly what makes this a SD_0 statement. It is not the assertion that “all small divisors are units, so all analytic problems disappear”: proposition 11.1 shows what is lost, and example 3.10 shows that coefficient-valuation bounds alone are insufficient at rank > 1 .

Lemma 9.3 (Divisible displacement: cancelling a common infinitesimal divisor). *Let $0 \neq c \in \mathfrak{m}_\Gamma$, $A \in X^2\mathcal{O}_\Gamma[X]$ and*

$$T(X) = (1 + c)X + cA(X) = X + cV(X), \quad V = X + A. \quad (9.2)$$

Then $1 + c$ is not a root of unity, and the normalized formal conjugacy of T to $(1 + c)X$, together with its inverse, has all coefficients supported in one positive-support monoid generated by $\text{supp } c$ and the positive parts of the coefficients of A .

Proof. For a positive integer n the first nonzero term of $(1 + c)^n - 1$ is nc , all remaining terms having strictly larger valuation, so the multiplier is not a root of unity. For $n \geq 2$,

$$(1 + c) - (1 + c)^n = -c(n - 1)u_n(c), \quad u_n(0) = 1, \quad u_n \in \mathbb{Q}[c]. \quad (9.3)$$

For $j < n$ the coefficient $[X^n]T(X)^j$ contains at least one nonlinear term, hence is divisible by c as a polynomial in c and the coefficients of A . Substituting this and (9.3) into (9.1) cancels one factor c before any valuation estimate is made. The remaining division is by the nonzero ordinary integer $n - 1$ — a valuation unit in equal characteristic zero, which is where warning 2.7 enters — and by $u_n(c)$, whose inverse is supported in $\langle \text{supp } c \rangle$. Induction on n gives one common monoid, a finite union of well-ordered positive supports; inversion of H uses only polynomials in its coefficients. $\square$

This cancellation is the whole reason the exact threshold is read from a *return* map: multiplying all the small divisors in (9.1) together would lose the common factor present in every nonlinear numerator. Source 01 states this explicitly, and no other source performs this cancellation.

Lemma 9.4 (Roots on the unit shell). *Let $Q \in \mathcal{O}_\Gamma[X]$ with $Q(0) = 1$ and suppose $P = \bar{Q}$ has degree $M > 0$. Then Q has no root in $\mathfrak{m}_\Gamma$ and exactly M roots of valuation zero, counted with algebraic multiplicity; for each nonzero root $a \in \mathbb{C}$ of P , the roots of Q with residue a have total multiplicity equal to the multiplicity of a in P .*

Proof. A root of positive valuation is impossible because the constant term would strictly dominate. Factor in the algebraically closed K_Γ as $Q(X) = \prod_j (1 - X/\beta_j)$ with roots repeated by multiplicity; every $v(\beta_j) \leq 0$ so all factors are integral, a factor with $v(\beta_j) < 0$ reduces to 1, and one with $v(\beta_j) = 0$ reduces to $1 - X/\bar{\beta}_j$. Reducing this finite factorization proves all assertions; a simple root of P corresponds to exactly one simple root of Q in that residue class. $\square$

9.2 The exact maximal ball

Let

$$f(z) = \lambda z + \sum_{n=2}^d a_n z^n \in K_\Gamma[z], \quad d \geq 2, \quad a_d \neq 0, \quad v(\lambda) = 0, \quad (9.4)$$

with λ not a root of unity, and put $\eta(f) = \max_{a_n \neq 0} (-v(a_n)/(n-1))$, $\alpha = \bar{\lambda}$. All maxima are finite maxima over nonzero polynomial coefficients.

Theorem 9.5 (Exact finite-return linearization). *Define $r = r(f)$ as follows. If α is not a root of unity, set $r = \eta(f)$. If α has exact order $q \geq 1$ (tag RES_q), write*

$$g(z) = f^{\circ q}(z) = \lambda^q z + \sum_{n=2}^{d^q} b_n z^n, \quad c = \lambda^q - 1, \quad \delta = v(c) > 0, \quad (9.5)$$

and set

$$r = \max_{b_n \neq 0} \frac{\delta - v(b_n)}{n-1} \quad (9.6)$$

— read off the actual coefficients of the finite q -th return iterate, including all cancellations — in which case $r > \eta(f)$.

Let $H(z) = z + \sum_{n \geq 2} h_n z^n$ be the unique normalized formal solution of $H \circ f = \lambda H$. Then H and its formal inverse are strongly Hahn-evaluable on B_r and their evaluated maps are mutually inverse isometries of B_r ; in particular $f : B_r \rightarrow B_r$ is conjugate to multiplication by λ . More strongly, with $\rho = t^r$,

$$H_r(X) = \rho^{-1} H(\rho X) = X + \sum_{n \geq 2} h_n \rho^{n-1} X^n \quad (9.7)$$

and its inverse have a uniform positive-support certificate, so in particular $v(h_n) + (n-1)r \geq 0$ for $n \geq 2$ — though the certificate is strictly stronger than that inequality.

No strictly larger centered valuation ball admits an injective conjugacy of f to multiplication by λ . In the nonresonant-residue case the obstruction is a nonzero $p \in S_r$ with $f(p) = 0$; in the RES_q case it is a nonzero $p \in S_r$ of exact period q .

Proof. Nonresonant residue. Take $\rho = t^\eta$ and $F(X) = \rho^{-1} f(\rho X)$; its nonlinear coefficients have valuations $v(a_n) + (n-1)\eta \geq 0$ with at least one equal to zero, so $F \in \mathcal{O}_\Gamma[X]$ with nonlinear reduction. By lemmas 3.9 and 9.1 its normalized conjugacy and inverse are strongly summable mutually inverse isometries of $\mathfrak{m}_\Gamma$, and scaling back gives every existence and support assertion. For maximality apply lemma 9.4 to $Q(X) = F(X)/(\lambda X) = 1 + \sum_{n \geq 2} (a_n \rho^{n-1}/\lambda) X^{n-1}$, whose reduction is nonconstant: Q has a unit root u , and $p = \rho u \in S_\eta$ satisfies $f(p) = 0 = f(0)$. Any centered ball strictly containing B_η contains that whole shell, and an injective conjugacy there would give $\lambda H(p) = H(f(p)) = H(f(0)) = \lambda H(0)$.

The resonant threshold lies inside the injectivity scale. With α of order q , the multiplier λ^q has residue one and is not one because λ is not a root of unity, so $c \neq 0$ and $\delta > 0$. Scaling by $\sigma_0 = t^\eta$, the polynomial $F_\eta(X) = \sigma_0^{-1} f(\sigma_0 X)$ is integral with reduction of degree at least two; reduction commutes with polynomial iteration, so $\sigma_0^{-1} g(\sigma_0 X) = \bar{F}_\eta^{\circ q}(X)$ also has degree at least two, and some $n \geq 2$ satisfies $v(b_n) + (n-1)\eta = 0$. For that index $(\delta - v(b_n))/(n-1) = \eta + \delta/(n-1) > \eta$, so $r > \eta$. At the final scale $\rho = t^r$ the polynomial $F(X) = \rho^{-1} f(\rho X)$ has reduction exactly αX : every nonlinear coefficient acquires positive valuation. It maps $\mathfrak{m}_\Gamma$ to itself and preserves the residue direction of a unit up to multiplication by α .

Linearizing the return and recovering f . At $\rho = t^r$ write

$$G(X) = \rho^{-1}g(\rho X) = (1+c)X + cA(X), \quad A(X) = \sum_{n \geq 2} \frac{b_n \rho^{n-1}}{c} X^n \in X^2 \mathcal{O}_\Gamma[X], \quad (9.8)$$

integrality being exactly (9.6). Lemma 9.3 constructs the normalized conjugacy J of G to $(1+c)X$ with a uniform support certificate. It remains to see that J linearizes F and not merely its q -th iterate: in the formal power-series group $T = J \circ F \circ J^{-1}$ commutes with multiplication by $\mu = 1+c$ because F commutes with $G = F^{\circ q}$, so writing $T(X) = \sum_{n \geq 1} \tau_n X^n$ the commutation says $\tau_n(\mu^n - \mu) = 0$; since μ is not a root of unity all τ_n with $n \geq 2$ vanish, and the linear coefficient is λ , so $T(X) = \lambda X$ and J is the unique normalized conjugacy of F . Adjoining the positive supports of the finitely many coefficients of F to the certificate, all compositions are legitimate by lemma 3.9, and scaling by ρ gives the existence, isometry and coefficient assertions.

The exact-period obstruction. Define the integral polynomial

$$Q(X) = \frac{G(X) - X}{cX} = 1 + \sum_{n \geq 2} \frac{b_n \rho^{n-1}}{c} X^{n-1}. \quad (9.9)$$

At least one nonlinear coefficient of G attains equality in (9.6), so $P = \bar{Q}$ is nonconstant, and lemma 9.4 gives a unit root u ; then $p = \rho u \in S_r$ and $g(p) = p$. The least period k of p divides q ; every point of its finite orbit lies on S_r and reduction of the normalized orbit multiplies by α at each step, so if $a = \bar{u} \neq 0$ then $\alpha^k a = a$ and $q \mid k$, whence $k = q$. On any larger centered ball an injective conjugacy would give $H(p) = H(f^{\circ q}(p)) = \lambda^q H(p)$, and $\lambda^q \neq 1$ forces $H(p) = 0 = H(0)$. $\square$

Remark 9.6 (Exactly what is exact). Theorem 9.5 determines the maximal *centered valuation ball* only: not the full pointwise summability locus of the formal series (exceptional evaluations on the bounding shell are allowed), not a maximal non-ball invariant domain, and not every analytic continuation of the conjugacy. The formal-coefficient family lands in $\mathcal{F}_1$; theorem 10.1 is what reaches the fixed-polydisk row, and only under RES_q .

9.3 The shell polynomial and the closest cycles

Retain r, ρ, c, Q and put

$$P(X) = \bar{Q}(X), \quad M = \deg P = \max\{n-1 : v(b_n) + (n-1)r = \delta\}, \quad (9.10)$$

a finite ordinary complex polynomial with $P(0) = 1$: the initial polynomial on the first face of the return equation seen from the origin.

Theorem 9.7 (Resonant shell cycles). *(i) $f^{\circ q}(z) = z$ has exactly M nonzero solutions on S_r , counted with algebraic multiplicity; every such solution has exact period q ; and there are no nonzero periodic points in B_r .*

(ii) $P(\alpha X) = P(X)$, i.e. $P \in \mathbb{C}[X^q]$; in particular $q \mid M$.

(iii) If P is squarefree there is one simple point $p_a = \rho(a + u_a)$, $u_a \in \mathfrak{m}_\Gamma$, for every root a of P ; these form exactly M/q distinct cycles, and $f(p_a) = p_{\alpha a}$.

(iv) For each such simple point,

$$\text{res} \left(\frac{(f^{\circ q})'(p_a) - 1}{c} \right) = aP'(a), \quad v((f^{\circ q})'(p_a) - 1) = \delta. \quad (9.11)$$

Proof. Counting and residue classes follow from [lemma 9.4](#) applied to Q ; exact period q was proved in [theorem 9.5](#); a nonzero periodic point in B_r would be sent by the injective conjugacy to a nonzero periodic point of multiplication by λ , which has none. For the symmetry write $F = \rho^{-1}f(\rho X)$ and $G = F^{\circ q} = X + cW(X)$ with $W = XQ$. The identity $F \circ G = G \circ F$ gives $(F(X + cW(X)) - F(X))/c = W(F(X))$, an identity of integral polynomials whose reduction leaves only the first Taylor term on the left since $c \in \mathfrak{m}_\Gamma$; as $\bar{F} = \alpha X$ this is $\alpha \bar{W}(X) = \bar{W}(\alpha X)$, and substituting $\bar{W} = XP$ gives $P(\alpha X) = P(X)$, so only degrees divisible by q occur. When P is squarefree, uniqueness in each residue class gives $f(p_a) = p_{\alpha a}$, and the action of α on $\mathbb{C}^\times$ has orbits of length q . Finally $G'(X) = 1 + cW'(X)$, and at $p_a = \rho u$ with $\bar{u} = a$ the derivative of the unscaled iterate equals $G'(u)$, whose reduction is $\bar{W}'(u) = P(a) + aP'(a) = aP'(a) \neq 0$. $\square$

Remark 9.8 (Multiplicity is not the number of distinct points). If P has a multiple root, [theorem 9.7](#) gives a multiplicity count only. It does *not* say that several actual periodic points have collided: higher Hahn coefficients can split one repeated initial root into distinct points with the same residue direction, and in that situation $v((f^{\circ q})'(p) - 1) = \delta$ need not hold, because the displayed residue may vanish. [Example 13.2](#) exhibits the degeneracy concretely.

Corollary 9.9 (A finite obstruction certificate). *In the RES_q case the data q, c, r, P certify exact maximality without computing a single coefficient of the infinite conjugacy. At most $d^q - 1$ shell points occur with multiplicity, and their count is a positive multiple of q .*

Proof. The return polynomial has degree d^q and $M = \deg P \leq d^q - 1$; exactness and positivity are [theorems 9.5](#) and [9.7](#). $\square$

[Section B](#) states the accompanying four-step finite algebraic procedure and *exactly* what it is relative to.

9.4 A finite-return perturbation certificate

Theorem 9.10 (Stability above the first return face). *Let f satisfy the RES_q hypotheses of [theorem 9.5](#) with data q, c, δ, r, P , and let $\tilde{f}$ be another nonlinear polynomial with the same multiplier λ and return coefficients $\tilde{b}_n$ (pad either list with zeros). Suppose*

$$v(\tilde{b}_n - b_n) + (n - 1)r > \delta \quad \text{for every nonzero difference.} \quad (9.12)$$

Then $r(\tilde{f}) = r(f)$ and the shell polynomial of $\tilde{f}$ is also P . If P is squarefree and at least one coefficient changes, put

$$\epsilon = \min_{\tilde{b}_n \neq b_n} (v(\tilde{b}_n - b_n) + (n - 1)r - \delta) > 0; \quad (9.13)$$

the closest cycles have a canonical matching by residue direction, matched points satisfy

$$v(\tilde{p}_a - p_a) \geq r + \epsilon, \quad (9.14)$$

the matching intertwines f and $\tilde{f}$ on these finite cycle sets, and their leading multiplier ratios in [\(9.11\)](#) agree.

Proof. At $\rho = t^r$ the two normalized displacement polynomials are Q as in [\(9.9\)](#) and $\tilde{Q}$ with $\tilde{b}_n$ in place of b_n . The hypothesis says every coefficient of $\tilde{Q} - Q$ has positive valuation, so $\tilde{Q}$ is integral with the same nonconstant reduction P ; the inequalities defining the maximum hold at r for all $\tilde{b}_n$ and at least one remains an equality, so $r(\tilde{f}) = r$. For a simple root a of P let $u = p_a/\rho$ and

$\tilde{u} = \tilde{p}_a/\rho$, both of residue a . When $u \neq \tilde{u}$ the divided difference $(Q(u) - Q(\tilde{u}))/(\rho(u - \tilde{u}))$ is integral with nonzero residue $P'(a)$; moreover $Q(u) = 0$ and $Q(\tilde{u}) = (Q - \tilde{Q})(\tilde{u})$, of valuation at least ϵ , whence $v(u - \tilde{u}) \geq \epsilon$, which is (9.14) (it is automatic if the two roots coincide). The direction rule $a \mapsto \alpha a$ and theorem 9.7 give the rest. $\square$

Warning 9.11 (The certificate is sufficient, not necessary; and it is about the return). A perturbation may change the first return face and nevertheless preserve its maximal slope, so (9.12) is sufficient only. It is also a condition on the *return* polynomial: it must not be silently replaced by the same inequalities on the original coefficients of f . Where coefficients are available only to a specified precision, theorem 9.10 says which strict inequalities certify that the computed first face is already correct; at a tie or a possible cancellation more precision may be necessary, and treating a truncated zero as an exact zero would invalidate the maximum.

10 Common-domain lifting: one leading coefficient, two instantiations

The support arguments of section 9 establish more than formal existence but less than ordinary analyticity of every coefficient function. Under RES_q a further argument supplies exactly that regularity, and it lands in the fixed-polydisk row of section 3.2.

Theorem 10.1 (Coherent realization of resonant linearization). *Under the RES_q hypotheses of theorem 9.5, the scaled conjugacy H_r of (9.7) has an expansion*

$$H_r(X) = \sum_{\gamma \in \mathbf{M}} H_\gamma(X) t^\gamma \quad \text{in } \mathcal{O}(D)((t^\Gamma)) \quad (10.1)$$

for a single ordinary complex disk D centered at zero, with $\mathbf{M}$ a positive-support monoid, $H_0(0) = 0$, $H'_0(0) = 1$, and $H_\gamma \in X^2 \mathcal{O}(D)$ for $\gamma > 0$. One may take any sufficiently small disk on which the shell polynomial P has no zero, and the leading coefficient is

$$H_0(X) = X \exp \left(\int_0^X \left(\frac{1}{sP(s)} - \frac{1}{s} \right) ds \right), \quad (10.2)$$

whose integrand is holomorphic at zero. After an ordinary shrinking, the formal inverse of H_r also has all Hahn coefficient functions holomorphic on one common ordinary neighborhood.

Proof. Use $G = X + cV$ from (9.8), so $V = X + A = XQ$ and $V_0 := \bar{V} = XP(X)$; all coefficients of the normalized conjugacy lie in a common monoid $\mathbf{M}$ by lemma 9.3. Taylor expansion of the formal identity $H_r(X + cV) = (1 + c)H_r(X)$, followed by cancellation of c , gives

$$VH'_r + \sum_{k \geq 2} \frac{c^{k-1}}{k!} V^k H_r^{(k)} = H_r. \quad (10.3)$$

This is first read coefficientwise in X , where only finitely many terms contribute at each degree because V has order one; it may also be regrouped at a fixed Hahn exponent, because the term of index k uses at least $k - 1$ positive generators from c , so lemma 3.4 makes that regrouping finite. At exponent zero, (10.3) becomes

$$V_0 H'_0 = H_0. \quad (10.4)$$

Choose an ordinary disk D centered at zero on which P has no zero; since $P(0) = 1$ the function $1/(XP(X)) - 1/X$ has a removable singularity at zero, and (10.2) defines the unique solution of (10.4) with $H'_0(0) = 1$, with H_0/X nonvanishing on D .

Now let $\gamma > 0$ in the well-ordered monoid $\mathbf{M}$ and assume H_β constructed on this same disk for every $\beta < \gamma$. Isolating the terms containing H_γ in (10.3) gives

$$LH_\gamma := V_0 H'_\gamma - H_\gamma = R_\gamma, \quad R_\gamma = -[t^\gamma] \left((V - V_0) H'_r + \sum_{k \geq 2} \frac{c^{k-1}}{k!} V^k H_r^{(k)} \right). \quad (10.5)$$

Every appearance of a coefficient of H_r on the right is multiplied by a positive exponent, so only H_β with $\beta < \gamma$ occurs, and lemma 3.4 makes R_γ a *finite* expression in earlier coefficient functions and their derivatives — hence holomorphic on the same D , with no new shrinking. Moreover $V - V_0$ vanishes to order at least two, since the coefficient of X in V is exactly one, and the terms with $k \geq 2$ likewise, so $R_\gamma \in X^2 \mathcal{O}(D)$. Variation of the constant in (10.4) gives the unique normalized solution

$$H_\gamma(X) = H_0(X) \int_0^X \frac{R_\gamma(s)}{V_0(s)H_0(s)} ds, \quad (10.6)$$

whose apparent singularity at zero is removable (numerator of order at least two, denominator of order exactly two) and whose denominator is nonzero elsewhere on D ; thus $H_\gamma \in X^2 \mathcal{O}(D)$. Transfinite induction on $\mathbf{M}$ constructs all coefficients on one fixed disk, and they have the same formal Taylor expansions as the unique solution of (9.1), so this analytic family is the regrouping of the H_r already constructed, not a different conjugacy. For the inverse, first shrink to a neighborhood on which H_0 is biholomorphic; the exponent-zero coefficient of the inverse is H_0^{-1} , and at any positive exponent the coefficient equation for composition is linear in the unknown with coefficient $H'_0(H_0^{-1}(Y))$, which never vanishes on a sufficiently small fixed neighborhood, all other terms depending on earlier exponents through finitely many ordinary derivatives and products. $\square$

Remark 10.2 (Why the differential recursion is genuinely triangular). It would be *incorrect* to justify (10.5) by saying that $(k-1)v(c) \rightarrow +\infty$: that can fail to be cofinal in an arbitrary-rank value group (warning 3.6). The correct statement is that at a *specified* exponent γ infinitely many decompositions into generators from one well-ordered positive support are impossible, which simultaneously controls the Taylor index, the Hahn products and the number of earlier coefficients entering the right side. The construction is transfinite, but each individual coefficient equation has finite dependence.

Warning 10.3 (No uniform norm bound on the coefficient family). **Theorem 10.1** asserts *no* uniform ordinary norm bound on (H_γ) : the coefficient functions may be arbitrarily large with arbitrarily large derivatives. What is uniform is their ordinary domain of holomorphy and the Hahn support certificate. The same disclaimer is made independently by sources 02, 03, 05 and 06 for their own coefficient families.

10.1 The same formula, twice, and why both instantiations survive

Proposition 10.4 (The leading-coefficient exponential integral). *Let U be a simply connected ordinary domain containing 0, and let $V_0 \in \mathcal{O}(U)$ satisfy $V_0(0) = 0$, $V'_0(0) = \mu \neq 0$ and $V_0(w) \neq 0$ for $w \in U \setminus \{0\}$. The unique solution of $V_0 h'_0 = \mu h_0$ on U with $h_0(0) = 0$, $h'_0(0) = 1$ is*

$$h_0(w) = w \exp \left(\int_0^w \left(\frac{\mu}{V_0(\zeta)} - \frac{1}{\zeta} \right) d\zeta \right), \quad (10.7)$$

whose integrand is holomorphic at 0 and whose quotient h_0/w is nonvanishing on U .

Proof. $\mu/V_0 - 1/w$ has a removable singularity at 0 because $(V_0/w)(0) = \mu$; on a simply connected U its primitive vanishing at 0 exists and is unique, and exponentiating gives a solution with the stated normalization, unique because the logarithmic derivative determines h_0 up to a constant fixed at 0. $\square$

Formula (10.7) is *one* formula. It is source 01's (10.2) with $V_0 = XP(X)$ and $\mu = 1$, and it is source 04's leading Koenigs coordinate with $V_0 = V$ the Euler polynomial and $\mu = V'(0)$, and both sources prove it by the same transfinite variation-of-constants recursion — source 01's (10.6), source 04's (14.6). It is therefore printed once, per [convention 1.1\(i\)](#). What does *not* merge is the pair of instantiations, and both survive:

- under RES_q , $V_0 = XP(X)$ where P is the *shell polynomial* of (9.10), an ordinary polynomial determined by the first face of the return equation. Its nonzero roots are the closest periodic directions in the valued field, and its *absence* of zeros on a small ordinary disk is what permits (10.4) to be solved there. One finite polynomial thus plays two different roles, and (10.2) is the precise bridge. This identification of P is unavailable under PAR.
- under PAR, $V_0 = V$ is the perturbing field itself and μ is its derivative at 0, which is what makes the *global* factorization [corollary 14.9](#) and the monodromy exponents [lemma 14.7](#) available. That factorization is unavailable under RES_q , where V_0 is not the input.

Concretely, source 01's three regimes give

$$H_0(X) = \begin{cases} X(1 + X^2)^{-1/2}, & \sigma_0 < \delta/2, \\ X(1 - X^2)^{-1/6}(1 + 2X^2)^{-1/3}, & \sigma_0 = \delta/2, \\ X(1 - 2X^4)^{-1/4}, & \sigma_0 > \delta/2, \end{cases} \quad (10.8)$$

with branches equal to one at zero (see [theorem 13.1](#)); differentiating their logarithms verifies $XP(X)H'_0(X) = H_0(X)$ directly. Source 04's instantiation gives $h_0 = w/(1 + w)$ for $V = w + w^2$ and $h_0 = w(1 + w^q)^{-1/q}$ for $V = w(1 + w^q)$; see [section 14](#).

11 Nonresonant Hahn linearization need not be coherent

The extra conclusion of [theorem 10.1](#) cannot be imported into [lemma 9.1](#). The obstruction is ordinary small-divisor divergence in the exponent-zero coefficient, *not* a failure of Hahn summability.

Proposition 11.1 (A divergent ordinary linearizer survives on the monad). *There is an ordinary complex polynomial $f(X) = \alpha X + X^2$ with α not a root of unity whose normalized formal conjugacy is strongly Hahn-evaluable and isometric on $\mathfrak{m}_\Gamma$ in every nontrivial workspace K_Γ , yet does not lie in the radius-free ring $\mathcal{A}_1 = \mathbb{C}\{X\}((t^\Gamma))$ and so has no common-domain representative as a normalized formal conjugacy.*

Proof. Choose $\alpha = e^{2\pi i\theta}$ with θ irrational and non-Brjuno — for instance [theorem 8.9\(e\)](#). By Yoccoz's quadratic theorem the germ $\alpha X + X^2$ is analytically linearizable exactly when θ is Brjuno [13], so its unique normalized formal linearizer $H \in \mathbb{C}[[X]]$ has zero ordinary radius of convergence. All its coefficients lie in the constant field, so its Hahn support is $\{0\}$ and [lemma 3.9](#) makes H and its formal inverse strongly evaluable isometries of $\mathfrak{m}_\Gamma$, the exact maximal ball being B_0 by [theorem 9.5](#). But the coefficient at Hahn exponent zero is the divergent ordinary series H itself, so the family is not in $\mathcal{A}_1$, let alone in the fixed-polydisk or common-domain germ rings. $\square$

Lemma 3.12 rules out an evasion by change of representation: a different common-domain family cannot represent the divergent conjugacy as the same function on all infinitesimals. Likewise a normalized common-domain conjugacy satisfying the same functional identity would satisfy it formally, and uniqueness in (9.1) would force the same divergent exponent-zero coefficient.

Warning 11.2 (What [proposition 11.1](#) does and does not say). There is no conflict with Yoccoz’s theorem: an ordinary divergent series can be Hahn-summable at an infinitesimal argument because each successive power introduces a new positive exponent. That is a change of analytic category, not a proof of classical analytic linearizability. Moreover the coherent conclusion of [theorem 10.1](#) is **resonant-only**, and [proposition 11.1](#) establishes failure of a *universal* common-domain assertion in the nonresonant case — it is explicitly **not** an if-and-only-if classification, and not every nonresonant case is noncoherent.

12 Workspace invariance and the full surcomplex class

Theorem 12.1 (Workspace invariance for the finite-return package). *Let $\Gamma \subseteq \Gamma'$ be divisible ordered abelian groups and view $K_\Gamma \subset K_{\Gamma'}$. For $f \in K_\Gamma[z]$ satisfying [theorem 9.5](#), the threshold r , the normalized formal conjugacy and the shell polynomial are unchanged by the extension. The conjugacy and its inverse strongly evaluate at every point of $B_r(K_{\Gamma'})$, including points absent from K_Γ . The finite shell root set, algebraic multiplicities, exact periods and multiplier formulas gain no new points and do not change, and the common-domain coefficient functions of [theorem 10.1](#) are also unchanged.*

Proof. The embedding preserves valuation, residue and all finite coefficient arithmetic, so the return polynomial and the finite maximum (9.6) are identical, and (9.1) gives the same formal coefficients in both fields. For a new point $z \in B_r(K_{\Gamma'})$, the normalized input z/t^r has positive well-ordered support in Γ' , whose union with the constructed certificate is still well ordered, so [lemma 3.9](#) applies unchanged. Every relevant root is a root of a polynomial over the algebraically closed K_Γ , which already factors completely into linear factors in $K_\Gamma[z]$, and the same finite factorization remains valid in $K_{\Gamma'}[z]$: no additional roots, no change of multiplicity. The remaining assertions concern the same formal coefficients, the same ordinary residue polynomial and the same uniquely normalized differential recursion. $\square$

Corollary 12.2 (Transfer to the surcomplex numbers). *Let $f \in \mathbf{SC}[z] = \mathbf{No}[i][z]$ be a nonlinear polynomial fixing zero, with unit multiplier for the natural Hahn valuation and that multiplier not a root of unity. Then the formulas and conclusions of [theorems 9.5, 9.7, 9.10](#) and [10.1](#) hold on the corresponding class-sized ball in $\mathbf{SC}$, interpreted pointwise by set-sized Hahn sums, and the threshold and finite shell cycles are independent of the workspace used to calculate them.*

Proof. Place the finitely many coefficients of f in one divisible workspace and construct its sequence of formal conjugacy coefficients there. For a particular argument, enlarge the workspace to include it and apply [theorem 12.1](#); two choices agree in their common larger workspace and so define one class function pointwise. The finite obstructing root set already lies in the initial algebraically closed workspace. No sum in this construction is indexed by a proper class. $\square$

[Corollary 12.2](#) may be read in a class theory such as NBG or with definable classes in ZFC; it requires no proper-class polynomial ring as a set and does not replace Hahn summation by fine-topological convergence. The analogous statements for the Hamiltonian package are [corollary 16.9](#), and for the several-variable linearizer [proposition 4.14](#).

13 Cancellation: a tuned cubic and a two-parameter phase diagram

The exact threshold (9.6) is read off the *actual* return coefficients b_n , including the cancellations that occur when $f^{\circ q}$ is composed. This subsection exhibits a family in which that distinction is strictly consequential: any bound depending only on $a = \max_{n \geq 2} |a_n|^{1/(n-1)}$, q and $|1 - \lambda^q|$ — the shape of the general lower radius of the rank-one antecedent [22, Thm. 1.1] — throws the improvement away.

13.1 A tangent binomial, for calibration

For $f(z) = (1+c)z + az^m$ with $0 \neq c \in \mathfrak{m}_\Gamma$, $a \neq 0$, $m \geq 2$, one has $q = 1$ and $r = (v(c) - v(a))/(m-1)$. The nonzero fixed points are exactly $p^{m-1} = -c/a$, all $m-1$ of them simple and on S_r , and at each the multiplier is known *exactly*, not merely to leading order:

$$f'(p) = 1 + c + map^{m-1} = 1 - (m-1)c. \quad (13.1)$$

With $A = \text{res}(ap^{m-1}/c)$ the shell polynomial is $P(X) = 1 + AX^{m-1}$ and $H_0(X) = X(1 + AX^{m-1})^{-1/(m-1)}$ on the ordinary branch normalized to one at zero. This already shows why a degree-independent small-divisor loss can be wasteful.

13.2 A cubic with two suppressed return terms

Let $\delta > 0$, $e = t^\delta$, and

$$f_e(z) = (-1 + e)z + z^2 - z^3. \quad (13.2)$$

Its residue multiplier has order $q = 2$, $\eta(f_e) = 0$, and $c = \lambda^2 - 1 = e(e-2)$ has valuation δ . The nonlinearity of the second iterate is not guessed from the original quadratic term; its exact coefficients are

n	b_n in $f_e^{\circ 2}$	n	b_n in $f_e^{\circ 2}$
2	$e(e-1)$	6	$6(e-1)$
3	$-e(e-2)(e-1)$	7	$-3(e-2)$
4	$-e(3e-4)$	8	-3
5	$3e^2 - 9e + 4$	9	1

At $e = 0$ the return displacement factors as

$$f_0^{\circ 2}(z) - z = z^5(z^2 - 2z + 2)(z^2 - z + 2), \quad (13.3)$$

so that the cubic *and* quartic terms have both disappeared; their actual valuations, not the separate sizes of the terms that cancelled, are what enter (9.6). The degree-five coefficient is a unit and the exact maximum is

$$r(f_e) = \delta/4. \quad (13.4)$$

For $\rho = t^{\delta/4}$ one gets $P(X) = 1 - 2X^4$: four simple closest period-two points in two cycles, with residue directions satisfying $a^4 = 1/2$; since $aP'(a) = -4$,

$$(f_e^{\circ 2})'(p_a) = 1 - 4c + o_v(c) = 1 + 8e + o_v(e), \quad (13.5)$$

where $R = o_v(c)$ means $v(R) > v(c)$, and $H_0(X) = X(1 - 2X^4)^{-1/4}$.

For comparison, the untuned quadratic $(-1 + e)z + z^2$ has threshold $\delta/2$ and shell polynomial $1 + X^2$. The cubic (13.2) therefore admits an *exactly larger* linearization ball with the same multiplier and the same basic coefficient scale $\eta = 0$; the difference is produced entirely by cancellation in a finite return.

13.3 The two-parameter transition

The cancellation can be broken on a second valuation scale. Let

$$f_{e,s}(z) = (-1 + e)z + z^2 + (-1 + s)z^3, \quad e = t^\delta, \quad s = t^{\sigma_0}, \quad \delta, \sigma_0 > 0, \quad (13.6)$$

with no Archimedean relationship assumed between the two positive exponents.

Theorem 13.1 (Cancellation phase diagram). *For (13.6),*

$$r(f_{e,s}) = \max\left\{\frac{\delta - \sigma_0}{2}, \frac{\delta}{4}\right\}, \quad (13.7)$$

and the shell polynomial, cycle count and leading return-multiplier ratios are

Regime	$P(X)$	Two-cycles	$aP'(a)$
$\sigma_0 < \delta/2$	$1 + X^2$	one	-2
$\sigma_0 = \delta/2$	$1 + X^2 - 2X^4 = (1 - X^2)(1 + 2X^2)$	two	-6, -3
$\sigma_0 > \delta/2$	$1 - 2X^4$	two	-4

Every listed initial root is simple. In the middle row the ratio -6 occurs for $a^2 = 1$ and -3 for $a^2 = -1/2$. The three leading coherent coordinates are (10.8).

Proof. For the general cubic $f(z) = (-1 + e)z + z^2 + bz^3$, direct composition gives

$$\begin{aligned} b_2 &= e(e - 1), & b_3 &= (e - 1)(be^2 - 2be + 2b + 2), \\ b_4 &= 3be^2 - 4be + b + 1, & b_5 &= b(3be^2 - 6be + 3b + 3e - 1), \\ b_6 &= b(6be - 5b + 1), & b_7 &= 3b^2(be - b + 1), \\ b_8 &= 3b^3, & b_9 &= b^4. \end{aligned} \quad (13.8)$$

Substituting $b = -1 + s$, the degree-three coefficient has leading terms $-2e - 2s$ and the degree-four coefficient has leading terms $4e + s$; because e and s have leading coefficient one, no cancellation occurs between those pairs even when $\delta = \sigma_0$. Hence $v(b_2) = \delta$, $v(b_3) = v(b_4) = \min\{\delta, \sigma_0\}$ and $v(b_n) = 0$ for $5 \leq n \leq 9$, so the maximum in (9.6) is the larger of $(\delta - \min\{\delta, \sigma_0\})/2$ and $\delta/4$, which is (13.7). If $\sigma_0 < \delta/2$, take $r = (\delta - \sigma_0)/2$: only the normalized degree-three return coefficient has nonzero residue in $Q - 1$, and that residue is one, so $P = 1 + X^2$. If $\sigma_0 > \delta/2$, take $r = \delta/4$: only degree five survives, with residue $4/(-2) = -2$, so $P = 1 - 2X^4$. At equality both terms survive and $P = 1 + X^2 - 2X^4$. All three are squarefree, so the cycle counts follow from theorem 9.7, and reduction of $XP'(X)$ at the roots gives -2, -4 and, in the middle case, $2a^2 - 8a^4 = -6$ for $a^2 = 1$ and -3 for $a^2 = -1/2$. $\square$

Example 13.2 (A repeated initial root at the transition). Replace s in the equality regime by $\kappa t^{\delta/2}$ with $\kappa \in \mathbb{C}^\times$. The same calculation gives $P_\kappa(X) = 1 + \kappa X^2 - 2X^4$, and as a polynomial in $Y = X^2$,

$$\text{Disc}_Y(1 + \kappa Y - 2Y^2) = \kappa^2 + 8.$$

For $\kappa^2 + 8 \neq 0$ the four shell points are simple in distinct residue directions. For $\kappa^2 + 8 = 0$ the initial polynomial has repeated roots — an *initial-direction degeneracy*, and not by itself an assertion of an actual multiple periodic point (remark 9.8); determining the finer splitting there requires higher Hahn terms of the return equation. The exact threshold remains $\delta/4$, since the degree-four term of P_κ cannot vanish.

Example 13.3 (A genuinely higher-rank example). Let $\Gamma = \mathbb{Q} + \mathbb{Q}\Omega$ with $\Omega > n$ for every ordinary positive integer n , and

$$f(z) = (1 + t^\Omega)z + t^3z^2 + z^4. \quad (13.9)$$

Here $q = 1$, $\eta = 0$ and $r = \max\{\Omega - 3, \Omega/3\} = \Omega - 3$. At $\rho = t^{\Omega-3}$ the shell polynomial is $P(X) = 1 + X$, and there is one closest nonzero fixed point $p = \rho(-1 + u)$, $u \in \mathfrak{m}_\Gamma$, with $f'(p) = 1 - t^\Omega + o_v(t^\Omega)$.

The other two nonzero fixed points are much farther from zero: all three solve $z^3 + t^3z + t^\Omega = 0$, and factoring off p gives $z^3 + t^3z + t^\Omega = (z - p)(z^2 + pz + p^2 + t^3)$; scaling the quadratic factor by $z = t^{3/2}Y$ and dividing by t^3 gives reduction $Y^2 + 1$, because $v(p) = \Omega - 3$ dominates every ordinary rational. Those two roots therefore have valuation $3/2$ and normalized residue directions $\pm i$.

This example *cannot* be faithfully described by a real-valued valuation preserving the order of all of Γ : sending Ω to a finite real number would fail to preserve $\Omega > n$ for all n . The exact calculation uses only ordered-group comparisons and [lemma 3.4](#). Finally, the same threshold persists if t^Ω in the linear term is replaced by $c = t^\Omega(1 + \sum_{j \geq 2} t^{2-1/j})$, a legitimate Hahn coefficient whose positive support is well ordered although infinitely many exponents lie below 2 before the shift by Ω : the leading valuation remains Ω and the normalized shell polynomial remains $1 + X$. The theorem is therefore *not* restricted to coefficients with discrete or finitely generated exponent support. Both halves of this example are specific to source 01.

14 Parabolic dynamics I: algebraicity and hidden monodromy

Everything in this section is under PAR and $\text{SD}_\emptyset$ ([warning 2.8](#)): the leading multiplier is exactly 1 perturbed by an infinitesimal, and no divisor condition is imposed anywhere. Here Γ is divisible and set-sized — divisibility is load-bearing, because it is what keeps the resonant rescaling $s = t^{\delta/q}$ inside the chosen workspace — and $\varepsilon = t^\delta$ with $\delta > 0$ is a single Hahn monomial, **not** a small nonzero complex number. Ordinary powers of ε form the Hahn subring $\mathbb{C}[[\varepsilon]]$, and every $O(\varepsilon^k)$ below means *divisibility* by ε^k in a formal series ring, never an analytic estimate uniform in a numerical parameter.

14.1 The common-domain coefficient ring and the algebraicity base

Fix an ordinary connected open $U \subset \mathbb{C}$ and work in $\mathcal{A}(U) = \mathcal{O}(U)((t^\Gamma))$, the fixed-polydisk row of [section 3.2](#) in one variable: one well-ordered exponent support, every coefficient holomorphic on that *same* U , derivatives and primitives coefficientwise with no estimate uniform in the exponent. A series with nowhere-zero leading coefficient on U is invertible in $\mathcal{A}(U)$ ([lemma 14.2](#)). Write $\mathcal{A}^{>0}(U)$ for the positive-valuation part. The algebraicity base is

$$\mathcal{B} = \mathbb{C}(w)((t^\Gamma)), \quad (14.1)$$

Hahn series with *arbitrary rational-function* coefficients.

Warning 14.1 (The base field is deliberately large). $\mathcal{B}$ is **not** $K_\Gamma(w)$. Its elements may have coefficient functions whose extra poles vary with the Hahn exponent, so there is no common pole-free subdomain for all of $\mathcal{B}$, and proving transcendence over $\mathcal{B}$ is strictly stronger than proving it over $K_\Gamma(w)$. A merge must not weaken [\(14.1\)](#) to $K_\Gamma(w)$. The mechanism that makes the larger base available is that an ordinary deck transformation fixes *every rational coefficient function separately*, so no degree bound, no finite support and no common bound on pole sets is needed.

The ambient field for algebraicity is

$$\mathcal{L} = \mathcal{M}(\tilde{X})((t^\Gamma)), \quad \mathcal{B} \hookrightarrow \mathcal{L}, \quad (14.2)$$

where $X = \mathbb{C} \setminus Z(V)$, $p: \tilde{X} \rightarrow X$ is its *ordinary* universal cover and rational functions are pulled back along p . A deck transformation acts on each coefficient by ordinary analytic continuation and leaves Hahn exponents fixed, hence is a strong field automorphism of $\mathcal{L}$ fixing $\mathcal{B}$ pointwise. No covering-space topology on $\mathbf{No}[i]$ is introduced, and no topological convergence in the surcomplex plane is used.

Lemma 14.2 (A sufficient unit criterion). *If $f = t^{\lambda_0}(a + u)$ with $a \in \mathcal{O}(U)$ nowhere zero and $u \in \mathcal{A}^{>0}(U)$, then f is invertible in $\mathcal{A}(U)$ with $f^{-1} = t^{-\lambda_0}a^{-1} \sum_{n \geq 0} (-u/a)^n$, and every coefficient of the inverse is holomorphic on the original U .*

Proof. The family is summable by lemma 3.4, multiplication by $1 + u/a$ telescopes coefficientwise to 1, and powers of a^{-1} are holomorphic on U , so no new singularity appears. $\square$

No converse characterization of the units of $\mathcal{A}(U)$ is claimed; only the stated criterion is used. Every nonzero series is invertible in $\mathcal{M}(U)((t^\Gamma))$.

14.2 Common-domain Hahn linearization

Theorem 14.3 (Common-domain Hahn linearization). *Let $U \subset \mathbb{C}$ be simply connected with $0 \in U$, let $V \in \mathcal{O}(U)$ satisfy $V(0) = 0$, $V'(0) = \mu \neq 0$ and $V(w) \neq 0$ on $U \setminus \{0\}$, and let $E \in \mathcal{A}^{>0}(U)$ satisfy $E(0) = 0$. Put*

$$F(w) = w + t^\delta(V(w) + E(w)), \quad \lambda = F'(0), \quad \mathbf{M} = \langle \{\delta\} \cup \text{supp } E \rangle. \quad (14.3)$$

There is a unique $H \in \mathcal{A}(U)$ with

$$H \circ F = \lambda H, \quad H(0) = 0, \quad H'(0) = 1, \quad (14.4)$$

its support is contained in $\mathbf{M}$, every coefficient is holomorphic on U , and its coefficient at exponent zero is h_0 of (10.7). All constructions commute with enlargement of the set-sized value group.

Proof. Theorem 4.3 gives $\log C_F = A\partial_w$ with $A \in \mathcal{A}^{>0}(U)$. Every word in $\Delta = t^\delta(V + E)$ has an exponent which, after removing one initial δ , lies in $\mathbf{M}$, so $A = t^\delta W$ with $W = V + W_+$, $\text{supp } W \subseteq \mathbf{M}$, W_+ of positive valuation and all coefficients holomorphic on U . Put $c = \text{Log } \lambda$ and $\alpha_c = t^{-\delta}c$; lemma 4.5 gives $W(0) = 0$ and $W'(0) = \alpha_c$, whose leading coefficient is μ . Coefficientwise division by w is possible since $W(0) = 0$, and the leading coefficient of W/w is V/w , nowhere zero on U including at 0, so W/w is invertible in $\mathcal{A}(U)$ by lemma 14.2. Hence

$$R(w) := \frac{\alpha_c}{W(w)} - \frac{1}{w} = \frac{\alpha_c/(W(w)/w) - 1}{w} \quad (14.5)$$

belongs to $\mathcal{A}(U)$, its numerator vanishing coefficientwise at 0 because $(W/w)(0) = \alpha_c$, with support in $\mathbf{M}$. Every coefficient of R has a unique primitive vanishing at 0 since U is simply connected; write $B(w) = \int_0^w R$, $B = B_0 + B_+$ with B_0 the coefficient at exponent zero, and set

$$H(w) = we^{B_0(w)} \sum_{n \geq 0} \frac{B_+(w)^n}{n!}, \quad (14.6)$$

whose exponential is certified by [lemma 3.4](#) and preserves the support bound M , using the same U at every stage. Then $H(0) = 0$, $H'(0) = 1$ and $H'/H = \alpha_c/W$ away from 0, with the cross-multiplied identity everywhere, so $DH = cH$; as D fixes Hahn constants, exponentiation gives $C_F H = \text{Exp}_H(c)H = \lambda H$, and the exponent-zero part of (14.6) is (10.7). For uniqueness, if $\tilde{H}$ satisfies (14.4) then $N^r \tilde{H} = (\lambda - 1)^r \tilde{H}$ so $D\tilde{H} = c\tilde{H}$; the quotient $\tilde{H}/H = (\tilde{H}/w)/(H/w)$ lies in $\mathcal{A}(U)$ because the leading coefficient of H/w is the nowhere-zero e^{B_0} , the differential equations force its derivative to vanish ($\mathcal{A}(U)$ being an integral domain for connected U), every coefficient is therefore constant, and its value at 0 is 1. Finally all outputs are formed from input supports by finite words, coefficientwise differentiation and uniquely normalized integration, none of which depends on the ambient group beyond containing the indicated supports. $\square$

Warning 14.4 (What kind of embedding was proved). [Theorem 14.3](#) constructs a strong logarithm and solves a coefficientwise ordinary differential equation. It does **not** assert convergence of the series at a nonzero ordinary complex value of t^δ , nor analytic embedding of every numerical near-identity map in an autonomous flow; those are different questions, and the Gelfreich–Vieiro reference is cited precisely to mark that line [7]. Nor does it assert that the global evaluated map $H^\#$ is injective on all of $U_\Gamma^\#$; injectivity is proved only on the infinitesimal disk of the resonant model ([theorem 14.22](#)). There is also no ordinary invariant-domain hypothesis: C_F denotes the Taylor operation (4.1), not a numerical self-map of U , and at a point $a + \eta \in U_\Gamma^\#$ the positive displacement preserves the standard part.

Corollary 14.5 (Universal leading profile at a resonant scale). *Let $q \geq 1$, $a \in \mathbb{C}^*$, and let $\phi(z) = az^{q+1} + \sum_{n>q+1} a_n z^n$ be an ordinary convergent germ. Choose $c_0 \in \mathbb{C}^*$ with $ac_0^q = 1$, set $s = c_0 t^{\delta/q}$, and let $f(z) = (1 + \varepsilon)z + \phi^\#(z)$ at infinitesimal z . After $z = sw$, the normalized Hahn coordinate on any simply connected ordinary U containing 0 but no root of $1 + w^q$ has leading coefficient*

$$w(1 + w^q)^{-1/q}, \quad (14.7)$$

with the branch normalized at 0.

Proof. Taylor evaluation of the convergent germ at sw is strongly meaningful for every finite w , and direct rescaling gives

$$s^{-1}f(sw) = w + \varepsilon \left(w + w^{q+1} + \sum_{n>q+1} a_n c_0^{n-1} t^{\delta(n-q-1)/q} w^n \right),$$

whose final sum has positive Hahn support and polynomial coefficients on such a U . Apply [theorem 14.3](#) with $V = w(1 + w^q)$; integration of $1/V - 1/w$ gives $-q^{-1} \log(1 + w^q)$. $\square$

[Corollary 14.5](#) asserts a leading profile and a common-domain construction. It explicitly does **not** extend the exact monodromy formula or the sharp disk theorem below to arbitrary higher-order perturbations.

14.3 Polynomial Euler maps and their exact logarithmic residues

Specialize to

$$G_\varepsilon(w) = w + \varepsilon V(w), \quad V \in \mathbb{C}[w], \quad V(0) = 0, \quad V'(0) = \mu \neq 0, \quad (14.8)$$

so [theorem 14.3](#) with $E = 0$ constructs $H_\varepsilon \in \mathcal{O}(U)[[\varepsilon]]$. The polynomial hypothesis supplies structure the general theorem does not have.

Proposition 14.6 (The generator gains no new pole). *The logarithmic generator of (14.8) has the form*

$$\log C_{G_\varepsilon} = \varepsilon W_\varepsilon(w) \partial_w, \quad W_\varepsilon = V B_\varepsilon, \quad B_\varepsilon \in 1 + \varepsilon \mathbb{C}[w][[\varepsilon]]. \quad (14.9)$$

Setting $\alpha_\varepsilon = \text{Log}(1 + \mu\varepsilon)/\varepsilon$ and $\Omega_\varepsilon = \alpha_\varepsilon/W_\varepsilon$, every coefficient of Ω_ε is a rational function of w with poles only at zeros of V , of order at most the multiplicity in V .

Proof. For a polynomial p , $(C_{G_\varepsilon} - I)p = \sum_{k \geq 1} \varepsilon^k V^k p^{(k)}/k!$, every coefficient of which is polynomial and divisible by V ; the same holds of each $N^r w$, and at a fixed power of ε only finitely many r occur, so the logarithm has polynomial coefficients all divisible by V , its first term being εV . The inverse of B_ε has polynomial coefficients by the formal geometric series, so Ω_ε is a formal series of polynomials divided by the single polynomial V . $\square$

The divisor assertion is stronger than rationality of each coefficient: higher terms of the modified vector field cannot introduce *new* spatial pole locations into the logarithmic derivative of the coordinate. Finite expansion of the operator logarithm gives, for the record,

$$W_\varepsilon = V - \frac{\varepsilon}{2} V V' + \varepsilon^2 \left(\frac{1}{12} V^2 V'' + \frac{1}{3} V (V')^2 \right) + O(\varepsilon^3), \quad (14.10)$$

obtained from $Nw = \varepsilon V$, $N^2 w = \varepsilon^2 V V' + \frac{\varepsilon^3}{2} V^2 V'' + O(\varepsilon^4)$, $N^3 w = \varepsilon^3 (V (V')^2 + V^2 V'') + O(\varepsilon^4)$ substituted in $N - N^2/2 + N^3/3$. **No assertion about all coefficients is inferred from this truncation;** the exact identity [lemma 14.7](#) is proved separately.

Lemma 14.7 (Exact residue at a simple fixed point). *At a simple zero a of V , with $\nu_a = V'(a)$,*

$$\text{Res}_{w=a}(\Omega_\varepsilon(w) dw) = \rho_a(\varepsilon) = \frac{\text{Log}(1 + \mu\varepsilon)}{\text{Log}(1 + \nu_a \varepsilon)}, \quad \rho_0 = 1. \quad (14.11)$$

Proof. [Lemma 4.5](#) at a gives $\varepsilon W'_\varepsilon(a) = \text{Log } G'_\varepsilon(a) = \text{Log}(1 + \nu_a \varepsilon)$. Since a is a simple zero of V , the factorization $W_\varepsilon = V B_\varepsilon$ makes it a simple zero over $\mathbb{C}[[\varepsilon]]$ as well, so the residue of $\alpha_\varepsilon/W_\varepsilon$ is $\alpha_\varepsilon/W'_\varepsilon(a)$. $\square$

These are *coefficientwise ordinary* residues of a rational differential. They are not local residues at displaced surcomplex poles; the fixed points of this polynomial family are exactly the ordinary zeros of V , so no displacement ambiguity arises.

Proposition 14.8 (The first correction is an explicit logarithm). *On the common domain U ,*

$$H_\varepsilon = h_0 \left[1 + \frac{\mu\varepsilon}{2} \log\left(\frac{V}{\mu h_0}\right) + O(\varepsilon^2) \right], \quad (14.12)$$

the quotient $V/(\mu h_0)$ being holomorphic, nowhere zero and equal to 1 at 0 after removing the common zero, so that its logarithm is uniquely fixed by vanishing at 0.

Proof. Since $\alpha_\varepsilon = \mu - \mu^2 \varepsilon/2 + O(\varepsilon^2)$, (14.10) gives $\Omega_\varepsilon = \mu/V + \frac{\mu\varepsilon}{2} (V' - \mu)/V + O(\varepsilon^2)$, whose coefficient at ε is holomorphic at 0 and equals $\frac{\mu}{2} (\log(V/(\mu h_0)))'$ because $h'_0/h_0 = \mu/V$. Integrate with the normalization at 0 and exponentiate. $\square$

Corollary 14.9 (Separation of the branching factors). *Suppose V has degree $d \geq 2$ with distinct zeros $0, a_1, \dots, a_{d-1}$. Partial fractions taken separately at each ε coefficient give*

$$\Omega_\varepsilon(w) = Q_\varepsilon(w) + \frac{1}{w} + \sum_{j=1}^{d-1} \frac{\rho_{a_j}(\varepsilon)}{w - a_j}, \quad Q_\varepsilon \in \varepsilon^2 \mathbb{C}[w][[\varepsilon]], \quad (14.13)$$

and with $P_\varepsilon(w) = \int_0^w Q_\varepsilon$, on branches normalized near 0,

$$H_\varepsilon(w) = w \operatorname{Exp}_H(P_\varepsilon(w)) \prod_{j=1}^{d-1} (1 - w/a_j)^{\rho_{a_j}(\varepsilon)}, \quad P_\varepsilon \in \varepsilon^2 \mathbb{C}[w][[\varepsilon]], \quad P_\varepsilon(0) = 0. \quad (14.14)$$

The factor $\operatorname{Exp}_H(P_\varepsilon)$ and its inverse have polynomial coefficients and belong to $\mathcal{B}$; in particular all nontrivial coefficientwise monodromy is carried by the displayed power factors.

Proof. The coefficients at ε^0 and ε^1 have numerator degree smaller than d by (14.10) and [proposition 14.8](#), so $Q_\varepsilon \in \varepsilon^2 \mathbb{C}[w][[\varepsilon]]$. For $\rho = \rho_0 + \rho_+$ define the power on a chosen ordinary logarithm branch by $(1 - w/a)^\rho = \exp(\rho_0 \log(1 - w/a)) \operatorname{Exp}_H(\rho_+ \log(1 - w/a))$. The logarithmic derivative of the right side of (14.14) is (14.13) and its value and derivative at 0 have the required normalization, so uniqueness in [theorem 14.3](#) proves the identity; at a fixed power of ε the exponential of P_ε uses only finitely many products of polynomials. $\square$

14.4 The torsion test and the finite-cover obstruction

Lemma 14.10 (An infinitesimal phase has no torsion). *For a finite Hahn constant $c = c_0 + c_+$ with $c_0 \in \mathbb{C}$ and $v(c_+) > 0$,*

$$\operatorname{Exp}_H(c) = 1 \iff c_+ = 0 \text{ and } c_0 \in 2\pi i\mathbb{Z}.$$

Consequently, for $\rho \in \mathbb{C} + \mathfrak{m}_\Gamma$, the element $\operatorname{Exp}_H(2\pi i\rho)$ has finite multiplicative order if and only if ρ is an ordinary rational number.

Proof. The coefficient at exponent zero in $\operatorname{Exp}_H(c)$ is e^{c_0} , so equality to 1 first requires $c_0 \in 2\pi i\mathbb{Z}$; if $c_+ \neq 0$ then $\operatorname{Exp}_H(c_+) - 1 = c_+ + c_+^2/2 + \dots$ has the same leading term as c_+ , all higher powers having strictly larger valuation, so it cannot vanish. The last assertion follows from $\operatorname{Exp}_H(2\pi i\rho)^m = \operatorname{Exp}_H(2\pi im\rho)$. $\square$

This single lemma converts “the exponent has a nonzero infinitesimal part” into “no finite branched cover works”.

Theorem 14.11 (Exact monodromy and obstruction to finite ramification). *At every simple zero a of V , continuation of H_ε around a positive simple loop about a is multiplication by*

$$M_a = \operatorname{Exp}_H(2\pi i\rho_a), \quad \rho_a = \frac{\operatorname{Log}(1 + \mu\varepsilon)}{\operatorname{Log}(1 + V'(a)\varepsilon)}. \quad (14.15)$$

The multiplier has finite order exactly when $V'(a) = \mu$, in which case it is 1. When $V'(a) \neq \mu$, no finite ordinary branched cover at a makes the continued coordinate single-valued.

Proof. Near a the rational-coefficient differential decomposes as $\Omega_\varepsilon dw = \rho_a dw/(w-a) +$ a holomorphic-coefficient differential; a positive circuit changes a coefficientwise primitive by $2\pi i\rho_a$, so exponentiation multiplies H_ε by M_a . Writing $\nu = V'(a)$, formal division of the two logarithms gives

$$\rho_a = \frac{\mu}{\nu} + \frac{\mu(\nu - \mu)}{2\nu}\varepsilon + O(\varepsilon^2), \quad (14.16)$$

so if $\nu \neq \mu$ the coefficient of ε is nonzero and [lemma 14.10](#) makes M_a of infinite order, while if $\nu = \mu$ the ratio is exactly 1. A finite branched cover has a positive integer ramification index m at a point above a , and in suitable local coordinates a circuit upstairs projects to m circuits downstairs, so its monodromy multiplier is $M_a^m \neq 1$ in the infinite-order case. This applies in particular to the covers used to make an ordinary algebraic function single-valued. $\square$

The obstruction is already visible at order ε^1 whenever $\nu \neq \mu$: it is not necessary to inspect infinitely many unknown correction terms to decide whether finite ramification will work.

Lemma 14.12 (A nonlinear polynomial cannot have one slope at all its zeros). *If a polynomial of degree $d \geq 2$ has d distinct zeros then $\sum_{V(a)=0} 1/V'(a) = 0$. In particular, if $V'(0) = \mu \neq 0$, not all slopes at its zeros can equal μ .*

Proof. Partial fractions give $1/V(w) = \sum_a (V'(a)(w-a))^{-1}$; the coefficient of $1/w$ at infinity on the left is zero because $d \geq 2$, and on the right it is the displayed sum. If every derivative were μ the sum would be $d/\mu \neq 0$. $\square$

14.5 The Hahn-algebraicity dichotomy

Lemma 14.13 (Reduction of an algebraic Hahn element). *If $Y \in \mathcal{L}$ has valuation zero with leading coefficient y_0 and Y is algebraic over $\mathcal{B}$, then y_0 is algebraic over $\mathbb{C}(w)$.*

Proof. Take a nonzero relation $\sum_{j=0}^m b_j Y^j = 0$ with $b_j \in \mathcal{B}$, let γ be the least valuation among the nonzero b_j , and multiply by $t^{-\gamma}$: all coefficients then have nonnegative valuation and at least one has nonzero coefficient at zero, and taking the coefficient at zero yields a nonzero polynomial over $\mathbb{C}(w)$ annihilating y_0 — multiplication reduces correctly because all series in the normalized relation have nonnegative valuation. $\square$

Lemma 14.14 (A higher-order logarithmic pole forces transcendence). *Let r be a rational function with a pole of order at least two at a . Any nonzero local solution of $y'/y = r$ is transcendental over $\mathbb{C}(w)$.*

Proof. On a small punctured sector, integration gives $y(w) = u(w)(w-a)^\beta \exp(\sum_{j=1}^{k-1} c_j(w-a)^{-j})$ with $c_{k-1} \neq 0$, u holomorphic and nonzero and $k \geq 2$ the pole order. Along a suitable ray ending at a the real part of the leading expression $c_{k-1}(w-a)^{-(k-1)}$ is positive and dominates the lower terms, so $|y(w)|$ grows faster than every power of $|w-a|^{-1}$ along that ray. An algebraic function cannot: after dividing a polynomial relation by its leading coefficient, its coefficients are meromorphic and hence bounded by a fixed power of $|w-a|^{-1}$ on a small punctured disk, and the elementary bound $|y| \leq 1 + \max_{j < m} |b_j|$ for a root of $y^m + \sum_{j < m} b_j y^j$ gives a polynomial growth bound for every root. $\square$

Theorem 14.15 (Polynomial Euler maps: the algebraicity dichotomy). *Let $V \in \mathbb{C}[w]$ with $V(0) = 0$, $V'(0) = \mu \neq 0$. On every simply connected ordinary U containing 0 and no other zero of V , equation*

(14.18) below has a unique solution $H_\varepsilon \in \mathcal{O}(U)[[\varepsilon]]$, whose coefficients continue to the universal cover of $X = \mathbb{C} \setminus Z(V)$, and

$$H_\varepsilon \text{ is algebraic over } \mathcal{B} = \mathbb{C}(w)((t^\Gamma)) \iff \deg V = 1, \quad (14.17)$$

where

$$H_\varepsilon(w + \varepsilon V(w)) = (1 + \mu\varepsilon)H_\varepsilon(w), \quad H_\varepsilon(0) = 0, \quad H'_\varepsilon(0) = 1, \quad (14.18)$$

composition being Taylor substitution in the positive Hahn displacement $\varepsilon V(w)$. At a simple zero a of V , positive continuation once around a multiplies H_ε by M_a of (14.15). If V is nonlinear and all its zeros are simple, at least one zero has infinite-order monodromy, which no finite branched cover removes.

Proof. Existence, uniqueness and the common-domain assertion are theorem 14.3 with $E = 0$; continuation and the monodromy formula are proposition 14.6, lemma 14.7, and theorem 14.11. If V is linear it is μw and $H_\varepsilon = w$ solves the equation, an element of $\mathcal{B}$. Assume $\deg V \geq 2$.

Route 1: a growth bound (needed for multiple zeros). If V has a multiple zero then μ/V has a pole of order at least two there, and $h'_0/h_0 = \mu/V$, so lemma 14.14 makes h_0 transcendental over $\mathbb{C}(w)$; since H_ε has valuation zero in $\mathcal{L}$, lemma 14.13 excludes algebraicity over $\mathcal{B}$.

Route 2: an infinite monodromy orbit (needed for simple zeros). If every zero is simple, lemma 14.12 supplies a zero a with $V'(a) \neq \mu$. Let τ_a be the corresponding deck action on $\mathcal{L}$; it fixes $\mathcal{B}$ and satisfies $\tau_a^n(H_\varepsilon) = M_a^n H_\varepsilon$ for $n \geq 0$. By theorem 14.11 these elements are all distinct, $H_\varepsilon \neq 0$ and M_a of infinite order; if H_ε were algebraic over $\mathcal{B}$ they would all be roots of one nonzero polynomial over $\mathcal{B}$, impossible in a field. $\square$

Remark 14.16 (Two routes, neither substituting for the other). Source 04 states explicitly that its two proof branches — the elementary growth bound for algebraic functions, and the infinite-monodromy-orbit argument — meet in the resonant model but do not logically substitute for one another. Both are therefore kept in theorem 14.15: route 1 is what covers multiple zeros, where there is no simple-pole residue to work with, and route 2 is what covers simple zeros, where μ/V has only first-order poles and lemma 14.14 does not apply. A proof discussing only simple-pole residues would miss example 14.20 entirely, even though the dichotomy remains true there.

The argument works over the larger base $\mathcal{B}$ precisely because ordinary deck transformations fix every rational coefficient function separately (warning 14.1).

Proposition 14.17 (Classification of algebraic leading coordinates). *For a nonlinear polynomial V as above, h_0 is algebraic over $\mathbb{C}(w)$ if and only if all zeros of V are simple and*

$$\frac{\mu}{V'(a)} \in \mathbb{Q} \quad \text{for every zero } a \text{ of } V, \quad (14.19)$$

in which case

$$h_0(w) = w \prod_{a \neq 0} (1 - w/a)^{\mu/V'(a)}. \quad (14.20)$$

Whenever these conditions hold, the first correction h_1 in $H_\varepsilon = h_0 + \varepsilon h_1 + \dots$ is already transcendental over $\mathbb{C}(w)$.

Proof. Multiple zeros are excluded by lemma 14.14. With all zeros simple, the leading part of (14.14) is (14.20); if all exponents are rational, taking a common denominator gives a polynomial relation, so h_0 is algebraic. Conversely continuation of an algebraic function gives a finite orbit, so

the local multipliers of (14.20) must have finite order, which for an ordinary complex exponent is rationality. For the last assertion choose a with $\nu = V'(a) \neq \mu$ by lemma 14.12, put $r_0 = \mu/\nu \in \mathbb{Q}$ and a positive integer m with $mr_0 \in \mathbb{Z}$; after m circuits about a the function h_0 returns to itself, while by proposition 14.8 the same continuation changes h_1 by $2\pi i m \frac{\mu}{2}(1 - r_0)h_0 \neq 0$. Repeated blocks of m circuits therefore produce infinitely many distinct branches of h_1 , and an algebraic function has only finitely many. $\square$

Example 14.18 (Rational leading coordinate, logarithmic correction). For $V = w + w^2$, $h_0 = w/(1+w)$ and $h_1 = \frac{w}{1+w} \log(1+w)$. At the fixed point -1 the exact exponent is $\text{Log}(1+\varepsilon)/\text{Log}(1-\varepsilon) = -1 + \varepsilon + O(\varepsilon^2)$, so the leading multiplier is 1 while the full multiplier begins $1 + 2\pi i \varepsilon + O(\varepsilon^2)$: the obstruction is present even without any branching in the leading function. A rational leading coordinate does not even predict the *branching type* of the first correction.

Example 14.19 (One trivial nonzero-root multiplier). For $V = w(1-w)(1-2w)$ one has $\mu = 1$, $V'(1) = 1$, $V'(1/2) = -1/2$, and

$$h_0 = \frac{w(1-w)}{(1-2w)^2}, \quad h_1 = \frac{3}{2}h_0 \log(1-2w).$$

The exact multiplier at 1 is 1, whereas at $1/2$ it is $1 + 3\pi i \varepsilon + O(\varepsilon^2)$ and has infinite order. Thus theorem 14.11 does *not* claim branching at every nonzero fixed point; lemma 14.12 guarantees at least one obstruction, which suffices for transcendence over the whole base.

Example 14.20 (Why multiple zeros need a separate argument). For $V = w(1+w)^2$, $\mu = 1$ and $h_0(w) = \frac{w}{1+w} \exp(\frac{1}{1+w} - 1)$. The pole of μ/V at -1 has order two, so the leading solution has an essential singularity and is already transcendental — remark 14.16.

14.6 The resonant model: exact scale, isometry, continuation

Fix an ordinary integer $q \geq 1$ and let $f(z) = (1+\varepsilon)z + z^{q+1}$ with $\varepsilon = t^\delta$, $s = t^{\delta/q}$ (divisibility of Γ placing s in K_Γ). Under $z = sw$,

$$g(w) = s^{-1}f(sw) = (1+\varepsilon)w + \varepsilon w^{q+1} = w + \varepsilon V_q(w), \quad V_q(w) = w(1+w^q). \quad (14.21)$$

Let Ψ_ε be its normalized common-domain coordinate; $U = \{|w| < 1\}$ is one admissible domain, and larger simply connected slit domains avoiding the roots of $1+w^q$ are also possible. Since $g(\zeta w) = \zeta g(w)$ for a q -th root of unity ζ , uniqueness gives $\Psi_\varepsilon(\zeta w) = \zeta \Psi_\varepsilon(w)$ and hence

$$\Psi_\varepsilon(w) = \sum_{m \geq 0} b_m(\varepsilon) w^{qm+1}, \quad b_0 = 1, \quad b_m \in \mathbb{C}[[\varepsilon]], \quad (14.22)$$

initially as the Taylor series of each ordinary coefficient function; its evaluation as a Hahn family at a specific w is a separate question. The leading solution is

$$\Psi_0(w) = w(1+w^q)^{-1/q}, \quad b_m(0) = (-1)^m \frac{(1/q)_m}{m!} \neq 0, \quad (14.23)$$

with $(x)_m = x(x+1)\cdots(x+m-1)$.

Proposition 14.21 (Exact finite recurrence). *With $\lambda = 1 + \varepsilon$, for $m \geq 1$*

$$b_m = - \frac{\sum_{\substack{0 \leq j < m \\ m-j \leq qj+1}} b_j \binom{qj+1}{m-j} \lambda^{qj+1-(m-j)} \varepsilon^{m-j}}{\lambda^{qm+1} - \lambda}, \quad (14.24)$$

which uniquely constructs $b_m \in \mathbb{C}[[\varepsilon]]$ and has leading recurrence $qm b_m(0) = -(q(m-1)+1)b_{m-1}(0)$.

Proof. In $g(w)^{qj+1} = w^{qj+1}(\lambda + \varepsilon w^q)^{qj+1}$ the coefficient of w^{qm+1} is the binomial expression displayed; the term $j = m$ contributes $\lambda^{qm+1}b_m$ and the right side of $\Psi_\varepsilon \circ g = \lambda\Psi_\varepsilon$ contributes λb_m . The denominator is $qm\varepsilon + O(\varepsilon^2)$ while every numerator term is divisible by ε , so induction gives $b_m \in \mathbb{C}[[\varepsilon]]$; only $j = m - 1$ survives after division by ε and reduction modulo ε , which also checks (14.23) directly. $\square$

The normalized formal coordinate of the original map is

$$h(z) = s\Psi_\varepsilon(z/s) = \sum_{m \geq 0} \varepsilon^{-m} b_m(\varepsilon) z^{qm+1}, \quad (14.25)$$

a formal power series in z over K_Γ — *not* a Hahn family already asserted summable at every z — whose nonzero coefficients have the exact valuations $v(\varepsilon^{-m} b_m(\varepsilon)) = -m\delta$.

Theorem 14.22 (Sharp domain and valuative isometry). *The family of Taylor terms in (14.25) is strongly summable at $z \in K_\Gamma$ if and only if $z = 0$ or*

$$v(z) > \delta/q. \quad (14.26)$$

On $D = \{v(z) > \delta/q\}$ the resulting function is a bijection $h : D \rightarrow D$ satisfying $h \circ f = \lambda h$ and

$$v(h(x) - h(y)) = v(x - y) \quad (x, y \in D). \quad (14.27)$$

No strictly larger centered valuative ball containing D admits an injective zero-fixing conjugacy to multiplication by λ .

Proof. Let $z \neq 0$, $\beta = v(z)$: the m -th nonzero Taylor term has valuation exactly

$$v(\varepsilon^{-m} b_m z^{qm+1}) = \beta + m(q\beta - \delta). \quad (14.28)$$

Two distinct failure mechanisms. If $q\beta = \delta$, every term has a nonzero coefficient at the *same* exponent β , so *local finiteness* fails. If $q\beta < \delta$, the displayed leading exponents form a strictly descending sequence in the union of supports, so *well-ordering* fails. This proves necessity with no convergence terminology. If $q\beta > \delta$, put $w = z/s$ with $v(w) > 0$: all coefficients b_m have nonnegative ε -support, and a term of spatial degree $qm + 1$ contributes a word of at least that length from $\text{supp } w$ together with a nonnegative number of δ 's, so lemma 3.4 makes the double family strongly summable, and multiplication by s preserves this.

For invertibility, formal reversion of $\Psi_\varepsilon = w + O(w^{q+1})$ gives $\Theta_\varepsilon \in \mathbb{C}[[\varepsilon]][[w]]$ with linear coefficient 1 and no constant term, requiring no division except by a unit; the same positive-word argument evaluates it at every $w \in \mathfrak{m}_\Gamma$, and the formal identities become strong identities there, both maps preserving $\mathfrak{m}_\Gamma$ because their difference from the identity has strictly greater valuation at a nonzero infinitesimal input. For the isometry put $u = x/s$, $v_1 = y/s$:

$$\Psi_\varepsilon(u) - \Psi_\varepsilon(v_1) = (u - v_1) \left[1 + \sum_{m \geq 1} b_m(\varepsilon) \sum_{j=0}^{qm} u^{qm-j} v_1^j \right],$$

whose bracket is a strong sum equal to 1 plus a positive-valuation element, each nonconstant summand having valuation at least $q \min\{v(u), v(v_1)\} > 0$ when both inputs are nonzero (a zero input is handled with the vanishing terms omitted). Finally every nonzero solution of $z^q = -\varepsilon$ is a fixed point of f of valuation δ/q , lying in K_Γ because its constant factor is a complex q -th root of -1 ; any strictly larger centered ball containing D contains all elements at valuation δ/q , hence such a fixed point in addition to 0, and an injective conjugacy k would satisfy $k(z) = \lambda k(z)$ at every fixed point, mapping both to 0 since $\lambda \neq 1$. $\square$

Warning 14.23 (The maximality statement is about centered balls). [Theorem 14.22](#) concerns centered valuative balls defined by a valuation threshold, under either boundary convention. It does **not** exclude continuation to a non-ball domain avoiding the obstructing fixed points, and does not assert that every coefficientwise continuation is injective. Those distinctions matter at the critical scale, which is exactly where [section 14.7](#) operates.

14.7 Continuation at a scale where the Taylor family fails

Let $w_0 \in U$ be an ordinary nonzero point. At $z = sw_0$ we have $v(z) = \delta/q$, so the Taylor family [\(14.25\)](#) is *not* strongly summable. On the other hand

$$s\Psi_\varepsilon(w_0) = s \sum_{n \geq 0} \varepsilon^n \Psi_n(w_0) \quad (14.29)$$

is a valid Hahn value, and more generally it is defined at $z = s(w_0 + \eta)$ for positive η by [\(3.4\)](#). There is no contradiction, and the distinction is exactly where the monodromy lives: in [\(14.29\)](#) each coefficient function Ψ_n is first an ordinary holomorphic function obtained by ordinary integration and continuation, and summation in the Hahn exponent comes *afterward*; in [\(14.25\)](#) summation in spatial Taylor degree is required to be strongly summable as a family of Hahn numbers, and it fails precisely on this boundary. The boundary value is a *coefficientwise analytic continuation*, not an illegal regrouping of a strong sum; on the inner disk both procedures are admissible and agree by finite-word Taylor identities.

For $V_q = w(1 + w^q)$ every nonzero zero satisfies $a^q = -1$ and $V'_q(a) = -q$, so all these zeros share the exact exponent

$$\rho_q(\varepsilon) = \frac{\text{Log}(1 + \varepsilon)}{\text{Log}(1 - q\varepsilon)} = -\frac{1}{q} + A_q\varepsilon + B_q\varepsilon^2 + O(\varepsilon^3), \quad (14.30)$$

$$A_q = \frac{q+1}{2q}, \quad B_q = \frac{(q+1)(q-4)}{12q}, \quad C_q = \frac{(q+1)(2q+1)}{12q}, \quad (14.31)$$

the ratio being legitimate because numerator and denominator both start in degree one with the denominator's coefficient $-q \neq 0$.

Theorem 14.24 (All-orders factorization and finite-cover obstruction for the model). *The rescaled coordinate has the all-orders form*

$$\Psi_\varepsilon(w) = w(1 + w^q)^{\rho_q(\varepsilon)} \text{Exp}_H(P_\varepsilon(w)), \quad P_\varepsilon \in \varepsilon^2\mathbb{C}[w][[\varepsilon]], \quad P_\varepsilon(0) = 0. \quad (14.32)$$

At every nonzero fixed point the monodromy multiplier $M_q = \text{Exp}_H(2\pi i \rho_q)$ satisfies

$$M_q^q = 1 + \pi i(q+1)\varepsilon + O(\varepsilon^2), \quad (14.33)$$

and has infinite order. The leading coordinate becomes single-valued after finite ramification, but no finite branched cover makes Ψ_ε single-valued at these points.

Proof. Apply [corollary 14.9](#): all nonzero-root exponents coincide and $\prod_{a^q=-1} (1 - w/a) = 1 + w^q$, the branches of the logarithms agreeing near 0 where each is normalized to vanish, so their sum is $\log(1 + w^q)$; this gives [\(14.32\)](#) with the same polynomial part. The coefficient at zero in $q\rho_q$ is -1 and the coefficient at ε is $(q+1)/2$, so exponentiation gives [\(14.33\)](#); no positive power is 1 by [lemma 14.10](#) since $A_q \neq 0$. The leading exponent $-1/q$ is rational, so a q -fold local ramification removes its branching, and [theorem 14.11](#) shows that the positive part of the full exponent survives every finite ramification. $\square$

A circuit repeated q times thus returns the leading branch to itself and still changes the full answer by a nonzero infinitesimal amount: the monodromy group records information lost under passage to the standard part. This is the phenomenon whose linear-ODE ancestor is already recorded by the companion report *Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers* (docs/surcomplex/differential-equations/); see [section 23.1](#) for the exact inheritance.

Proposition 14.25 (Explicit second-order expansion). *With $L(w) = \log(1 + w^q)$ normalized by $L(0) = 0$, on any admissible common domain*

$$\Psi_\varepsilon(w) = w(1 + w^q)^{-1/q} \left[1 + A_q \varepsilon L(w) + \varepsilon^2 \left(B_q L(w) + \frac{A_q^2}{2} L(w)^2 - C_q w^q \right) + O(\varepsilon^3) \right], \quad (14.34)$$

equivalently $P_\varepsilon(w) = -C_q \varepsilon^2 w^q + O(\varepsilon^3)$.

Proof. With $V = V_q$ and $\mu = 1$, expansion using (14.10) gives $[\varepsilon^2]\Omega_\varepsilon = (1/3 - V'/4 - (V')^2/12 - VV''/12)/V$; substituting $V = w(1 + w^q)$ and collecting powers of w^q yields

$$[\varepsilon^2]\Omega_\varepsilon = -\frac{(q+1)(2q+1)}{12} w^{q-1} + \frac{(q+1)(q-4)}{12} \frac{w^{q-1}}{1 + w^q},$$

whose first summand is the polynomial part with normalized primitive $-C_q w^q$ and whose second has normalized primitive $B_q L(w)$; the coefficient at ε integrates to $A_q L(w)$ and the coefficient at zero gives Ψ_0 . $\square$

For $q = 1$ this reads

$$\Psi_\varepsilon(w) = \frac{w}{1+w} \left[1 + \varepsilon \log(1+w) + \frac{\varepsilon^2}{2} (\log^2(1+w) - \log(1+w) - w) + O(\varepsilon^3) \right]. \quad (14.35)$$

For $q = 4$ the coefficient B_q happens to vanish, but the first-order logarithm and hence the infinite-order monodromy do not: an accidental cancellation at one higher order cannot repair the obstruction.

Remark 14.26 (Three representations of one inner-disk coordinate). The formal Taylor series (14.25) supplies exact valuations and the sharp disk; the common-domain coefficient series $\Psi_\varepsilon = \sum \varepsilon^n \Psi_n$ supplies continuation at the rescaled boundary; and the factorization (14.32) isolates monodromy in a single Hahn constant with a single-valued polynomial-coefficient factor left over. They answer different questions: to evaluate very close to 0 the recurrence (14.24) is efficient and directly summable; to determine branching near a boundary fixed point the exact ratio of logarithms avoids computing many individual Taylor coefficients; and to see why algebraicity fails even when the leading function is algebraic, the first correction or the monodromy orbit suffices.

Warning 14.27 (Transcendental, but differentially algebraic). Transcendence in [theorem 14.15](#) is relative to the specific base $\mathcal{B}$. Source 04 volunteers the counterweight: H_ε is *differentially* algebraic in a very explicit way, satisfying the first-order linear equation $H'_\varepsilon = \Omega_\varepsilon H_\varepsilon$ with $\Omega_\varepsilon \in \mathcal{B}$. So “transcendental” must not be upgraded to “differentially transcendental”. Nor is it a claim that any individual value $H_\varepsilon(w_0)$ lies outside $\mathbf{No}[i]$; the statement concerns a *function* in an ambient meromorphic Hahn field and is explicitly consistent with $\mathbf{No}[i]$ being algebraically closed.

Everything above stays true of H_ε , and the sentences above are not retracted by what follows. What the companion report *Tail-Spans and Differential Transcendence in Surreal and Surcomplex Hahn Fields* (docs/surreal/tail-spans-and-differential-transcendence/) shows is that the

refusal is about H_ε and not about the workspace: its Theorem `tail:thm:mixed` exhibits explicit functions $F_A(z, t) = \sum_{n \in A} t^n \sqrt{1 - z/2^n}$, with the branch equal to 1 at $z = 0$, one for each infinite $A \subseteq \mathbb{N}_{>0}$, all of whose mixed jets under ∂_z and $t d/dt$ are algebraically independent over $\mathbb{C}(z)((t^\mathbb{Q}))$ — the $\Gamma = \mathbb{Q}$ case of a base of the same shape as $\mathcal{B}$ — so that these functions satisfy no algebraic differential equation of any finite order over that field. Its Theorem `tail:thm:continuumanalytic` does this for continuum many such functions on the single fixed disk $|z| < 1$, with no shrinking of the disk, matching the fixed-domain discipline of [section 15](#). Those functions are not solutions of [equation \(14.18\)](#) and nothing about them transfers to H_ε ; the pointer records that “transcendental but differentially algebraic” is a property of H_ε , not a ceiling on what a base of this shape permits. Its Remark `tail:rem:dynupgrade` states the same relation from the other side.

15 Parabolic dynamics II: time forms, periods, and moduli

This section is also under PAR and $\text{SD}_\emptyset$, and its maps are near-identity, hence unipotent: there are *no* eigenvalue ratios and *no* Diophantine hypotheses ([warning 2.8](#)). The single defining choice is the coefficient algebra: everything happens in $\mathcal{A} = \mathcal{O}(D)((t^\Gamma))$ for *one* fixed ordinary connected $D \subset \mathbb{C}$, the fixed-polydisk row of [section 3.2](#), and **no shrinking of D is permitted in any theorem**. The Hahn monomials are constants, not functions of z ; the only derivation is ∂_z applied coefficientwise. Write also $\mathcal{B} = \mathcal{M}(D)((t^\Gamma))$, a field, used only for division steps; $\mathcal{A}$ itself is a commutative integral domain and generally not a field, with $K = K_\Gamma$ embedded as the constants. All integrals below are *ordinary* complex line integrals taken coefficient by coefficient over ordinary curves in D .

Warning 15.1 (These periods are not surcomplex contour integrals). A period below is an ordinary coefficientwise contour integral. It is **not** a surcomplex contour integral, **not** a residue at an arbitrary surcomplex point, and **not** a fine-topological notion. Source 05 cites the companion report *Contours and Stokes: Jordan separation, Cauchy theory, and Stokes calculus on the surcomplex plane* ([docs/surcomplex/contours-and-stokes/](#)) as background and deliberately does *not* use surcomplex contours; it carries a remark forbidding exactly that identification. Furthermore, if one allows an unrelated choice of primitive on each monad one has enlarged the function class and can destroy the global obstruction: the theorems below are about a single element of $\mathcal{O}(D)((t^\Gamma))$, not a collection of unrelated infinitesimal germs.

15.1 Primitives, time forms, and pullback exactness

Lemma 15.2 (Support-preserving primitives). *For $g \in \mathcal{A}$, a primitive $P \in \mathcal{A}$ with $P' = g$ exists if and only if $\oint_\gamma g(z) dz = 0$ (coefficientwise) for every ordinary closed piecewise C^1 curve in D . If it exists, the primitive with $P(z_*) = 0$ is unique and introduces no new support exponents.*

Proof. Necessity holds coefficientwise by the ordinary fundamental theorem for line integrals. Conversely define each coefficient primitive by path integration from z_* ; vanishing periods make this path-independent, and on an ordinary disk the result is an ordinary holomorphic primitive, hence holomorphic on all of D . Use zero for a coefficient that was zero, so the support is a subset of $\text{supp } g$. Two primitives differ by an element of K , removed by the normalization. $\square$

Assume now that $W = W_F = X_F z$ has a nowhere-zero leading coefficient. By [lemma 14.2](#) the time form

$$\Omega_F = \frac{dz}{W_F}$$

lies in $\mathcal{A} dz$: “holomorphic” refers to all its ordinary coefficient functions, and its *valuation may be negative*. Its significance is elementary: in a local primitive coordinate P with $P' = 1/W$ the vector field $W\partial_z$ becomes a unit translation field; the issue is whether such local primitives fit together globally, and whether one can compare the primitives of two maps without leaving the common coefficient domain. For $H = z + h \in \mathcal{G}_+(D)$ the pullback of a one-form is $H^*(g dz) = (g \circ H)H' dz$.

Lemma 15.3 (Explicit exactness of an infinitesimal pullback). *For $g \in \mathcal{A}$ and $h \in \mathcal{A}_{>0}$ the series*

$$Q_h(g) = \sum_{n \geq 1} \frac{h^n}{n!} g^{(n-1)} \quad (15.1)$$

is strongly summable and satisfies

$$(z + h)^*(g dz) - g dz = dQ_h(g). \quad (15.2)$$

Consequently every coefficientwise period is unchanged by a near-identity pullback.

Proof. For $S = \text{supp } h$, the support in (15.1) lies in $\text{supp } g + (S^* \setminus \{0\})$ with finitely many contributions per coefficient, so termwise differentiation is legitimate and gives

$$(Q_h(g))' = \sum_{n \geq 1} \frac{h^n}{n!} g^{(n)} + h' \sum_{n \geq 1} \frac{h^{n-1}}{(n-1)!} g^{(n-1)} = g(z+h) - g(z) + h'g(z+h) = g(z+h)(1+h') - g(z),$$

which is (15.2). Exact coefficient forms have zero ordinary periods by lemma 15.2. $\square$

Lemma 15.3 is a support-controlled version of the familiar fact that an infinitesimal change of coordinate acts trivially on one-dimensional holomorphic de Rham cohomology. It is included as infrastructure; the homotopy principle itself is *not* a novelty claim, and there is no assertion that $s \mapsto z + sh$ is a fine-continuous path in the surreal topology — formula (15.1) is the construction and its support certificate is what licenses it.

Proposition 15.4 (Time forms under conjugacy). *If W_{F_1}, W_{F_2} are invertible in $\mathcal{A}$ then for $H \in \mathcal{G}_+(D)$, $H \circ F_1 = F_2 \circ H$ iff $H^*\Omega_{F_2} = \Omega_{F_1}$; in particular conjugate maps have identical time-form periods on every ordinary closed curve in D .*

Proof. Equality of time forms reads $H'/W_{F_2}(H) = 1/W_{F_1}$, equivalent to proposition 4.7; period invariance is lemma 15.3. $\square$

15.2 The discrete equation as a primitive problem

Let $F \in \mathcal{G}_+(D)$ be nonidentity, $W = W_F$, $\mathcal{D} = W\partial_z$, and

$$E(X) = \frac{e^X - 1}{X} = \sum_{n \geq 0} \frac{X^n}{(n+1)!}, \quad B(X) = \frac{X}{e^X - 1} = \sum_{n \geq 0} \frac{B_n}{n!} X^n, \quad (15.3)$$

with $B_0 = 1$, $B_1 = -1/2$. Substitution of $\mathcal{D}$ into either series is strongly summable, and these are inverse automorphisms of the underlying strongly linear space $\mathcal{A}$; *no* multiplicativity of $B(\mathcal{D})$ or $E(\mathcal{D})$ is asserted. Write $\Delta_F \psi = \psi \circ F - \psi$.

Theorem 15.5 (Discrete equation as a primitive problem). *For nonidentity $F \in \mathcal{G}_+(D)$,*

$$\ker \Delta_F = K, \quad \Delta_F(\mathcal{A}) = W \partial_z \mathcal{A}. \quad (15.4)$$

*For $g \in \mathcal{A}$ the equation $\psi \circ F - \psi = g$ has a solution $\psi \in \mathcal{A}$ if and only if the meromorphic Hahn quotient g/W belongs to $\mathcal{A}$ **and***

$$\oint_{\gamma} \frac{g(z)}{W(z)} dz = 0 \quad \text{for every ordinary closed curve } \gamma \subset D. \quad (15.5)$$

If $P' = g/W$, all solutions are $\psi = B(\mathcal{D})P + C$ with $C \in K$. When the leading coefficient of W is nowhere zero, the membership $g/W \in \mathcal{A}$ is automatic.

Proof. By [theorem 4.3](#) and the scalar identities in (15.3),

$$\Delta_F = e^{\mathcal{D}} - I = \mathcal{D}E(\mathcal{D}) = E(\mathcal{D})\mathcal{D}, \quad (15.6)$$

and $E(\mathcal{D})B(\mathcal{D}) = I = B(\mathcal{D})E(\mathcal{D})$ follows coefficientwise: for each input $\psi \in \mathcal{A}$, apply [corollary 3.5](#) with $B = \text{supp } \psi$ and $S = \text{supp } W$ to justify every rearrangement. Thus the image equals $\mathcal{D}\mathcal{A}$. The kernel of $\mathcal{D}$ is K since $W \neq 0$, $\mathcal{A}$ is a domain and $\ker \partial_z = K$, and both $E(\mathcal{D})$ and its inverse act as the identity on constants, so (15.6) also gives $\ker \Delta_F = K$. Since division by W is available in $\mathcal{B}$, membership in $W \partial_z \mathcal{A}$ is equivalent to g/W being the derivative of an element of $\mathcal{A}$, which is [lemma 15.2](#). Finally $\Delta_F B(\mathcal{D})P = \mathcal{D}P = WP' = g$, and the kernel statement gives uniqueness up to K ; the last assertion is [lemma 14.2](#). $\square$

Remark 15.6 (Why the uncorrected forcing supplies the periods). One might instead multiply by $B(\mathcal{D})$ and integrate $B(\mathcal{D})g/W$. This gives the same obstruction: for $n \geq 1$, $\mathcal{D}^n g/W = (\mathcal{D}^{n-1}g)'$ is exact, so all Bernoulli correction terms have zero periods. The factorization (15.6) makes the uncorrected quotient g/W and its obstruction especially transparent.

Corollary 15.7 (Exact global Abel criterion). *Suppose the leading coefficient of W_F is nowhere zero. Then $\psi \circ F = \psi + 1$ has a solution in $\mathcal{A}$ if and only if all the periods of Ω_F vanish, in which case any primitive $P' = 1/W_F$ already satisfies the equation and all solutions differ from P by a constant in K .*

Proof. Apply [theorem 15.5](#) with $g = 1$; also $\mathcal{D}P = 1$ and $\mathcal{D}^2P = 0$, so $e^{\mathcal{D}}P = P + 1$, while the Bernoulli formula gives $B(\mathcal{D})P = P - 1/2$, differing only by a constant. $\square$

Warning 15.8 (An Abel function is not a globally injective coordinate). The identity $P' = 1/W_F$ guarantees a nonvanishing derivative on the halo under the leading-unit hypothesis, but it does **not** make P globally injective on a multiply covered domain: even in ordinary complex analysis a locally univalent primitive need not be globally univalent. [Corollary 15.7](#) is an Abel-function theorem, not an unconditional global conjugacy with translation on a subset of K .

Theorem 15.9 (Complete finite-dimensional discrete obstruction). *Let $D_r = \mathbb{C} \setminus \{a_1, \dots, a_r\}$ with distinct ordinary points and positively oriented small ordinary circles γ_j about a_j enclosing no other puncture, and put*

$$(\mathcal{P}_F g)_j = \frac{1}{2\pi i} \oint_{\gamma_j} \frac{g}{W_F} dz. \quad (15.7)$$

If $F \in \mathcal{G}_+(D_r)$ has a nowhere-zero leading displacement coefficient, then

$$0 \longrightarrow K \longrightarrow \mathcal{A} \xrightarrow{\Delta_F} \mathcal{A} \xrightarrow{\mathcal{P}_F} K^r \longrightarrow 0 \quad (15.8)$$

is an exact sequence of K -vector spaces, with explicit right inverse

$$(q_1, \dots, q_r) \mapsto W_F(z) \sum_{j=1}^r \frac{q_j}{z - a_j}, \quad (15.9)$$

so that every forcing decomposes as

$$g = \Delta_F \psi + W_F \sum_{j=1}^r \frac{(\mathcal{P}_F g)_j}{z - a_j}, \quad (15.10)$$

with ψ determined up to a constant once the second term is fixed.

Proof. The initial kernel statement is [theorem 15.5](#). Every ordinary closed curve in D_r has, against a holomorphic one-form, the same periods as the integer combination of small puncture circles given by its winding numbers: remove small disks about the punctures inside a large disk containing the curve and cut the remaining planar region along arcs joining the boundary components, so that the cut region is simply connected and the integrals along the two sides of each cut cancel. Applying this ordinary argument coefficientwise shows that vanishing of the r components of (15.7) is equivalent to (15.5), so middle exactness follows from [theorem 15.5](#). The differential $dz/(z - a_j)$ has integral $2\pi i$ around γ_j and zero around the other chosen circles, so (15.9) is a right inverse; subtracting it from g and applying exactness gives (15.10). $\square$

No identification of the infinite-dimensional $\mathcal{A}$ with $K \otimes_{\mathbb{C}} \mathcal{O}(D_r)$ is used: the finite-dimensional quotient is computed directly by coefficientwise periods and explicit representatives.

Example 15.10 (A local obstruction where the leading coefficient vanishes). The quotient condition in [theorem 15.5](#) must not be dropped outside the nonsingular stratum. On a disk containing 0 take $W = \varepsilon z^2$ and $F = z/(1 - \varepsilon z)$ with $\varepsilon = t^\delta$, $\delta > 0$; this F is the exponential of $W\partial_z$ in $\mathcal{G}_+(D)$. The equation $\psi \circ F - \psi = 1$ has *no* solution in $\mathcal{A}$, because $1/W = \varepsilon^{-1}z^{-2}$ is not holomorphic at 0 — equivalently F fixes 0 and evaluating the equation there would give $0 = 1$. On a punctured disk, by contrast, $-1/(\varepsilon z)$ is an Abel function. This separates a local divisibility obstruction from the global period obstruction.

15.3 The complete global conjugacy classification

Fix $\delta > 0$, $\varepsilon = t^\delta$, and $a \in \mathcal{O}(D)$ nowhere zero, and define the nonsingular stratum

$$\mathcal{G}_{\delta,a}(D) = \{F \in \mathcal{G}_+(D) : F = z + \varepsilon a + r, \ v(r) > \delta\} = \{F : W_F = \varepsilon(a + u), \ u \in \mathcal{A}_{>0}\}, \quad (15.11)$$

with normalized time form

$$\beta_F = \varepsilon \Omega_F = b_F(z) dz, \quad b_F = a^{-1} + \text{positive Hahn terms}. \quad (15.12)$$

The leading data (δ, a) are a conjugacy invariant under $\mathcal{G}_+(D)$: the identity $W_2(H) = H'W_1$ of [proposition 4.7](#) changes neither the leading coefficient nor the valuation, because $H = z + \text{positive}$ and $H' = 1 + \text{positive}$. It is therefore natural to classify one stratum at a time.

Lemma 15.11 (Implicit displacement lemma). *Let $b = b_0 + u \in \mathcal{A}_{\geq 0}$ with $b_0 \in \mathcal{O}(D)$ nowhere zero and $u \in \mathcal{A}_{>0}$. For each $p \in \mathcal{A}_{>0}$ there is a unique $h \in \mathcal{A}_{>0}$ satisfying*

$$bh + \frac{b'}{2!}h^2 + \frac{b''}{3!}h^3 + \dots = p. \quad (15.13)$$

Every coefficient of h is holomorphic on D ; if $S = \text{supp } u \cup \text{supp } p$ then $\text{supp } h \subseteq S^ \setminus \{0\}$; and if $p(z_*) = 0$ then $h(z_*) = 0$.*

Proof. If $S = \emptyset$ take $h = 0$. Otherwise let $M = S^*$, well ordered by lemma 3.4; set the coefficient of h at 0 to zero and determine h_γ for $\gamma \in M \setminus \{0\}$ in increasing order. The coefficient of h_γ on the left of (15.13) is precisely b_0 ; any other linear term comes from $u_\alpha h_\nu$ with $\alpha > 0$ and $\nu = \gamma - \alpha < \gamma$, and in a term with $n \geq 2$ factors of h every factor exponent is positive and strictly smaller than γ . The coefficient is a finite sum (expand the factors into words in S and apply lemma 3.4), so the recursion has the form $h_\gamma = b_0^{-1}(p_\gamma - R_\gamma)$ with R_γ a finite expression in earlier coefficients and ordinary derivatives of b , defining a holomorphic function on D . For unrestricted uniqueness, the difference of the two left sides for two solutions $h_1, h_2 \in \mathcal{A}_{>0}$ factors as $(h_1 - h_2)(b_0 + v)$ with $v \in \mathcal{A}_{>0}$ — factor each difference of powers, the resulting sum being strongly summable over the finite union of the positive supports of u, h_1, h_2 — and the second factor is a unit by lemma 14.2. At z_* , evaluation gives the same scalar implicit equation over K with linear leading coefficient $b_0(z_*) \neq 0$, so if $p(z_*) = 0$ then zero is a solution and the same factorization shows it is the only positive one. $\square$

Warning 15.12 (Where the nowhere-zero hypothesis does its work). Lemma 15.11 is exactly the point at which the classification proof stops applying when a has zeros. Its common-domain conclusion is stronger than solving a separate germ problem at each point: *every stage divides only by the same nowhere-zero ordinary function b_0* and uses derivatives of functions already holomorphic on D , so no radius is lost at any Hahn exponent. Source 05 states outright that the zeros case needs local zero divisors, multiplicities and singular normal-form obstructions combined with global periods, and that its classification is not claimed there.

Theorem 15.13 (Complete period invariant and marked conjugacy). *For $F_1, F_2 \in \mathcal{G}_{\delta,a}(D)$ with a nowhere zero, the following are equivalent:*

- (i) *there is $H \in \mathcal{G}_+(D)$ with $H \circ F_1 = F_2 \circ H$;*
- (ii) *$\oint_\gamma \Omega_{F_1} = \oint_\gamma \Omega_{F_2}$ for every ordinary closed piecewise C^1 curve $\gamma \subset D$;*
- (iii) *$\beta_{F_1} - \beta_{F_2} = dp$ for some $p \in \mathcal{A}_{>0}$.*

When they hold, for every specified $z_ \in D$ there is a unique conjugacy H as in (i) with $H(z_*) = z_*$, obtained by taking the normalized primitive $p(z_*) = 0$ in (iii) and solving*

$$\sum_{n \geq 1} \frac{b_{F_2}^{(n-1)}}{n!} h^n = p, \quad H = z + h. \quad (15.14)$$

No shrinking of D is permitted anywhere in this statement.

Proof. (i) $\Rightarrow$ (ii) is proposition 15.4. Multiplication by the nonzero constant ε shows (ii) equivalent to the vanishing of all periods of $\beta_{F_1} - \beta_{F_2}$; both normalized time forms have the same leading coefficient a^{-1} , so their difference has positive support, and the normalized primitive of lemma 15.2, when it exists, also has positive support — this is the equivalence of (ii) and (iii) including the normalization of p . Assume (iii): the leading coefficient of b_{F_2} is a^{-1} , nowhere zero, so lemma 15.11 gives a unique positive h in (15.14), and lemma 15.3 gives $H^* \beta_{F_2} - \beta_{F_2} = dp = \beta_{F_1} - \beta_{F_2}$, i.e. $H^* \Omega_{F_2} = \Omega_{F_1}$, so H is a conjugacy by proposition 15.4; its inverse lies in $\mathcal{G}_+(D)$ by theorem 4.3 and $H(z_*) = z_*$ by lemma 15.11. If $\tilde{H} = z + \tilde{h}$ is any conjugacy fixing z_* , the same pullback identity yields $d(Q_{\tilde{h}}(b_{F_2}) - p) = 0$, so that expression is a constant in K with value zero at z_* , whence $\tilde{h}$ satisfies the same implicit equation and uniqueness in lemma 15.11 gives $\tilde{H} = H$. $\square$

Corollary 15.14 (Simply connected domains). *If D is simply connected, any two maps in $\mathcal{G}_{\delta,a}(D)$ are conjugate by a unique near-identity change of variable fixing a prescribed ordinary basepoint, and each is conjugate to $\exp(\varepsilon a \partial_z)z$.*

Proof. Every ordinary holomorphic coefficient one-form on D has a primitive, so (iii) of [theorem 15.13](#) holds. $\square$

For $a = 1$ the reference map is $z + \varepsilon$, and this is a genuine global near-identity conjugacy to that translation on $D^\#$; for general a the reference map is the indicated vector-field exponential and no univalence of a primitive of dz/a is required.

Warning 15.15 (Three hypotheses that do actual work). a must not vanish, because inversion and the implicit displacement recursion divide by a or a^{-1} ([warning 15.12](#)). The conjugating map must have identity ordinary part, because otherwise it can act nontrivially on ordinary homology and on the leading form — source 05 records that extending the classification to a larger group would require the ordinary automorphism action on the leading form and its periods, and that equal period vectors in a fixed basis would then no longer be the correct equivalence relation by themselves. And all coefficient functions must be defined on the same D : replacing global functions by independent germs removes the meaning of a single period comparison. The theorem is not claimed after any of these hypotheses is erased.

15.4 Explicit moduli

Theorem 15.16 (An explicit moduli space). *On $D_r = \mathbb{C} \setminus \{a_1, \dots, a_r\}$, for the fixed stratum (δ, a) define*

$$q_j(F) = \frac{1}{2\pi i} \oint_{\gamma_j} \left(\beta_F - \frac{dz}{a(z)} \right) \in \mathfrak{m}_K. \quad (15.15)$$

Then $F \mapsto (q_1(F), \dots, q_r(F))$ induces a bijection

$$\mathcal{G}_{\delta,a}(D_r)/\text{conjugacy by } \mathcal{G}_+(D_r) \longleftrightarrow \mathfrak{m}_K^r, \quad (15.16)$$

and for $\mathbf{q} \in \mathfrak{m}_K^r$ a representative is

$$b_{\mathbf{q}}(z) = \frac{1}{a(z)} + \sum_{j=1}^r \frac{q_j}{z - a_j}, \quad W_{\mathbf{q}} = \frac{\varepsilon}{b_{\mathbf{q}}}, \quad F_{\mathbf{q}} = \exp(W_{\mathbf{q}} \partial_z) z. \quad (15.17)$$

No countability, finite generation or rank-one condition is imposed on the supports of the q_j .

Proof. The finite union of the supports of the q_j is positive and well ordered, so $b_{\mathbf{q}} \in \mathcal{A}_{\geq 0}$ with nowhere-zero leading coefficient a^{-1} , invertible by [lemma 14.2](#); hence $W_{\mathbf{q}} = \varepsilon(a + \text{positive})$ and $F_{\mathbf{q}}$ lies in the stated stratum with normalized time form exactly $b_{\mathbf{q}} dz$, and the simple-pole representatives give $q_j(F_{\mathbf{q}}) = q_j$. Conjugate maps have the same vector by period invariance; conversely equal vectors give equal periods around each puncture, hence, as in [theorem 15.9](#), equality of every ordinary period, and [theorem 15.13](#) supplies a conjugacy. $\square$

The parametrization is algebraic over the Hahn constants in its displayed one-form representatives. Its basis is the *specified* puncture loops; arbitrary changes of ordinary coordinates are not quotiented out.

15.5 Nonconjugate dynamics beyond every one-scale jet

Fix $\delta > 0$, $\varepsilon = t^\delta$, and define the *flat ideal*

$$J_\delta = \{0\} \cup \{c \in K : v(c) > n\delta \text{ for every integer } n \geq 1\}, \quad (15.18)$$

an ideal of the valuation ring $K_{\geq 0}$, nonzero *precisely* when some $\eta \in \Gamma$ exceeds all the $n\delta$ (the forward implication by taking the valuation of a nonzero member, the reverse from $t^\eta \in J_\delta$). In particular $J_\delta = 0$ when the multiples of δ are cofinal, as in any Archimedean value group. For coefficient functions the corresponding ideal is

$$I_\delta(D) = \bigcap_{n \geq 1} \varepsilon^n \mathcal{A}_{\geq 0} = \{0\} \cup \{f \in \mathcal{A} : v(f) > n\delta \ \forall n \geq 1\}, \quad (15.19)$$

the equivalence of weak and strict inequalities following by replacing n with $n + 1$.

Warning 15.17 (The ideals must be formed in $\mathcal{A}_{\geq 0}$). It is essential that (15.18), (15.19) are formed in $K_{\geq 0}$ and $\mathcal{A}_{\geq 0}$, **not** in K or $\mathcal{A}$, where ε is a *unit* and its powers generate everything. Consequently the map to one-scale jets

$$\mathcal{A}_{\geq 0} \longrightarrow \varprojlim_n \mathcal{A}_{\geq 0} / \varepsilon^n \mathcal{A}_{\geq 0} \quad (15.20)$$

has kernel exactly $I_\delta(D)$, and in the noncofinal case it loses nonzero information. The next theorem shows it can lose a complete family of global conjugacy obstructions, not merely some irrelevant constants.

Theorem 15.18 (Flatly indistinguishable but globally nonconjugate). *Let $r \geq 1$, $D_r = \mathbb{C} \setminus \{a_1, \dots, a_r\}$, and suppose $J_\delta \neq 0$. For $\mathbf{c} \in J_\delta^r$ put*

$$R_{\mathbf{c}}(z) = \sum_{j=1}^r \frac{c_j}{z - a_j}, \quad W_{\mathbf{c}} = \frac{\varepsilon}{1 + \varepsilon R_{\mathbf{c}}}, \quad F_{\mathbf{c}} = \exp(W_{\mathbf{c}} \partial_z) z. \quad (15.21)$$

All these maps lie in $\mathcal{G}_{\delta,1}(D_r)$ and satisfy:

(a) $F_{\mathbf{0}} = z + \varepsilon$ and $F_{\mathbf{c}} - F_{\mathbf{0}} \in I_\delta(D_r)$ for every $\mathbf{c}$; if $\mathbf{c} \neq \mathbf{0}$ and $\eta = \min_{c_j \neq 0} v(c_j)$ then exactly

$$v(F_{\mathbf{c}} - F_{\mathbf{0}}) = 2\delta + \eta; \quad (15.22)$$

(b) *their time forms are exactly*

$$\Omega_{F_{\mathbf{c}}} = \varepsilon^{-1} dz + \sum_{j=1}^r \frac{c_j dz}{z - a_j}; \quad (15.23)$$

(c) $F_{\mathbf{c}}$ and $F_{\mathbf{d}}$ are conjugate by an element of $\mathcal{G}_+(D_r)$ if and only if $\mathbf{c} = \mathbf{d}$;

(d) a global Abel function in $\mathcal{A}$ exists for $F_{\mathbf{c}}$ if and only if $\mathbf{c} = \mathbf{0}$.

Proof. The finite union of the supports of the c_j is well ordered and positive, so $R_{\mathbf{c}} \in \mathcal{A}_{>0}$, and the denominator in (15.21) has leading coefficient 1 and is a unit, so $W_{\mathbf{c}} = \varepsilon +$ terms of valuation greater than δ ; theorem 4.3 places $F_{\mathbf{c}}$ in the stated stratum, and for $\mathbf{c} = \mathbf{0}$ the vector field is constant with exponential exactly $z + \varepsilon$. If $\mathbf{c} \neq \mathbf{0}$ then $v(R_{\mathbf{c}}) = \eta$, since a nontrivial linear combination of the $(z - a_j)^{-1}$ cannot vanish identically — ordinary residues at the punctures recover its coefficients. The geometric expansion gives

$$W_{\mathbf{c}} - \varepsilon = -\varepsilon^2 R_{\mathbf{c}} + \varepsilon^3 R_{\mathbf{c}}^2 - \dots, \quad v(W_{\mathbf{c}} - \varepsilon) = 2\delta + \eta. \quad (15.24)$$

Put $V = W_{\mathbf{c}} - \varepsilon$ and $\lambda_0 = 2\delta + \eta$. In the exponential expansion of $(\varepsilon \partial_z + V \partial_z)^n z$ for $n \geq 2$, the word with no factor $V \partial_z$ vanishes because $(\varepsilon \partial_z)^2 z = 0$, and every other word has valuation at least $\lambda_0 + (n - 1)\delta > \lambda_0$ since derivatives do not lower Hahn valuation; all such words are summable by lemma 3.4. So the leading difference comes from V itself, proving (15.22), and $\eta > n\delta$ for all n puts the difference in (15.19). The logarithm of $F_{\mathbf{c}}$ is $W_{\mathbf{c}}$ by the inverse identity in theorem 4.3, and taking its reciprocal gives (15.23); hence $(2\pi i)^{-1} \oint_{\gamma_j} \Omega_{F_{\mathbf{c}}} = c_j$, and for the normalized form the corresponding value is εc_j . Equality of periods is therefore equivalent to $\mathbf{c} = \mathbf{d}$, so theorem 15.13 proves (c) and corollary 15.7 proves (d). $\square$

Remark 15.19 (A failure of one-scale information, not of Hahn equality). All the maps of [theorem 15.18](#) have the same image under (15.20). They do *not* have the same full Hahn coefficients: the missing coefficients occur at exponents larger than every $n\delta$. The result therefore contradicts neither uniqueness of Hahn expansions nor separation by arbitrary value-group cutoffs. It shows exactly why replacing arbitrary-rank support control by a single “small-parameter completion” can erase a global obstruction. This is the interpretive reading source 05 attaches to its own theorem, and is a framing argument supported by the theorem rather than itself a theorem.

Example 15.20 (A concrete surcomplex instance). Take $\Gamma = \mathbb{Q} \oplus \mathbb{Q}$ lexicographically, $\delta = (0, 1)$, $\eta = (1, 0)$, so that $\eta > n\delta$ for every integer $n \geq 1$ because comparison is decided by the first coordinate; realize this inside the surreals as $\mathbb{Q}\omega + \mathbb{Q}$ via $(q, r) \mapsto q\omega + r$. Under (3.1), $\varepsilon = \omega^{-1}$ and $c = t^\eta = \omega^{-\omega}$. On $D = \mathbb{C}^*$ the field

$$W_c(z) = \frac{\omega^{-1}}{1 + \omega^{-(\omega+1)}/z} \quad (15.25)$$

defines an explicit surcomplex map $F_c = \exp(W_c \partial_z)z$ with the same truncation as $z + \omega^{-1}$ modulo every positive integer power of ω^{-1} , yet $(2\pi i)^{-1} \oint_{|z|=1} \Omega_{F_c} = \omega^{-\omega} \neq 0$: it is not globally near-identity conjugate to that translation and has no single-valued Hahn-coherent Abel function on the punctured plane. This is a global effect at a strictly smaller *scale*, and is explicitly **not** a claim about ordinary exponentially small asymptotics of complex numbers — $\omega^{-\omega}$ is an exact Hahn monomial in a rank-two workspace. A nonzero ordinary multiple of it already gives a different member of the family, and the full parameter space J_δ is much larger.

15.6 Worked parabolic examples and the iterative residue

For a map written in powers of one positive monomial, $F = z + \varepsilon v_1 + \varepsilon^2 b + \varepsilon^3 c + O(\varepsilon^4)$ with $v_1, b, c \in \mathcal{O}(D)$, direct expansion of (4.4) gives

$$W_F = \varepsilon v_1 + \varepsilon^2 \left(b - \frac{1}{2} v_1 v_1' \right) + \varepsilon^3 \left(c - \frac{1}{2} (v_1 b' + b v_1') + \frac{1}{12} v_1^2 v_1'' + \frac{1}{3} v_1 (v_1')^2 \right) + O(\varepsilon^4), \quad (15.26)$$

obtained from $\Delta z = h$, $\Delta^2 z = h h' + \frac{1}{2} h^2 h'' + \dots$, $\Delta^3 z = h (h')^2 + h^2 h'' + \dots$ with $h = F - z$ and the logarithm coefficients $1, -\frac{1}{2}, \frac{1}{3}$. Here $O(\varepsilon^4)$ means membership in $\varepsilon^4 \mathcal{O}(D)[[\varepsilon]]$ and is used only in examples already specified as one-parameter series; it is never a substitute for arbitrary-rank support conditions. If v_1 is nowhere zero, the time form begins

$$\Omega_F = \left[\frac{1}{\varepsilon v_1} + \frac{v_1'}{2v_1} - \frac{b}{v_1^2} + O(\varepsilon) \right] dz, \quad (15.27)$$

so already at the next order the ordinary logarithmic derivative of the leading vector field can create a nonzero period.

Proposition 15.21 (Exact residue of the Euler time form). *On $D = \mathbb{C}^*$ let $F_p(z) = z + \varepsilon z^{p+1}$, $p \geq 1$. Then*

$$W_{F_p}(z) = \sum_{n \geq 1} c_{p,n} \varepsilon^n z^{np+1}, \quad c_{p,1} = 1, \quad c_{p,2} = -\frac{p+1}{2}, \quad (15.28)$$

$$\frac{1}{W_{F_p}(z)} = \varepsilon^{-1} z^{-p-1} + \frac{p+1}{2} z^{-1} + \sum_{n \geq 1} d_{p,n} \varepsilon^n z^{np-1}, \quad (15.29)$$

so that

$$\operatorname{Res}_{z=0} \Omega_{F_p} = \frac{p+1}{2} \quad (15.30)$$

exactly, with no omitted positive-valuation correction. Hence there is no global Abel function for F_p in $\mathcal{O}(\mathbb{C}^*)((t^\Gamma))$.

Proof. Taylor substitution by F_p preserves the grading in which $\varepsilon^n z^{np+1}$ has grade n , since $(z + \varepsilon z^{p+1})^{kp+1} = z^{kp+1}(1 + \varepsilon z^p)^{kp+1}$; every term of the finite difference and its iterates therefore has the displayed form, and taking the logarithm preserves it, the first two coefficients following from (15.26). Factor $W_{F_p} = \varepsilon z^{p+1} C_p(\varepsilon z^p)$ with $C_p(x) = 1 - (p+1)x/2 + \dots$ and invert C_p to obtain (15.29); for $n \geq 1$ the exponent $np - 1$ is nonnegative, so none of the last terms has a z^{-1} coefficient, and (15.30) is exact. The period about 0 is nonzero, so corollary 15.7 forbids a global Abel function. $\square$

The leading field εz^{p+1} itself has zero period on $\mathbb{C}^*$ and its exponential has Abel function $-1/(p\varepsilon z^p)$; so passing from that exact flow to its Euler map changes the global obstruction even though their first displacements agree. For $p = 1$,

$$W_{F_1} = \varepsilon z^2 - \varepsilon^2 z^3 + \frac{3}{2}\varepsilon^3 z^4 - \frac{8}{3}\varepsilon^4 z^5 + \frac{31}{6}\varepsilon^5 z^6 - \frac{157}{15}\varepsilon^6 z^7 + O(\varepsilon^7). \quad (15.31)$$

Warning 15.22 (These coefficients are known, and nothing is claimed about their convergence). The coefficients (15.31) agree, after multiplication by the step size, with the standard modified equation in Hairer's worked example [24]; they are **known examples, not new coefficients**, and are not proposed as discoveries. Nothing is asserted about convergence of the ordinary step-size series in a real step size; Hairer's warning that it generally diverges is cited and accepted. The constant (15.30) is the familiar iterative-residue phenomenon, and the assertion here is its exact interpretation in the stated common-domain Hahn category, not the discovery of that classical invariant. The *same* degree-six expansion appears independently in source 03 as the iterative logarithm of $z + tz^2$ (section 22.2); it is one calculation and is printed once, in (15.31).

Corollary 15.23 (A global coherent normal form for the Euler family). F_p is conjugate on $(\mathbb{C}^*)^\#$, by a near-identity Hahn-coherent change of variable, to the time-one map of

$$V_p(z) = \frac{\varepsilon z^{p+1}}{1 + \frac{p+1}{2}\varepsilon z^p}, \quad (15.32)$$

the conjugacy fixing $z = 1$ being unique. It is **not** a conjugacy of F_p to the time-one map of εz^{p+1} .

Proof. The time form of (15.32) is $\varepsilon^{-1} z^{-p-1} dz + \frac{p+1}{2} dz/z$, with the same period as (15.29); their difference consists of nonnegative powers of z coefficientwise, hence is exact on $\mathbb{C}^*$, so theorem 15.13 applies. The last assertion holds because the leading vector-field flow has zero period while F_p does not. $\square$

On any simply connected slit domain in $\mathbb{C}^*$ the normal form has the local Abel function $-1/(p\varepsilon z^p) + \frac{p+1}{2} \log z$, whose logarithmic monodromy is precisely the obstruction to a single-valued global Abel function on $\mathbb{C}^*$.

Example 15.24 (A difference equation with an explicit Bernoulli correction). Take $W = \varepsilon z$, so $F = e^\varepsilon z$, on $D = \mathbb{C}^*$. The forcing $g = z^m$ with nonzero integer m has zero period after division by W , and $\psi(z) = z^m/(e^{m\varepsilon} - 1)$ is a solution; the denominator has valuation δ so its reciprocal is a legitimate Laurent–Hahn constant, and expanding gives $\psi(z) = (\frac{1}{m\varepsilon} - \frac{1}{2} + \frac{m\varepsilon}{12} - \dots)z^m$, exactly

the Bernoulli solver of [theorem 15.5](#) applied to $P = z^m/(m\varepsilon)$. For the constant forcing $g = 1$, by contrast, $g dz/W = \varepsilon^{-1} dz/z$ has nonzero period and no global solution exists, displaying the one-dimensional cokernel for the punctured plane.

Example 15.25 (Local straightening does not erase global periods). $F(z) = (1+\varepsilon)z$ and $G(z) = e^\varepsilon z$ have the same leading displacement εz on $\mathbb{C}^*$, with logarithmic vector fields $\log(1+\varepsilon)z$ and εz and periods $2\pi i/\log(1+\varepsilon)$ and $2\pi i/\varepsilon$, which differ — so there is no global near-identity conjugacy there. On a simply connected domain with a branch of $\log z$ there is one:

$$H(z) = z \exp\left(\left[\frac{\varepsilon}{\log(1+\varepsilon)} - 1\right] \log z\right),$$

the bracketed coefficient having positive valuation, so the exponential is a strongly summable Hahn expansion with coefficients holomorphic on the chosen branch domain, and direct substitution gives $H \circ F = G \circ H$. The failure to make that formula single-valued on $\mathbb{C}^*$ is exactly what the period theorem detects.

15.7 Ordinary slow time on one fixed holomorphic flow chart

[Theorem 4.3](#) concerns an infinitesimal vector field. One may also ask what happens at the larger time scale on which the leading ordinary vector field moves a finite distance; a direct exponential is generally not a strongly summable Hahn expression there. The correct construction solves the leading ordinary flow first and then lifts all positive Hahn coefficients along it. Write

$$W = \varepsilon X, \quad X(z) = a(z) + \sum_{\beta \in S} t^\beta X_\beta(z), \quad (15.33)$$

with $S \subset \Gamma_{>0}$ well ordered and all coefficient functions holomorphic on D ; in this subsection a may have zeros. Let $\Sigma \subset \mathbb{C}$ be an ordinary disk centered at 0, $D_0 \subset D$ an ordinary open domain, and assume a holomorphic ordinary flow chart

$$\phi : \Sigma \times D_0 \rightarrow D, \quad \partial_s \phi(s, z) = a(\phi(s, z)), \quad \phi(0, z) = z \quad (15.34)$$

is specified, with $J(s, z) = \partial_z \phi(s, z)$ nowhere vanishing on that chart.

Theorem 15.26 (Common-chart Hahn lifting of the slow flow). *There is exactly one series*

$$\Psi(s, z) = \phi(s, z) + \sum_{\gamma \in S^* \setminus \{0\}} t^\gamma \psi_\gamma(s, z) \quad (15.35)$$

with coefficients holomorphic on the same product domain $\Sigma \times D_0$ satisfying

$$\partial_s \Psi = X(\Psi), \quad \Psi(0, z) = z, \quad (15.36)$$

$X(\Psi)$ meaning Taylor expansion at the ordinary point $\phi(s, z)$. Uniqueness holds among all positive-Hahn perturbations of ϕ on this product domain, not only those with the displayed support bound. Each coefficient is obtained by one ordinary integral,

$$\psi_\gamma(s, z) = J(s, z) \int_0^s J(u, z)^{-1} R_\gamma(u, z) du, \quad (15.37)$$

with R_γ a finite expression in previously determined coefficients and in ordinary derivatives of the X_β and a along ϕ . Finally Taylor evaluation at $s = \varepsilon$ is strongly summable and

$$\Psi(\varepsilon, z) = \exp(W \partial_z) z \quad (z \in D_0^\#), \quad (15.38)$$

and more generally, for $c \in K$ with $c = 0$ or $v(c) + \delta > 0$, evaluation at $s = c\varepsilon$ is defined and gives $\Psi(c\varepsilon, z) = \exp(cW \partial_z) z$.

Proof. Let $M = S^*$ and seek $q = \Psi - \phi$ with support in $M \setminus \{0\}$. Expanding $X(\phi + q) = a(\phi) + a'(\phi)q + \sum_{n \geq 2} a^{(n)}(\phi)q^n/n! + \sum_{\beta \in S} t^\beta \sum_{n \geq 0} X_\beta^{(n)}(\phi)q^n/n!$ — all compositions with ϕ being holomorphic on the specified product because its image lies in D — the coefficient equation at $\gamma \in M \setminus \{0\}$ is

$$\partial_s \psi_\gamma = a'(\phi) \psi_\gamma + R_\gamma, \quad \psi_\gamma(0, z) = 0, \quad (15.39)$$

every factor in R_γ involving a positive exponent smaller than γ except for specified coefficients of X , with finitely many contributions by lemma 3.4 exactly as in lemma 15.11. Differentiating (15.34) in z gives $\partial_s J = a'(\phi)J$, $J(0, z) = 1$, so variation of constants solves (15.39) by (15.37); since J is nowhere zero and Σ is simply connected the integral is independent of its path in Σ and is jointly holomorphic in (s, z) , locally by integrating the holomorphic power series in s with coefficients holomorphic in z and patching the uniquely normalized primitives. There is therefore *no* successive domain shrinkage and *no* requirement of a uniform bound on these coefficients as γ varies. The support lies in M and the expansion of $X(\Psi)$ is strongly summable by lemma 3.4, so the construction solves (15.36). For uniqueness, at the least exponent where two positive-Hahn perturbations of ϕ satisfying that equation differ, their difference solves the homogeneous version of (15.39) with zero initial condition and is zero; this argument does not require either support to lie in M . For the last assertion expand every coefficient at $s = 0$ and substitute $s = \varepsilon$: the possible exponents lie in the monoid generated by $S \cup \{\delta\}$ and a fixed exponent has finitely many word representations, so the evaluation is strongly summable. Repeated differentiation of (15.36) gives $\partial_s^n \Psi(0, z) = (X\partial_z)^n z$, whence $\Psi(\varepsilon, z) = \sum_n \varepsilon^n (X\partial_z)^n z / n! = \exp(W\partial_z)z$. For $s = c\varepsilon \in \mathfrak{m}_K$ replace $\{\delta\}$ by the well-ordered positive support of $c\varepsilon$. $\square$

Warning 15.27 (Exactly what the slow-time theorem licenses). At an ordinary time $s_0 \in \Sigma$, evaluating (15.35) produces an element with leading ordinary part $\phi(s_0, z)$ and positive Hahn corrections, the ordinary coefficient integrals having *already been performed*: that is a separate analytic operation from an unregrouped Hahn exponential. For example, with $X = z^2$ and $W = \varepsilon z^2$ the slow flow is the ordinary rational function $\phi(s, z) = z/(1 - sz)$ on any product chart avoiding $sz = 1$, whereas at a nonzero ordinary s the raw formal terms of $\exp(sz^2\partial_z)z = \sum_{n \geq 0} s^n z^{n+1}$ all have Hahn exponent 0 with infinitely many nonzero, and are *not* a strongly summable Hahn family. The rational function comes from ordinary complex analysis on its chart, not from declaring that family a Hahn sum. Theorem 15.26 therefore licenses one particular passage to the time scale ε^{-1} — solve the leading ordinary motion, then integrate each positive correction along it — and licenses **no** arbitrary exchange of ordinary with Hahn summation, **no** global continuation past an ordinary singularity of the supplied chart, and **no** interpretation of an arbitrary surreal-valued time as an actual continuous trajectory. It is also explicitly *not* analytic convergence of the modified series in a real step size.

16 Hamiltonian normal forms, integrability, and certified domains

This section is under $\neg$ FR and, according to the case, SD_0 or SD_{fr} . The frequencies are *ordinary complex constants*, never surreal or Hahn quantities; Γ needs neither divisibility nor algebraic closure; and the two coefficient algebras

$$\mathcal{P}_n = \mathbb{C}[z_1, \dots, z_n], \quad \mathcal{E}_n = \mathcal{O}(\mathbb{C}^n), \quad \mathcal{H}_\Gamma(\mathcal{P}_n), \quad \mathcal{H}_\Gamma(\mathcal{E}_n) \quad (16.1)$$

are the load-bearing distinction: **no** uniform norm bound as γ varies in the entire case, **no** uniform degree bound in the polynomial case, and results about actual finite-halo evaluation are **not**

transferred to $\mathcal{H}_\Gamma(\mathbb{C}\{z\})$, whose coefficients are unrelated convergent germs. In $2d$ phase variables $q_1, \dots, q_d, p_1, \dots, p_d$ use the fixed sign convention

$$\{F, G\} = \sum_{j=1}^d (F_{q_j} G_{p_j} - F_{p_j} G_{q_j}), \quad \text{ad}_\chi F = \{\chi, F\}, \quad (16.2)$$

and write $I_j = q_j p_j$,

$$H_0 = \sum_{j=1}^d \omega_j I_j, \quad \omega \cdot k \neq 0 \quad (0 \neq k \in \mathbb{Z}^d). \quad (16.3)$$

16.1 Supported substitutions and Lie maps

Proposition 16.1 (Positive near-identity maps). *Let $U \in \mathcal{H}_\Gamma(\mathcal{E}_n)_{>0}^n$. Then $\Phi(z) = z + U(z)$ has a unique inverse $z + V(z)$ with $V \in \mathcal{H}_\Gamma(\mathcal{E}_n)_{>0}^n$; both are bijections of $\mathcal{O}_\Gamma^n$ preserving standard parts; and if U has polynomial coefficients, so does V .*

Proof. Let S be the union of the supports of the components of U and $\mathbf{M} = M(S)$. The inverse equation is $V(z) = -U(z + V(z))$; expanding the right side by Taylor's formula, its term independent of V is $-U(z)$ and every other term has at least one positive input from U and at least one from V , so at exponent $\gamma > 0$ all occurrences of coefficients of V have exponents strictly smaller than γ , and recursion on the well-ordered $\mathbf{M}$ determines V_γ uniquely — including at limit positions of $\mathbf{M}$, where *no topological limit is requested*. At a fixed exponent only finitely many Taylor terms and input words occur by lemma 3.4, their coefficients using only derivatives, products and finite sums of the coefficients of U , hence entire, or polynomial in the polynomial case. This gives a right inverse; the same construction for a left inverse, with associativity of supported substitution or the least-exponent uniqueness argument, shows the two agree, and proposition 3.11 gives the actual bijections. $\square$

Proposition 16.2 (Supported Hamiltonian exponential). *If $\chi \in \mathcal{H}_\Gamma(\mathcal{E}_{2d})_{>0}$ then for every $F \in \mathcal{H}_\Gamma(\mathcal{E}_{2d})$ the family $(\text{ad}_\chi^n F/n!)_{n \geq 0}$ is strongly summable; its sum $T_\chi F = e^{\text{ad}_\chi} F$ is an algebra and Poisson automorphism with inverse $T_{-\chi}$; and with $\Phi = (T_\chi q_1, \dots, T_\chi q_d, T_\chi p_1, \dots, T_\chi p_d)$ one has $T_\chi F = F \circ \Phi$, the realized Φ being a symplectic bijection of the finite halo. Polynomial coefficients are preserved.*

Proof. With $A = \text{supp } F$ and $S = \text{supp } \chi$, each term of $\text{ad}_\chi^n F$ has support in $A + nS$, differentiation does not change Hahn weights, and a fixed exponent receives contributions for finitely many words in S attached to an exponent in A : lemma 3.4 gives strong summability and every regrouping used below. Leibniz gives $\text{ad}_\chi^n(FG) = \sum_j \binom{n}{j} (\text{ad}_\chi^j F)(\text{ad}_\chi^{n-j} G)$, so after division by $n!$ and supported summation T_χ is multiplicative; the Jacobi identity gives the identical binomial rule for the bracket, so $T_\chi\{F, G\} = \{T_\chi F, T_\chi G\}$; and the binomial identity for $\exp(\mathcal{E})\exp(-\mathcal{E})$ gives the inverse. For an ordinary entire function the formal chain rule for the derivation ad_χ identifies $T_\chi F$ with Taylor substitution by Φ — check this at each power of an auxiliary formal scalar multiplying χ , where it is the usual formal flow identity, then evaluate that scalar at 1 using the proved word summability — and applying it coefficientwise gives the assertion for general F . The coordinate Poisson relations are preserved, proving symplecticity, and $\Phi = \text{id} + \text{positive terms}$, so propositions 3.11 and 16.1 give the actual maps. $\square$

Definition 16.3 (The finite-expression certificate). Label each input perturbation coefficient occurrence by its positive Hahn exponent. A term in a normalizing coefficient is built by a *finite expression tree* whose leaves are labeled inputs and whose internal operations are Poisson brackets, linear projections, scalar division, or fixed linear solution operators; its weight is the sum of its leaf weights. For a fixed total weight there are finitely many leaf words by [lemma 3.4](#); every genuinely nonlinear internal step has at least two positive inputs, linear operators only decorating those steps or the leaves, so a fixed leaf word admits only finitely many relevant tree shapes. This is a finite certificate for every coefficient even when the set of smaller weights is infinite, and it is also the reason a later reordering of weights under rescaling is legitimate ([theorem 16.21](#)).

16.2 The homological operator and a sharp divisor threshold

Define $LF = \{F, H_0\}$, so that

$$L(q^\alpha p^\beta) = \omega \cdot (\alpha - \beta) q^\alpha p^\beta \quad (16.4)$$

and $\neg\text{FR}$ makes the kernel exactly the monomials with $\alpha = \beta$. For a Taylor expansion at the origin set

$$\Pi\left(\sum_{\alpha,\beta} c_{\alpha\beta} q^\alpha p^\beta\right) = \sum_{\alpha} c_{\alpha\alpha} I^\alpha, \quad \Pi + Q_\Pi = 1. \quad (16.5)$$

Lemma 16.4 (Entire action projection). Π maps $\mathcal{E}_{2d}$ to functions $g(I)$ with $g \in \mathcal{E}_d$, uniquely; it preserves polynomial coefficients, does not decrease phase vanishing order, and commutes with L . On $Q_\Pi \mathcal{P}_{2d}$ the inverse L^{-1} is always defined and polynomial, with no arithmetic estimate.

Proof. For entire f , Cauchy's estimate on a phase polydisk of radius R gives $|c_{\alpha\beta}| \leq C_R R^{-|\alpha|-|\beta|}$, so $|c_{\alpha\alpha}| \leq C_R (R^2)^{-|\alpha|}$; as R is arbitrary the action series is entire, and distinct action monomials are distinct phase monomials, giving uniqueness. The remaining assertions follow termwise, and for a polynomial the inverse divides finitely many nonresonant coefficients by nonzero complex constants. $\square$

One may also realize Π by averaging over the ordinary torus action $q_j \mapsto e^{i\theta_j} q_j$, $p_j \mapsto e^{-i\theta_j} p_j$; no surcomplex contour integration is needed for that observation.

Theorem 16.5 (Sharp homological criterion). *For nonresonant ω the following are equivalent: (i) $\Theta(\omega) < \infty$; (ii) the coefficientwise L^{-1} maps every nonresonant entire function to an entire function; (iii) the coefficientwise L^{-1} maps every nonresonant convergent germ at the origin to a convergent germ, with a possible loss of radius. If these fail, there is a nonresonant entire F of phase order at least three for which $L^{-1}F$ has no positive radius of convergence.*

Proof. If $M_D \leq CA^D$ and the coefficients of F obey $|f_{\alpha\beta}| \leq C_R R^{-D}$, those of $L^{-1}F$ obey $|(L^{-1}F)_{\alpha\beta}| \leq CC_R (A/R)^D$; the number of monomials of total degree D in $2d$ variables is $\binom{D+2d-1}{2d-1}$, which grows polynomially, so this bound proves absolute convergence on every polydisk of radius less than R/A — one admissible R for a germ, arbitrarily large R for an entire function. Conversely if $\Theta = \infty$, select distinct degrees $D_j \rightarrow \infty$ with $D_j \geq \max(j, 3)$ and nonresonant monomials $m_j = q^{\alpha_j} p^{\beta_j}$ of those degrees with $\delta_j = |\omega \cdot (\alpha_j - \beta_j)|$ and $\delta_j^{-1/D_j} \geq j^4$, and put

$$F = \sum_j \delta_j^{1/2} m_j. \quad (16.6)$$

On a polydisk of radius R the j -th term has modulus at most $(R/j^2)^{D_j}$ and the tail is bounded by a geometric series since $D_j \geq j$, so F is entire; in the putative inverse the absolute coefficient of m_j is $\delta_j^{-1/2}$, whose D_j -th root is at least j^2 , violating every Cauchy bound on a positive-radius polydisk. Since F is itself a germ, both (ii) and (iii) fail, and its order and nonresonance were built into the selection. $\square$

Warning 16.6 (What Θ measures, and what it must not be advertised as). A polynomial lower bound on $|\omega \cdot k|$ implies $\Theta = 1$ but is far stronger than necessary here. Finite exponential growth is the exact condition for this *single coefficientwise homological operator* to preserve all entire functions. The nonlinear Hahn theorem below applies that operator repeatedly and requires *no* uniform norm estimate across Hahn layers; this is precisely why the criterion must **not** be advertised as an ordinary complex-analytic convergence criterion of Bruno or KAM type. Sources 06 repeats this disclaimer in three separate places, and the merge preserves it.

16.3 Arbitrary-support Hamiltonian normalization

Theorem 16.7 (Exact positive-Hahn normal form). *Let ω satisfy (16.3) and let*

$$H = H_0 + R, \quad R = \sum_{s \in S} t^s R_s(q, p), \quad S \subseteq \Gamma_{>0} \text{ well ordered, } \text{ord}_0 R_s \geq 3. \quad (16.7)$$

Assume either (a) every R_s is a polynomial, with no restriction on ω beyond nonresonance (tag SD_0), or (b) every R_s is entire and $\Theta(\omega) < \infty$ (tag SD_{fr}). Then there is a unique positive generator χ in the corresponding coefficient algebra with

$$\Pi\chi = 0, \quad \text{ord}_0 \chi_\gamma \geq 3, \quad e^{\text{ad}_\chi} H = H_0 + B(I), \quad (16.8)$$

where B has positive Hahn support and coefficients of action vanishing order at least two, and $\text{supp } \chi, \text{supp } B \subseteq M(S) \setminus \{0\}$. The coordinate map $\Phi = e^{\text{ad}_\chi}(q, p)$ and its inverse are symplectic bijections of $\mathcal{O}_\Gamma^{2d}$, fixing the origin, tangent to the identity there, preserving standard parts everywhere, and satisfying the actual identity

$$H^\# \circ \Phi^\# = (H_0 + B(I))^\#. \quad (16.9)$$

All statements commute with enlargement of the Hahn workspace.

Proof. Expanding the Lie exponential,

$$e^{\text{ad}_\chi}(H_0 + R) = H_0 + R + L\chi + \mathcal{B}(\chi, R), \quad \mathcal{B}(\chi, R) = \sum_{n \geq 1} \frac{\text{ad}_\chi^n R}{n!} + \sum_{n \geq 2} \frac{\text{ad}_\chi^n H_0}{n!}, \quad (16.10)$$

every term of $\mathcal{B}$ involving at least two positive factors, counting occurrences of R and χ . Since $\Pi L = 0$ and $\Pi\chi = 0$, the nonresonant normal-form equation is exactly

$$\chi = -L^{-1}Q_\Pi(R + \mathcal{B}(\chi, R)). \quad (16.11)$$

Put $\mathbf{M} = M(S)$: at weight $\gamma \in \mathbf{M} \setminus \{0\}$ the coefficient of $\mathcal{B}$ depends only on χ_μ with $0 < \mu < \gamma$, since an occurrence of χ_γ would be accompanied by another positive factor and so have total weight greater than γ . Well-founded recursion on $\mathbf{M}$ therefore determines

$$\chi_\gamma = -L^{-1}Q_\Pi(R_\gamma + [t^\gamma]\mathcal{B}(\chi_{<\gamma}, R)), \quad (16.12)$$

a formula that remains meaningful when γ has infinitely many predecessors, because lemma 3.4 allows only finitely many of them to participate in a word of total weight γ . More explicitly, each coefficient is a finite sum of the labeled trees of definition 16.3: their leaves are input coefficients R_s , their total weight is γ , and their internal nonlinear nodes have at least two positive inputs, so the finitely many words of total weight γ bound the number and choice of leaves and there are finitely many tree shapes and bracket orderings. Hence every coefficient uses finitely many applications of derivatives, multiplication, Q_Π and L^{-1} : the claimed support bound holds and no unproved infinite operation occurs within a Hahn coefficient. In case (a) these operations preserve polynomials by lemma 16.4; in case (b) they preserve entire functions by theorem 16.5. Every application of $L^{-1}Q_\Pi$ gives $\Pi\chi_\gamma = 0$, and the estimate $\text{ord}_0\{F, G\} \geq \text{ord}_0 F + \text{ord}_0 G - 2$ shows inductively that each χ_γ has order at least three while terms in $\mathcal{B}$ have order at least four; hence $B = \Pi(R + \mathcal{B}(\chi, R))$ has phase order at least four, since a resonant monomial has even phase degree and R has order at least three, and by lemma 16.4 it is an entire or polynomial function of the actions at every Hahn weight, of action order at least two. Equation (16.11) removes the nonresonant part of (16.10), so (16.8) holds as a Hahn identity, and proposition 16.2 defines the exponential. For uniqueness, at the least exponent γ of the difference of two positive gauge-normalized generators, all terms coming from $\mathcal{B}$ have weight strictly greater than γ , leaving $L(\chi_\gamma - \tilde{\chi}_\gamma) = 0$ with a nonresonant difference, on which L is injective by (16.4); this also rules out additional support outside M . Finally propositions 3.11 and 16.2 give the symplectic maps, their inverse, and (16.9), and no construction used an exponent outside $M(S)$. $\square$

Remark 16.8 (What has become exact). In a classical analytic problem the generator can be formal in the phase variables with zero radius of convergence. In case (a) each Hahn coefficient of the generator is instead a *finite polynomial*: its coefficients may be enormous ordinary complex numbers, but they are finite constants at their Hahn layer, and an arbitrarily large ordinary number cannot cancel the positive valuation of t^γ (remark 2.3). Evaluation on the finite halo is therefore controlled by supports rather than by a numerical norm of the normalizing coefficients. In case (b) infinitely many phase monomials occur within a single Hahn layer, so their ordinary convergence must be settled *before* Hahn summability can be invoked; the homological threshold does exactly that, and requires no common numerical bound on the resulting entire coefficients because the common domain is already all of $\mathbb{C}^{2d}$.

Corollary 16.9 (Finite surcomplex phase space). *For polynomial layers, and for entire layers satisfying the threshold, theorem 16.7 realizes the normalization at every finite point of $\mathbf{No}[i]^{2d}$, and the maps obtained in different set-sized workspaces agree on their common domain.*

Proof. Given a phase tuple, enlarge the workspace to contain all of its exponents and those of the input and apply theorem 16.7; its coefficient formulas and the Taylor evaluation are unchanged by the enlargement. $\square$

Warning 16.10 (Uniqueness has a gauge). The unique object in theorem 16.7 is the *single exponential* generator with $\Pi\chi = 0$, and the normal form it produces in that gauge. Arbitrary symplectic changes of variables normalizing a Hamiltonian need **not** be unique: in particular transformations generated by functions of the actions can preserve a normal form. Source 06 declines to suppress this freedom by an unstated uniqueness claim, and section D distinguishes the single-exponential gauge recursion from the different algorithm that composes successive exponentials without taking a logarithm.

16.4 Exact integrability and the whole Poisson centralizer

Write $T = e^{\text{ad}_X}$, $N = TH$, and $J_j = T^{-1}I_j = e^{-\text{ad}_X}I_j$.

Theorem 16.11 (Exact integrability). *The $J_1, \dots, J_d$ are entire-coefficient Hahn functions — with polynomial coefficients in case (a) — and*

$$\{H, J_j\} = 0, \quad \{J_j, J_k\} = 0, \quad H = N(J_1, \dots, J_d). \quad (16.13)$$

They are functionally independent wherever, in the normalized coordinates, none of the pairs (q_j, p_j) is $(0, 0)$. These hold as realized identities on the finite halo.

Proof. The I_j commute with one another and with every function of the actions, including N ; applying the Poisson automorphism T^{-1} gives the first two identities and the algebra substitution T^{-1} applied to $N(I)$ the third. The d differentials $dI_j = p_j dq_j + q_j dp_j$ have disjoint coordinate pairs and are linearly independent exactly when each is nonzero; pullback by the invertible differential of Φ^{-1} preserves that rank, and [proposition 3.11](#) respects all these identities. $\square$

Theorem 16.12 (Poisson centralizer). *In $\mathcal{H}_\Gamma(\mathcal{E}_{2d})$,*

$$\text{Cent}(H) := \{F : \{F, H\} = 0\} = \{G(J_1, \dots, J_d) : G \in \mathcal{H}_\Gamma(\mathcal{E}_d)\}, \quad (16.14)$$

*the representing G being unique. Its Hahn support need **not** be positive. Thus (16.14) describes all first integrals that admit an entire-coefficient Hahn representation on the finite halo.*

Proof. It suffices to classify the centralizer of $N(I) = H_0 + B(I)$, since T is a Poisson automorphism. Let $\tilde{F} = TF$ with $\{\tilde{F}, N\} = 0$ and apply Π coefficientwise: by [lemma 16.4](#), $\Pi\tilde{F} = G(I)$ for a unique $G \in \mathcal{H}_\Gamma(\mathcal{E}_d)$, and every action function commutes with N , so $U := \tilde{F} - \Pi\tilde{F}$ is coefficientwise nonresonant and also commutes with N . If $U \neq 0$, let a_0 be the least exponent of its support: the coefficient at t^{a_0} in $\{U, B\}$ is zero, all exponents of B being positive and all of U at least a_0 , so the coefficient at t^{a_0} in $\{U, N\} = 0$ is $LU_{a_0} = 0$ with U_{a_0} nonresonant and nonzero, contradicting (16.4). Hence $U = 0$ and $\tilde{F} = G(I)$, so $F = G(J)$. Conversely every $G(J)$ commutes with H ; its definition is legitimate even for arbitrary well-ordered support of G — write $J = I + \Delta$ with positive Δ , use the ordinary Taylor expansion of each entire coefficient of G at I , and apply [lemma 3.4](#). Uniqueness follows by applying T or by the injectivity of $g(I) \mapsto g$; faithfulness of [proposition 3.11](#) means no additional identity appears on passing to finite-halo functions. $\square$

Warning 16.13 (The coefficient class is essential). [Theorem 16.12](#) does **not** classify first integrals defined only near a particular point, multivalued functions, or arbitrary functions that happen to be locally constant in a fine topology, and does not say that all first integrals have polynomial coefficients. The entire-coefficient class is large enough to contain arbitrary entire functions of the actions and rigid enough for the leading nonresonant coefficient argument to work; that is exactly the balance the theorem exploits.

Proposition 16.14 (Exact ordinary-time evolution). *Let $\Omega_j(I) = \partial N / \partial I_j = \omega_j + \Delta_j(I)$ with $v(\Delta_j) > 0$ (positive Hahn support). For ordinary complex τ put*

$$q_j(\tau) = e^{\tau\omega_j} \text{Exp}_H(\tau\Delta_j(I))q_j(0), \quad p_j(\tau) = e^{-\tau\omega_j} \text{Exp}_H(-\tau\Delta_j(I))p_j(0). \quad (16.15)$$

These give an exact flow of N on the finite halo for every $\tau \in \mathbb{C}$; conjugation by Φ gives the flow of H ; the coefficients depend holomorphically on τ ; and the flow law holds as an identity of Hahn-coherent functions.

Proof. The actions are constant along (16.15) because the two factors for each j are inverse, so $\Delta_j(I)$ is constant along the curve, and coefficientwise differentiation in ordinary time gives $\dot{q}_j = \Omega_j(I)q_j = \partial N/\partial p_j$ and $\dot{p}_j = -\Omega_j(I)p_j = -\partial N/\partial q_j$. The Hahn exponentials are strongly summable by positivity, each coefficient is entire in τ and the phase variables, and their constant Hahn terms are nonzero ordinary exponentials, so finite phase points remain finite; the exponential identities, justified coefficientwise by finite Hahn regroupings, give the flow law, and symplectic conjugacy transports the equations to H . $\square$

Warning 16.15 (This is a coefficientwise holomorphic-time identity). [Proposition 16.14](#) is not an assertion about continuous ordinary-time paths in the fine topology of the proper-class field. No infinite surreal time, no global surreal exponential of an arbitrary surcomplex argument, and no intrinsic coefficient-field derivation is used ([warning 3.2](#)). “Analytic” in this section means exactly the entire-coefficient Taylor-compatible realization of [proposition 3.11](#).

16.5 Failure for entire layers: an exact converse

Theorem 16.16 (Sharp universal normalization criterion). *For fixed nonresonant ω the following are equivalent: (i) $\Theta(\omega) < \infty$; (ii) every positive Hahn perturbation with entire coefficients of phase order at least three admits the entire-coefficient gauge-normalized generator and finite-halo realization of [theorem 16.7](#), for every set-sized ordered value group; (iii) every one-parameter Hamiltonian $H_0 + \varepsilon_0 F$ with F entire of order at least three admits a symplectic formal change $\Psi = \text{id} + O(\varepsilon_0)$ with convergent-germ coefficients at the origin transforming it into a function of the actions. If (i) fails, failure of (iii) is already forced at the coefficient of ε_0 , and there is no positive gauge-normalized entire Hahn generator for the same input in any workspace containing its scale.*

Proof. (i) $\Rightarrow$ (ii) is [theorem 16.7](#); (ii) $\Rightarrow$ (iii) follows by taking $\Gamma = \mathbb{Z}$ and $\varepsilon_0 = t$. Suppose $\Theta = \infty$ and take the entire nonresonant F of [theorem 16.5](#). If a symplectic change as in (iii) exists, write $\Psi(z) = z + \varepsilon_0 V(z) + O(\varepsilon_0^2)$ with V a holomorphic vector-field germ; the first-order symplectic condition says V is a symplectic vector field, so its contraction with the constant symplectic form is a closed holomorphic one-form, and a closed holomorphic one-form germ is exact (a primitive is given on a small star-shaped polydisk by the radial homotopy integral). Hence, with the sign matching (16.2), there is a holomorphic germ χ_1 with $V(G) = \{\chi_1, G\}$ for every holomorphic germ G . The coefficient of ε_0 in $(H_0 + \varepsilon_0 F) \circ \Psi$ is $F + L\chi_1$, and if the transformed Hamiltonian is a function of the actions its nonresonant part must vanish: $L(Q_\Pi \chi_1) = -F$. The projection Q_Π preserves convergent germs by the same Cauchy estimate as in [lemma 16.4](#), and coefficientwise injectivity of L on nonresonant series forces $Q_\Pi \chi_1 = -L^{-1}F$, which is not a convergent germ. For the final assertion, a putative positive gauge-normalized entire Hahn generator for $H_0 + t^\eta F$ cannot have a least nonzero coefficient below η (its normal-form equation there would be $L\chi_\eta = 0$, impossible in the gauge), and at weight η the nonlinear terms have larger weight, so again $\chi_\eta = -L^{-1}F$, not entire. This argument does not depend on the rank or divisibility of the enlarged value group. $\square$

Proposition 16.17 (An explicit super-Liouville obstruction). *Define integers by*

$$a_1 = 1, \quad a_{n+1} = 10^{4a_n}, \quad Q_n = 10^{a_n}, \quad \alpha = \sum_{n \geq 1} 10^{-a_n}, \quad (16.16)$$

put $P_n/Q_n = \sum_{j=1}^n 10^{-a_j}$ (so $P_n \in \mathbb{Z}$) and $D_n = P_n + Q_n < 2Q_n$. Then

$$0 < \alpha Q_n - P_n < 2Q_n 10^{-a_{n+1}} = 2Q_n 10^{-Q_n^4} < e^{-D_n^2}, \quad (16.17)$$

α is irrational, and $\omega = (1, \alpha)$ is nonresonant. In two degrees of freedom,

$$F(q_1, q_2, p_1, p_2) = \sum_{n \geq 1} e^{-D_n^2/2} q_1^{P_n} p_2^{Q_n}, \quad H = I_1 + \alpha I_2 + \varepsilon_0 F, \quad (16.18)$$

where F is entire of phase order at least three ($D_1 = 11$), the Hamiltonian has no symplectic normalization $\text{id} + O(\varepsilon_0)$ with convergent-germ coefficients to an action-only normal form, and its first nonresonant homological solution has zero radius of convergence.

Proof. The first bound in (16.17) dominates the very rapidly decreasing tail by twice its first term; for the last, $Q_n \geq 10$ and $\log(2Q_n) - Q_n^4 \log 10 < -4Q_n^2 < -D_n^2$. If $\alpha = A/B$ were rational then $\alpha Q_n - P_n$ would be at least $1/B$ for every n , contradicting (16.17); hence ω is nonresonant. On a phase polydisk of radius R the n -th monomial of F is bounded by $e^{-D_n^2/2} R^{D_n}$ and the distinct integers D_n tend to infinity, so these bounds are summable for every finite R and F is entire. By (16.4) the homological divisor of the n -th monomial is $P_n - \alpha Q_n$, so the absolute coefficient after division exceeds $e^{D_n^2/2}$ by (16.17) and its D_n -th root tends to infinity, excluding every positive analytic radius; the first-order argument of theorem 16.16 then applies. $\square$

Remark 16.18 (The obstruction is one homological inverse, not a long iteration). Both F in (16.18) and its formal homological solution depend only on q_1 and p_2 , and any two functions of those variables have zero Poisson bracket. Therefore in the purely formal phase category $\chi = -\varepsilon_0 L^{-1} F$ removes the perturbation with *no* higher Lie corrections at all. The obstruction is thus not divergence of a long nonlinear iteration; it is already the failure of a single homological inverse to preserve the specified analytic coefficient class. This observation is specific to source 06.

Remark 16.19 (Why polynomial layers escape the same example). Assign the same monomials distinct increasing positive Hahn weights, for instance $\hat{H} = H_0 + \sum_{n \geq 1} t^n e^{-D_n^2/2} q_1^{P_n} p_2^{Q_n}$: then every Hahn layer is a polynomial and theorem 16.7(a) applies, despite the *same* frequency vector, because dividing an individual coefficient by its tiny ordinary divisor still gives an ordinary complex number rather than a failure of a phase Taylor series within that layer. This is no contradiction: $\hat{H}$ is a *different* Hamiltonian, with a different distribution of monomials among the infinitesimal scales, and source 06 records explicitly that “which scale carries which monomial” is an input choice with mathematical consequences and not a physical fact. The divergent formal generator of proposition 16.17 also remains evaluable at infinitesimal phase points — its ordinary complex Taylor coefficients are summed against positive phase supports by lemma 3.4 — so the counterexample does not exclude normalization on an infinitesimal monad; its obstruction is to the ordinary analytic coefficient *germs* required by the global finite-halo theorem. This is the Hamiltonian instance of proposition 6.14.

16.6 Support-and-degree control beyond the finite halo

Assume polynomial coefficients in (16.7), split each nonzero coefficient into its finitely many monomials, and regard a monomial occurrence as an input *label* $\ell = (\gamma, \alpha, \beta)$ with coefficient $c_\ell \neq 0$ and $d_\ell = |\alpha| + |\beta| \geq 3$. The label set is a set; infinitely many labels are allowed. For $\rho \in \Gamma$ put

$$w_\rho(\ell) = \gamma + (d_\ell - 2)\rho, \quad D_\rho = (t^\rho \mathcal{O}_\Gamma)^{2d}. \quad (16.19)$$

Definition 16.20 (Admissible rescaling). The input monomial family is *admissible at* ρ if (1) $w_\rho(\ell) > 0$ for every label; (2) the set $W_\rho = \{w_\rho(\ell)\}$ is well ordered; and (3) each weight in W_ρ comes from only finitely many labels. These are exactly strong summability of the individually rescaled monomials into a positive Hahn series with polynomial coefficients.

The point $\rho = 0$ is always admissible; a *negative* ρ gives a domain containing phase points of *infinite* magnitude. For infinitely many input monomials, condition (1) alone implies neither (2) nor (3) (examples 16.25 and 16.26). The Hamiltonian rescaling operator is

$$\mathcal{S}_\rho F(z) = t^{-2\rho} F(t^\rho z), \quad (16.20)$$

which is a Poisson *Lie* algebra homomorphism wherever its monomial expansion is summable,

$$\{\mathcal{S}_\rho F, \mathcal{S}_\rho G\} = \mathcal{S}_\rho \{F, G\}, \quad (16.21)$$

but is **not** a homomorphism for ordinary products: the factor $t^{-2\rho}$ is exactly what compensates the two phase derivatives in a bracket. It fixes H_0 , preserves resonant monomials, and commutes with L and Π .

Theorem 16.21 (Compatible support-certified domains). *Suppose the polynomial input of theorem 16.7 is admissible at ρ . Then its canonical normalizing map and inverse have strongly summable monomial realizations on D_ρ and are mutually inverse symplectic bijections of that domain, and the Hamiltonian identity and the commuting action identities hold there. If the normalizing construction is performed independently after rescaling at two admissible values, the realized maps agree on the overlap after undoing the scalings: these domains carry one compatible normalization, not a choice of unrelated local conjugacies.*

Proof. Admissibility makes $H_\rho := \mathcal{S}_\rho H = H_0 + \sum_\ell c_\ell t^{w_\rho(\ell)} q^{\alpha_\ell} p^{\beta_\ell}$ a positive Hahn perturbation with polynomial coefficients, so theorem 16.7(a) gives χ_ρ, Φ_ρ on the finite halo, and undoing the phase scaling gives

$$\Psi_\rho(z) = t^\rho \Phi_\rho(t^{-\rho} z) \quad \text{of } D_\rho, \quad (16.22)$$

symplectic in the original bracket because an equal scalar dilation in all phase coordinates is conformally symplectic and the dilation and its inverse cancel. That this is the *same* normalization uses the labeled trees of definition 16.3 with each input coefficient expanded into its monomial labels: every nonzero monomial contribution to a generator or Hamiltonian coefficient with leaves $\ell_1, \dots, \ell_r$ has phase degree

$$2 + \sum_{j=1}^r (d_{\ell_j} - 2), \quad (16.23)$$

and every contribution to a nonlinear coordinate coefficient has phase degree

$$1 + \sum_{j=1}^r (d_{\ell_j} - 2), \quad (16.24)$$

both inductively, since a Poisson bracket subtracts two from the sum of phase degrees while L, L^{-1} and Π preserve degree, the coordinate construction starting at degree one. Under Hamiltonian scaling the exponent of the tree in (16.23) becomes precisely $\sum_j \gamma_{\ell_j} + \rho \sum_j (d_{\ell_j} - 2) = \sum_j w_\rho(\ell_j)$, and for coordinates the same sum is obtained using $t^{-\rho} \Phi(t^\rho z)$. Since W_ρ is positive and well ordered, lemma 3.4 gives finitely many words at a fixed rescaled weight, the finite-fiber hypothesis gives finitely many choices of original labels per word, and there are then finitely many trees: the rescaled generator, Hamiltonian and coordinate expressions are strongly summable *before* any regrouping. Hence (16.21) applies to every finite tree and may then be summed; the rescaled generator is $\mathcal{S}_\rho \chi$, satisfying the same nonresonant gauge and normal-form equation as χ_ρ , and uniqueness identifies them, so on coordinates $\Phi_\rho(z) = t^{-\rho} \Phi(t^\rho z)$. The same argument applies to the inverse exponential

and to the actions, and Taylor evaluation at finite rescaled points is [proposition 3.11](#). For two admissible rescalings both maps arise from the same labeled trees, and at a point of the overlap those tree evaluations are strongly summable by each certificate with every regrouping agreeing coefficientwise — which proves compatibility *without* assuming that rescaling preserves the order of the original Hahn exponents; in general it does not. $\square$

Warning 16.22 (Scope of the extension). [Theorem 16.21](#) extends the *polynomial*-coefficient normalizer, Hamiltonian and constructed actions only. It does **not** license evaluating an arbitrary entire function at an infinite phase or action argument; a centralizer statement on D_ρ must instead be made in the corresponding *rescaled* entire-coefficient algebra, and transporting all original entire coefficients to infinite arguments without a summability certificate would be an unjustified enlargement of the function class. The theorem gives *guaranteed*, not maximal, domains.

Corollary 16.23 (A degree-dependent radius). *Let $H = H_0 + t^\eta P_m$ with $\eta > 0$ and P_m homogeneous of phase degree $m \geq 3$, and write $\chi = \sum_{n \geq 1} t^{n\eta} \chi_n$, $\Phi = z + \sum_{n \geq 1} t^{n\eta} \Phi_n$. Every nonzero χ_n is homogeneous of degree $2 + n(m - 2)$ and every nonzero component of Φ_n has degree $1 + n(m - 2)$, and the normalizer and its inverse are defined on every D_ρ with*

$$\eta + (m - 2)\rho > 0. \quad (16.25)$$

Proof. Every tree at exponent $n\eta$ has exactly n leaves, an ordered abelian group being torsion-free and η the only input weight; apply [\(16.23\)–\(16.24\)](#). The finitely many input monomials all have the same shifted weight $\eta + (m - 2)\rho$, whose positivity supplies all three admissibility conditions. $\square$

16.7 A sharp cubic boundary, and two independent summability failures

Example 16.24 (The cubic boundary, with a coordinate-independent obstruction). In one degree of freedom take

$$H(q, p) = qp + \varepsilon_0 q^2 p, \quad \varepsilon_0 = t^\eta, \quad \eta > 0, \quad (16.26)$$

where there is no arithmetic issue at all because $\omega_1 = 1$. With $\chi = -\varepsilon_0 q^2 p$ one has $\text{ad}_\chi q = \varepsilon_0 q^2$, $\text{ad}_\chi p = -2\varepsilon_0 qp$, and the coordinate exponentials give

$$\Phi(q, p) = \left(\frac{q}{1 - \varepsilon_0 q}, p(1 - \varepsilon_0 q)^2 \right), \quad H \circ \Phi = qp, \quad (16.27)$$

verifiable also by direct rational substitution; its Jacobian determinant is one, so it is symplectic, and $\Pi_\chi = 0$ makes it the canonical normalizer of [theorem 16.7](#). For $\eta + \rho > 0$ the expansion $q/(1 - \varepsilon_0 q) = \sum_{n \geq 0} \varepsilon_0^n q^{n+1}$ is strongly summable on D_ρ ; at the boundary point $q = \varepsilon_0^{-1}$, $p = 0$, every term of that family has exponent $-\eta$ and the rational function has a *pole*. Hence replacing the strict inequality in [\(16.25\)](#) by a non-strict one cannot be valid in general.

There is also a *coordinate-independent* obstruction on the whole boundary polydisk $D_{-\eta}$: from $H_p = q(1 + \varepsilon_0 q)$ and $H_q = p(1 + 2\varepsilon_0 q)$, [\(16.26\)](#) has exactly two critical points there, $(0, 0)$ and $(-\varepsilon_0^{-1}, 0)$, whereas qp has exactly one. An everywhere invertible symplectic change on that domain with $H \circ \Psi = qp$ would give a bijection between these critical-point sets by the chain rule, which is impossible. The boundary failure is therefore not merely a defect in one geometric-series proof.

Example 16.25 (Positive shifted weights, but no well-order). Over $\Gamma = \mathbb{Q}$ consider the valid polynomial-coefficient perturbation $R(q, p) = \sum_{n \geq 2} t^{n+1/n} q^{n+2}$, whose original exponents strictly increase. At $\rho = -1$ the shifted weights are $1/n$, all strictly positive, but forming a *descending* sequence; at the rescaled point $q = 1$ the rescaled perturbation would be $\sum_{n \geq 2} t^{1/n}$, which is not a Hahn series. Positivity alone is therefore insufficient even to *define* the rescaled input.

Example 16.26 (Positive well-ordered weights, but infinite fibers). Take $R(q, p) = \sum_{n \geq 2} t^{n+1} q^{n+2}$. At $\rho = -1$ all shifted weights equal 1: the weight set is positive and well ordered, but infinitely many labels contribute to that one weight. The rescaled expression is $t \sum_{n \geq 2} q^{n+2}$, whose evaluation at the ordinary point $q = 1$ is defined neither by finite coefficient addition nor by ordinary convergence; it cannot supply a polynomial-coefficient Hahn perturbation on the whole rescaled finite halo.

Examples 16.25 and 16.26 show why none of the three conditions of definition 16.20 can be dropped from a general theorem of this form. They do **not** assert that finite fibers are necessary for every possible analytic resummation in some larger, differently specified class.

16.8 A genuinely multiscale worked example

Take $\omega = (1, \sqrt{2})$, $\kappa = 2 - \sqrt{2}$, and $A = q_1^2 p_2$, $B_h = p_1^2 q_2$, $C = I_1 I_2$. For arbitrary positive scales $\eta, \theta \in \Gamma$ put

$$H = I_1 + \sqrt{2} I_2 + u(A + B_h) + vC, \quad u = t^\eta, \quad v = t^\theta, \quad (16.28)$$

nonresonance following from the irrationality of $\sqrt{2}$. The two scales need *not* be Archimedean-comparable; the lexicographic example of warning 3.6 allows $n\eta < \theta$ for every n . Retaining u, v as independent *labels*, and writing $O_{\text{lab}}(3)$ for terms involving at least three input labels — **not** terms above a valuation cutoff — direct homological calculation gives

$$\chi = \frac{u}{\kappa}(B_h - A) + \frac{uv}{\kappa^2}(A - B_h)(2I_2 - I_1) + O_{\text{lab}}(3), \quad (16.29)$$

$$N = I_1 + \sqrt{2} I_2 + v I_1 I_2 + \frac{u^2}{\kappa}(I_1^2 - 4I_1 I_2) + O_{\text{lab}}(3), \quad (16.30)$$

the coefficients of v , u^2 and v^2 in χ vanishing and those of uv and v^2 in the displayed normal form vanishing. The key bracket is $\{A, B_h\} = 4I_1 I_2 - I_1^2$; since $LA = \kappa A$ and $LB_h = -\kappa B_h$ the first generator is $u(B_h - A)/\kappa$, and the second normal term is $\frac{1}{2}\Pi\{u(B_h - A)/\kappa, u(A + B_h)\} = \frac{u^2}{\kappa}(I_1^2 - 4I_1 I_2)$. Also $\{A, C\} = A(2I_2 - I_1)$ and $\{B_h, C\} = -B_h(2I_2 - I_1)$, which gives the mixed generator by $-L^{-1}Q_\Pi$; these identities fix both the sign convention and the placement of the inverse transformation.

To first label degree the coordinate map and actions are

$$\begin{aligned} \Phi(q_1) &= q_1 - \frac{2u}{\kappa} p_1 q_2 + O_{\text{lab}}(2), & \Phi(p_1) &= p_1 - \frac{2u}{\kappa} q_1 p_2 + O_{\text{lab}}(2), \\ \Phi(q_2) &= q_2 + \frac{u}{\kappa} q_1^2 + O_{\text{lab}}(2), & \Phi(p_2) &= p_2 + \frac{u}{\kappa} p_1^2 + O_{\text{lab}}(2), \\ J_1 &= I_1 + \frac{2u}{\kappa}(A + B_h) + O_{\text{lab}}(2), & J_2 &= I_2 - \frac{u}{\kappa}(A + B_h) + O_{\text{lab}}(2), \end{aligned}$$

the full supported series satisfying $H = N(J)$ and commuting exactly, the displayed remainders serving only to show a manageable initial segment. There are two input degrees — the u terms cubic, the v term quartic — so every generator or Hamiltonian coefficient with label $u^a v^b$ is homogeneous of phase degree $2 + a + 2b$ while the corresponding coordinate coefficient has degree $1 + a + 2b$, and a sufficient domain certificate is simply

$$\eta + \rho > 0, \quad \theta + 2\rho > 0. \quad (16.31)$$

The input label set being finite, positivity already supplies well-ordering and finite fibers here. In a higher-rank group, writing these as valuation inequalities rather than numerical radii avoids

introducing a nonexistent real-valued absolute value. If two label weights coincide, for instance $2\eta = \theta$, their terms are added at that Hahn exponent and the construction remains valid; label-degree truncation and Hahn-weight truncation are still different operations, and the verification code deliberately uses the former.

16.9 A discrete one-variable parallel

The same support mechanism is visible in a simpler setting. It is included as a bridge to the established non-Archimedean literature [22], *not* as a claim that polynomial perturbative linearization is new in rank one.

Proposition 16.27 (Polynomial-layer linearization). *Let $\lambda \in \mathbb{C}^\times$ not be a root of unity and let $f(z) = \lambda z + R(z)$ with $R \in \mathcal{H}_\Gamma(\mathcal{P}_1)_{>0}$ and $\text{ord}_0 R_s \geq 2$. There is a unique $h(z) = z + U(z)$ with positive polynomial Hahn coefficients, $\text{ord}_0 U_\gamma \geq 2$, satisfying*

$$f \circ h = h \circ (\lambda \text{ id}), \quad (16.32)$$

it is a bijection of the finite halo, and (16.32) holds there as an actual identity. No bound on $|\lambda^n - \lambda|^{-1}$ is required.

Proof. On z^n with $n \geq 2$ the operator $L_\lambda U = U(\lambda z) - \lambda U(z)$ is multiplication by the nonzero constant $\lambda^n - \lambda$, hence invertible on polynomials of order at least two, and (16.32) becomes $U = L_\lambda^{-1} R(z + U)$. At a fixed Hahn exponent every nonlinear occurrence of U on the right is accompanied by a positive coefficient of R , hence uses strictly smaller exponents of U , so the supported recursion and finite-expression argument of proposition 16.1 apply; each coefficient remains a polynomial, the least-difference argument proves uniqueness, and propositions 3.11 and 16.1 give the actual conjugacy. $\square$

For $f(z) = \lambda z + \varepsilon_0 z^2$, writing $h(z) = \sum_{n \geq 1} b_n \varepsilon_0^{n-1} z^n$ with $b_1 = 1$ gives the exact recurrence

$$(\lambda^n - \lambda) b_n = \sum_{j=1}^{n-1} b_j b_{n-j} \quad (n \geq 2), \quad (16.33)$$

with one phase monomial per Hahn layer; at finite z it is the *support*, not the complex growth of the b_n , that makes this a valid Hahn identity. This is the same quadratic Schröder recursion that sources 02, 03, 04 and 05 each also compute, in four different normalizations; section 22.1 prints it once with the translation between them.

17 Quasi-periodic divisors: a finite flag for a surreal frequency

Everything up to this point has been *germ and fixed-point* dynamics: a map F with a distinguished fixed point, its multiplier, its return iterates, its periods, or a Hamiltonian at the origin of a finite phase halo. Source 07 is the other classical half of small-divisor theory. Its object is a *vector field on a real torus* whose frequency vector has surreal components, its divisors are the infinitely many scalars $k \cdot \Upsilon$ indexed by $k \in \mathbb{Z}^d$, and its equation is the cohomological equation. None of the six germ sources contains a resonance lattice, a cohomological equation, a Fourier divisor or a conjugacy to a constant vector field; conversely nothing in this part concerns a multiplier, a return iterate or a period. What the two halves share is the machinery of section 3 — the set-sized workspace, strong

summability, the positive-support word lemma, halo evaluation — which is therefore not restated here, and the architecture of [remark 2.2](#), which is instantiated a fourth time in [section 18](#).

The headline of this part has no counterpart in the other six. A frequency vector with d surreal components can carry a transfinite hierarchy of scales, and there are infinitely many Fourier divisors; yet only **at most** d distinct divisor valuations are arithmetically visible, and a canonical descending flag of integer lattices identifies them. That finite reduction is what turns an apparently transfinite problem into a finite one, and it is the reason this material occupies three sections rather than a remark.

17.1 The torus, the strip, and the notation of this part

Before anything else, read [warnings 2.9](#) to [2.11](#). They record that this report now carries *two* frequency vectors of different kinds, that κ_Υ below is an element of Γ and not a rate, that Υ is never the exponent Ω of [example 3.7](#) nor the time form Ω_F of [section 15](#), and that the finitely many visible divisor valuations are not a spectrum in any sense this collection uses.

Fix $d \geq 1$ and write $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$. Fix $\rho > 0$ and let

$$\mathbb{T}_\rho^d = \{z \in \mathbb{C}^d / (2\pi\mathbb{Z})^d : |\operatorname{Im} z_j| < \rho \text{ for all } j\}, \quad \mathcal{O}(\mathbb{T}_\rho^d) \quad (17.1)$$

be the *open* complex strip of half-width ρ around the real torus and the algebra of $2\pi\mathbb{Z}^d$ -periodic functions holomorphic on it. Functions in $\mathcal{O}(\mathbb{T}_\rho^d)$ need not be bounded at the boundary of the strip. With $|k| = |k_1| + \cdots + |k_d|$, write

$$f(\theta) = \sum_{k \in \mathbb{Z}^d} \widehat{f}(k) e^{ik \cdot \theta}, \quad \|f\|_r = \sum_{k \in \mathbb{Z}^d} |\widehat{f}(k)| e^{r|k|} \quad (0 < r < \rho). \quad (17.2)$$

A periodic holomorphic function lies in $\mathcal{O}(\mathbb{T}_\rho^d)$ exactly when all these seminorms are finite: boundedness on a narrower closed strip and Cauchy's Fourier estimate give exponential decay there, and inserting an intermediate strip gives the weighted ℓ^1 bound; conversely those bounds give locally uniform convergence of the Fourier series and its termwise derivatives on every narrower strip. Hence $\mathcal{O}(\mathbb{T}_\rho^d)$ is closed under finite sums, products and all orders of differentiation, and Fourier projection onto *any* subset of $\mathbb{Z}^d$ preserves it, since it only deletes terms from $\|f\|_r$.

The torus derivatives $\partial/\partial\theta_j$ act coefficientwise and annihilate K_Γ : they are external derivations, and [warning 3.2](#) applies verbatim — they are not the Berarducci–Mantova derivation, and $\partial_\theta \sin \theta = \cos \theta$ here says nothing about whether a surreal y solves $\partial^2 y + y = 0$ for that internal derivation. Source 07 states this disclaimer independently and in its own words; it is the same disclaimer, printed once.

17.2 The fixed-strip coefficient category

Definition 17.1 (The common-strip Hahn algebra). $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d)) = \mathcal{O}(\mathbb{T}_\rho^d)((t^\Gamma))$ is the Hahn algebra over the coefficient algebra [\(17.1\)](#): elements are $f = \sum_{\beta \in B} t^\beta f_\beta$ with B well ordered and every $f_\beta \in \mathcal{O}(\mathbb{T}_\rho^d)$. Fourier coefficients and the torus mean are taken coefficientwise,

$$\widehat{f}(k) = \sum_{\beta} t^\beta \widehat{f_\beta}(k) \in K_\Gamma, \quad \operatorname{av}(f) = \widehat{f}(0), \quad \mathbf{Q}_0 f = f - \operatorname{av}(f),$$

and $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))_{>0}$ denotes positive support, zero included. For a vector the support is the union of the component supports and the valuation their minimum.

This is the *fixed-domain* row of [section 3.2](#) in its torus instantiation: one and the same ρ serves every Hahn exponent β , and separate analytic germs for separate coefficients do not suffice. The distinction is stated once, in [convention 3.3](#), and is not re-erected here; source 07 imports it from the collection's catalogue explicitly, as a hypothesis of each of its main theorems rather than as a convention. As in the entire row of [section 3.2](#), **no** bound on $\|f_\beta\|_r$ uniform in β is imposed — a deliberate choice, and the reason nothing here specializes t to a nonzero real number ([section 19.7](#)).

Elements of $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ are genuine functions on infinitesimal thickenings: for ordinary $z_0 \in \mathbb{T}_\rho^d$ and $\xi \in \mathfrak{m}_\Gamma^d$, [\(3.4\)](#) applies verbatim and [proposition 3.11](#) makes $f \mapsto f^\#(z_0 + \xi)$ a faithful K_Γ -algebra homomorphism into **SC**. No continuity from an ordinary torus into the fine topology of **No**[i] is claimed.

Lemma 17.2 (Subexponential Fourier multipliers). *Let $m : \mathbb{Z}^d \rightarrow \mathbb{C}$ satisfy $|m(k)| \leq C_\varepsilon e^{\varepsilon|k|}$ for every $\varepsilon > 0$. Then $\widehat{Tf}(k) = m(k)\widehat{f}(k)$ maps $\mathcal{O}(\mathbb{T}_\rho^d)$ into itself, and so does any finite product of such multipliers with polynomial functions of k .*

Proof. For $r < r' < \rho$ apply the estimate with $\varepsilon = r' - r$ to get $\|Tf\|_r \leq C_{r'-r}\|f\|_{r'}$. A fixed polynomial in k is bounded by a constant times $e^{\eta|k|}$ for every $\eta > 0$; split an arbitrary exponent budget among finitely many factors. $\square$

Lemma 17.3 (Taylor substitution on the torus, with its Lipschitz estimate). *Let $h \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d$ have positive valuation. Then*

$$S_h f = f \circ (\text{id} + h) := \sum_{\beta} t^\beta \sum_{\alpha \in \mathbb{N}^d} \frac{\partial^\alpha f_\beta}{\alpha!} h^\alpha \quad (17.3)$$

is well defined in $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$, with $\text{supp}(S_h f) \subseteq \text{supp}(f) + C^$ for $C \supseteq \text{supp } h$ well ordered and positive. Substitution is a ring homomorphism, obeys the chain rule, and is associative for infinitesimal substitutions; these are the torus instances of [proposition 4.1](#) and are not reproved. In addition, for positive-valuation h, g ,*

$$v(S_h f) \geq v(f), \quad v(S_h f - S_g f) \geq v(f) + v(h - g). \quad (17.4)$$

Proof. Well-definedness, the support bound and the algebraic identities are [lemma 3.4](#), [corollary 3.5](#), and [proposition 4.1](#) applied with $U = \mathbb{T}_\rho^d$: at each exponent the word lemma permits only finitely many Taylor monomials and finitely many shifted word pairs in $\mathcal{W}_{\text{supp } f, C}(\gamma)$, and every resulting coefficient is a finite sum of products of derivatives of functions in $\mathcal{O}(\mathbb{T}_\rho^d)$, hence lies in $\mathcal{O}(\mathbb{T}_\rho^d)$. For the second estimate in [\(17.4\)](#) subtract corresponding Taylor monomials and telescope the products: every surviving term carries one factor from $h - g$ and all its other factors have positive valuation, and the multi-index-zero terms cancel. $\square$

[Equation \(17.4\)](#) is the one part of the torus substitution calculus that is not already in [proposition 4.1](#); it is what forces uniqueness in [theorem 19.1](#) and controls the remainder in [theorem 19.4](#).

Warning 17.4 (Two summations, never interchanged). The Fourier series of an ordinary coefficient is an ordinary complex analytic sum; the series in t^Γ is a Hahn sum. Every argument below groups all Fourier modes *inside each analytic coefficient* first. It is nowhere asserted that the unrestricted two-index family of Fourier and Hahn terms is Hahn-summable, and indeed a family with infinitely many nonzero terms at one Hahn exponent violates [definition 3.1](#) however well its complex coefficients converge. This is the torus form of the discipline of [definition 3.1](#).

17.3 The flag, and one support for every reciprocal divisor

Let $\Upsilon \in F_\Gamma^d \setminus \{0\}$ and collect the component Conway/Hahn normal forms into

$$\Upsilon = \sum_{\beta \in S} t^\beta \varpi_\beta, \quad \varpi_\beta \in \mathbb{R}^d \setminus \{0\}, \quad S \text{ well ordered}, \quad (17.5)$$

so the *exact resonance lattice* is

$$\mathbf{N}_\Upsilon = \{k \in \mathbb{Z}^d : k \cdot \Upsilon = 0\} = \bigcap_{\beta \in S} \ker(k \mapsto k \cdot \varpi_\beta). \quad (17.6)$$

This is *exact vanishing of a Hahn scalar*, never smallness; that is the content of the tag QR.

Set $\mathbf{L}_0 = \mathbb{Z}^d$. While $\mathbf{L}_{j-1} \neq \mathbf{N}_\Upsilon$, let γ_j be the *least* exponent of S whose coefficient form does not vanish identically on $\mathbf{L}_{j-1}$, and put

$$\mathbf{a}_j(k) = k \cdot \varpi_{\gamma_j}, \quad \mathbf{L}_j = \{k \in \mathbf{L}_{j-1} : \mathbf{a}_j(k) = 0\}, \quad \Sigma_j = \mathbf{L}_{j-1} \setminus \mathbf{L}_j. \quad (17.7)$$

The construction stops as soon as the current lattice equals $\mathbf{N}_\Upsilon$. The Σ_j are the *nonresonant strata*.

Theorem 17.5 (Finite flag and one common reciprocal support). *The construction (17.7) terminates after*

$$r \leq d \quad (17.8)$$

steps, and:

- (i) *each $\mathbf{L}_j$ is a saturated subgroup of $\mathbb{Z}^d$ whose rank strictly decreases at each step, $\gamma_1 < \dots < \gamma_r$, $\mathbf{L}_r = \mathbf{N}_\Upsilon$, and the strata partition $\mathbb{Z}^d \setminus \mathbf{N}_\Upsilon$;*
- (ii) *for $k \in \Sigma_j$,*

$$k \cdot \Upsilon = t^{\gamma_j} \mathbf{a}_j(k)(1 + \varrho_k), \quad v(\varrho_k) > 0, \quad v(k \cdot \Upsilon) = \gamma_j, \quad (17.9)$$

with $\varrho_k = 0$ permitted; in particular the divisor valuation takes at most d distinct values on $\mathbb{Z}^d \setminus \mathbf{N}_\Upsilon$;

- (iii) *writing*

$$\mathbf{E}_j = \{\beta - \gamma_j : \beta \in S, \beta > \gamma_j\}, \quad \mathbf{E} = \bigcup_{j=1}^r \mathbf{E}_j, \quad \mathbf{T}_\Upsilon = \bigcup_{j=1}^r (-\gamma_j + \mathbf{E}^*), \quad (17.10)$$

the set $\mathbf{E}$ is well ordered and positive, $\mathbf{T}_\Upsilon$ is well ordered, and

$$\text{supp}((k \cdot \Upsilon)^{-1}) \subseteq \mathbf{T}_\Upsilon \quad \text{for every } k \notin \mathbf{N}_\Upsilon. \quad (17.11)$$

With $\kappa_\Upsilon = \gamma_r$ one has $\mathbf{T}_\Upsilon \subseteq [-\kappa_\Upsilon, +\infty)$.

Proof. Each $\mathbf{L}_j$ is a common kernel of real additive homomorphisms on $\mathbb{Z}^d$ and is therefore saturated: if a nonzero integer multiple of k lies in the kernel, so does k . At a nonterminal step $\mathbf{a}_j$ is a nonzero homomorphism on the finitely generated free abelian group $\mathbf{L}_{j-1}$ with torsion-free nonzero image, so $\text{rank } \mathbf{L}_j < \text{rank } \mathbf{L}_{j-1}$; at most d such decreases occur, which is (17.8).

By the minimal choice of γ_j , every coefficient form below γ_j vanishes on $\mathbf{L}_{j-1}$, and the form at γ_j vanishes on $\mathbf{L}_j$; hence the next selected exponent is strictly larger. Termination occurs exactly when every coefficient form vanishes on the current lattice, i.e. at $\mathbf{N}_\Upsilon$. The same observation identifies the leading term for $k \in \Sigma_j$: explicitly

$$\varrho_k = \sum_{\substack{\beta \in S \\ \beta > \gamma_j}} \frac{k \cdot \varpi_\beta}{\mathbf{a}_j(k)} t^{\beta - \gamma_j}, \quad (17.12)$$

whose support lies in E_j and is positive. This is (i) and (ii).

A tail of a well-ordered set translated by a fixed element is well ordered, and a finite union of well-ordered subsets of a linear order is well ordered; this gives E . [Lemma 3.4](#) then gives E^* and T_Y . Finally

$$\frac{1}{k \cdot Y} = \frac{t^{-\gamma_j}}{a_j(k)} \sum_{n \geq 0} (-\varrho_k)^n \quad (k \in \Sigma_j) \quad (17.13)$$

is a Hahn identity by [lemma 3.4](#) — not a limit of partial sums ([definition 3.1](#) and [warning 3.6](#)) — with the asserted support. Every element of E^* is nonnegative and $\gamma_j \leq \kappa_Y$, giving the lower bound. $\square$

Remark 17.6 (r counts visible scales, not monomials). The number r counts *arithmetically visible* scales, not the monomials of Y . Exponents of S whose forms vanish on the current lattice are skipped, possibly across a long transfinite interval. The theorem does **not** say that only d monomials of Y matter to the eventual solution: every later coefficient can still move higher Hahn coefficients of the inverse, through (17.12). What is finite is the list of divisor valuations and the list of real forms on which all small-divisor arithmetic is tested.

Proposition 17.7 (Coefficient structure of the inverse). *Fix $\eta \in \Gamma$ and a stratum Σ_j . On that stratum the function $k \mapsto [t^\eta](k \cdot Y)^{-1}$ is a finite sum of expressions*

$$c \frac{\prod_{\ell=1}^n (k \cdot \varpi_{\beta_\ell})}{a_j(k)^{n+1}}, \quad \sum_{\ell=1}^n (\beta_\ell - \gamma_j) = \eta + \gamma_j, \quad \beta_\ell > \gamma_j, \quad (17.14)$$

with c real and empty products allowed at $n = 0$. The number of terms and the denominator powers depend on η but are finite for each fixed η .

Proof. Expand (17.13) and extract the coefficient. All letters lie in E_j , and [lemma 3.4](#) permits finitely many words of all lengths at a fixed weight; $\beta \mapsto \beta - \gamma_j$ is injective, so the coefficient choices are also finitely many. $\square$

Remark 17.8 (A common support is not summability over the modes). The family $((k \cdot Y)^{-1})_{k \notin N_Y}$ is in general **not** a Hahn-summable family: infinitely many leading terms may occupy the exponent $-\gamma_j$. [Theorem 17.5](#) supplies a *common support*, not a Hahn sum over k , exactly as [definition 3.1](#) requires. [Proposition 17.7](#) is what permits ordinary Fourier summation at each Hahn coefficient; that is the additional analytic step, and it is where arithmetic enters.

Proposition 17.9 (An arithmetic-free valuation bound). *Let $C_Y = Y_1 \partial_{\theta_1} + \cdots + Y_d \partial_{\theta_d}$. If $h \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ has no Fourier modes in N_Y and $C_Y h = g$, then $v(h) \geq v(g) - \kappa_Y$. No small-divisor hypothesis is used.*

Proof. For $h \neq 0$ put $\beta = v(h)$. The nonzero h_β has a nonzero Fourier coefficient at some $k \notin N_Y$, so $v(\widehat{h}(k)) = \beta$ and $v(\widehat{g}(k)) = \beta + v(k \cdot Y) \leq \beta + \kappa_Y$ by (17.9). Every Fourier coefficient of g has valuation at least $v(g)$; rearrange. The zero case is immediate. $\square$

18 The fixed-strip solvability threshold, and an arbitrary-order lifting obstruction

18.1 The stratified condition, and the category table instantiated a second time

Definition 18.1 (Stratified subexponential condition SD_{ss}). Υ satisfies (SD_{ss}) if for each flag stratum Σ_j and every real $\varepsilon > 0$ there is a real $C_{j,\varepsilon} > 0$ with

$$\frac{1}{|\mathfrak{a}_j(k)|} \leq C_{j,\varepsilon} e^{\varepsilon|k|} \quad (k \in \Sigma_j), \quad (18.1)$$

equivalently $\limsup_{|k| \rightarrow \infty, k \in \Sigma_j} \log^+(1/|\mathfrak{a}_j(k)|)/|k| = 0$.

Remark 18.2 (The one invariant, a fourth divisor family, and two instantiations of one table). [Remark 2.2](#) defines one quantity — the exponential rate of the reciprocals of a graded family of nonzero ordinary complex divisors — and evaluates it on three families. (SD_{ss}) is that same quantity on a *fourth* family, evaluated stratum by stratum:

Divisor family	Grading	Rate
$\lambda^n - \lambda$ (drifted multiplier)	n	σ (SD_{dr})
$\lambda^\alpha - \lambda_j$ (exact multiplier)	$ \alpha $	τ (SD_{ex})
$\omega \cdot (\alpha - \beta)$ (ordinary frequency)	$ \alpha + \beta $	$\log \Theta$ (SD_{fr})
$\mathfrak{a}_j(k) = k \cdot \varpi_{\gamma_j}$ (leading real forms of Υ , on Σ_j)	$ k $	0 required on every j (SD_{ss})

So (SD_{ss}) is the vanishing-rate condition, imposed on each of the finitely many strata; it is the exact analogue of $\tau = 0$ and of $\sigma = 0$, and it is *not* the analogue of $\tau < \infty$. That is consistent with the category table of [remark 2.2](#), which is stated there once and is instantiated twice in this report:

Row of remark 2.2	Germ instantiation, divisors $\lambda^\alpha - \lambda_j$ or $\lambda^q - 1$	Quasi-periodic instantiation, divisors $k \cdot \Upsilon$
one fixed unchanged domain, rate 0	fixed polydisk $\mathcal{O}(\mathbb{D}_R^d)((t^\Gamma))$, $\tau = 0$ (row one of theorem 6.1)	fixed strip $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$, (SD_{ss}) (theorem 18.3)
polynomial / formal coefficients, no arithmetic	theorem 6.1 rows four and five; SD_0	not established here: source 07 proves no polynomial or formal row for the torus, and none is claimed
radius-free germs / entire coefficients, rate finite	theorem 6.1 rows two and three; theorem 16.5	not established here; the fixed strip is the only category source 07 classifies
common-domain germs, the sixth category (second table of remark 2.2)	$\tau < \infty$, radius $Re^{-\tau}$, for λI_d (theorem 7.13); $\tau < \infty$, radius at least $Re^{-r_{\text{ang}}\tau}$, for nonscalar Λ (theorem 7.23); $\sigma = 0$ under drift (theorem 8.5)	not established here

Only the first row is instantiated on both sides. Reading across the other two rows, in either direction, would assert theorems no source proves.

18.2 Universal solvability on a fixed strip

Let $\Pi_{\mathbf{N}_\Upsilon}$ denote coefficientwise Fourier projection onto $\mathbf{N}_\Upsilon$; it preserves $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ because arbitrary Fourier projection preserves $\mathcal{O}(\mathbb{T}_\rho^d)$.

Theorem 18.3 (Universal fixed-strip solvability). *For a nonzero $\Upsilon \in F_\Gamma^d$ the following are equivalent.*

- (i) Υ satisfies (SD_{ss}).
- (ii) For every $\rho > 0$ and every $f \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ with $\Pi_{\mathbf{N}_\Upsilon} f = 0$ there is a unique $u \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ with $\mathcal{C}_\Upsilon u = f$ and $\Pi_{\mathbf{N}_\Upsilon} u = 0$.

When these hold, the solution operator is

$$\widehat{\mathcal{G}_\Upsilon f}(k) = \begin{cases} \widehat{f}(k)/(i k \cdot \Upsilon), & k \notin \mathbf{N}_\Upsilon, \\ 0, & k \in \mathbf{N}_\Upsilon, \end{cases} \quad (18.2)$$

and it satisfies

$$\text{supp}(\mathcal{G}_\Upsilon f) \subseteq \text{supp}(f) + \mathbf{T}_\Upsilon, \quad v(\mathcal{G}_\Upsilon f) \geq v(f) - \kappa_\Upsilon. \quad (18.3)$$

Without the normalization the kernel consists exactly of the elements of $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ with Fourier support inside $\mathbf{N}_\Upsilon$. If (SD_{ss}) fails, there is already an ordinary real analytic mean-compatible forcing, supported at a single Hahn exponent and on one stratum, for which a forced coefficient of any putative solution is not a distribution on the real torus.

Sufficiency. For $\eta \in \mathbf{T}_\Upsilon$ define the scalar multiplier $m_\eta(k) = [t^\eta](i k \cdot \Upsilon)^{-1}$ for $k \notin \mathbf{N}_\Upsilon$ and $m_\eta(k) = 0$ on $\mathbf{N}_\Upsilon$. On each stratum [proposition 17.7](#) writes m_η as a finite sum of expressions with polynomial numerator in k and a finite power of $\mathbf{a}_j(k)$ in the denominator; (SD_{ss}) and [lemma 17.2](#) make each a subexponential multiplier. There are finitely many strata — this is where (17.8) is used — and projection onto a stratum does not affect the estimate, so m_η defines $G_\eta : \mathcal{O}(\mathbb{T}_\rho^d) \rightarrow \mathcal{O}(\mathbb{T}_\rho^d)$.

Write $f = \sum_{\beta \in B} t^\beta f_\beta$ and put

$$u_\gamma = \sum_{\substack{\beta \in B, \eta \in \mathbf{T}_\Upsilon \\ \beta + \eta = \gamma}} G_\eta f_\beta. \quad (18.4)$$

By [lemma 3.4](#) this sum is finite at each γ and $B + \mathbf{T}_\Upsilon$ is well ordered, so $u = \sum_\gamma t^\gamma u_\gamma$ lies in $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$: the common domain and the common support are *proved*, not inferred from the modewise formula. Fourier extraction gives (18.2), and multiplying modewise by $i k \cdot \Upsilon$ gives the equation. Uniqueness and the kernel description follow from the same modewise calculation, K_Γ being a field. The support estimate is the construction and $\mathbf{T}_\Upsilon \geq -\kappa_\Upsilon$ gives the valuation estimate. $\square$

Necessity. Suppose (SD_{ss}) fails on Σ_j . Choose $\varepsilon > 0$ and modes $k_n \in \Sigma_j$, no two equal up to sign, with $|k_n| \rightarrow \infty$ and $|\mathbf{a}_j(k_n)|^{-1} > e^{2\varepsilon|k_n|}$; this is possible after decreasing ε if necessary. The real function

$$f(\theta) = 2 \sum_{n \geq 1} e^{-\varepsilon|k_n|} \cos(k_n \cdot \theta) \quad (18.5)$$

lies in $\mathcal{O}(\mathbb{T}_\varepsilon^d)$, has no resonant modes and sits at Hahn exponent zero. Any solution must have $\widehat{u}(k_n) = e^{-\varepsilon|k_n|}/(i k_n \cdot \Upsilon)$, so by (17.9) the coefficient of $t^{-\gamma_j}$ has Fourier coefficients of modulus greater than $e^{\varepsilon|k_n|}$. Fourier coefficients of a distribution on a compact torus grow at most polynomially — apply its finite-order continuity estimate to the Fourier characters — so that forced coefficient is not a distribution, and in particular not holomorphic on any strip of positive width. Adding resonant modes cannot alter these nonresonant coefficients. $\square$

Corollary 18.4 (The divisor loss κ_Υ is attained). *For any $k \in \Sigma_r$ and any $s \in \Gamma$ the solution for $f = t^s e^{ik \cdot \theta}$ has valuation $s - \kappa_\Upsilon$. Hence the valuation loss in (18.3) cannot be improved uniformly.*

Proof. Its unique nonzero Fourier mode is divided by a scalar of valuation κ_Υ . $\square$

Remark 18.5 (A stratified Diophantine sufficient condition). A convenient sufficient hypothesis is a stratified Diophantine bound $|\mathbf{a}_j(k)| \geq c_j(1 + |k|)^{-\tau_j}$ on Σ_j . A term of length n in (17.14) then has a multiplier growing at most polynomially with exponent $n + (n + 1)\tau_j$, so every individual coefficient operator has explicit finite loss estimates between narrower-strip seminorms. It is **not** asserted that those losses stay bounded as η varies. The fixed-strip theorem is available precisely because the category demands a common open domain and not a single bounded analytic norm for every coefficient — the same mechanism as in [warning 10.3](#).

Corollary 18.6 (Higher tails do not change the arithmetic test). *If $v(\tilde{\Upsilon} - \Upsilon) > \kappa_\Upsilon$ then $\tilde{\Upsilon}$ has the same flag, the same leading real forms, and — whenever $\mathbf{N}_\Upsilon = \{0\}$ — the same exact resonance lattice; in that nonresonant case $\tilde{\Upsilon}$ satisfies (SD_{ss}) exactly when Υ does.*

Proof. No coefficient at or below κ_Υ changes. Each nonzero integer vector already has a nonvanishing leading divisor coefficient at one of those levels, so no resonance can appear and its leading form is unchanged. $\square$

The nonresonance qualification is not decoration: when $\mathbf{N}_\Upsilon \neq \{0\}$ a new higher tail can split some exact resonances and introduce additional flag levels, and the criterion must then be applied to the extended flag.

18.3 Arbitrarily long analytic jets with no analytic lift

The next construction is a failure that no valuation-only argument can detect, and it lives entirely at integer exponents: the obstruction is not caused by exotic uncountable supports.

Define integers and a real number by

$$b_1 = 2, \quad b_{j+1} = 10^{2b_j}, \quad \mathfrak{l} = \sum_{j \geq 1} 10^{-b_j}, \quad (18.6)$$

and put $Q_j = 10^{b_j}$, $P_j = Q_j \sum_{\ell \leq j} 10^{-b_\ell}$, $\mathfrak{d}_j = Q_j \mathfrak{l} - P_j$, $k_j = (-P_j, Q_j)$, $K_j = |k_j|$. Here P_j is an integer with $0 < P_j < Q_j$ and

$$0 < \mathfrak{d}_j \leq 2Q_j 10^{-Q_j^2}, \quad Q_j \leq K_j < 2Q_j, \quad (18.7)$$

by bounding the tail of the rapidly increasing sequence of decimal places; the same estimate proves $\mathfrak{l}$ irrational, since a fixed denominator q would force $|\mathfrak{d}_j| \geq 1/q$. Consider the infinitesimally detuned frequency and operator

$$\Upsilon = (1, \mathfrak{l} + t), \quad \mathcal{C}_\Upsilon = \partial_x + (\mathfrak{l} + t)\partial_y = L_0 + t\partial_y, \quad L_0 = \partial_x + \mathfrak{l}\partial_y. \quad (18.8)$$

The real leading vector $(1, \mathfrak{l})$ is already nonresonant, so there is one flag level, at $\gamma_1 = 0$, and (SD_{ss}) fails very strongly.

Theorem 18.7 (Arbitrary finite analytic lifting, but no full lift). *For every integer $N \geq 1$ and every $\rho > 0$ there is an ordinary real entire periodic f_N such that*

- (i) $\mathcal{C}_\Upsilon u = f_N$ has a mean-zero solution modulo t^{N+1} with coefficients $u_0, \dots, u_N \in \mathcal{O}(\mathbb{T}_\rho^d)$;
- (ii) $u_0, \dots, u_{N-1}$ are entire periodic functions;
- (iii) the forced coefficient u_{N+1} is not a distribution on the real torus, so there is no solution in $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_{\rho'}^2))$ for any $\rho' > 0$, and none after an order-preserving extension of the value group.

One may take

$$f_N(x, y) = 2 \sum_{j \geq 1} \mathfrak{d}_j^{N+1} e^{-\rho K_j} \cos(k_j \cdot (x, y)). \quad (18.9)$$

Proof. By (18.7) the Fourier coefficients in (18.9) decay faster than e^{-RK_j} for every fixed $R > 0$, so f_N is entire, real and of zero mean. At the mode k_j the divisor is $\mathfrak{d}_j + Q_j t$; expanding its inverse as a Hahn geometric series (legitimate by lemma 3.4) gives

$$\widehat{u}_n(k_j) = \frac{(-1)^n}{i} Q_j^n \mathfrak{d}_j^{N-n} e^{-\rho K_j} \quad (n \geq 0), \quad (18.10)$$

with complex conjugates at $-k_j$ and no other nonzero modes; set every mean to zero. For $n < N$ the remaining positive power of $\mathfrak{d}_j$ gives superexponential decay, so u_n is entire. For $n = N$ the coefficients are the polynomial factor Q_j^N times $e^{-\rho K_j}$, whose weighted Fourier sums are finite on every strip of width $r < \rho$, so $u_N \in \mathcal{O}(\mathbb{T}_\rho^d)$. The scalar identities give $L_0 u_0 = f_N$ and $L_0 u_n = -\partial_y u_{n-1}$ for $1 \leq n \leq N$, which is the finite-jet statement in $\mathcal{O}(\mathbb{T}_\rho^d)[[t]]/(t^{N+1})$. At the next coefficient

$$|\widehat{u_{N+1}}(k_j)| = Q_j^{N+1} \mathfrak{d}_j^{-1} e^{-\rho K_j} \geq \frac{1}{2} Q_j^N 10^{Q_j^2} e^{-2\rho Q_j},$$

which exceeds every polynomial in K_j along the sequence and so is not a distributional Fourier coefficient sequence.

For the value-group clause: at each nonzero Fourier mode the scalar equation has the *unique* solution $\widehat{f}_N(k)/(i k \cdot \Upsilon)$ in the Hahn field, and the inverse of that divisor is the same geometric series in any larger ordered-group Hahn field. Therefore the coefficient of t^{N+1} remains exactly (18.10), and admitting negative or noninteger Hahn exponents in a proposed solution changes nothing; extra constants affect only the zero Fourier mode. $\square$

Warning 18.8 (The robustness under value-group extension is argued, not assumed). This report's standing hazard is the temptation to assert that some norm or estimate is invariant under enlarging Γ . Theorem 18.7(iii) does **not** do that. It cites no invariance of any norm across a value-group extension; it argues from *uniqueness of the modewise Hahn inverse* in the larger field, which is an algebraic statement about a field, together with the fact that the same geometric expansion computes it. Source 07 states the argument in exactly that form, and the corresponding invariance in the positive direction is proposition 4.14, which is likewise proved rather than assumed.

This is a finite-versus-infinite lifting obstruction in a precise analytic ring. It does not say that the scalar algebraic inverses fail to exist — they all exist — but that infinitely many modewise inverses do not assemble into even one analytic coefficient at a specified order. The same operator (18.8) works for every N ; only the entire forcing changes. In particular **no test consisting of a fixed finite number of analytic coefficient equations can replace the arithmetic hypothesis** in a general lifting theorem. The construction adapts the classical Liouville mechanism [20], which is not claimed as new; what is offered is the arbitrarily prescribed order at which it first bites.

19 Torus normal form: conjugacy to a constant vector field

We pass from the scalar equation to vector fields. Multiplying the whole field by a scalar monomial normalizes

$$\gamma_1 = 0, \quad 0 \leq \kappa_\Upsilon = \gamma_r, \quad (19.1)$$

which subtracts γ_1 from both the frequency and the perturbation valuations, so the eventual condition $v(\mathcal{V}) > \kappa_\Upsilon$ is unchanged when translated back. Throughout this section Γ is nontrivial, $\mathbf{N}_\Upsilon = \{0\}$ (tag $\neg\text{QR}$: no exact resonances) and (SD_{ss}) holds. The operator $\mathcal{G}_\Upsilon$ then inverts $\mathcal{C}_\Upsilon$ on all mean-zero elements and is applied componentwise. Let

$$\mathcal{X}(\theta) = \Upsilon + \mathcal{V}(\theta), \quad \mathcal{V} \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d, \quad s_0 = v(\mathcal{V}) > \kappa_\Upsilon. \quad (19.2)$$

Theorem 19.1 (Hahn-analytic constant normal form). *Under the preceding assumptions there is a unique pair*

$$h \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d, \quad \mathbf{c} \in K_\Gamma^d, \quad v(h) > 0, \quad \text{av}(h) = 0,$$

such that, with $\Phi_h = \text{id} + h$,

$$D\Phi_h(\theta) (\Upsilon + \mathbf{c}) = \mathcal{X}(\Phi_h(\theta)). \quad (19.3)$$

Moreover:

- (i) $v(h) \geq s_0 - \kappa_\Upsilon$ and $v(\mathbf{c}) \geq s_0$;
- (ii) with $B = \text{supp } \mathcal{V}$ and

$$P_0 = \bigcup_{j=1}^r (B - \gamma_j), \quad P = E \cup P_0, \quad M = P^*, \quad (19.4)$$

- these are positive generators in a well-ordered set, and $\text{supp } h \subseteq M \setminus \{0\}$, $\text{supp } \mathbf{c} \subseteq B + M$;
- (iii) Φ_h has an infinitesimal Taylor inverse on $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$, and all nonzero integer divisors of $\Upsilon + \mathbf{c}$ have the same leading real coefficients and the same valuations as those of Υ ;
- (iv) if $\mathcal{V}$ has real values coefficientwise on the real torus, then h and $\mathbf{c}$ are real in the same sense, with $\mathbf{c} \in F_\Gamma^d$.

The case $\mathcal{V} = 0$ is included, with $h = \mathbf{c} = 0$.

19.1 Reduction to a fixed-point equation

Taking the torus mean of (19.3) and using $\text{av}(Dh) = 0$ gives

$$\mathbf{c} = \mathbf{c}(h) := \text{av}(\mathcal{V} \circ (\text{id} + h)), \quad (19.5)$$

and the remaining equation becomes

$$h = \mathcal{T}(h) := \mathcal{G}_\Upsilon \left(Q_0(\mathcal{V} \circ (\text{id} + h)) - Dh \mathbf{c}(h) \right). \quad (19.6)$$

The argument of $\mathcal{G}_\Upsilon$ has zero mean, since $\mathbf{c}(h)$ is constant and every column of Dh has zero mean, so the inverse is used only where it is defined.

All terms of $B - \gamma_j$ are positive because $s_0 > \kappa_\Upsilon \geq \gamma_j$. The sets in (19.4) are well ordered and lemma 3.4 applies. If h has positive support in M then lemma 17.3 gives $\text{supp}(\mathcal{V} \circ (\text{id} + h)) \subseteq B + M$ and the same for $\mathbf{c}(h)$; $Dh \mathbf{c}(h)$ adds M once more; and applying an inverse coefficient at $-\gamma_j + E^*$ shifts these supports into $B - \gamma_j + M \subseteq M$. Every resulting term contains at least one letter from P_0 , so no constant exponent appears. This proves preservation of a support certificate. Existence still requires proof that infinitely many iteration steps assemble into a Hahn series.

19.2 Existence by marked support words

Existence in theorem 19.1. Put $h^{(0)} = 0$ and $h^{(n+1)} = \mathcal{T}(h^{(n)})$. Each iterate lies in $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d$, has zero mean and positive support in $\mathbf{M}$, its coefficients obtained by finitely many analytic operations at each exponent.

Call a generator in P_0 a *marked* letter and a generator in $\mathbf{E}$ *unmarked*; an exponent lying in both keeps both labels. There are only finitely many origins for a given letter — j ranges over finitely many values and the translation determines the original exponent of B or S — so the finite-word conclusion of lemma 3.4 is unchanged by the labels.

A term of $\mathcal{T}(h)$ consists of a coefficient of $\mathcal{V}$, an inverse coefficient, and finitely many factors of h or its derivatives. The combined exponent of the $\mathcal{V}$ coefficient and the leading shift $-\gamma_j$ of that inverse supplies one marked letter $\beta - \gamma_j$; the higher inverse terms supply unmarked letters in $\mathbf{E}$; all other letters come from the factors of h . By induction, the difference

$$h^{(n+1)} - h^{(n)} \tag{19.7}$$

can have a nonzero coefficient only at an exponent representable by a word with at least $n + 1$ *marked* letters. For $n = 0$ there is at least one $\mathcal{V}$ factor. For the induction step, subtract the two Taylor expressions and telescope the finite products at each coefficient: every difference term contains a factor from $h^{(n)} - h^{(n-1)}$ or its derivative together with one additional $\mathcal{V}$ coefficient. The term $Dh \mathfrak{c}(h)$ has the same property, by writing its difference as $D(h - g)\mathfrak{c}(h) + Dg(\mathfrak{c}(h) - \mathfrak{c}(g))$ and expanding the second difference by the same telescoping. Applying $\mathcal{G}_\Gamma$ creates one new marked shift and unmarked inverse-tail letters, exactly as above.

At a fixed exponent γ , lemma 3.4 permits only finitely many labelled word representations, so their numbers of marked letters are bounded; hence (19.7) has zero coefficient at γ for all large n . The coefficients $[t^\gamma]h^{(n)}$ therefore stabilize *exactly*, not approximately. The union of the difference supports lies in the well-ordered $\mathbf{M}$ and each exponent occurs in finitely many differences, so the differences form a Hahn-summable family in the sense of definition 3.1; let h be their sum. Each stabilized coefficient lies in $\mathcal{O}(\mathbb{T}_\rho^d)$; no new analytic limit has been taken. At a fixed exponent $\mathcal{T}(h)$ uses only finitely many certified word decompositions and finitely many stabilized input coefficients, so passing to the stabilized coefficients gives $h = \mathcal{T}(h)$. Define $\mathfrak{c}$ by (19.5); (19.6) and (19.5) give (19.3).

Since substitution does not lower valuation, $v(\mathfrak{c}) \geq s_0$; the right side inside $\mathcal{G}_\Gamma$ has valuation at least s_0 because Dh has positive valuation; and proposition 17.9 or (18.3) gives $v(h) \geq s_0 - \kappa_\Gamma$. The support statements are the construction. $\square$

Warning 19.2 (The word argument is the existence step, and the obvious substitute is invalid). The weaker estimate $v(h^{(n+1)} - h^{(n)}) \geq (n + 1)(s_0 - \kappa_\Gamma)$ does **not** suffice in a general ordered group: that sequence of lower bounds need not be cofinal. With $\Gamma = \mathbb{Z} + \mathbb{Z}\Omega$ and $v(h) = 1$ the bounds never pass Ω , so a sequential valuation argument would prove nothing at or beyond that exponent, and example 19.9 exhibits a solution with coefficients exactly there. No Banach fixed-point theorem and no sequential completeness of $\mathbf{No}[i]$ is used. This is the same refusal as warning 3.6, reached independently by source 07 and implemented by a *labelled* refinement of the word certificate — counting marked letters — which is the one mechanism in this report that corollary 3.5 does not already supply.

19.3 Uniqueness and the inverse coordinate map

Uniqueness in [theorem 19.1](#). Let h, g be positive-valuation mean-zero solutions with $h \neq g$ and put $\delta = v(h - g) > 0$; their drifts are forced by (19.5). By (17.4), $v(\mathbf{c}(h) - \mathbf{c}(g)) \geq s_0 + \delta$, and subtracting the expressions inside (19.6) gives valuation at least $s_0 + \delta$, since $\mathbf{c}(h)$ has valuation at least s_0 and differentiation does not lower valuation. Hence $v(\mathcal{T}(h) - \mathcal{T}(g)) \geq s_0 + \delta - \kappa_\Upsilon > \delta$, contradicting $h - g = \mathcal{T}(h) - \mathcal{T}(g)$. Uniqueness holds among *all* positive-valuation mean-zero solutions, not only those carrying the constructed support certificate. $\square$

Lemma 19.3 (Infinitesimal coordinate inverses on the torus). *Every $\Phi_h = \text{id} + h$ with $v(h) > 0$ has a unique inverse $\text{id} + g$ of positive valuation in the Taylor sense, inducing an automorphism of $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$.*

Proof. Solve $g = -h \circ (\text{id} + g)$ by iteration from zero: the same marked-word stabilization, now marking each occurrence of a coefficient of h , gives coefficient stabilization inside the positive monoid generated by $\text{supp } h$. Uniqueness follows from $v(h \circ (\text{id} + g_1) - h \circ (\text{id} + g_2)) \geq v(h) + v(g_1 - g_2) > v(g_1 - g_2)$, which is (17.4). For two-sidedness write $\Psi \circ \Phi_h = \text{id} + q$; associativity gives $\Phi_h \circ (\text{id} + q) = \Phi_h$, hence $q + h \circ (\text{id} + q) - h = 0$, and if $q \neq 0$ the second difference has valuation strictly greater than $v(q)$, which is impossible. This is the torus counterpart of [lemma 4.2](#) and [proposition 4.1](#), proved here by the labelled word certificate rather than by the triangular recursion, because the unknown is a full Hahn-analytic coefficient family rather than a Taylor coefficient list. $\square$

Completion of [theorem 19.1](#). [Lemma 19.3](#) gives the inverse assertion. Since $v(\mathbf{c}) \geq s_0 > \kappa_\Upsilon$, [corollary 18.6](#) applies and no leading coefficient of a nonzero integer divisor changes; for complex data the same leading real coefficients persist although some higher coefficients can be complex. Complex conjugation together with $k \mapsto -k$ preserves all scalar Fourier inversions for a real frequency, and Taylor substitution, mean and differentiation preserve coefficientwise reality, so real $\mathcal{V}$ gives real $h, \mathbf{c}$; alternatively reality follows from uniqueness. $\square$

19.4 Necessity, and the sharp scale boundary

Theorem 19.4 (Non-linearizable perturbations beyond any prescribed valuation). *Assume $\gamma_1 = 0$, $\mathbf{N}_\Upsilon = \{0\}$ and Γ nontrivial. If (SD_{ss}) fails, then for every prescribed $H \in \Gamma$ there are $s > H$, a real $\rho > 0$ and a real perturbation $\mathcal{V} \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d$ with $v(\mathcal{V}) = s$ such that $\Upsilon + \mathcal{V}$ admits no positive-valuation mean-zero Hahn-analytic conjugacy to a constant vector field, even on a narrower positive-width strip. One may require $s > 2\kappa_\Upsilon$ and take $\mathcal{V}$ with a single Hahn exponent.*

Proof. Choose $s > \max\{H, 2\kappa_\Upsilon\}$, possible in a nontrivial ordered abelian group. Take the real f of (18.5) on a failing stratum Σ_j and put $\mathcal{V} = t^s f e_1$. Suppose $\Phi_h = \text{id} + h$ were such a conjugacy on some strip; restrict to a common narrower strip if necessary. Taking means still forces $\mathbf{c} = \text{av}(\mathcal{V} \circ \Phi_h)$ with $v(\mathbf{c}) \geq s$, and since $v(h) > 0$ the equation $\mathcal{C}_\Upsilon h = \mathbf{Q}_0(\mathcal{V} \circ \Phi_h) - D h \mathbf{c}$ has right side of valuation at least s , so [proposition 17.9](#) gives $v(h) \geq s - \kappa_\Upsilon$. Subtracting the uncomposed forcing, write $\mathcal{C}_\Upsilon h = t^s f e_1 + R$ with $R = \mathbf{Q}_0(\mathcal{V} \circ \Phi_h - \mathcal{V}) - D h \mathbf{c}$; Taylor telescoping (17.4) gives $v(R) \geq s + v(h) \geq 2s - \kappa_\Upsilon$. At every nonzero Fourier mode the scalar equation therefore gives

$$\hat{h}(k) = \frac{t^s \hat{f}(k) e_1}{i k \cdot \Upsilon} + \frac{\hat{R}(k)}{i k \cdot \Upsilon}, \quad v\left(\frac{\hat{R}(k)}{i k \cdot \Upsilon}\right) \geq 2s - 2\kappa_\Upsilon > s.$$

For $k = k_n$ in the failing stratum the coefficient of $t^{s-\gamma_j}$ in the first component of h is therefore forced to be $e^{-\varepsilon|k_n|}/(i\mathbf{a}_j(k_n))$ — here $s - \gamma_j \leq s$, so the remainder cannot contribute — and those coefficients grow faster than $e^{\varepsilon|k_n|}$ and cannot belong to a distribution, exactly as in the necessity proof of [theorem 18.3](#). $\square$

Corollary 19.5 (Exact universal infinitesimal normal-form criterion). *For normalized Υ with $N_\Upsilon = \{0\}$ over nontrivial Γ , (SD_{ss}) is equivalent to the assertion that for every $\rho > 0$ every $\mathcal{V} \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d$ of valuation greater than κ_Υ admits the normalized conjugacy of [theorem 19.1](#). When (SD_{ss}) fails, restricting perturbations to arbitrarily higher valuations does **not** rescue a universal theorem.*

Proposition 19.6 (Failure exactly at the threshold, with no zero of the slow speed). *Let $\kappa_\Upsilon > 0$ lie in Γ and take*

$$\Upsilon = (1, t^{\kappa_\Upsilon}), \quad \mathcal{V}(\theta_1, \theta_2) = (0, \tfrac{1}{2}t^{\kappa_\Upsilon} \cos \theta_2).$$

Then Υ satisfies (SD_{ss}) and its last flag level is κ_Υ , yet this perturbation of valuation exactly κ_Υ admits no positive-valuation mean-zero conjugacy to a constant vector field.

Proof. The two strata have leading forms k_1 and k_2 , so (SD_{ss}) is immediate. Suppose a conjugacy existed. The mean equation forces $\mathbf{c}_1 = 0$; since substitution by a positive-valuation h does not change the coefficient at exponent κ_Υ of $\mathcal{V}$, that coefficient is $(0, \frac{1}{2} \cos \theta_2)$, of mean zero, so $v(\mathbf{c}_2) > \kappa_\Upsilon$ (with $\mathbf{c}_2 = 0$ allowed). At exponent κ_Υ in the second component of [\(19.3\)](#), $t^{\kappa_\Upsilon} \partial_2 h_2$ contributes nothing because $[t^0]h_2 = 0$, and neither do the drift or $Dh\mathbf{c}$. What remains is $\partial_1([t^{\kappa_\Upsilon}]h_2) = \frac{1}{2} \cos \theta_2$; averaging in θ_1 makes the left side zero and leaves the right side unchanged. $\square$

The slow component of this field is $t^{\kappa_\Upsilon}(1 + \frac{1}{2} \cos \theta_2)$, whose leading real factor is everywhere positive: the example does **not** rely on introducing a stationary point. It says that a perturbation of the *same* size as a slow frequency may require a coordinate correction of order one rather than an infinitesimal one. The strict threshold $v(\mathcal{V}) > \kappa_\Upsilon$ is therefore sharp *as a uniform sufficient bound*; particular perturbations at the boundary can of course still be linearizable.

19.5 A canonical invariant density

The claim here concerns coefficientwise integration of analytic functions on the torus. It is **not** a countably additive **No**-valued probability measure on arbitrary subsets, and not a real-time ergodic theorem; [warning 15.1](#) and [warning 16.15](#) record the same refusal elsewhere in this report. For what the strong and the coefficientwise notions of Hahn countable additivity give here, see `meas:rem:dynamics` in `docs/surreal/hahn-valued-measures-and-probability/`.

Lemma 19.7 (Formal change of variables). *For $\Phi_h = \text{id} + h$ with $v(h) > 0$ and every $f \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$,*

$$\text{av}((f \circ \Phi_h) \det D\Phi_h) = \text{av}(f). \quad (19.8)$$

Proof. Introduce an ordinary formal auxiliary variable z and $\Phi_z = \text{id} + zh$. At each Hahn exponent Taylor substitution and the determinant produce a *polynomial* in z , there being finitely many support words at that exponent ([lemma 3.4](#)). With $A = D\Phi_z$, the finite determinant identities and the chain rule give

$$\frac{d}{dz}((f \circ \Phi_z) \det A) = \text{div}_\theta((f \circ \Phi_z) \text{adj}(A)h) : \quad (19.9)$$

use $A \operatorname{adj}(A) = (\det A)I$ for the derivative of f , the derivative-of-determinant formula for the Dh terms, and the elementary Piola identity — the remaining cofactor derivatives cancel in pairs because mixed partials of Φ_z commute. Each coefficient on the right of (19.9) is a divergence of a periodic analytic vector field, hence of mean zero, so the mean on the left is independent of z ; evaluate the coefficient polynomials at $z = 0$ and $z = 1$. There is no infinite integration in z and no ordinary convergence of a Hahn sum is assumed. $\square$

Theorem 19.8 (Invariant density and its uniqueness). *Let $\mathcal{X}$, Φ_h and $\mathfrak{c}$ be as in theorem 19.1 and put $\Psi = \Phi_h^{-1}$. Then*

$$\mu_{\mathcal{X}} = \det D\Psi \in 1 + \mathcal{H}_{\Gamma}(\mathcal{O}(\mathbb{T}_{\rho}^d))_{>0} \quad (19.10)$$

is the unique density in $\mathcal{H}_{\Gamma}(\mathcal{O}(\mathbb{T}_{\rho}^d))$ with $\operatorname{div}(\mu_{\mathcal{X}}\mathcal{X}) = 0$ and $\operatorname{av}(\mu_{\mathcal{X}}) = 1$, and the K_{Γ} -linear functional

$$I_{\mathcal{X}}(f) = \operatorname{av}(f \circ \Phi_h) = \operatorname{av}(f\mu_{\mathcal{X}}) \quad (19.11)$$

satisfies $I_{\mathcal{X}}(1) = 1$ and $I_{\mathcal{X}}(\mathcal{X} \cdot \nabla f) = 0$. For real data $\mu_{\mathcal{X}}$ is real, with positive surcomplex values on the real torus, being one plus an infinitesimal.

Proof. The inverse is infinitesimal, so its Jacobian determinant is one plus an element of positive valuation, and the chain rule gives $(\mu_{\mathcal{X}} \circ \Phi_h) \det D\Phi_h = 1$. Applying lemma 19.7 to $f\mu_{\mathcal{X}}$ gives the equality in (19.11), and $f = 1$ gives the normalization. Conjugacy and the chain rule yield $(\mathcal{X} \cdot \nabla f) \circ \Phi_h = (\Upsilon + \mathfrak{c}) \cdot \nabla(f \circ \Phi_h)$, whose mean is zero; coefficientwise integration by parts against every Fourier character gives $\operatorname{div}(\mu_{\mathcal{X}}\mathcal{X}) = 0$. For uniqueness let $\tilde{\mu}$ be a normalized solution and set $g = (\tilde{\mu} \circ \Phi_h) \det D\Phi_h$; the same Jacobian identities transport its divergence equation to $(\Upsilon + \mathfrak{c}) \cdot \nabla g = 0$, and since $\Upsilon + \mathfrak{c}$ has no nonzero integer resonances (theorem 19.1(iii)), Fourier uniqueness makes g constant in θ . Then (19.8) gives $\operatorname{av}(g) = \operatorname{av}(\tilde{\mu}) = 1$, so $g = 1$ and $\tilde{\mu} = \mu_{\mathcal{X}}$. Reality and pointwise positivity follow from theorem 19.1(iv) and the positive-valuation form of (19.10). $\square$

The identity $I_{\mathcal{X}}(\mathcal{X} \cdot \nabla f) = 0$ is the precise invariance assertion. No real-time flow taking values continuously in the fine topology of $\mathbf{No}[i]$ is constructed, and no ergodic theorem for such a flow is claimed.

19.6 Worked examples with genuinely different scales

Example 19.9 (Three visible scales and a reciprocal support of order type ω^2). Take $\Gamma = \mathbb{Z} + \mathbb{Z}\Omega$ with Ω greater than every ordinary positive integer — the same use of the symbol Ω as in examples 3.7 and 13.3, an element of Γ and not a frequency (warning 2.9) — and

$$\Upsilon = (1, t, t^{\Omega}). \quad (19.12)$$

The flag is $\mathbb{Z}^3 \supset \{k_1 = 0\} \supset \{k_1 = k_2 = 0\} \supset \{0\}$ with levels $0, 1, \Omega$ and leading forms k_1, k_2, k_3 on their strata. Each nonzero leading form is a nonzero integer, so its reciprocal has modulus at most one and $(\mathbf{SD}_{\text{ss}})$ holds with no delicate ordinary arithmetic at all. Nevertheless the mode $k = (1, 1, 1)$ already has

$$\frac{1}{1+t+t^{\Omega}} = \sum_{n,m \geq 0} (-1)^{n+m} \binom{n+m}{n} t^{n\Omega+m}, \quad (19.13)$$

whose support has order type ω^2 — the ordinal ω , not a frequency — since every exponent of the $n = 0$ layer precedes every exponent of the $n = 1$ layer, and so on. Formula (19.13) follows

by grouping words according to their numbers of letters 1 and Ω , and is confirmed by the finite recurrence $C_{n,m} + C_{n-1,m} + C_{n,m-1} = \delta_{(n,m),(0,0)}$ for $C_{n,m} = (-1)^{n+m} \binom{n+m}{n}$ (negative index meaning zero), which the verification program checks.

On the second stratum $(k_2 t + k_3 t^\Omega)^{-1} = \frac{t^{-1}}{k_2} \sum_{n \geq 0} (-k_3/k_2)^n t^{n(\Omega-1)}$, and on the last stratum the inverse is simply $t^{-\Omega}/k_3$: a large support coexists with a three-level divisor classification. The forcing $f = t^{\Omega+1}(\sin \theta_1 + \sin \theta_2 + \sin \theta_3)$ has exact mean-zero solution

$$u = -t^{\Omega+1} \cos \theta_1 - t^\Omega \cos \theta_2 - t \cos \theta_3, \quad (19.14)$$

of valuation $1 = v(f) - \Omega$, so the last scale really does control the worst inverse loss ([corollary 18.4](#)). A vector perturbation of valuation $\Omega + 1$ falls under [theorem 19.1](#) with a coordinate correction of valuation at least one. This is also the example that exposes the invalidity of a sequential contraction proof ([warning 19.2](#)): corrections of order $1, 2, 3, \dots$ never pass Ω , while the coefficient-stabilization proof constructs the exact transformation including its coefficients at and beyond that scale.

Example 19.10 (Real arithmetic on the leading stratum, and an arbitrary tail). Take $\Upsilon = (1, \sqrt{2}, t)$. Its first stratum is all (p, q, r) with $(p, q) \neq (0, 0)$, with leading form $p + \sqrt{2}q$; its second is $(0, 0, r)$ with $r \neq 0$, at level one. Since $|p + \sqrt{2}q|^{-1} = |p - \sqrt{2}q|/|p^2 - 2q^2| \leq |p| + \sqrt{2}|q|$ for $(p, q) \neq (0, 0)$, the first stratum is Diophantine and the second has the integer leading form r ; the universal conjugacy threshold is $v(\mathcal{V}) > 1$. One may now append an arbitrary set-supported real vector tail of valuation greater than one: it can have transfinite support and change every higher coefficient of the solution, yet by [corollary 18.6](#) it changes neither the arithmetic test nor the threshold. This separates ordinary approximation properties from surreal support complexity.

Example 19.11 (An explicit slow-circle normal form). Let $s > \kappa_\Upsilon > 0$ and consider $\mathcal{X}(\theta_1, \theta_2) = (1, t^{\kappa_\Upsilon} + t^s \sin \theta_2)$, and put $\varrho = t^{s-\kappa_\Upsilon}$. The mean-zero conjugacy has the form $\Phi_h = (\theta_1, \theta_2 + h(\theta_2))$ and carries $\mathcal{X}$ to $(1, \nu)$, where

$$(1 + h'(\theta))\nu = t^{\kappa_\Upsilon} + t^s \sin(\theta + h(\theta)), \quad (19.15)$$

and the new frequency is exactly

$$\nu = t^{\kappa_\Upsilon} \sqrt{1 - \varrho^2} = t^{\kappa_\Upsilon} \left(1 - \frac{1}{2}\varrho^2 - \frac{1}{8}\varrho^4 - \frac{1}{16}\varrho^6 - \dots\right), \quad (19.16)$$

the square root being the Hahn binomial branch with constant term one. To see this, use the inverse coordinate in the second variable: the inverse equation gives $\Psi'(\theta) = \nu t^{-\kappa_\Upsilon} (1 + \varrho \sin \theta)^{-1}$, whose mean is one, so $t^{\kappa_\Upsilon}/\nu = \text{av}((1 + \varrho \sin \theta)^{-1}) = \sum_{m \geq 0} \binom{2m}{m} \varrho^{2m} 4^{-m} = (1 - \varrho^2)^{-1/2}$, using the geometric expansion, the vanishing of odd sine moments and $\text{av}(\sin^{2m} \theta) = \binom{2m}{m} 4^{-m}$. All of these are Hahn identities in the positive monomial ϱ . The first terms of the unique mean-zero h are

$$h(\theta) = -\varrho \cos \theta - \frac{\varrho^2}{4} \sin 2\theta + \varrho^3 \left(\frac{1}{12} \cos 3\theta - \frac{1}{4} \cos \theta \right) + O_H(\varrho^4), \quad (19.17)$$

where $O_H(\varrho^4)$ means that the remaining integer powers of ϱ have exponent at least four and is **not** an assertion about convergence in a real parameter. In particular the lower bound $v(h) \geq s - \kappa_\Upsilon$ of [theorem 19.1\(i\)](#) is attained, and the first frequency drift occurs later, at valuation $2s - \kappa_\Upsilon$. The invariant density of [theorem 19.8](#) is equally explicit,

$$\mu_{\mathcal{X}}(\theta_2) = \frac{\sqrt{1 - \varrho^2}}{1 + \varrho \sin \theta_2}, \quad (19.18)$$

with mean one, making $\mu_{\mathcal{X}} \cdot (t^{\kappa_\Upsilon} + t^s \sin \theta_2)$ independent of θ_2 . The verification program reconstructs h and ν independently by exact Fourier arithmetic and checks (19.15) through degree ten in ϱ .

19.7 What this part does not reach

Three delimitations belong here rather than only in [section 26](#), because they bound the reading of every statement above.

First, all of it is a *coefficientwise Hahn* theory. A formal series $\sum_{n \geq 0} t^n h_n(\theta)$ lies in $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ whenever every h_n is holomorphic on the same strip, *however fast* $\|h_n\|_r$ grows — faster than every exponential in n is permitted — so substituting a nonzero real number for t need not be meaningful. Nothing here improves classical real analytic KAM or Brjuno arithmetic: those theorems must control Fourier decay and the accumulation of losses through infinitely many nonlinear steps simultaneously [19, 21], whereas (SD_{ss}) only guarantees that each *individual* coefficient operation stays holomorphic, the Hahn support argument handling how the coefficients are assembled. This is the torus instance of **N55**, **N56** and [warning 26.1](#).

Second, the torus derivatives are external derivations annihilating K_Γ ([warning 3.2](#)); nothing above speaks to internal surreal differential equations, and source 07 says so in order to keep itself separate from the companion report on differential equations ([section 23.1](#)).

Third, the nonlinear theorem requires $\mathbf{N}_\Upsilon = \{0\}$. The linear theorem allows $\mathbf{N}_\Upsilon \neq \{0\}$, but the resonant nonlinear case is open, and source 07 warns explicitly that replacing the mean projection by $\Pi_{\mathbf{N}_\Upsilon}$ is *not* justified, since the retained normal form is then no longer a constant vector field and creates additional terms in the conjugacy equation ([question 25.9](#)). The flag itself is finite but not automatically computable: detecting exact integer relations among arbitrary named reals may be unavailable, so existence of the flag is separated from decidability of its entries ([question 25.11](#)).

20 Finite-order multipliers and resonant splitting

Nonresonance fails for multipliers of finite order. There is nevertheless an exact normal form retaining precisely the resonant dynamics, with *no* small-divisor estimates anywhere (tag SD₀). Note that the hypothesis $\Lambda^q = I$ here is the multiplier *itself* of finite order, which is neither RES_q — where λ is explicitly not a root of unity — nor mere MR ([warning 2.1](#)).

Theorem 20.1 (Same-domain finite-order decomposition). *Let Λ be diagonal unitary with $\Lambda^q = I$ for an integer $q \geq 1$, and let $F = \Lambda z + f$ with $f \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{D}_R^d))_+^d$ vanishing coefficientwise to order at least two. There exist a normalized positive coordinate change H and a positive vector field Y vanishing to order at least two, both on the same $\mathbb{D}_R^d$, with*

$$H \circ F \circ H^{-1} = \Lambda \circ \Phi_Y, \quad Y(\Lambda z) = \Lambda Y(z), \quad (20.1)$$

the rotation and flow in (20.1) commuting. Every monomial $z^\alpha e_j$ of Y satisfies $\lambda^\alpha = \lambda_j$. All coefficients have finite support-word certificates from $\text{supp } f$, and the construction commutes with workspace enlargement. Moreover

$$F^{\circ q} = \text{id} \iff Y = 0 \iff F \text{ is positively conjugate to } \Lambda. \quad (20.2)$$

The same statement holds for radius-free germs, entire coefficients, polynomials and formal coefficients.

Proof. The reduction of $P = F^{\circ q}$ is the identity, it fixes zero and has exact Jacobian I there, so $X = \log C_P$ is a positive vector field whose coefficients vanish to order at least two ([theorem 4.3](#)). Since F commutes with P , its pullback commutes with $\log C_P$, so F preserves X and commutes

with all its admissible ordinary flow times. Set $A_f = F \circ \Phi_{-X/q}$: the factors commute, so $A_f^{\circ q} = P \circ \Phi_{-X} = \text{id}$, and its ordinary reduction is Λ . Define the finite average

$$H(z) = \frac{1}{q} \sum_{j=0}^{q-1} \Lambda^{-j} A_f^{\circ j}(z). \quad (20.3)$$

The leading term of every summand after multiplication by Λ^{-j} is z , so $H = \text{id} + \text{positive}$ with value zero and derivative I at zero, with a positive inverse on the same coefficient domain; reindexing the finite sum and using $A_f^q = \text{id}$, $\Lambda^q = I$ gives $H \circ A_f = \Lambda H$. With $(H_*X)(w) = dH(H^{-1}(w))a(H^{-1}(w))$ for $X = a \cdot \partial$, Taylor substitution shows H_*X is positive of the same analytic category and that $H \circ \Phi_X \circ H^{-1} = \Phi_{H_*X}$, either by the coefficientwise chain rule or by conjugating pullback operators. Put $Y = \frac{1}{q} H_*X$: since A_f preserves X , the conjugated map Λ preserves Y , which is exactly $Y(\Lambda z) = \Lambda Y(z)$ and gives the monomial resonance condition; and $F = A_f \circ \Phi_{X/q}$ gives (20.1) with its commuting factors. The operations used are finite compositions with the ordinary rotation, positive logarithms and exponentials, substitution inverses and the finite sum (20.3), each preserving the original coefficient domain and the generated support monoid. Finally $P = \text{id}$ iff $X = 0$ iff $Y = 0$; if $Y = 0$ the displayed normal form is a conjugacy to Λ , and conversely such a conjugacy forces $F^q = \text{id}$. $\square$

In one variable with λ of exact order q , the resonant vector field has the form $Y(z) = \sum_{n \geq 2, n \equiv 1 \pmod{q}} a_n z^n \partial_z$, interpreted coefficientwise when the a_n are themselves Hahn series; the coefficient *family*, rather than an unrestricted rearrangement in n , remains the defining object. For $q = 1$ the theorem reduces to [theorem 4.3](#).

Warning 20.2 (Credit and non-uniqueness). The finite-factor/flow splitting follows the classical semisimple–unipotent pattern [6], and neither the finite averaging idea nor that abstract pattern is claimed new; what is established here is that the entire construction is valid for arbitrary positive Hahn support and retains the same ordinary coefficient domain. Uniqueness of resonant normal forms is **not** claimed: further coordinate changes commuting with Λ can transform Y , and formula (20.3) specifies the particular normalized coordinate used. For resonant multipliers of *infinite* order no universal analytic Poincaré–Dulac statement is proved; see [question 25.3](#).

21 Critical scales: four different things that fail

Four sources each exhibit a scale at which something stops working, and the four failures are *not* the same failure. Collecting them is what prevents a reader from concluding that the report has one scale obstruction with four proofs.

Proposition 21.1 (Quadratic scaling and the boundary of the halo theorem). *For every non-root-of-unity $\lambda \in \mathbb{C}^*$ and every $\eta > 0$, the map $F(z) = \lambda z + t^\eta z^2$ has a normalized conjugacy with polynomial Hahn coefficients*

$$H(z) = z + \sum_{n \geq 1} b_n t^{\eta n} z^{n+1}, \quad (21.1)$$

which evaluates on every finite halo irrespective of the growth of the b_n . For every nonzero ordinary w , substituting $z = t^{-\eta} w$ into the individual terms of (21.1) is not a strongly summable family. If λ is a non-Brjuno irrational rotation, the ordinary series $w + \sum_{n \geq 1} b_n w^{n+1}$ has zero radius of convergence as well.

Proof. Let $L_\lambda(w) = w + \sum_{n \geq 1} b_n w^{n+1}$ be the unique formal solution of $L_\lambda(\lambda w + w^2) = \lambda L_\lambda(w)$, whose coefficients nonresonance gives recursively; substituting $w = t^\eta z$ and dividing by t^η gives exactly (21.1) with $H \circ F = \lambda H$. Each Hahn coefficient is polynomial, the support is a subset of $\{n\eta : n \geq 1\}$, and evaluation at a finite point is justified by lemma 3.4 with no condition on the ordinary magnitudes. Infinitely many b_n are nonzero: otherwise L_λ would be a nonconstant polynomial and the degrees of $L_\lambda \circ Q_\lambda$ and λL_λ would be $2 \deg L_\lambda$ and $\deg L_\lambda$. At the critical scale every nonzero summand becomes $b_n t^{n\eta} (t^{-\eta} w)^{n+1} = t^{-\eta} b_n w^{n+1}$, so infinitely many members share the Hahn exponent $-\eta$, violating definition 3.1: this is a failure of **local finiteness**, not of a real-valued norm estimate. Finally, if L_λ had a positive ordinary radius it would be a local holomorphic coordinate with derivative one, analytically linearizing Q_λ , contradicting Brjuno–Yoccoz for a non-Brjuno rotation. $\square$

Even for a Brjuno rotation, ordinary convergence of the coefficient series would not by itself make the repeated-weight family a strong Hahn sum: it would be an additional ordinary analytic summation convention. The non-Brjuno example rules out even that local analytic reinterpretation of the formal linearizer. No assertion is made about sectorial or resurgent summation under additional hypotheses.

Proposition 21.2 (A precise rescaled-germ obstruction). *For the same family, $H_t(z) = t^{-1} h_\lambda(tz)$ with $h_\lambda(w) = w + \sum_{n \geq 2} b_n w^n$ the ordinary normalized formal linearizer of $p_\lambda(w) = \lambda w + w^2$, the change of variable H_t and its inverse are well defined by Hahn sums on the full set*

$$\{z \in K_\Gamma : v(tz) > 0\} \quad (21.2)$$

and give a conjugacy there. Rescaling to the critical scale $z = t^{-1}w$ gives $tH_t(t^{-1}w) = h_\lambda(w)$, and this formal germ is the Taylor germ of a common-domain Hahn-holomorphic section at $w = 0$ if and only if $\text{rad}(h_\lambda) > 0$.

Proof. $H_t(z) = t^{-1} h_\lambda(tz)$ is coefficientwise algebra, and $tF_t(z) = p_\lambda(tz)$ then gives $H_t \circ F_t = \lambda H_t$ by formal substitution. At any argument with $tz \in \mathfrak{m}_\Gamma$ the ordinary formal series $h_\lambda(tz)$ is Hahn summable — scalar coefficients of valuation zero, powers of a positive-support argument, and lemma 3.4 — as is its ordinary formal inverse, and both preserve the infinitesimal monad because their linear term is the identity and higher terms have strictly higher valuation; multiplying by t^{-1} preserves (21.2). For the last assertion, suppose a section $A(w) = \sum_{\gamma \geq 0} t^\gamma a_\gamma(w)$ with all a_γ holomorphic on one $\mathbb{D}_r$ has Taylor series $h_\lambda(w)$ in $K_\Gamma[[w]]$: the coefficient at Hahn exponent zero has ordinary Taylor series h_λ , which must therefore converge at least on $\mathbb{D}_r$, all other coefficient germs having zero Taylor series. Conversely, if h_λ converges on a disk, the section with only exponent-zero coefficient h_λ is the desired one. $\square$

For $t = \omega^{-1}$ the domain (21.2) includes every finite surcomplex number and many infinite ones, and excludes exactly the critical scale where tz has nonzero ordinary standard part. This is consistent with the familiar non-Archimedean linearization phenomenon over a trivially valued coefficient field, and is not presented as a replacement for the ultrametric results of Lindahl [22]. Even when h_λ converges ordinarily, substituting a nonzero ordinary w directly into the displayed Hahn family is *not* a Hahn summation: infinitely many nonzero terms would occupy exponent zero. One must first use the separately justified ordinary holomorphic germ and then its canonical Taylor lift, and this is where an illegitimate exchange of ordinary with Hahn summation would occur.

Warning 21.3 (The four critical-scale phenomena are distinct). (1) Proposition 21.1 (source 03): at the critical scale *local finiteness* fails — infinitely many terms occupy one exponent — and this is what stops the finite-halo theorems from being all-scale theorems.

- (2) [Theorem 14.22](#) (source 04): the sharp disk $v(z) > \delta/q$ has **two** distinct failure mechanisms, local finiteness *at* equality and *well-ordering* strictly below ((14.28)), and the boundary nevertheless carries a legitimate coefficientwise continuation ([section 14.7](#)) — so here the scale is where one summation stops and a *different* legal operation begins.
- (3) [Proposition 21.2](#) (source 02): the failure is at the level of *representability*, an exact iff between positive ordinary radius of the rescaled germ and existence of a common-domain section, plus the observation that a coefficientwise-entire section need not be globally analytic on all of $\mathbf{No}[i]$ under every spatial rescaling.
- (4) [Example 16.24](#) (source 06): the boundary polydisk carries *two* independent obstructions — a pole of an explicit rational normalizer, and a coordinate-independent *critical-point count* — so the failure is not a defect of one geometric-series proof, and [examples 16.25](#) and [16.26](#) separate well-ordering from finite fibers as independent requirements.

Sources 02 and 05 each add the further remark that the existence of a Hahn conjugacy does *not* imply ordinary holomorphic dependence on t near $t = 0$: for a divergent h_λ the formal dependence encoded in (21.1) need not converge as an ordinary two-variable series in (t, z) ; and source 05's [warning 15.27](#) gives the positive counterpart, one licensed passage to the scale ε^{-1} .

22 Worked examples and exact coefficient calculations

22.1 The quadratic Schröder recursion, once, in six normalizations

Four of sources 01–07 compute the coefficients of the normalized formal linearizer of an ordinary quadratic, and sources 08 and 09 compute them again (below). They are one calculation; the translation between their normalizations is the following.

For the ordinary equation $L_\lambda(\lambda w + w^2) = \lambda L_\lambda(w)$ with $L_\lambda(w) = w + \sum_{n \geq 1} b_n w^{n+1}$, the first coefficients are

$$\begin{aligned} b_1 &= -\frac{1}{\lambda(\lambda-1)}, & b_2 &= \frac{2}{\lambda(\lambda-1)^2(\lambda+1)}, \\ b_3 &= -\frac{5\lambda^2+1}{\lambda^2(\lambda-1)^3(\lambda+1)(\lambda^2+\lambda+1)}, \\ b_4 &= \frac{2(7\lambda^3+2\lambda+3)}{\lambda(\lambda-1)^4(\lambda+1)^2(\lambda^2+1)(\lambda^2+\lambda+1)}, \end{aligned} \tag{22.1}$$

obtained by equating successive powers of w , the denominators at stage n arising from $\lambda^{n+1} - \lambda$. In the shifted indexing $h_\lambda(w) = w + \sum_{n \geq 2} \beta_n w^n$, the same recursion reads

$$\beta_1 = 1, \quad \beta_n = -\frac{1}{\lambda^n - \lambda} \sum_{m=\lceil n/2 \rceil}^{n-1} \beta_m \binom{m}{n-m} \lambda^{2m-n} \quad (n \geq 2), \tag{22.2}$$

the binomial being the coefficient of w^n in $(\lambda w + w^2)^m$, with $\beta_2 = -1/(\lambda^2 - \lambda)$ and $\beta_3 = 2\lambda/((\lambda^2 - \lambda)(\lambda^3 - \lambda))$; in the normalization $h(z) = \sum_{n \geq 1} b_n \varepsilon_0^{n-1} z^n$ of [section 16.9](#) it reads (16.33). Multiplying $b_n z^{n+1}$ by $t^{n\eta}$ turns these ordinary formal data into the strongly summable surcomplex linearizer of (21.1), *even when the ordinary series has zero radius* — which is the whole point of [proposition 21.1](#). None of these is a convergence assertion: they are coefficient identities.

Sources 08 and 09 compute the same recursion twice more. Source 08 works in this report's

convention, with $F = \lambda z + t^\eta z^2$ and $H = z + \sum_m t^{m\eta} u_m$, and finds

$$u_1 = -\frac{z^2}{\lambda^2 - \lambda}, \quad u_2 = \frac{2\lambda z^3}{(\lambda^2 - \lambda)(\lambda^3 - \lambda)}, \quad u_3 = -\frac{(5\lambda^3 + \lambda)z^4}{(\lambda^2 - \lambda)(\lambda^3 - \lambda)(\lambda^4 - \lambda)},$$

which is $u_m = b_m z^{m+1}$ for the first three b_m of (22.1). Source 09 works in the forward convention of [convention 7.1\(iii\)](#). Its coordinate $\tilde{H} = z + \sum_{m \geq 1} c_m t^{m\eta} z^{m+1}$ is the inverse of (21.1), so its c_m are the coefficients of the compositional inverse of L_λ , with $c_0 = 1$ and $c_m = \sum_{r+s=m-1} c_r c_s / (\lambda^{m+1} - \lambda)$. Thus $c_1 = 1/(\lambda^2 - \lambda) = -b_1$, $c_2 = 2c_1/(\lambda^3 - \lambda)$ and $c_3 = (2c_2 + c_1^2)/(\lambda^4 - \lambda)$. Neither source asserts any estimate for the growth of these scalars in m . Source 09 adds that at this input every tree vertex has weight one. The fixed-complexity factor of [theorem 7.5](#) then becomes n^n , and no ordinary convergence follows ([remark 7.7](#)).

22.2 The iterative logarithm of a quadratic perturbation

Take $\Gamma = \mathbb{Q}$, write $t = t^1$, and consider $F(z) = z + tz^2$ with entire polynomial coefficients. Directly from (4.4),

$$X_F = \left(tz^2 - t^2 z^3 + \frac{3}{2} t^3 z^4 - \frac{8}{3} t^4 z^5 + \frac{31}{6} t^5 z^6 - \frac{157}{15} t^6 z^7 + O(t^7) \right) \partial_z, \quad (22.3)$$

where $O(t^7)$ refers only to this single-parameter formal expansion. Exponentiating half of this vector field produces the unique positive half-iterate of F , with polynomial coefficients at every Hahn weight, and the ordinary size and growth of the rational coefficients in (22.3) have no bearing on strong summability in t . For a general map, with a bookkeeping parameter ϱ and $F_\varrho = z + \varrho u$, expanding the logarithm through degree three gives

$$X_{F_\varrho} z = \varrho u - \frac{\varrho^2}{2} (Du)u + \varrho^3 \left(\frac{1}{12} D^2 u(u, u) + \frac{1}{3} (Du)^2 u \right) + O(\varrho^4), \quad (22.4)$$

since $\Delta^2 z = \varrho^2 (Du)u + \frac{1}{2} \varrho^3 D^2 u(u, u) + O(\varrho^4)$ and $\Delta^3 z = \varrho^3 (D^2 u(u, u) + (Du)^2 u) + O(\varrho^4)$; with $u = z^2$ this gives the first three terms of (22.3). [Equation \(22.3\)](#) is (15.31) with $\varepsilon \mapsto t$ and $p = 1$; it is the same calculation, recorded once in each of the two notations its sources use and identified here ([warning 15.22](#)).

22.3 A complete centralizer and a persistent double fixed point

For $X = tz^2 \partial_z$ the flow has the especially simple expression

$$\Phi_c(z) = \frac{z}{1 - ctz} = z + \sum_{n \geq 1} (ct)^n z^{n+1}, \quad c = 0 \text{ or } v(c) > -1, \quad (22.5)$$

the last series being a strongly summable expansion on finite halos. The time-one map here is $z/(1 - tz)$, *not* the quadratic map of [section 22.2](#). By [theorem 4.12](#) the full positive centralizer of this time-one map on the entire-coefficient halo consists exactly of (22.5), and for every nonzero admissible time $\Phi_c(z) - z = ctz^2/(1 - ctz)$, whose factor $ct/(1 - ctz)$ is a unit in the Hahn coefficient algebra. Hence the fixed-point ideal is *exactly* (z^2) for every such time: the point at zero retains its scheme-theoretic double structure, not merely its location — which is what [theorem 4.10](#) buys over a zero-set statement ([remark 4.11](#)).

22.4 Two independent perturbation scales, and a resonant splitting

Let t and u represent t^1 and t^Ω in $\Gamma = \mathbb{Q} + \mathbb{Q}\Omega$, and consider the map with multiplier -1 , $F(z) = -z + tz^2 + uz^3$, illustrating [theorem 20.1](#) with $q = 2$. The averaging coordinate and the resonant vector field begin

$$\begin{aligned} H(z) &= z - \frac{t}{2}z^2 + \frac{t^2}{2}z^3 - \frac{5tu}{4}z^4 + O((t, u)^3), \\ Y(z) &= \left(-uz^3 - t^2z^3 - \frac{3}{2}u^2z^5 - \frac{25}{4}t^2uz^5 - \frac{7}{2}u^3z^7 + O((t, u)^4)\right)\partial_z, \end{aligned} \quad (22.6)$$

and $HF = (-\Phi_Y)H$. Every displayed monomial of Y has odd coordinate degree, exactly as $Y(-z) = -Y(z)$ requires.

Warning 22.1 ($O((t, u)^N)$ is a label cutoff, not a valuation cutoff). In [\(22.6\)](#), $O((t, u)^N)$ means discarded monomials of *total parameter degree* at least N in an auxiliary two-parameter jet algebra. It is **not** a valuation cutoff: for example t^{100} has much smaller Hahn valuation than u , although its total parameter degree is larger. The same caution applies to $O_{\text{lab}}(\cdot)$ in [section 16.8](#) and to *every* verification program in [section 24](#), all of which truncate by parameter or label degree. Sources 03 and 06 each state this independently, and source 06 adds the sharpest form: if $\theta > n\eta$ for every n , infinitely many powers of u lie below the first v layer, and no finite label truncation pretends to compute that entire valuation initial segment.

The verification program of source 03 also uses the two-dimensional identity-reducing map $G(x, y) = (x + tx^2 + uy, y + txy + ux)$, checking at total parameter depth three the logarithm, the inverse, the exponential, the Leibniz identity and the fixed-point factorization [\(4.11\)](#). That example includes infinitesimal *linear* terms, which are permitted in the identity-reducing flow theorem [theorem 4.3](#) even though they are excluded from the exact-multiplier linearization theorem [theorem 6.1](#) — an instance of the tag distinction of [section 2.3](#) showing up in a computation.

22.5 An arbitrary-support example beyond a finite grid

Use S and Γ from [example 3.7](#) and let $F = \Lambda z + \sum_{\sigma \in S} t^\sigma f_\sigma(z)$ with each f_σ vanishing to order at least two. Under a valid analytic hypothesis from [theorem 6.1](#), the higher-rank coefficients obey

$$\begin{aligned} h_\Omega &= -\mathcal{L}^{-1}f_\Omega, \\ h_{\Omega+1} &= \mathcal{L}^{-1}\left(D(\mathcal{L}^{-1}f_\Omega)(\Lambda z)f_1(z) + D(\mathcal{L}^{-1}f_1)(\Lambda z)f_\Omega(z)\right). \end{aligned} \quad (22.7)$$

Only the word (Ω) contributes in the first line, and only $(\Omega, 1)$ and $(1, \Omega)$ can contribute in the second. Infinitely many input coefficients at rational weights below Ω enter neither formula. This is a direct use of finite word dependence ([proposition 6.2](#)), not a rank-one approximation to the value group.

23 Relation to the companion reports in this collection

Each entry names the report by title and repository path. Companion reports are cited this way and never by their internal labels.

23.1 Extends: differential equations over the surreals

Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers (docs/surcomplex/differential-equations/) already contains, in its section on monodromy as an ordinary-domain construction, the phenomenon that [section 14](#) builds on. Its example takes the scalar equation $\partial_z Y = ((\lambda + \varepsilon)/z)Y$, obtains the monodromy $M = e^{2\pi i \lambda} \text{Exp}(2\pi i \varepsilon)$, and concludes that for $\lambda \in \mathbb{Z}$ the *ordinary* monodromy is 1 while the full monodromy is not, with leading change exactly $2\pi i \varepsilon$: reduction to the ordinary coefficient can erase a genuine global obstruction.

Inheritance. The two reports stay separate — fundamental matrices of linear coherent ODEs are not conjugacies of iterated maps, and the algebraicity dichotomy [theorem 14.15](#) over $\mathbb{C}(w)((t^\Gamma))$ has no counterpart there. But that example is *the* antecedent of the monodromy exponent ρ_a of [\(14.15\)](#), and is cited as such at the point where ρ_a is introduced. What is new here is not the invisibility phenomenon but the package around it: the torsion lemma [lemma 14.10](#), the finite-cover obstruction [theorem 14.11](#), the classification of algebraic leading coordinates [proposition 14.17](#) with its already-transcendental first correction, and the dichotomy itself. That report’s cross-category warning is also adopted verbatim, once, as [warning 3.2](#), in place of re-deriving the coordinate-versus-surreal-derivation distinction ten times.

Source 07 separates itself from that report by name. Its torus derivatives $\partial/\partial\theta_j$ are external derivations annihilating K_Γ , and it says in as many words that this convention “keeps the present theory separate from the repository’s internal differential-equation report”. The cohomological equation [\(18.2\)](#) is therefore not a surreal differential equation for the Berarducci–Mantova derivation, and no theorem of either report transfers to the other. That is the same boundary [warning 3.2](#) draws for the germ half, stated once and applying to both. Source 07 also records that a targeted repository code search for “cohomological” returned no match, while stating that this is not an exhaustive audit of every manuscript or every synonym.

23.2 Adjacent: surcomplex analysis

Surcomplex analysis: holomorphic functions on $K = \mathbf{No}[i]$ (docs/surcomplex/analysis/) is the foundation that fixes the no-topological-convergence discipline and states the Neumann support material. All ten sources restate that discipline and all ten state the support lemma. Sources 02, 03, 05 and 09 each reproduce a minimal-bad-sequence Higman proof, and source 08 reduces the lemma to Poonen’s positive-support theorem.

Inheritance. Inherited, not restated. Exactly one proof of the support lemma is kept — [lemma 3.4](#) with [section A](#), the cleanest finite-word form — and the summability discipline is cited to that report in [definition 3.1](#). The other statements, one per source, are deleted. This is also what keeps the present report honest about which parts of its machinery are classical.

23.3 Adjacent: infinitesimal analytic geometry

Infinitesimal Analytic Geometry over the Surcomplex Numbers (docs/surcomplex/analytic-geometry/) owns the four-ring table and its separating witness. Sources 02, 03 and 04 each re-erect that separation from scratch, with source 02’s $\sum_{m \geq 1} t^m/(1 - mz)$ and source 04’s $\sum_{n \geq 1} t^{n\delta}/(1 - nw)$ being that report’s witness with $\eta = 1$ and $\eta = \delta$.

Inheritance. Neither the ring table nor the witness is reprinted. Every theorem here states its ring in that report’s names ([convention 3.3](#)), and exactly two rows are added — entire coefficients and polynomial coefficients — with the explicit note that they are new *rows*, not new rings ([section 3.2](#)).

Several sources already say they are continuing that distinction; this merge makes the continuation literal.

23.4 Adjacent: contours and Stokes

Contours and Stokes: Jordan separation, Cauchy theory, and Stokes calculus on the surcomplex plane (`docs/surcomplex/contours-and-stokes/`) is background for [section 15](#) and is deliberately *not* used there: every integral in that section is an ordinary complex line integral taken coefficient by coefficient over ordinary curves in a fixed $D \subset \mathbb{C}$. [Warning 15.1](#) states the resulting prohibition, which source 05 carries as a remark of its own.

23.5 Keep separate, with a cross-note: rank-one Berkovich geometry

Rank-One Non-Archimedean Analytic Geometry inside the Surcomplex Numbers (`docs/surcomplex/rank-one-berkovich/`) is the one report in the collection where convergence is genuine convergence: it works inside a *fixed rank-one* $\mathbb{C}((t^{\mathbb{R}}))$ with $|a| = \exp(-v(a))$ and spherical completeness, where the absolute value is a real-valued one and the field is complete and algebraically closed.

Cross-note, and it is load-bearing. This report forbids that reading throughout. It works at arbitrary valuation rank, with a deliberately rank-two example ([example 13.3](#)), and cites Lindahl’s rank-one non-Archimedean linearization as an antecedent it does not reprove. A Lindahl-type ultrametric disk radius and source 01’s maximal centered valuation ball are *different categories* ([warning 2.4](#)), and the exact-ball and shell-periodicity conclusions of [theorems 9.5](#) and [9.7](#) must **not** be transported to $\mathbb{C}_p$: source 01 says so itself and cites p -adic quadratic-like examples whose bounding sphere carries no periodic point ([warning 2.7](#)). Without this note a reader will read the two reports’ “disks” as the same object. No result of this report and no result of that one is transferred in either direction.

24 Computational record and reproducibility

The ten verification programs are preserved unmodified in `code/` and their recorded outputs in `data/`. They supply *finite exact symbolic* checks. None of them proves an infinite statement in this report. Separately, the repository’s Lean modules `Surreal/HahnSeries/Neumann.lean` and `Surreal/HahnSeries/Ancestry.lean` prove the finite-word support lemma, the corrected input-relative combinatorial ancestry bounds, and the counterexample to an input-independent bound. This does not yet formalize the corollary’s full operator-summability assertion or the analytic theorems built on it. The exact coverage is recorded in `docs/FORMALIZATION.md`.

24.1 What each suite checks, and what it does not

Source	Checks	Content	Explicit disclaimer
01	54	Nine coefficients of the general cubic second iterate against a closed form; the tuned factorization (13.3) and the degree-3/degree-4 cancellation at $e = 0$; the three phase regimes with <i>integer</i> exponent representatives $\delta = 8$, $\sigma_0 = 2, 4, 6$ (radii $3, 2, 2$), their thresholds, shell polynomials and squarefreeness; multiplier ratios by polynomial remainder; the three coherent leading coefficients through their <i>logarithmic derivatives</i> (deliberately avoiding a numeric branch choice of fractional powers); Schröder recurrence to degree 9 for $\lambda = 3/2, -4/3, 5/4, 1 + i$; the tangent binomial identity (13.1) for $m = 2..7$; $\text{Disc}_Y(1 + \kappa Y - 2Y^2) = \kappa^2 + 8$; integrality and residues of $h_2..h_9$ for the normalized tuned cubic at $e = t^4$, $r = 1$, including $[X^5]H_0 = 1/2$ and $[X^9]H_0 = 5/8$. All exact/symbolic, no floating point.	Does not establish the arbitrary-rank support lemma, algebraic closure, maximality for every polynomial, the transfinite coherent lifting, or novelty; not an implementation of general Hahn fields; not proof-assistant verification.
02	1,109	Reciprocal-jet formula (8.4) versus the inverse recurrence (133) and truncated product of a denominator with its reciprocal (133), both for $2 \leq n \leq 20$, $0 \leq k \leq 6$; ordinary quadratic Schröder identity through degree 12 (13); Hahn-scaled quadratic identity (156); nonlinear recursion with multiplier drift and cubic forcing, $F(z) = (\lambda + t)z + t(z^2 + z^3)$, checked after each parameter step through t^4 and spatial degree ten (154); two-variable multiplicative-detuning reciprocal jets with exact unit phases $(3 + 4i)/5$ and $(5 + 12i)/13$, total degrees 2..9, both components, jet orders 0..4 (520). Exact in $\mathbb{Q}(i)$ via <code>fractions.Fraction</code> ; test multiplier $\lambda = 3/5 + 4i/5$, not a root of unity because a root of unity in $\mathbb{Q}(i)$ is an algebraic integer.	Tests algebraic identities, signs, normalizations and truncation only. Does not prove Hahn summability, well-founded recursion, domain preservation, the small-divisor limsup, the exact radii, or novelty. The multivariable test verifies nonzero denominators only in its finite range and does not certify nonresonance for all multi-indices.

Source	Checks	Content	Explicit disclaimer
03	21	$\exp(\log F) = F$ and two-sided inversion; half and third iterates; time addition; Leibniz; identification of the logarithm with its coordinate vector field; the two-variable fixed-point matrix factorization (4.11); normalized nonresonant conjugacy, its inverse and the degree bound (6.13); the finite-order averaging and resonant flow identities. One-parameter flow checks through degree six; two-parameter multivariable, nonresonant and resonant checks through total parameter degree three. Sparse exact jet algebra over $\mathbb{Q}(i)$ with a symbolic λ for the four coefficients (22.1).	Finite polynomial jet identities truncated by <i>total parameter degree</i>, explicitly not a Hahn valuation cutoff in a higher-rank group (warning 22.1). Does not test arbitrary well-ordered supports, analytic radius claims, continued-fraction asymptotics, the universal quantifiers, or priority. No Lean verification.
04	2,764	Koenigs recurrence (14.24) and the leading Pochhammer coefficients for $q = 1.8$, $m \leq 24$, through ε^8; the <i>directly substituted</i> Schröder residual through ε^9 (an independent test, not merely agreement with a displayed formula); the first and second coefficient functions through w^{24q+1}; the logarithm ratio ρ_q of (14.30) at orders 0, 1, 2 for $q = 1.8$; the derivation identity $\log C_g(p) = (\log C_g(w))p'$ for $q = 1.6$, $p = w^n$, $0 \leq n \leq 6$, through ε^6; generator coefficients and fixed-point derivatives, reduced using $w^q = -1$ to avoid approximate complex roots. Standard library only, <code>fractions.Fraction</code>, no floating point.	No finite test proves Neumann summability, existence on an arbitrary ordinary domain, compatibility for arbitrary value groups, or the algebraicity dichotomy. Not a symbolic surcomplex CAS and not a numerical evaluator at arbitrary surreal inputs.

Source	Checks	Content	Explicit disclaimer
05	245	Euler families $p = 1, 2, 3$: exp of log, the Julia identity corollary 4.6 in the form $W_F(F) = F'W_F$, inverse via negative time, the homogeneous grading coefficientwise, residues equal $(p + 1)/2$ at degree one and zero elsewhere; a mixed-coefficient map (round trip, log-equals-vector-derivation, Leibniz, Bernoulli difference solver); the flow group law (times 2 and 3 compose to 5); the unique-root formula; an explicit conjugacy with pushforward of the vector field, pullback of the normalized time form and the basepoint marking $h_k(1) = 0$; untruncated symbolic identities for the flat family ($1/W_c = 1/\varepsilon + c/z$, residue c , leading correction $-1/z$); and 50 lexicographic order comparisons. Exact in $\mathbb{Q}(z)[\varepsilon]/(\varepsilon^8)$.	Transcription and algebra checking only. Proves nothing about arbitrary-rank support finiteness, global primitive existence, the classification, or the slow-time theorem. Fifty of the 245 assertions are finite lexicographic illustrations for $n = 1, \dots, 50$; the all-n inequality is proved in the text, not inferred from them. The verifier also loses one top coefficient on division by ε , so reciprocal and pullback comparisons are checked only through degree $N - 1$.
06	119	Exact arithmetic in $\mathbb{Q}(\sqrt{2})$, with a hand-rolled bivariate label series ring truncated at total label degree four: the single-exponential gauge recursion for (16.28) ; resonance of every normal coefficient and the nonresonant gauge of every generator coefficient; the four displayed coefficients; all 16 symplectic coordinate brackets; four inverse Lie maps; direct substitution $H(\Phi) = N$ and $J(\Phi) = I$; the two first integrals, their involution, and one nonlinear function of the actions being an integral; 51 degree-defect identities; the cubic boundary map (16.27) ; the support counterexample's finite-sample order pattern; and (16.33) for $\lambda = 2$ through $n = 10$.	Truncates by <i>total label degree</i> in two independent labels, not by Hahn valuation, and only to degree four. Explicitly not a proof of arbitrary-support summability, nor of theorems 16.5 and 16.16 , nor of theorem 16.12 , nor of the Liouville construction proposition 16.17 .

Source	Checks	Content	Explicit disclaimer
07	—	No printed total; two named checks, every assertion exact. The slow-circle conjugacy example 19.11 is reconstructed recursively from $(1 + h')\nu = 1 + \varrho \sin(\theta + h)$ with $\nu(0) = 1$: at each order the mean is determined first, giving the new coefficient of ν, then the remaining zero-mean Fourier polynomial is integrated. It asserts at every order that the mean is real and that h is Hermitian, that the full conjugacy residual vanishes through degree ten, that $\nu^2 = 1 - \varrho^2$ coefficientwise through the same degree, and that the three displayed terms of (19.17) are correct; the recorded frequency coefficients are $1, 0, -\frac{1}{2}, 0, -\frac{1}{8}, 0, -\frac{1}{16}, 0, -\frac{5}{128}, 0, -\frac{7}{256}$. It then checks 117 instances ($n \in 0..8$, $m \in 0..12$) of the coefficient recurrence for (19.13), and classifies all 1,330 nonzero integer modes in $[-5, 5]^3$ into the three strata of example 19.9, obtaining 1,210/110/10. Exact $\mathbb{Q}(i)$ arithmetic by pairs of <code>fractions.Fraction</code>; every check is an <code>assert</code>; no floating point and no tolerance anywhere.	Finite algebraic checks only. They verify neither the irrationality of (18.6), nor any infinite-mode subexponential estimate, nor the word lemma, nor the infinite-support conjugacy theorem; those rest on the proofs in the text. No Lean or Wolfram verification of any infinite theorem is claimed.

Source	Checks	Content	Explicit disclaimer
08	—	No single total; the counts are different units and are not added. Exact arithmetic in $\mathbb{Q}(i)$ with SymPy 1.14 and the test multiplier $\lambda = (3 + 4i)/5$, which is unimodular and not a root of unity. Two independent finite constructions — the positive-parameter operator recursion and a direct solve in increasing total coordinate degree — are compared, and the conjugacy identity and both inverse identities are checked. One variable at parameter degree 4 and coordinate degree 11 (7 component identities). Two variables with mixed monomials at parameter degree 3 and coordinate degree 7 (14 component identities). The cyclic drifting construction (8.17) and its inverse at parameter degree 5 and coordinate degree 13, with the fold $u = v = t$ keeping every colliding contribution (6 polynomial identities).	Does not prove holomorphy on a common domain or strong summability over arbitrary ordered groups. The test multiplier is a convenient exact value, not an approximation to a positive-rate example.
08, cont.		The first nonlinear parameter block against its closed reciprocal formula (60 exact scalar equalities). Divisible-subtree packing on 8,566 tree-weight configurations (sizes 1–6, local weights 1 or 2, $q = 1, \dots, 10$; 85,660 inequalities). Strict-chain packing on 14,892 chains (total weight 1–16, length at most 6; 119,136 inequalities).	Finite ranges only. Does not prove the asymptotic small-divisor rate, the existence of irrationals of prescribed rate, or noncancellation along an infinite convergent subsequence. No novelty, no peer review, no Lean.
09	—	No single total; status PASS. Standard library, <code>fractions.Fraction</code> . Packing: all plane rooted skeletons with at most 5 vertices and positive weights of total at most 11 (8,514 weighted skeletons), every relevant q : 83,996 instances of lemma 7.3, each with an independent construction of the disjoint blocks; 45 aggregate plane-tree count bounds; the zero-weight counterexample. Clustering: 33,030 pairs of a reduced rational test value of θ (denominator at most 60) and a total weight $n \leq 30$, 7,075 of them with a nonempty marked set, checking the determinant, divisibility and proportional-error identities of lemma 7.2.	The rational tests infer nothing about an irrational limsup. Does not prove the support lemma, the general packing inequality, or any small-divisor rate.

Source	Checks	Content	Explicit disclaimer
09, cont.		Formal algebra: six exact rational cases (multipliers 2, 2/3, -2; one-label geometric data and two colliding integer labels) through parameter order 4 and Taylor degree 8. They compare three independently organized constructions: Taylor/parameter recursion for $\check{H}$, the plane-skeleton formula (7.13) with explicit slot multiplicities, and an operator geometric expansion for H . That gives 270 coefficient comparisons, 2,478 tree-label-weight terms and 36 groups of identities, including both inverse identities and both conventions.	The test multipliers are <i>not</i> on the unit circle. They check tree factors, signs, multiplicities and composition only, and the unit-circle radius estimate is not tested. Does not prove convergence on a disk, workspace invariance, or novelty. No Lean code.
10	—	No single total; status PASS. Standard library only; exact $\mathbb{Q}(i)$ arithmetic by pairs of <code>fractions.Fraction</code> . Contraction lemma lemma 7.26: all 2,730 recursively labelled colored internal-tree configurations with at most three internal vertices, arities two or three, two colors, every leaf-color count and both output colors, and every nonempty marked subset, giving 18,466 checks of the nonnegative contracted leaf vector, the character identity, the degree identity, disjointness of blocks and the degree budget (7.21); and 1,000 pseudorandom configurations with four to nine internal vertices from a fixed seed. Algebra: the nonscalar unit multipliers $(3 + 4i)/5$ and $(5 + 12i)/13$ with $F_1 = \frac{3+4i}{5}x + t(x^2 + 2xy) + t^2y^3$, $F_2 = \frac{5+12i}{13}y + t(x^2 - y^2) + t^2x^2y$; the forward coordinate by one recursion and the inverse by an independently organized residual recursion, through parameter degree 3 and total spatial degree 6; both component conjugacy identities and both two-sided inverse identities (8 component identities), and the 21 nonzero divisors used by the forward recursion. Each coordinate has 23 nonzero monomials at this truncation.	Detects transcription, sign, color, multiplicity and truncation errors only. Does not establish lemma 7.24, the limsup of theorem 7.27, analytic convergence, arbitrary-support summability or originality. The random configurations are samples, not an enumeration; nonresonance of the test multipliers is checked only in the finite range used. No Lean code.

24.2 Independently verified facts about the suites

The following were checked directly against the delivered files and by running the code, and override any contrary statement in a source README.

- (1) **All ten suites pass.** Run on Python 3.14.4 with SymPy 1.14.0 where SymPy is used, every one exits 0; sources 08 and 09 were run, on a copy, when they were integrated. Reported totals where a script prints one: 54 (01), 1,109 (02), 21 (03), 2,764 (04), 245 (05), 119 (06); source 07 prints no total. Source 04's count reproduced the recorded 2,764 exactly; source 05 reproduced its recorded transcript line for line; source 03 reproduced its four recorded example expansions byte for byte; and the regenerated JSON of sources 06 and 07 matched the shipped ones up to line endings only.
- (2) **Four of the ten scripts rewrite their own evidence.** Source 03's `verify.py` writes `data/verification.json` by default; source 04's writes `data/verification.json` and `data/coefficients.csv`; source 06's writes `data/verification.json`, and does so only after all checks pass; source 07's takes `--output` with `default verification.json` resolved against the current working directory and writes it with an unconditional `write_text` — no existence check, no backup, no dry run — and the invocation documented in its own README is exactly the one that overwrites the delivered record. Source 01's script prints to stdout, but its Makefile's `verify` target redirects that stdout into `verification.txt` in the same directory, overwriting the delivered record. **Every suite must therefore be run on a copy, never in the delivered tree:** a byte-identity check performed after an in-place run compares two equally modified copies and passes while proving nothing.
- (3) **Every suite prints its own scope disclaimer.** That is a real and unusual discipline in this material, and those disclaimers are exactly the limitations carried in [section 26](#).
- (4) Source 04's README lists `SHA256SUMS.txt` among its package contents, and no such file exists in the delivered archive. This is a packaging gap, recorded rather than repaired.
- (5) Source 03's recorded console transcript names its report path as `/mnt/data/surcomplex_dynamics/data/verification.json`, i.e. it was recorded in a different container from the delivered one. The check contents match the article's prose regardless.
- (6) Source 07's README likewise lists `SHA256SUMS.txt` among its package contents, and no such file exists in the delivered archive. Like item (4) this is recorded rather than repaired.
- (7) **The finiteness bound of [theorem 17.5](#) was checked independently of source 07.** Modelling $\Upsilon_i = \sum_j c_{ij} t^{\gamma_j}$ with $\gamma_1 < \gamma_2 < \dots$ gives $k \cdot \Upsilon = \sum_j L_j(k) t^{\gamma_j}$ with $L_j(k) = \sum_i k_i c_{ij}$, so $v(k \cdot \Upsilon)$ is the least γ_j with $L_j(k) \neq 0$ and $\{k : \text{level} \geq \gamma_{m+1}\} = \ker L_1 \cap \dots \cap \ker L_m$, whose saturation has rank $d - \text{rank}(L_1, \dots, L_m)$. A new level appears exactly when that rank increases, so the kernel rank strictly drops, and it can drop at most d times: that is both the bound and the flag. Brute-force testing over rational coefficient matrices with deliberate resonances forced in 40 per cent of the cases gave **zero violations of the bound in 400 random trials**, and a worked instance at $d = 3$ had kernel ranks $2 \rightarrow 1 \rightarrow 0$ with achieved levels $\{0, 1, 3\}$ — three levels, attaining the bound, with two indices skipped exactly where the rank did not move ([remark 17.6](#)). The theorem and its stated mechanism both hold. This is an independent check of a finite mechanism, not a proof of any infinite statement.
- (8) **Sources 08 and 09 reproduce and do not overwrite.** Re-run on a copy with Python 3.14.4 and SymPy 1.14.0, source 08's program produced a record that agrees with `data/08-exact-and-drifting-multipliers-verification.json` in every field except the elapsed time and the Python version; the shipped run used Python 3.13.5. Source 09's regenerated record agrees with `data/09-single-loss-linearization-verification.json` field for field. Both scripts are overwrite-safe. Without `--output` they print only, and they refuse

an existing output path: source 08 unless `--overwrite` is given, source 09 always. Source 08's docstring names the pre-placement path `data/verification.json`. Source 09's delivered record sat at its package root, and its Makefile calls the script by its delivered name `verify.py`. The placed `code/08-exact-and-drifting-multipliers-build.sh` and the pdf target of `code/09-single-loss-linearization-Makefile` build the two standalone source manuscripts, which are not shipped in this directory. Neither package ships or promises a checksum manifest.

- (9) **Source 10 reproduces and does not overwrite.** Its script `code/10-nonscalar-common-domain-verify.py` takes no arguments, writes no file and prints ten lines to standard output. Run on a copy with Python 3.14.4 it exits 0 in under two seconds, and its output agrees line for line with `data/10-nonscalar-common-domain-verification_results.txt` (apart from the platform's line endings). The counts quoted in the table above are the ones printed. The delivered package also held the source manuscript, its PDF, a README and a checksum manifest; the manifest was checked against the delivered files, all six entries matching, and none of those four files is shipped here. Its provenance note is shipped as `10-nonscalar-common-domain-SOURCES.md`; it describes this report as it stood at source 10's pin, before the nonscalar subsections were added.

24.3 Building this report

The report is standalone LaTeX with an internal bibliography and no external `.bib` file, graphics, shell escape or network access:

```
latexmk -pdf -interaction=nonstopmode article.tex
```

Reproducing the ten suites, each *on a copy*, requires Python 3.10 or later and, for five of them, SymPy; the pinned versions recorded by the sources are `sympy==1.14.0`. Source 10's script needs only the standard library of Python 3.9 or later.

25 Open questions

The questions below are exactly those the sources leave open, with the categories in which they are open stated explicitly. Sources 08 and 09 settle part of two of them, [questions 25.1](#) and [25.2](#). Both are kept, re-scoped, with the settled part and its source recorded. The two sources add no new question; their remaining directions are folded into [questions 25.1](#), [25.7](#) and [25.8](#). Source 10 settles most of what sources 08 and 09 left of [question 25.1](#); the question is kept once more, re-scoped to what remains. Source 10 adds no new question either; its remaining directions are folded into [questions 25.1](#) and [25.8](#).

Question 25.1 (Common-domain germs at a finite positive rate: the optimal radius for a nonscalar multiplier of angular rank at least two). This is [question 8.20](#), and [warning 8.21](#) is its status. It is settled negatively in the *drifted* category SD_{dr} by [theorems 8.2](#) and [8.5](#), and over $\mathbb{Z}$ as well by [theorem 8.15](#). In the exact-multiplier category SD_{ex} it is settled positively for a scalar $\Lambda = \lambda I_d$ by [theorem 7.13](#) (sources 08 and 09). As recorded at their pins, it was **open** for a nonscalar diagonal unitary Λ . Source 03's remark applied there: an exact characterization would need to control interactions across unbounded word depths rather than a single homological inverse. For a scalar multiplier that control is [theorem 7.5](#). For a nonscalar one the divisors $\lambda^\beta - \lambda_j$ are not indexed by a

single integer, and what was missing was a vector analogue of the tree bound, or a counterexample to one (section 7.6). At the pins of sources 01–07 the whole exact-multiplier case was open.

Status. Source 10 supplies a vector analogue with loss $\min(r_{\text{ang}}, m)\tau$ at complexity m (theorem 7.27) and settles the question positively for every nonscalar diagonal unitary Λ : when $\tau < \infty$, every input has a common-domain linearizer, of radius at least $Re^{-r_{\text{ang}}\tau}$ (theorem 7.23), and when $\tau = \infty$ not every input does (theorem 7.34). The largest universal radius R_{univ} satisfies $Re^{-r_{\text{ang}}\tau} \leq R_{\text{univ}} \leq Re^{-\tau}$ and equals $Re^{-\tau}$ when $r_{\text{ang}} = 1$. Still **open**: the value of R_{univ} when $r_{\text{ang}} \geq 2$ and $0 < \tau < \infty$. A sharper accounting might reduce the factor r_{ang} ; a matching lower example would have to show several exponential rates saturating their charges at once, and surviving the cancellations of an actual conjugacy, which large individual tree factors do not give (section 7.13). Outside the question as posed, source 10 also leaves drifting multipliers in several variables, resonant spectra, Jordan blocks, nonunitary spectra and infinite coordinate dimension untreated.

Question 25.2 (The fixed cyclic value group; *settled* in one variable). As recorded at the pins of sources 01–07: the negative direction of theorem 8.2 is established over $\Gamma_* = \mathbb{Q} + \sqrt{2}\mathbb{Q}$, hence for the universal class of workspaces, and it embeds in any surcomplex workspace containing that ordered subgroup. It is *not* asserted for a fixed cyclic value group such as $\mathbb{Z}$, where different parameter blocks can collide at one Hahn exponent. Resolving the effect of those collisions is a separate problem, not a consequence of the two-independent-exponent proof, and it concerns the existence of *some* smaller common neighborhood for every input, not the weaker statement about retaining the original disk.

Status. Source 08 settles the question, negatively, in the one-variable setting in which it is posed. Over $\mathbb{Z}$, with the single Hahn monomial of (8.17), the colliding blocks cannot cancel the highest pole (proposition 8.14), and the coefficient radii are exactly $Re^{-m\sigma}$ (theorem 8.15). The universal common-domain property holds over any fixed nonzero Γ exactly when $\sigma = 0$ (corollary 8.17). Still **open** is the several-variable multiplicative-detuning collapse theorem 8.7 over a fixed cyclic group. Source 08 does not address it, and it remains asserted only over Γ_* .

Question 25.3 (Resonant multipliers of infinite order). For a diagonal unitary multiplier with resonances but *without* finite order, find a support-controlled analytic normal form with precise necessary and sufficient coefficient-category hypotheses. Nonresonant monomials can be treated by the homological inverse, but resonant terms create a coupled normal-form problem and remove the global inverse used in (6.5), and a finite cyclic averaging operator is no longer available. The Hamiltonian analogue is open too: for resonant ω the first reduction retains a nontrivial resonant Hamiltonian algebra, the gauge equation changes because L^{-1} has a larger kernel, and whether the dynamics of that retained algebra can itself be integrated under effective support conditions is not answered by the nonresonant centralizer proof.

Question 25.4 (Higher-dimensional centralizers). In several variables, classification of the positive centralizer reduces exactly to classification of the commuting positive vector fields, and the one-variable meromorphic-ratio argument of theorem 4.12 cannot classify those higher-dimensional Lie centralizers; additional first integrals or commuting geometric structures can occur. Independently, source 05 records that in several variables there is no single reciprocal time one-form encoding a vector field and commuting fields need not be constant multiples, so the period normal form does not transport either.

Question 25.5 (Moving fixed points, and several variables, for the parabolic theory). Characterize the monodromy of $w + \varepsilon V(w) + \varepsilon^2 R(w)$ when its fixed points move. An analogue of the local derivative identity lemma 4.5 at moving formal fixed points is a natural intermediate target; translating it into a common-domain continuation theorem would also require controlling those

moving singularities. The Euler family *avoids* that difficulty rather than concealing it. In several variables the logarithm-of-substitution construction still suggests a vector field, but the scalar quotient α_c/W has *no* direct replacement, and resonant homological equations with noncommuting monodromy would replace the one-dimensional integration argument.

Question 25.6 (The period classification where the leading coefficient vanishes). When a has zeros, [theorem 15.5](#) still controls the discrete equation, but the conjugacy classification needs more: local zero divisors, multiplicities and singular normal-form obstructions must be combined with global periods. [Warning 15.12](#) identifies the precise point where the proof stops. Separately, extending [theorem 15.13](#) to conjugators with nonidentity ordinary part requires the ordinary automorphism action on the leading form and on its periods, after which equal period vectors in a fixed basis would no longer be the right equivalence relation by themselves.

Question 25.7 (Maximal domains at infinite scale). [Theorem 16.21](#) gives *guaranteed*, not maximal, domains. [Example 16.24](#) detects a sharp boundary through both a pole and an extra critical point; a general procedure combining support certificates with algebraic or analytic obstructions could yield substantially better maximal-domain information for finitely presented perturbations. Relatedly, for entire layers failing the universal threshold, particular inputs may still normalize because their coefficients vanish at the dangerous monomials; an input-specific necessary and sufficient condition would have to remain stable under the nonlinear brackets created by normalization, and a first-order divisibility test alone need not be enough. Source 08 poses the exact-multiplier form of the same question. The radius $Re^{-\tau}$ of [theorem 7.13](#) is a universal guarantee, not a formula for each input, and polynomial inputs already do much better. Determining input-specific improvements from vanishing coefficients or algebraic structure is a separate question, and any criterion must survive the nonlinear degree interactions of the conjugacy equation.

Question 25.8 (Ordinary analytic realization). Which of the Hahn normal forms constructed here admit compatible *ordinary* analytic realizations? Ordinary analytic specialization of a Hahn or formal parameter requires growth estimates or an additional summation theory, and this report supplies neither. Sectorial analytic moduli of ordinary parabolic germs are not removed, and the analytic embedding obstruction for ordinary parabolic germs is not resolved. Likewise, adding weighted coefficient norms, Gevrey bounds in the parameter, or ordinary convergence in (u, v) produces strictly stronger problems whose arithmetic thresholds could differ from the thresholds established here; [\(5.7\)](#) is offered only as a starting point. Sources 08 and 09 add the depth dimension. Their estimates [propositions 7.11](#) and [7.19](#) let the constants depend arbitrarily on the support word, so they give no uniform-depth or Gevrey bounds across depth, and no joint analytic convergence in a complex parameter and the coordinate. Source 09 locates the missing information in the constants $W_{\gamma,m}$ and $B_{\gamma}(r)$ and in the polynomial factors of [\(7.14\)](#). Restrictions on them could support Gevrey or other weighted estimates, but none follows from support finiteness alone. In the nonscalar case the prefactor $C_{m,\varepsilon,\Lambda}$ of [theorem 7.27](#) is not even explicit, and no bound uniform in m is available (source 10).

Question 25.9 (The resonant quasi-periodic normal form). [Theorem 18.3](#) allows QR, but the constant normal form [theorem 19.1](#) requires $N_{\Upsilon} = \{0\}$. In the resonant case a natural target is a normal form retaining the Fourier modes in N_{Υ} . Source 07 warns explicitly that simply replacing the mean projection Q_0 by $\Pi_{N_{\Upsilon}}$ in the proof is **not** justified: the retained normal form is then no longer a constant vector field and creates additional terms in the conjugacy equation. Products and substitutions interact with those modes, and higher perturbations may split the remaining resonances, so the extended flag of [corollary 18.6](#) would have to be carried through the nonlinear

argument. This is the quasi-periodic counterpart of [question 25.3](#) and is open for the same structural reason — a larger kernel for the inverse operator — but it is a different problem on a different divisor family.

Question 25.10 (Quantitative coefficient-growth classes on the torus). One may add bounds linking $\|f_\beta\|_r$ to a weight on the support. The argument of [section 19](#) does *not* prove preservation of such a weighted class, because inverse denominator powers and derivative losses grow along support words ([remark 18.5](#)). Identifying hypotheses that preserve these weights, and that would permit a meaningful specialization of selected monomials to small real parameters, is open; that is exactly where genuine non-linear convergence arithmetic would have to re-enter, and it is the torus form of [question 25.8](#).

Question 25.11 (Effective arithmetic certificates for the flag). [Theorem 17.5](#) makes the flag finite but not automatically computable from arbitrary real coefficients: detecting an exact integer relation among arbitrary named real numbers may itself be unavailable. For rational and suitable algebraic coefficient data, explicit lattice and approximation certificates can make the construction effective. The theorems as stated separate *existence* of a finite flag from *decidability* of its entries and its arithmetic bounds, and no complexity claim is made. This is the same distinction as **N64** and [section B](#) draw for the finite-return certificate.

26 Limitations and non-claims

This section is the consolidated record required by [convention 1.1\(iv\)](#). It carries **one hundred and twenty-three** distinct limitations stated by the ten sources — eleven from source 01, nine from source 02, ten from source 03, fourteen from source 04, twelve from source 05, twelve from source 06, ten from source 07, sixteen from source 08, twelve from source 09 and seventeen from source 10 — with the originating source named in each case. None has been merged away, softened, or absorbed into a stronger statement. Where two sources state overlapping limitations, both wordings are preserved because their scopes differ.

26.1 Status, refereeing, and formalization

The following items record the status of the preserved source packages. For the later, partial repository formalization of the support results, see [section 24](#).

- N1.** (01) An AI-assisted, unrefereed research draft; not refereed and not formally verified. Its PDF metadata records the author as “Research manuscript prepared with ChatGPT”. Priority is explicitly not certified and no named published conjecture is claimed solved.
- N2.** (02) An AI-assisted, unrefereed research draft; not refereed, not proof-assistant verified. **No Lean theorem corresponding to the new analytic results was compiled** as part of the package.
- N3.** (03) Self-described AI-assisted, unrefereed, not formally verified. Its literature search is described as targeted and auditable but *not* a priority determination; no named published conjecture is claimed solved; independent expert review of both the proofs and the literature comparison is called appropriate.
- N4.** (04) An unrefereed, AI-generated research draft whose proofs are “offered for scrutiny”, not proofs checked by a proof assistant. No Lean, Wolfram or other proof-assistant formalization is included or claimed.

- N5.** (05) Not independently refereed and not formally verified; none of the central claims is an independently checked Lean theorem.
- N6.** (06) Not peer-reviewed and not proof-assistant verified; no Lean or other formalization is supplied or claimed.

26.2 Priority and novelty

- N7.** (01) The repository and literature comparisons are both explicitly *targeted*: three catalogue README files plus a search for “linearization”, not a line-by-line audit of every research package, and the novelty assessment is a “proposed-originality assessment, not an exhaustive priority claim”.
- N8.** (02) Priority is not independently verified. Only three theorems — here [theorems 8.2](#), [8.5](#) and [8.7](#) — are named as candidate-new, and the source explicitly concedes that “a specialist could identify an existing formal-family theorem that subsumes part of the present formulation; that possibility should be checked before any publication claim”. The support lemma, the homological inverse, formal nonresonant conjugacy, the quadratic scaling trick and the centralizer argument have classical antecedents and are not offered as independent discoveries.
- N9.** (02) The repository inspection is explicitly *not* a line-by-line audit of every manuscript or every Lean file; only the catalogue and the opening foundation definitions were read.
- N10.** (03) Classical credit is stated plainly and is not claimed as new: formal logarithms, flow embedding and semisimple–unipotent splitting [\[6\]](#); the small-divisor problem and formal-versus-analytic linearization [\[11, 13, 14\]](#); and the support lemmas [\[2, 3, 4\]](#).
- N11.** (04) Priority is *not* claimed: the literature search is described as targeted and bounded and “did not certify priority”; no named published open problem is claimed solved; novelty is “to the author’s present knowledge”.
- N12.** (04) Acknowledged precedents, not contributions: Neumann’s support lemma, formal embedding and formal logarithms, and rank-one ultrametric radius theory with the quadratic disk scale [\[22\]](#). In that source’s own words, “we do not present a disk estimate alone as a new theorem of classical dynamics”.
- N13.** (04) The repository inspection was “a targeted inspection, not an exhaustive line-by-line audit” at one frozen revision.
- N14.** (05) Novelty is assessed by a *targeted*, not exhaustive, literature search; the statements are called candidate new results; an unlocated result may exist in the literature on formal flows, deformation of one-forms, or generalized series; and no named published conjecture is claimed settled.
- N15.** (06) Priority is not established: the contributions are labelled “proposed contributions with proofs”, not certified bibliographic firsts, on the basis of a non-exhaustive literature search; no named published open conjecture is claimed solved. Classical formal Birkhoff normalization, non-Archimedean linearization and general formal-sum Lie exponentiation are credited rather than claimed [\[25, 26, 22, 8\]](#).
- N16.** (04) For the Neumann 1949 reference, the publisher’s direct PDF endpoint was not accessible in the session; the metadata and the standard lemma were cross-checked instead.
- N17.** (04) Packaging gap: the source README lists `SHA256SUMS.txt` among the archive contents, and no such file exists in the delivered archive (see [section 24.2\(4\)](#)).

26.3 Scope of the hypotheses

- N18. (01) [Theorem 9.5](#) and its package cover nonlinear polynomials in *one* variable, with unit multiplier not a root of unity, over complex Hahn fields of arbitrary ordered-group rank, in equal characteristic zero. Explicitly *not* covered: general infinite Hahn-analytic power series — where a common support certificate, a finite or attained dominant face, and analytic composition would each need separate proof; actual root-of-unity multipliers; attracting or repelling multipliers; and higher-dimensional simultaneous resonances. Valuation-expanding periodic points of polynomials $q^{-1}P$ with $v(q) > 0$ are studied in the report [docs/surcomplex/expanding-polynomial-dynamics/](#); their multipliers are Hahn elements of negative valuation, not ordinary repelling complex multipliers, and that report linearizes nothing (`epd:prop:multiplier`, `epd:rem:expansion`).
- N19. (01) Equal characteristic zero is a *genuine* hypothesis, not cosmetic: the divisor cancellation [\(9.3\)](#) leaves a factor $n - 1$ in the denominator, a valuation unit here but not in mixed characteristic. The formula and the shell-periodicity conclusion must **not** be transported to $\mathbb{C}_p$; [\[23, Cor. C\]](#) gives p -adic quadratic-like examples whose linearization bounding sphere contains no periodic point ([warning 2.7](#)).
- N20. (02) Resonant multipliers are excluded *entirely* from [section 8](#); resonant normal forms need a different target equation, not the linear normal form used there.
- N21. (02) The several-variable results are stated only for diagonal unitary L and *equal-radius* polydisks; no optimal general Reinhardt-domain claim is made or needed.
- N22. (03) The finite-order theorem [theorem 20.1](#) requires $\Lambda^q = I$, not merely the presence of resonances, and resonant normal forms are *not* claimed unique: further Λ -commuting changes can transform Y .
- N23. (04) The most consequential hypotheses of [section 14](#) are not cosmetic: the simple zero at the normalization point permits the regular quotient [\(14.5\)](#); a common ordinary domain permits coefficientwise primitives without an uncontrolled sequence of shrinking radii; positive support makes the operator logarithm strongly meaningful; the polynomial Euler form fixes all zeros and gives the exact factor $W_\varepsilon = VB_\varepsilon$, whereas a general perturbation can move zeros or create different coefficient pole patterns; and divisibility of Γ is used to keep the resonant rescaling inside the chosen workspace.
- N24. (05) [Theorems 15.13](#) and [15.16](#) require the leading coefficient a to be *nowhere zero* on D . The unit hypothesis in [lemma 15.11](#) is exactly where the proof stops applying, and the zeros case needs local zero divisors, multiplicities and singular normal-form obstructions combined with global periods.
- N25. (05) Conjugators are restricted to identity ordinary part. A conjugacy with nonidentity ordinary part can change the leading vector field and the homology basis, so equal period vectors in a fixed basis would no longer be the right equivalence relation; that extension is not done.
- N26. (05) No converse characterization of the units of $\mathcal{A}$ is claimed, only the sufficient criterion [lemma 14.2](#).
- N27. (05) The $\mathcal{A}$ -membership condition $g/W \in \mathcal{A}$ in [theorem 15.5](#) is indispensable outside the nonsingular stratum; period vanishing alone is insufficient there ([example 15.10](#)).
- N28. (06) Nonresonant frequencies only. The gauge equation changes for resonant ω , because the homological inverse then has a larger kernel; the resonant case is listed as open ([question 25.3](#)).
- N29. (06) No result is claimed for Hahn series with unrelated convergent-germ coefficients, nor for infinite-dimensional Hamiltonian systems.
- N30. (06) Uniqueness is *gauge-relative*: only the single exponential generator with $\Pi\chi = 0$ and

the normal form it produces are unique; general normalizing symplectic changes are not ([warning 16.10](#)).

26.4 Exactly what “exact” and “analytic” mean

- N31. (01) “Exact” in [theorem 9.5](#) means the maximal *centered valuation ball* only — not the full pointwise summability locus of the series (exceptional evaluations on the bounding shell are allowed), not a maximal non-ball invariant domain, and not every analytic continuation of the conjugacy.
- N32. (01) The bounding shell S_r is an algebraic valuation-shell expression: *not* a topological boundary (valuation balls are clopen) and *not* an ordered-modulus circle. The common-domain disk D is *not* identified with B_r : all points of the normalized monad have ordinary residue zero while D contains many nonzero ordinary points ([warning 2.4](#)).
- N33. (01) [Theorem 10.1](#) gives *no* uniform ordinary norm bound on the coefficient family (H_γ) ; these may be arbitrarily large with arbitrarily large derivatives. Only the domain of holomorphy and the support certificate are uniform ([warning 10.3](#)).
- N34. (01) The coherent conclusion is *resonant-only*, and [proposition 11.1](#) shows failure of a *universal* common-domain statement in the nonresonant case — explicitly **not** an if-and-only-if classification, and not every nonresonant case is noncoherent.
- N35. (01) The stability certificate [theorem 9.10](#) is sufficient, *not* necessary: a perturbation may change the first return face and still preserve its maximal slope.
- N36. (02) The positive theorem gives *no* uniform bound on $\|h_\gamma\|_r$ as γ varies, no norm, no Gevrey or weighted coefficient estimates, and no ordinary holomorphic dependence on t near $t = 0$; those are strictly stronger problems whose arithmetic thresholds could differ, and [\(5.7\)](#) is offered only as a starting point.
- N37. (03) “Finite dependence” bounds the number of input *coefficient functions* and analytic operators at a specified weight. It is *not* an unconditional finite-time algorithm: one application of $\mathcal{L}^{-1}$ to an arbitrary analytic function can involve infinitely many of its ordinary Taylor coefficients, and only polynomial input makes each requested coefficient a finite exact algebraic calculation ([warning 6.12](#)).
- N38. (04) Transcendence is relative to the specific base $\mathbb{C}(w)((t^\Gamma))$. The source volunteers the counterweight that H_ε is *differentially algebraic* in a very explicit way, satisfying $H'_\varepsilon = \Omega_\varepsilon H_\varepsilon$ with $\Omega_\varepsilon \in \mathcal{B}$, so “transcendental” must not be upgraded to “differentially transcendental” ([warning 14.27](#)).
- N39. (04) “Transcendental” concerns a *function* in an ambient meromorphic Hahn field. It is not a claim that any individual value $H_\varepsilon(w_0)$ lies outside $\mathbf{No}[i]$, and is explicitly consistent with $\mathbf{No}[i]$ being algebraically closed.
- N40. (04) [Theorem 14.3](#) does *not* assert that the global evaluated map $H^\#$ is injective on all of $U^\#$; injectivity is proved only on the infinitesimal disk for the resonant model ([warning 14.4](#)).
- N41. (04) The sharp-domain maximality of [theorem 14.22](#) is about *centered valutive balls* under either boundary convention. It does not exclude continuation to a non-ball domain avoiding the obstructing fixed points, and does not assert that every coefficientwise continuation is injective ([warning 14.23](#)).
- N42. (05) An Abel function is not automatically a globally injective coordinate: a locally univalent primitive need not be globally univalent ([warning 15.8](#)).
- N43. (06) “Analytic” is a *defined* term meaning the entire-coefficient Taylor-compatible realization of [proposition 3.11](#). The source does not assert fine-topological continuity of nonconstant ordinary-time paths in $\mathbf{No}[i]$, and does not identify phase differentiation with an intrinsic

derivation of the surreal coefficient field ([warnings 3.2](#) and [16.15](#)).

- N44. (06) [Theorem 16.21](#) extends the polynomial-coefficient normalizer, Hamiltonian and constructed actions *only*. It does **not** license evaluating an arbitrary entire function at an infinite phase or action argument; a centralizer statement on D_ρ must be made in the rescaled entire-coefficient algebra ([warning 16.22](#)).
- N45. (06) Rescaling gives *guaranteed*, not maximal, domains. The cubic example shows a sharp boundary, but the general maximal-domain problem is left open ([question 25.7](#)).
- N46. (06) [Examples 16.25](#) and [16.26](#) show that none of the three admissibility conditions can be dropped from a theorem of this form. They do *not* assert that finite fibers are necessary for every resummation in some larger, differently specified class.

26.5 No global, all-scale, or topological claims

- N47. (02) No global conjugacy on all of $\mathbf{No}[i]$, no surreal-time flow, no class-sized analytic construction, and no coherence between arbitrary infinite-scale charts is claimed; [proposition 21.2](#) supplies a concrete critical-scale test that any such extension must pass.
- N48. (03) No all-scale theorem. The common-domain results concern finite halos and the germ/formal results the infinitesimal monad; [proposition 21.1](#) explicitly blocks automatic extension to infinite surcomplex arguments. No global analytic atlas on $\mathbf{No}[i]$, no ordinary contour theory on its fine topology, and no fine-topologically continuous complex-time flow is constructed.
- N49. (03) There is no assertion that every set-theoretic or algebraic conjugacy of evaluated functions must arise from a coefficient family; failure in one specified analytic category prohibits no set-theoretic conjugacy, and the formal-coefficient conjugacy always exists under nonresonance and can be evaluated on the monad ([proposition 6.14](#)). No interchange of an infinite ordinary coefficient sum with a non-locally-finite Hahn family is permitted.
- N50. (03) No claim of compatibility with the Berarducci–Mantova surreal derivation or with any global composition operator on $\mathbf{No}$; no claim that these coefficient categories fall outside every framework of Cluckers–Lipshitz analytic structures [28]; and no sectorial or resurgent summation is asserted.
- N51. (04) No global Julia or Fatou theory on the proper class $\mathbf{No}[i]$, no classification of arbitrary surcomplex dynamical systems, and no ordinary topological convergence in the full surcomplex plane is used anywhere. A global symbolic description of a different class of maps, polynomials $q^{-1}P$ with $v(q) > 0$ and simple split reduction, is in `docs/surcomplex/expanding-polynomial-dynamics/` (`epd:thm:fibers`, `epd:thm:topology`). It is not a Julia or Fatou theory either (`epd:cor:empty-affine`), and nothing there bears on the germ linearization of this report.
- N52. (04) The formal logarithm is *not* identified with a convergent autonomous flow at a numerical parameter; [7] is cited precisely to keep formal embedding and analytic embedding apart ([warning 14.4](#)).
- N53. (05) The slow-time theorem gives nothing past the ordinary singularities of the supplied flow chart, licenses no general exchange of ordinary with Hahn summation, and gives no interpretation of an arbitrary surreal time as an actual continuous trajectory ([warning 15.27](#)).
- N54. (05) No ordinary analytic specialization: growth estimates and a summation theory are neither supplied nor claimed; sectorial analytic moduli of ordinary parabolic germs are not removed; the analytic embedding obstruction for ordinary parabolic germs is not resolved; and which Hahn normal forms admit compatible ordinary analytic realizations is a distinct problem ([question 25.8](#)).

- N55. (06) No ordinary analytic convergence in a complex perturbation parameter is inferred: substituting $t^\eta = \varepsilon_0 \in \mathbb{C}^\times$ need not exist on the coefficient algebra and can turn a valid Hahn series into a divergent ordinary series.
- N56. (06) No fixed real-valued norm and no sequential completeness of K_Γ is assumed or available; all control is by supports, and the proper-class field $\mathbf{No}[i]$ is reached only through compatible set-sized workspaces, never as a single object.

26.6 Unresolved problems recorded as such

- N57. (02) The negative common-domain result *requires* multiplier drift $F'(0) = \lambda + u$ and two rationally independent positive exponents. The source does **not** settle the fixed-multiplier problem when $0 < \sigma < \infty$ — “eliminating drift removes the direct reciprocal-jet mechanism” — and does not claim the necessity direction for a fixed cyclic value group ([warning 2.6](#) and [question 25.2](#)). Both are source 02’s limitations and are kept as such. Source 08 later supplied the one-variable fixed-cyclic necessity, still with drift ([theorem 8.15](#)). Sources 08 and 09 settled the scalar fixed-multiplier case positively ([theorem 7.13](#)), and source 10 the nonscalar diagonal one, with radius at least $Re^{-r_{\text{ang}}\tau}$ ([theorem 7.23](#)).
- N58. (03) The intermediate regime $0 < \tau < \infty$ for *common-domain* germs is explicitly unresolved ([question 8.20](#)), and the radius bound ([6.12](#)) is a lower bound only, stated *not* to prove that radius collapse occurs ([warning 6.10](#)). This is source 03’s limitation and is kept as such. The regime has since been settled for a scalar multiplier by sources 08 and 09 ([theorem 7.13](#)), and for a nonscalar diagonal one by source 10 ([theorem 7.23](#)); only the optimal radius when $r_{\text{ang}} \geq 2$ remains open ([question 8.20](#)).
- N59. (03) No universal analytic Poincaré–Dulac statement for resonant multipliers of infinite order is proved ([question 25.3](#)).
- N60. (03) Higher-dimensional positive centralizers are reduced to commuting positive vector fields but *not* classified; the one-variable meromorphic-ratio argument does not extend ([question 25.4](#)).
- N61. (04) The moving-fixed-point case $w + \varepsilon V + \varepsilon^2 R$ is open and named as future work; the Euler family “avoids that additional difficulty rather than conceals it”. Several variables is open, the scalar quotient α_c/W having no direct replacement ([question 25.5](#)).
- N62. (05) Several variables: no single reciprocal time one-form encodes a vector field and commuting fields need not be constant multiples, so [theorem 4.12](#) and the finite-period normal form are explicitly *not* to be transported ([question 25.4](#)).

26.7 What the computations do and do not establish

- N63. (01) The 54 finite checks do not establish the arbitrary-rank support lemma, algebraic closure, maximality for every polynomial, the transfinite coherent lifting, or novelty; they are not an implementation of general Hahn fields and not proof-assistant verification.
- N64. (01) The finite procedure of [section B](#) is relative to oracles: equality and valuation of arbitrary Hahn expressions need not be effective, root-of-unity recognition for an arbitrary complex coefficient needs a presentation allowing that decision, d^q may be very large, and treating a truncated zero as an exact zero would invalidate the maximum. It is an exact characterization and a certificate, not a polynomial-time complexity theorem.
- N65. (02) The finite checks test algebraic identities, signs, normalizations and truncation only. They do *not* prove Hahn summability, well-founded recursion, ordinary analytic domain preservation, an infinite small-divisor limsup, the exact convergence radii, or novelty; and the

- multivariable test verifies nonzero denominators only in its finite range, not nonresonance for all multi-indices of its sample tuple.
- N66. (03) The 21 checks are finite polynomial jet identities truncated by *total parameter degree*, explicitly not a Hahn valuation cutoff in a higher-rank group. They do not test arbitrary well-ordered supports, analytic radius claims, continued-fraction asymptotics, the universal quantifiers, or priority, and no proof-assistant verification is claimed ([warning 22.1](#)).
 - N67. (04) The 2,764 exact assertions are finite consistency checks only. No finite test proves Neumann summability, existence on an arbitrary ordinary domain, compatibility for arbitrary value groups, or the algebraicity dichotomy; the program is neither a complete symbolic surcomplex CAS nor a numerical evaluator at arbitrary surreal inputs.
 - N68. (05) `verify.py` is transcription and algebra checking only: it proves nothing about arbitrary-rank support finiteness, global primitive existence, the classification, or the slow-time theorem. **Fifty of its 245 assertions are finite lexicographic illustrations for $n = 1, \dots, 50$; the all- n inequality is proved in the text, not inferred from them.**
 - N69. (05) The verifier loses one top coefficient on division by ε (its division routine sets the new top coefficient to zero), so reciprocal and pullback comparisons are checked only through degree $N - 1$.
 - N70. (06) The 119 checks truncate by *total label degree* in the two independent labels, not by Hahn valuation, and only to degree four. They are explicitly not a proof of arbitrary-support summability, nor of [theorems 16.5](#) and [16.16](#), nor of [theorem 16.12](#), nor of the Liouville construction [proposition 16.17](#).

26.8 The quasi-periodic source

- N71. (07) Self-declared as proposed original results with detailed proofs, unrefereed, and not formalized in any proof assistant. Its README and title page both state that the exact theorem package was not located in a *focused* literature review and that this does not establish priority; independent mathematical review is called for before correctness or originality is treated as settled. Its classical inputs — ordinary Fourier division with its subexponential criterion, the Liouville construction method, positive-support Hahn inversion, and the finite Jacobian identities — are named as classical and are not claimed [[19](#), [20](#), [2](#), [4](#)].
- N72. (07) A *coefficientwise Hahn* theorem only. It explicitly does **not** assert convergence after substituting a nonzero real parameter for a Hahn monomial, and does not improve classical real analytic KAM or Brjuno arithmetic [[19](#), [21](#)]. No bound uniform in the Hahn exponent on the analytic seminorms $\|f_\beta\|_r$ is supplied, and growth faster than every exponential in the exponent index is permitted ([section 19.7](#)).
- N73. (07) The torus derivatives are external derivations annihilating the Hahn scalar field, deliberately **not** the Berarducci–Mantova derivation on **No** [[27](#)], so nothing in [sections 17](#) to [19](#) speaks to internal surreal differential equations ([warning 3.2](#)).
- N74. (07) **A support bound is not summability.** The family of reciprocal divisors indexed by the Fourier modes is stated to be in general *not* Hahn-summable; only a common support is proved ([theorem 17.5\(iii\)](#), [remark 17.8](#)), with ordinary Fourier summation performed inside each Hahn coefficient and never interchanged with it ([warning 17.4](#)).
- N75. (07) No class-sized objects. Everything is a set-sized workspace, with the full surreal field entering only through the embedding [\(3.1\)](#); no continuity into the fine topology of **No**[i] and no class-indexed analytic limit is claimed. The invariant density [theorem 19.8](#) is coefficientwise integration of analytic functions — **not** a countably additive **No**-valued measure on arbitrary subsets and **not** a real-time ergodic theorem. The report `docs/surreal/`

- `hahn-valued-measures-and-probability/` shows that in the strong sense no invariant probability exists for an ordinary action all of whose orbits are infinite (`meas:thm:orbits`), and discusses this density there (`meas:rem:dynamics`).
- N76.** (07) The nonlinear constant normal form requires a trivial exact resonance lattice. The resonant case is left open, and the source warns that replacing the mean projection by the resonant projection is not justified, because the retained normal form is then no longer a constant vector field ([question 25.9](#)).
 - N77.** (07) The flag is finite but *not* automatically computable: detecting exact integer relations among arbitrary named reals may be unavailable, so existence of the flag is explicitly separated from decidability of its entries ([question 25.11](#)).
 - N78.** (07) Three extensions are listed as needing further work — quantitative coefficient-growth classes preserved under the construction, the resonant normal form, and effective arithmetic certificates — and the source states that these arise from the article and are *not* claimed to be catalogued open problems ([questions 25.9](#) to [25.11](#)).
 - N79.** (07) The finite checks verify neither the irrationality of the Liouville constant ([18.6](#)), nor any infinite-mode subexponential estimate, nor the word lemma, nor the infinite-support conjugacy theorem.
 - N80.** (07) An instance of this report’s standing hazard, but handled: the source cites *no* invariance of any norm across a value-group extension. The robustness of [theorem 18.7](#) under enlarging Γ is argued from uniqueness of the modewise Hahn inverse rather than assumed ([warning 18.8](#)).

26.9 Sources 08 and 09: exact and drifting multipliers

Source 08 states sixteen limitations and source 09 twelve. Where they overlap (status, priority, ordinary specialization, the nonscalar case), both wordings are kept, because source 09 states its own for one variable. Source 09’s scope limitation is re-scoped rather than dropped: it is the limitation of source 09, and after the merge the scalar several-variable case is covered by source 08. The nonscalar case, which both sources leave open, is covered by source 10 except for the optimal radius when $r_{\text{ang}} \geq 2$ ([section 26.10](#)); **N81** and **N97** are kept as sources 08 and 09 state them.

- N81.** (08) Only the scalar case λI_d of the exact-multiplier question is proved. The general nonscalar diagonal case is not resolved, and no assertion of success or failure is made there. The rational-clustering proof does not apply as written to the divisors $\lambda^\beta - \lambda_j$ ([section 7.6](#)). *Re-scoped after source 10’s merge:* the nonscalar case is [theorem 7.23](#), proved by a different argument, and only its optimal radius for $r_{\text{ang}} \geq 2$ is open (**N114**).
- N82.** (08) The cyclic theorem is settled only in the one-dimensional drifting setting in which the repository question was posed ([section 8.6](#)); the several-variable collapse over a fixed cyclic group is not addressed.
- N83.** (08) Neither theorem gives ordinary analytic convergence after t^s , or the cyclic parameter, is replaced by a nonzero complex number. There is no general specialization map from positive Hahn series to values at a nonzero complex parameter: after the replacement, infinitely many contributions may enter one ordinary coefficient and need not be summable.
- N84.** (08) Holomorphy is asserted on the *open* polydisk only. There is no boundedness and no continuation across its boundary, and the constants and polynomial exponents of [proposition 7.11](#) may grow without bound as the Hahn weight varies.
- N85.** (08) In the chain estimate the exponential constant is uniform, but the polynomial factor (n^{3r} in source 08’s form) is not uniform in the chain length. No estimate suitable for summing all depths in an ordinary complex parameter is provided.

- N86. (08) Sharpness is for the *coherent coefficient representation* only. There is no claim that an individual map admits no continuation, or that no differently defined conjugacy exists on a larger class of surreal points.
- N87. (08) Common-domain collapse means failure of a specified analytic representation on ordinary neighborhood halos. It does *not* mean the absence of every surcomplex linearization: evaluation on the monad survives ([proposition 6.14](#)). No globally Hahn-entire nonpolynomial map on all of $\mathbf{No}[i]$ is constructed.
- N88. (08) The unequal-radius statement, with polydisk radii $R_j e^{-\tau}$, is a coordinate-rescaling corollary, not a claim about optimal arbitrary Reinhardt domains.
- N89. (08) The results do not replace the classical Bruno–Yoccoz theory. No growth estimate for the quadratic scalars u_m as $m \rightarrow \infty$ is asserted. The bounds are deliberately not uniform in depth, and the classical tree-expansion and Bruno estimates control a different accumulation problem.
- N90. (08) All derivatives act on the ordinary coordinate functions and annihilate the Hahn coefficient field. Nothing is claimed about the Berarducci–Mantova derivation, the differentiation of ω , or internal surreal differential equations ([warning 3.2](#)).
- N91. (08) The executable checks do not prove the asymptotic small-divisor rate, the existence of irrationals of prescribed rate, holomorphy on a common domain, strong summability over arbitrary ordered groups, or the impossibility of cancellation along an infinite convergent subsequence. They establish no novelty and replace no peer review, and no Lean or other proof-assistant verification is included. The test multiplier $(3 + 4i)/5$ is a convenient exact value, not an approximation to a positive-rate example. The finite enumeration ranges are stated so that the record cannot be mistaken for the universal assertions.
- N92. (08) $Re^{-\tau}$ is a universal guarantee, not a formula for every particular input. Input-specific improvements are a different question.
- N93. (08) There is no uniform-depth, Gevrey, or joint analytic convergence result in a complex parameter and the coordinate. The estimates allow the constants to depend arbitrarily on a fixed support word.
- N94. (08) Priority is not certified, and the literature and repository comparisons are targeted. The repository was read at the pinned commit through selected files and line ranges, not line by line. [16] was inspected only at abstract and metadata level because its full text could not be retrieved, and [13] only at metadata level. A prior theorem on formal families, a refinement of the tree estimates, or an uninspected paper could subsume a specialization of the results.
- N95. (08) The lower-bound proof of [theorem 8.15](#) claims no termwise positivity. The error may contain many terms, and some of them may be large ([remark 8.16](#)).
- N96. (08) Status: proposed research results with proofs, not refereed or proof-assistant verified, and not claims of bibliographic priority. The open status of the two repository questions is relative to the pinned revision, and neither is called a famous published conjecture.
- N97. (09) One dynamical variable only; source 09 does not settle the several-variable case. “Several” in [corollary 7.20](#) refers to formal parameters, not dynamical variables. Source 09 also states that the one-dimensional constant $e^{-\tau}$ may not be imported unchanged into a nonscalar several-variable problem. *Re-scoped after the merge*: the scalar case λI_d in every d is source 08’s [theorem 7.13](#), and the nonscalar case remained open ([N81](#)) until source 10 ([theorem 7.23](#)), which gives the constant $e^{-\tau}$ when $r_{\text{ang}} = 1$ and $e^{-\tau_{\text{ang}}\tau}$ in general.
- N98. (09) The drifted multiplier and the fixed-cyclic-value-group question are not covered. Source 09 neither reproves the drift result nor extends it to the fixed cyclic setting; the latter is source 08’s [theorem 8.15](#).

- N99.** (09) Not covered: resonant multipliers, arbitrary infinite surcomplex multipliers, mixed characteristic, moving base points, and nonidentity ordinary nonlinear reductions.
- N100.** (09) No maximal noncircular continuation domain, no universal boundary regularity statement, and no classification of the inputs that happen to linearize when $\tau = \infty$. The optimality witness proves sharpness of the common *centered Taylor disk* only. It does not claim a natural boundary, or that continuation through every point of the convergence circle is impossible.
- N101.** (09) The formal support recursion is not claimed to be uniformly computable from arbitrary presentations of Γ , arbitrary real θ and arbitrary coefficient functions. The finite sets $\mathcal{W}_S(\gamma)$ and the constants C_a exist, but a finite algorithm needs effective access to them or suitable certificates.
- N102.** (09) $(C_a n)^m$ is not harmless uniformly in m . For $\lambda z + t^\eta z^2$ one has $m = n$ and the factor is n^n , so no ordinary analytic convergence follows. Specialization of t^η at a nonzero complex number is not justified; for a non-Bruno λ it produces a divergent linearizer.
- N103.** (09) There are no uniform coefficient-norm bounds: the constants and polynomial degrees need not be uniform in the exponent. The halo statements are coefficientwise. They assert no convergence of complex-parameter specializations and no convergence of partial sums in the fine topology of $\mathbf{No}[i]$.
- N104.** (09) The tests use the multipliers 2, $2/3$ and -2 , which are not on the unit circle. They check tree factors, signs, multiplicities and composition formulas only, not the unit-circle radius estimate. The program proves neither the support lemma, nor the general packing inequality, nor convergence on a disk, nor any small-divisor rate, nor workspace invariance, nor novelty. No Lean code is supplied. The four-layer Lean route the source sketches is an implementation plan, not a completed formalization, and no existing Lean declaration is certified to have the interface it needs.
- N105.** (09) Novelty was checked by a targeted comparison only, at the pinned commit and without a line-by-line audit of the repository. The tree method is credited to classical tree expansions [16, 18]. A specialist may identify a theorem in the parameter-dependent linearization literature that subsumes the result. The draft is AI-assisted and not peer reviewed.
- N106.** (09) “Unique” means unique among tangent-to-identity formal coordinates or their coherent coefficientwise analytic realizations, not among arbitrary set-theoretic maps.
- N107.** (09) Its two routes share the essential arithmetic lemma. They are not two independent proofs of it (section 7.4).
- N108.** (09) The result is not a claim that all small-divisor radius collapse is illusory. It identifies the role of exactness in preventing collapse, in one variable.

26.10 Source 10: the nonscalar diagonal multiplier

Source 10 states seventeen limitations, collected from its title page, body, README and provenance note. Where one overlaps a limitation of source 08 or 09 (status, priority, ordinary specialization, uniformity in depth), both wordings are kept, because source 10 states its own for a nonscalar multiplier.

- N109.** (10) Status: an AI-assisted, unrefereed research draft. Its new arguments have not been checked in a proof assistant or independently peer reviewed, and no Lean formalization, Wolfram verification or external peer review was performed or is implied. A successful typesetting run verifies document construction, not the mathematics.
- N110.** (10) The question it answers is a specific question in an unpublished repository report, this

- one (question 8.20), not a named published conjecture.
- N111.** (10) Priority is not certified. The literature comparison was targeted: the HTML text of the 13 September 2026 revision of [18] was inspected, especially its tree formulas, admissible cuts and divisor estimates, while [17] was seen only through its abstract and metadata, with no full-text audit. The rational-rank formulation was not located in that comparison. This is a bounded novelty assessment, not a bibliographic proof, and an earlier equivalent result under other terminology remains possible.
- N112.** (10) The repository comparison is bounded: the document catalogue and selected parts of this report's source at the pin (abstract and provenance, support and halo machinery, the homological operator and category classification, the depth-dependent radius theorem with its nonscalar limitation, the scalar tree bound, and the verification discussion). It is not an audit of every report, every Lean declaration, or every manuscript kept only in Git history.
- N113.** (10) Not claimed new: Hahn fields and the positive-support lemma (Higman's theorem imported, Neumann credited), formal homological inversion, rooted-tree methods and admissible-cut preservation [18, 17], surreal normal forms [5], Cauchy estimates, the root test, elementary continued fractions including the prescribed-rate family, and the algebraic-number estimate of example 7.36. The scalar common-domain theorem and the scalar fixed-complexity rate are already in this report and are not claimed; the general upper bound $Re^{-\tau}$ and the $\tau = \infty$ obstruction are first-weight obstructions already here, included only to complete the criterion.
- N114.** (10) For $r_{\text{ang}} > 1$ the factor r_{ang} is a proved bound, not an optimality assertion. The optimal universal radius lies between $Re^{-r_{\text{ang}}\tau}$ and $Re^{-\tau}$ and is undetermined. The proof does not show that several rate values can saturate their charges at once, or that saturation survives the cancellations of an actual conjugacy. The theorem answers the existence of a common radius, not the optimal general nonscalar radius.
- N115.** (10) Holomorphy is asserted on the *open* polydisk only; nothing is asserted about continuity up to its boundary.
- N116.** (10) Uniqueness is in the normalized formal positive-support class. No classification of nonnormalized, non-Hahn-coherent or purely set-theoretic conjugacies is given.
- N117.** (10) The prefactor $C_{m,\varepsilon,\Lambda}$ of theorem 7.27 comes from a subsequence argument: it is finite but has no computable formula. No bound uniform in m follows, and replacing m by n in (7.23) and dropping the prefactor is invalid.
- N118.** (10) Lemma 7.24 needs a finite family of vector sequences, and asserts nothing for a number of sequences growing with the height; and it needs heights growing linearly, since exponentially large heights would invalidate the polynomial Cramer bound.
- N119.** (10) "Sharp" concerns the largest universally guaranteed centered isotropic polydisk for the ordinary coefficient functions. A particular forcing may have much larger domains, and a coefficient may continue analytically outside its centered Taylor polydisk; the isotropic radius is not a statement about maximal analytic continuation.
- N120.** (10) Not covered: infinitesimally drifting multipliers, resonant spectra, nontrivial Jordan blocks, nonunitary spectra in analytic categories, and infinite-dimensional coordinate spaces. The support method is compatible with many of these, but its admissible-divisor and geometry arguments do not transfer automatically.
- N121.** (10) No ordinary analytic convergence is asserted after substituting a nonzero complex number for t . A radius bound for each coefficient gives no control of its size as the Hahn or parameter order grows. The result neither replaces a Bruno-type theorem for ordinary germs nor contradicts the known failures of classical analytic linearization.
- N122.** (10) All derivatives are ordinary coordinate derivatives extended coefficientwise; the

Berarducci–Mantova derivation is not used. Strong summation is an algebraic support operation: no convergence of partial sums in the fine topology of $\mathbf{No}[i]$ is assumed, and nothing is asserted about nonconstant continuous ordinary-time paths. The proof does not reconstruct the proper class $\mathbf{No}$ as one workspace.

- N123.** (10) The failure at $\tau = \infty$ is a failure of ordinary coefficient holomorphy on a common domain. It is not a failure of formal conjugacy at infinitesimal arguments, where a formal series of Taylor radius zero can still be evaluated ([proposition 6.14](#)).
- N124.** (10) The finite checks detect transcription, sign, color, multiplicity and truncation errors. They do not establish [lemma 7.24](#), the limsup of [theorem 7.27](#), analytic convergence, summability for arbitrary supports, or originality. The 1,000 larger configurations are reproducible samples, not an enumeration, and no global nonresonance or asymptotic arithmetic claim is inferred from the finite algebra test.
- N125.** (10) The new analytic theorem is not formalized in Lean. The existence of related support lemmas in the repository’s Lean code does not imply formal verification of it, and source 10’s three-interface formalization outline is a plan, not an implementation.

The list above contains 125 numbered items for the 123 distinct source-level limitations. The two extra entries arise because source 03’s limitation on resonant normal forms is recorded in two places — **N22**, the finite-order hypothesis $\Lambda^q = I$, and **N59**, the openness of the infinite-order case — and because source 04’s hypothesis audit is given its own entry (**N23**) in addition to its fourteen listed non-claims. The distribution of *distinct* source-level limitations is 11/9/10/14/12/12/10/16/12/17 across sources 01 through 10; the corresponding numbered-item counts are 11/9/11/15/12/12/10/16/12/17.

Warning 26.1 (The one sentence this report refuses). No statement of this report says that Hahn or surcomplex coefficients *abolish* the classical small-divisor problem. What is true is stated in [remarks 2.2](#) and [2.3](#): passing to positive Hahn support *separates* two questions that ordinary analytic convergence forces one to solve at once — algebraic dependence across perturbation orders, and ordinary analyticity within each coefficient — and the sharp thresholds describe exactly that separation. Support alone suffices for polynomial and formal coefficients at arbitrary rank ($\mathbf{SD}_0$); arithmetic enters exactly when an ordinary domain or entirety is demanded ($\mathbf{SD}_{\text{dr}}$, $\mathbf{SD}_{\text{ex}}$, $\mathbf{SD}_{\text{fr}}$, $\mathbf{SD}_{\text{ss}}$); and under PAR there is no divisor family at all ($\mathbf{SD}_{\emptyset}$). The quasi-periodic part makes the same point in its sharpest form: [proposition 17.9](#) is a genuine theorem with *no* arithmetic hypothesis whatever, and [theorem 18.3](#) shows that the moment an analytic strip is demanded the arithmetic becomes not merely sufficient but necessary. The two positions are complementary, not opposed, and stating them without their coefficient categories is what would make them look contradictory.

A A short proof of the word-order input

We record the special case of Higman’s lemma used in [lemma 3.4](#), so that the support proof can be audited without a separate combinatorial argument hidden in the notation. It is classical [3]. Sources 02 and 05 each give the same argument; this is the one copy.

Let S be well ordered. A word $(a_1, \dots, a_m)$ *embeds* in a word $(b_1, \dots, b_n)$ if there are indices $i_1 < \dots < i_m$ with $a_j \leq b_{i_j}$ for every j . A sequence of words is *bad* if no earlier word embeds in a later one. Suppose an infinite bad sequence exists. Choose a bad sequence $w_0, w_1, \dots$ minimally by word length at each stage: after choosing a prefix, choose a shortest next word among those admitting a bad infinite extension. The choices take place in a set, so ordinary choice suffices, and no w_n is empty.

Write $w_n = v_n a_n$, separating the last letter. Any infinite sequence in a well-ordered set has an

infinite nondecreasing subsequence — select a position attaining the minimum of the current infinite tail, then repeat on the tail strictly after that position; the successive minima cannot decrease. So choose $n_0 < n_1 < \dots$ with $a_{n_0} \leq a_{n_1} \leq \dots$ and consider

$$w_0, \dots, w_{n_0-1}, v_{n_0}, v_{n_1}, v_{n_2}, \dots$$

This is bad: an earlier w_i embedding in a later v_{n_j} would also embed in w_{n_j} , contradicting the original badness; if v_{n_i} embeds in v_{n_j} for $i < j$, appending the dominated last letters embeds w_{n_i} in w_{n_j} , again a contradiction; and comparisons within the unchanged prefix were already forbidden. The new bad sequence agrees with the chosen one until position n_0 and has the strictly shorter word v_{n_0} there, contradicting the minimal choice. Hence no bad sequence exists, which is the required word well-quasi-order.

When $S \subset \Gamma_{>0}$, a proper word embedding *strictly* raises the sum: either an unused positive letter is present in the larger word, or a matched letter is strictly larger. That observation converts the word theorem into both the well-ordering of S^* and the finiteness of the words of one weight, as claimed in [lemma 3.4](#).

B The finite certificate for the resonant threshold

For polynomial inputs with effective coefficient arithmetic, the geometric certificate of [corollary 9.9](#) can be computed without constructing H at all.

1. Determine $\alpha = \bar{\lambda}$. If it is not a root of unity, compute the finite maximum $\eta(f)$ and return $r = \eta(f)$.
2. If its order is q , compute the finite composition $g = f^{\circ q}$, $c = \lambda^q - 1$ and $\delta = v(c)$.
3. Compute every nonzero nonlinear return coefficient b_n *including actual cancellations*, and take the maximum [\(9.6\)](#).
4. Scale by $\rho = t^r$ and reduce [\(9.9\)](#) to obtain P . Its degree gives the shell multiplicity; factoring P determines the residue directions; and $XP'(X)$ gives the leading multiplier ratios at simple roots.

This is a finite algebraic procedure *relative to the required oracles*, and not a claim of universal computability for arbitrary surcomplex inputs: see **N64** of [section 26](#).

C A coefficientwise recipe for a certified linearizer coefficient

Suppose one wishes to compute h_γ for polynomial input in [theorem 6.1](#), with an effective presentation of the finite word set $\mathcal{W}_S(\gamma)$. The following specifies what must be certified.

Collect the finite letter set $D_S(\gamma)$ and the finite set of partial sums of all words of weight γ : only input coefficient polynomials at these letters can matter. The maximum word length gives the bound $L = \ell_S(\gamma)$. In [\(6.5\)](#) retain only $k = 0, \dots, L - 1$, and in each Taylor expansion of $\mathcal{N}_f$ retain only products whose support words can complete an allowed word of total weight γ ; polynomial degrees can be bounded in advance by [\(6.13\)](#). Apply $\mathcal{L}^{-1}$ to each retained monomial by the exact scalar division [\(5.2\)](#). Nonresonance of the finitely many divisors actually used must be established, or supplied as an exact symbolic condition. Sum the resulting finite expressions and check Schröder's equation at the requested weight; the word certificate proves that discarded expressions cannot affect it. This is a finite exact algorithm *relative to the support certificate and the polynomial coefficient data*.

For analytic input the same support certificate still applies, but $\mathcal{L}^{-1}$ is an operator on an entire coefficient function, not merely a finite list of Taylor coefficients; a radius certificate from [theorem 6.9](#) is then part of the analytic output. A finite jet calculation alone must *not* be described as computing the entire analytic coefficient, nor as proving its convergence ([warning 6.12](#)).

The parabolic counterpart, for [theorem 15.13](#), is explicit in the same sense. Given F_1, F_2 in one nonsingular stratum, form their logarithmic vector fields by (4.4), compute the normalized reciprocal forms $\beta_i = b_i dz$ by the geometric unit expansion of [lemma 14.2](#), and test the periods of $(b_1 - b_2) dz$; on a finitely punctured plane that is a finite list of Hahn periods, though computing an arbitrary coefficient integral is not assumed decidable. When the periods vanish, integrate coefficientwise from z_* to get p with $p' = b_1 - b_2$, $p(z_*) = 0$; with $b_2 = b_0 + u$ and S the positive union of the supports of p and u , the desired $h = H - z$ has support in $S^* \setminus \{0\}$ and, for each γ in that well-ordered set,

$$h_\gamma = b_0^{-1} \left(p_\gamma - [t^\gamma] \left[u h + \sum_{n \geq 2} \frac{b_2^{(n-1)}}{n!} h^n \right] \right), \quad (\text{C.1})$$

the bracket using only earlier coefficients of h . The number of operations at each exponent is finite, but the total support may be transfinite and need not have a finite presentation. For an ordinary single-parameter expansion $b_2 = b_0 + \varepsilon b_1 + \varepsilon^2 b_2^{[2]} + \dots$ and $p = \varepsilon p_1 + \varepsilon^2 p_2 + \dots$ the first coefficients are

$$h_1 = \frac{p_1}{b_0}, \quad h_2 = \frac{1}{b_0} \left(p_2 - b_1 h_1 - \frac{1}{2} b_0' h_1^2 \right),$$

displaying both sources of nonlinear correction: the tail of the target time form, and the change of its ordinary leading coefficient along the displacement. The support certificate is part of the result — it is what makes each coefficient defined even when the multiples of the smallest input exponent are not cofinal — and it is *not* a promise that arbitrary Hahn-series equality, primitive existence or period evaluation can be decided by a finite program.

D The single-exponential gauge recursion as an algorithm

For a finite collection of input labels $x_1, \dots, x_k$, an implementation of [theorem 16.7](#) can work over the ordinary formal label ring through a chosen total label degree N :

1. set $\chi = 0$ and retain $H = H_0 + R$ with independent labels attached to its input layers;
2. for $n = 1, \dots, N$ compute $e^{\text{ad}_\chi} H$ through total label degree n , using the current χ of label degree less than n ;
3. for each coefficient E_ν with $|\nu| = n$, append $-L^{-1} Q_\Pi E_\nu$ to the coefficient of x^ν in χ ;
4. recompute $e^{\text{ad}_\chi} H$ through the requested order, and compute coordinates by the same exponential and actions by the inverse exponential.

Adding a degree- n generator changes the degree- n transformed Hamiltonian only by $L\chi_n$, so step 3 is the *single-exponential gauge* recursion — **not** the different algorithm that composes a list of successive exponential maps without taking their logarithm ([warning 16.10](#)). For an infinite input label set this is a conceptual construction rather than a finite executable loop; homogeneous leaf degree still organizes the universal expression trees of [definition 16.3](#), and at a specified Hahn weight [lemma 3.4](#) permits only finitely many leaf words, hence finitely many homogeneous leaf degrees and finite expressions. That is the exact connection between the finite-label algorithm and the arbitrary-support proof. Total label degree is not a valuation cutoff ([warning 22.1](#)).

E Antecedent ledger

For audit, the external inputs and their exact roles. None is claimed as a contribution of this report.

Input	Role here
Neumann’s positive-support lemma [2], via Higman’s word well-quasi-order [3], with Poonen’s account of Hahn–Mal’cev–Neumann units [4]	Lemma 3.4 , corollary 3.5 , and section A : the certificate behind <i>every</i> infinite operator expression in this report. Classical; used as a black box for Higman’s lemma beyond the reproved special case.
Algebraic closure of a maximally complete Hahn field [4, Cor. 4]	Lemma 9.4 and theorem 12.1 only. Not needed for section 16 .
Conway normal forms and surreal normal-form arithmetic [5]	(3.1) and the workspace discipline. No set of all surreals is formed.
Formal logarithms, flow embedding, Jordan–Chevalley / semisimple–unipotent splitting [6]	The classical pattern behind theorems 4.3 and 20.1 . What is established here is the arbitrary-positive-Hahn-support, same-coefficient-domain implementation, not the pattern.
Formal interpolation of near-identity maps versus analytic embedding at a numerical parameter [7]	Cited exactly to mark the line that warning 14.4 draws.
Exponentiating strongly summable derivations of generalized-series algebras [8]; automorphisms of the surreals [9]	The general algebraic mechanism behind theorem 4.3 and proposition 16.2 . The existence of a formal logarithm is credited, not claimed.
Julia’s equation and differential transcendence of iterative logarithms [10]	Background for corollary 4.6 and section 15 . The parameter-Hahn construction here is a different category; the analytic embedding obstructions for ordinary parabolic germs are not removed.
Ordinary homological equation, small divisors, formal-versus-analytic linearization [11, 12]; the Brjuno–Yoccoz quadratic theorem [13, 14]	Section 5 ; and Brjuno–Yoccoz used as an <i>external</i> theorem, not reproved, in propositions 11.1 and 21.1 and remark 8.10 .
Regular dependence of ordinary linearization on the multiplier [15]; explicit tree expansions [16]; linear versus nonlinear arithmetic conditions in regularity classes [19]	Natural parameter-regularity context for the reciprocal jets (8.4). No Whitney estimates, no joint analytic parameter convergence, and no novelty claim for tree-organized formal coefficients.

Input	Role here
Davie's small-divisor estimates as discussed by Marmi [12]; tree expansions for one- and multi-dimensional linearization and their Bruno estimates [16, 18]; continued-fraction best approximation [12, 15]; Poonen's positive-support geometric-series theorem [4, Thm. 4.1]	The classical background of sections 7 and 8.6 . Sources 08 and 09 credit the counting and tree methods and claim only the fixed-complexity form theorem 7.5 , the sharp radius theorem 7.13 and the collision noncancellation proposition 8.14 . Source 08 inspected [16] at abstract and metadata level only, [18] in its version 2 of 13 September 2026, and [13] at metadata level. No priority is certified (N94 , N105).
Preservation of admissible cuts in operator trees [18, Prop. 5.6]; Carletti's tree formulas for Lagrange inversion and linearization over non-Archimedean formal-series structures [17]; integer minors, Cramer's rule and norms of algebraic integers	The classical background of sections 7.7 and 7.9 and example 7.36 . Source 10 claims only the rate count lemma 7.24 , the product bound theorem 7.27 and the common domain they give. It inspected the HTML text of [18] and only the abstract and metadata of [17]. No priority is certified (N111).
Equal-characteristic-zero ultrametric linearization, sharp disk radii, and p -adic bounding spheres without periodic points [22, 23]	The rank-one antecedent of section 9 and the source of the cross-note section 23.5 and the characteristic warning warning 2.7 .
Modified differential equations in backward error analysis [24]	The origin of the coefficients (15.31), which are known examples (warning 15.22); its warning that the ordinary step-size series generally diverges is cited and accepted.
Convergence and generic divergence of Birkhoff normal forms [25, 26]	Classical context for section 16 ; ordinary analytic convergence is never inferred from a Hahn identity.
Transseries as germs of surreal functions [27]; fields with analytic structure [28]; Hahn-meromorphic function theory [29]	Adjacent frameworks, kept distinct: see warning 3.2 and N50 of section 26 . No theorem is transferred between categories without a separate proof.
Linear versus nonlinear arithmetic conditions for torus vector fields [19]; the one-frequency cohomological equation and its Liouville obstructions [20]; the Brjuno condition through best approximations [21]	The classical small-divisor background of sections 18 and 19 : ordinary Fourier division and its subexponential criterion, and the Liouville mechanism adapted in theorem 18.7 . Neither the ingredient nor any strengthening of ordinary analytic KAM arithmetic is claimed (N72 , section 19.7).

Input	Role here
The motivating repository at its pinned snapshot [1]	Sources 01–06 cite it at commit 39f2be6667ade51bca2b45daa47e289d69c09764 and source 07 at aa846271b4dcae2c055b216126a87210292ec19b, for the coefficient-category distinction only. Sources 08 and 09 cite it at a3124af79f66b8b9c196d76b4cbc5ac3938907c4 for the statements of questions 25.1 and 25.2 and the constructions they reuse. Source 10 cites it at 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0 for question 8.20 , theorem 6.9 and the scalar results of section 7 , which it cites rather than reproves. None uses an unreferenced repository theorem as a premise. The inspections were targeted, not exhaustive (N9 , N13 , N71 , N94 , N105 , N111).

F Notation at a glance

Symbol or condition	Meaning and role
$\Gamma, K_\Gamma, t^\gamma$	Nonzero set-sized ordered abelian group; $K_\Gamma = \mathbb{C}((t^\Gamma))$; $t^\gamma = \omega^{-\gamma}$ for the surcomplex reading (3.1). Divisibility and algebraic closure are assumed only where flagged.
$v, \mathcal{O}_\Gamma, \mathfrak{m}_\Gamma$	$v = \min \text{supp}$ with <i>increasing</i> supports (warning 2.5); $v \geq 0$ and $v > 0$. Trivial on nonzero complex constants (remark 2.3).
$+, > 0$ subscript	All nonzero support exponents strictly positive. Not positivity of ordinary complex coefficients.
B_r, S_r, D_ρ	Valuation ball, shell and polydisk; larger threshold means <i>smaller</i> domain (section 2.4). S_r is not a topological boundary.
$\mathbb{D}_R, \mathbb{D}_R^d, D, U$	Ordinary disk, polydisk or open set; larger R means <i>larger</i> domain. A domain for coefficient functions only (warning 2.4).
$U_\Gamma^\#, D^\#, \mathfrak{m}_\Gamma^d$	Finite halo and infinitesimal monad; evaluation (3.4) is a faithful algebra homomorphism (proposition 3.11).
$\text{rad}, \text{rad}_{\text{iso}}$	Ordinary Taylor radius and isotropic Taylor radius of one coefficient function; larger is better.
$S^* = M(S), \mathcal{W}_S(\gamma), \ell_S(\gamma), D_S(\gamma)$	Additive monoid, ordered words of weight γ , maximum word length, occurring letters (lemma 3.4). $\ell_S(\gamma)$ is finite even when infinitely many support elements precede γ .
$\mathcal{W}_{B,S}(\gamma), \ell_{B,S}(\gamma), E_{B,S}(\gamma), D_{B,S}(\gamma)$	Input-exponent/word pairs reaching γ from B , their maximum word length (zero if there are no pairs), their starting exponents, and their positive letters. These are the relative data for arbitrary Hahn inputs in corollary 3.5 .
$\mathcal{B}_M$	Uniform-monoid support algebra (definition 3.8); the uniformity across Taylor degrees is load-bearing.
$\mathcal{L}, \mathcal{L}_L, L$	Homological operators (5.1), (16.4); $\mathcal{L}^{-1}$ preserves Hahn exponents and total degree.

Symbol or condition	Meaning and role
$\sigma(\lambda), \tau(\Lambda), \Theta(\omega)$	The three divisor rates: drifted, exact-multiplier, frequency (section 2.3). σ, τ are log-rates in $[0, \infty]$; $\Theta \in [1, \infty]$ is the exponential of a rate.
$\mathfrak{d}_n, \ x\ _{\mathbb{Z}}, q_*, C_a$	$ \lambda^n - 1 $, so that $ d_{n+1} = \mathfrak{d}_n$; the distance to the nearest integer; the cluster denominator of lemma 7.2 (neither the RES_q order nor a convergent); and the constant (7.5).
$\check{H} = H^{-1} = z + \check{h}$	The forward coordinate of source 09, $F \circ \check{H} = \check{H}(\lambda \cdot)$; source 09's H is this report's H^{-1} (convention 7.1). Source 10 writes it $G = z + g$, with $F \circ G = G \circ \Lambda$ (convention 7.22).
$r_{\text{ang}}(\Lambda)$	Rational angular rank $\dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}}(1, \theta_1, \dots, \theta_d) - 1$ of an exact diagonal multiplier (7.15); $1 \leq r_{\text{ang}} \leq d$ under NR. Not the valuation rank of Γ , not the flag length r , not any other r of this report (convention 7.22).
$\mathfrak{d}_v, \alpha_v, \beta_v, P(T), A_m^\Lambda(n), C_a(\Lambda)$	Divisor at a vertex of an admissible colored tree, its leaf vector, the contracted leaf vector of lemma 7.26, the tree's divisor product, its maximum at complexity m and weight n , and the constant (7.16). $A_m^\Lambda(n)$ is not $A_m(n)$ of corollary 7.6.
$h_{k,\ell}, h_m^{\text{cyc}}$	The coefficient of $u^k v^\ell$ in the two-parameter drifted linearizer, and that of t^m in the single-exponent one; $h_m^{\text{cyc}} = \sum_{\ell} h_{m-\ell,\ell}$ (8.20).
$\text{RES}_q, \text{MR/NR}, \text{FR}, \text{PAR}, \text{QR}$	The five resonance notions (section 2.2). A threshold proved under one is never quoted under another.
$\text{SD}_0, \text{SD}_{\text{dr}}, \text{SD}_{\text{ex}}, \text{SD}_{\text{fr}}, \text{SD}_{\emptyset}, \text{SD}_{\text{ss}}$	The six small-divisor notions (section 2.3).
$\Upsilon, \varpi_\beta, \text{N}_\Upsilon$	The <i>Hahn</i> frequency vector in F_Γ^d , its real coefficient vectors, and its exact resonance lattice $\{k : k \cdot \Upsilon = 0\}$. Never $\omega \in \mathbb{C}^d$ of section 16, and never the exponent Ω of example 3.7 (warning 2.9).
$\text{L}_j, \Sigma_j, \gamma_j, \mathfrak{a}_j, r$	The resonance lattices of the flag, its strata, its visible levels, its leading <i>real</i> forms and its length $r \leq d$ (theorem 17.5). L_j is not the multiplier Λ and not the homological $\mathcal{L}$; Σ_j is not the ordinary time disk Σ of theorem 15.26.
$\text{E}, \text{T}_\Upsilon, \kappa_\Upsilon$	The shifted tail generators, the common reciprocal support $\bigcup_j (-\gamma_j + \text{E}^*)$, and the last visible level γ_r : an <i>element of</i> Γ , never a rate (warning 2.9).
$\mathbb{T}_\rho^d, \mathcal{O}(\mathbb{T}_\rho^d), \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$	The fixed open complex strip, its periodic holomorphic functions, and the fixed-strip Hahn algebra: the torus instance of the fixed-domain row of section 3.2, with one ρ for every Hahn exponent and no uniform seminorm bound.
$\mathcal{C}_\Upsilon, \mathcal{G}_\Upsilon, \Pi_{\text{N}_\Upsilon}, \text{av}, \text{Q}_0$	The cohomological operator $\sum_j \Upsilon_j \partial_{\theta_j}$, its normalized inverse (18.2), resonant Fourier projection, coefficientwise torus mean, and mean-zero projection. Q_0 is not the Hamiltonian Q_Π .
$\mathcal{X}, \mathcal{V}, \Phi_h, \mathfrak{c}, \mu_{\mathcal{X}}$	The perturbed torus field $\Upsilon + \mathcal{V}$, its perturbation, the torus coordinate change $\text{id} + h$ (<i>not</i> the flow Φ_X or Φ_c), the constant frequency correction, and the invariant density (19.10).
r, q, c, δ, P, M	Finite-return threshold (9.6), residue order, $\lambda^q - 1$, $v(c)$, shell polynomial (9.10), its degree.

Symbol or condition	Meaning and role
$\varepsilon = t^\delta, s = t^{\delta/q}$	A single Hahn monomial of positive valuation; <i>not</i> a small complex number. All $O(\varepsilon^k)$ mean divisibility in a formal series ring.
X_F, W_F, Φ_X, Φ_c	Logarithmic vector field, its one-variable coefficient, time-one and time- c maps (theorem 4.3 and corollary 4.8).
$\Omega_F = dz/W_F,$ $\beta_F = \varepsilon\Omega_F$	Time form and its normalization; “holomorphic” refers to the coefficient functions, and the valuation may be negative.
$J_\delta, I_\delta(D)$	Flat ideal and one-scale-jet kernel, formed in $K_{\geq 0}$ and $\mathcal{A}_{\geq 0}$, <i>never</i> in K or $\mathcal{A}$ (warning 15.17).
$\mathcal{B} = \mathbb{C}(w)((t^\Gamma)), \mathcal{L}$	The algebraicity base with arbitrary rational-function coefficients (<i>not</i> $K_\Gamma(w)$; warning 14.1) and the meromorphic ambient field (14.2).
ρ_a, M_a, ρ_q	Exact monodromy exponent and multiplier (14.15), and the model’s common exponent (14.30).
$\Pi, Q_\Pi, \chi, J_j, \Theta(\omega)$	Action projection and its complement, the gauge-normalized generator, the commuting actions, and the frequency divisor invariant.
$w_\rho(\ell), \mathcal{S}_\rho$	Shifted monomial weight (16.19) and the Hamiltonian rescaling operator (16.20), a Poisson Lie but <i>not</i> a ring homomorphism.
$I_{\text{Fix}(F)}$	Ideal generated by $F_j - z_j$, compared by the explicit invertible matrix (4.10), not by its zero set.

G Per-theorem hypothesis audit

Statement	Indispensable hypotheses, and the boundary of the conclusion
lemma 3.4	Well-ordered <i>positive</i> support. No Archimedean hypothesis, no rank-one, no countability, no finitely generated grid. Gives finite words, not a valuation-contraction estimate (warning 3.6).
proposition 3.11	One common ordinary domain for all coefficients; positive support of the infinitesimal increment. Faithful, but polynomial/entire coefficients do not automatically evaluate at infinite surcomplex points.
theorem 4.3	Ordinary reduction the identity; positive perturbative support; arbitrary open U . No ordinary parameter convergence, no uniform coefficient bound.
theorem 4.10	Positive vector field; nonzero admissible time. Ideals compared by (4.11). Says nothing about maps whose reduction is not the identity.
theorem 4.12	$U \subset \mathbb{C}$ <i>connected</i> , one variable, $F \neq \text{id}$; admissible times $v(c) + \alpha > 0$. Not a classification in $d > 1$ (remark 4.13); the centralizer can <i>grow</i> under workspace extension (warning 4.15).
theorem 6.1	NR with the multiplier <i>exactly</i> Λ ; positive support; no degree-0 or degree-1 coordinate terms at any exponent; a fixed polydisk of finite positive radius in row one. Rows are not interchangeable; the sixth category is theorem 7.13 for $\Lambda = \lambda I_d$ and theorem 7.34 for nonscalar Λ , whose optimal radius for $r_{\text{ang}} \geq 2$ is question 8.20 .

Statement	Indispensable hypotheses, and the boundary of the conclusion
theorem 6.9	$\tau < \infty$. Lower bound only (warning 6.10).
theorem 7.5 and corollary 7.6	$\tau < \infty$ for the bound; <i>strictly positive</i> integer weights (warning 7.4); one irrational rotation. The polynomial factor $(C_a n)^m$ is harmless only at fixed m .
theorem 7.13	NR with the multiplier <i>exactly</i> the scalar λI_d ; $f_s(0) = 0$ and $Df_s(0) = 0$ coefficientwise; one common input polydisk; $\tau < \infty$. Open polydisk only, no uniform norm bound, and a universal rather than a per-input radius. Nonscalar Λ is not covered (section 7.6 and table 5); it is theorem 7.23 .
lemma 7.24 and corollary 7.25	A <i>fixed finite</i> family of integer vector sequences with heights $O(N)$ and positive limiting rates. No Diophantine or algebraicity hypothesis on the angles. Counts distinct rates, not vectors.
theorem 7.27 and corollary 7.28	NR; $\tau < \infty$; arities at least two, so every weight is positive (no infinitesimal linear term). Fixed m ; the prefactor is not explicit and not uniform in m .
theorems 7.23 and 7.34	NR with the multiplier <i>exactly</i> a diagonal unitary Λ ; $f_s(0) = 0$ and $Df_s(0) = 0$ coefficientwise; one common input polydisk; $\tau < \infty$. Open polydisk only; a universal radius, proved sharp for $r_{\text{ang}} = 1$; the optimal radius for $r_{\text{ang}} \geq 2$ is open (section 7.13).
corollary 7.32	Convergent-germ inputs; the radius is relative to the finitely many inputs at letters of $D_S(\gamma)$. Lower bound only.
propositions 7.17 and 7.19	One variable; the forward convention $\check{H} = H^{-1}$; otherwise the hypotheses of theorem 7.13 . Shares its arithmetic lemmas with route A.
theorem 6.11	Polynomial input. Holds for any invertible diagonal nonresonant Λ , without $ \lambda_j = 1$.
theorem 6.3	NR; drift permitted; well-ordered positive A ; $f_a(0) = 0$. No uniform bound, no Gevrey estimate, no dependence on t at $t = 0$.
theorems 8.2, 8.5 and 8.7	Multiplier drift required , plus two rationally independent positive exponents; several-variable version needs multiplicative detuning and equal-radius polydisks. As stated, these three are asserted over Γ_* . Source 08's theorem 8.15 removes the second exponent in one variable, over $\mathbb{Z}$, but still requires drift. Theorem 8.7 is not asserted for a fixed cyclic value group (warning 2.6 and question 25.2).
theorem 8.15 and corollary 8.17	Multiplier drift required ; one variable; value group $\mathbb{Z}$, or any fixed nonzero Γ by embedding; the explicit input (8.17). No termwise positivity is claimed (remark 8.16), and there is no ordinary specialization of t (N83).
theorems 9.5, 9.7 and 13.1	RES_q with λ <i>not</i> a root of unity; one variable; polynomial f ; <i>equal characteristic zero</i> (warning 2.7); divisible Γ for lemma 9.4 . “Exact” means the maximal centered valuation ball only.
theorem 10.1	RES_q . Gives one fixed ordinary disk on which P has no zero; no uniform norm bound on (H_γ) ; resonant-only, and not an iff (warning 11.2).

Statement	Indispensable hypotheses, and the boundary of the conclusion
theorem 9.10	Same multiplier λ ; inequalities on the <i>return</i> coefficients; squarefree P for the matching. Sufficient, not necessary (warning 9.11).
theorems 14.3 and 14.15	PAR; simply connected U containing 0; V nonvanishing on $U \setminus \{0\}$ with a <i>simple</i> zero at 0; polynomial Euler form for the dichotomy; divisible Γ . Transcendence over $\mathbb{C}(w)((t^\Gamma))$ only, and not differential transcendence (warning 14.27).
theorem 14.11	A <i>simple</i> zero a ; ordinary loops on the ordinary universal cover. Finite order exactly when $V'(a) = \mu$; the obstruction is visible already at order ε^1 .
theorem 14.22	PAR; the model $f = (1 + \varepsilon)z + z^{q+1}$; divisible Γ . Maximality for <i>centered</i> valuative balls; two distinct failure mechanisms at and below the threshold (warning 14.23).
theorem 15.5	$F \neq \text{id}$. Outside the nonsingular stratum the quotient g/W_F must still lie in $\mathcal{A}$; period vanishing alone is insufficient (example 15.10).
theorems 15.13 and 15.16	Same leading data (δ, a) with a <i>nowhere zero</i> ; near-identity conjugator on the <i>original</i> common domain; ordinary coefficientwise periods; the specified puncture-loop basis (warnings 15.12 and 15.15).
theorem 15.18	δ <i>noncofinal</i> , i.e. $J_\delta \neq 0$; jets formed in $\mathcal{A}_{\geq 0}$ (warning 15.17). Not a statement about ordinary exponentially small asymptotics.
theorem 15.26	A <i>specified</i> ordinary holomorphic flow chart with image in D and nowhere-vanishing variational factor J . Nothing past the chart's ordinary singularities (warning 15.27).
theorems 16.5 and 16.16	Nonresonant <i>ordinary complex</i> ω . Measures one coefficientwise operator; not a Bruno- or KAM-type convergence criterion (warning 16.6).
theorem 16.7	$\neg \text{FR}$; positive well-ordered S ; $\text{ord}_0 R_s \geq 3$; case (a) polynomial layers with no arithmetic condition, case (b) entire layers with $\Theta < \infty$. Uniqueness is gauge-relative (warning 16.10).
theorem 16.12	The <i>entire-coefficient</i> Hahn algebra. Does not classify local, multivalued or fine-topologically locally constant integrals (warning 16.13).
theorem 16.21	Polynomial input admissible at ρ : positivity, well-ordering <i>and</i> finite fibers, all three independent (examples 16.25 and 16.26). Guaranteed, not maximal, domains (warning 16.22).
theorem 20.1	$\Lambda^q = I$ — the multiplier itself of finite order, which is neither RES_q nor mere MR. Resonant normal forms not unique (warning 20.2).
theorem 17.5	Finite dimension d ; <i>one</i> well-ordered support for the whole frequency vector. Without finite d the kernel rank need not reach its minimum in finitely many steps; independent unbounded scalar choices do not supply a shared support. Gives a <i>common support</i> , not summability of the reciprocal family (remark 17.8); and r counts visible scales, not monomials of Υ (remark 17.6).

Statement	Indispensable hypotheses, and the boundary of the conclusion
proposition 17.9	No Fourier modes in $\mathbf{N}_\Gamma$. No divisor estimate of any kind ; this is the one quantitative statement of the quasi-periodic part that is arithmetic-free.
theorem 18.3	(SD _{ss}) on <i>every</i> nonresonant stratum; <i>all</i> input coefficients holomorphic on <i>one</i> open strip. Without the first, a forced coefficient can fail to be a distribution; without the second, independent germs need not give a common analytic domain (convention 3.3). The loss κ_Γ is attained (corollary 18.4); no loss bound uniform in the Hahn exponent is asserted (remark 18.5).
theorem 18.7	Integer exponents only; the explicit Liouville constant (18.6). The obstruction is to assembling infinitely many modewise inverses into <i>one</i> analytic coefficient — the scalar inverses all exist. Robustness under enlarging Γ is argued from uniqueness of the modewise Hahn inverse, not from any norm invariance (warning 18.8).
theorem 19.1	Γ nontrivial; $\neg\text{QR}$; (SD _{ss}); normalization $\gamma_1 = 0$; $v(\mathcal{V}) > \kappa_\Gamma$ <i>strictly</i> — proposition 19.6 fails at equality. Existence needs finitely many labelled support words per exponent; increasing valuations alone do not produce a Hahn sum (warning 19.2). Uniqueness needs the mean-zero normalization and positive valuation of h ; constant translations remove the normalization.
theorem 19.4 and corollary 19.5	(SD _{ss}) <i>fails</i> ; $\neg\text{QR}$; Γ nontrivial. Refutes a <i>universal</i> theorem at arbitrarily high valuation; says nothing about any particular perturbation.
theorem 19.8	Coefficientwise torus mean and finite Jacobian calculus, on the conjugacy of theorem 19.1 . No measure on arbitrary subsets and no real-time ergodic theorem follows (N75).

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